

## Project Data Page:





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## **DESIGN CALCULATION**

*In Accordance with ASME Section VIII Division 1*

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL15HN001Flash vessel7%.pvdb

Date of Analysis : Sep 18, 2026 1:58pm

**PV Elite 28 U2, April 2026**

## Vessel Design Summary:

### ASME Code, Section VIII Division 1, 2025

|                                                   |               |
|---------------------------------------------------|---------------|
| Diameter Spec : 1650.000 mm ID                    |               |
| Vessel Design Length, Tangent to Tangent          | 7900.00 mm    |
| Distance of Bottom Tangent above Grade            | 2300.00 mm    |
| Distance of Base above Grade                      | 300.00 mm     |
| Specified Datum Line Distance                     | 2000.00 mm    |
| Internal Design Temperature                       | 85 °C         |
| Internal Design Pressure                          | 2.500 MPa     |
| External Design Temperature                       | 85 °C         |
| Maximum Allowable Working Pressure                | 3 MPa         |
| Shop Test Pressure                                | 3.311 MPa     |
| Required Minimum Design Metal Temperature         | -40.0 °C      |
| Warmest Computed Minimum Design Metal Temperature | -40.0 °C      |
| Wind Design Code                                  | ASCE/SEI 7-22 |
| Earthquake Design Code                            | ASCE/SEI 7-22 |

## Materials of Construction:

| Component Type | Material   | Class | Thickness | UNS #  | Material Treatment Options |
|----------------|------------|-------|-----------|--------|----------------------------|
| Shell          | SA-516 70  | ...   | ...       | K02700 | +N+IT                      |
| Head           | SA-516 70  | ...   | ...       | K02700 | +N+IT                      |
| Skirt          | SA-285 C   | ...   | ...       | K02801 |                            |
| Skirt          | SA-516 70  | ...   | ...       | K02700 |                            |
| Nozzle         | SA-333 6   | ...   | ...       | K03006 | +IT                        |
| Nozzle         | SA-350 LF1 | 1     | ...       | K03009 | +IT                        |
| Nozzle         | SA-516 70  | ...   | ...       | K02700 | +IT                        |
| Re-Pad         | SA-516 70  | ...   | ...       | K02700 | +IT                        |
| Nozzle Flg     | SA-350 LF1 | 1     | ...       | K03009 | +IT                        |
| Basering       | SA-285 C   | ...   | ...       | K02801 |                            |
| Base Bolting   | SA-36      | ...   | ...       | K02600 |                            |

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized  
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

## Element Pressures and MAWP (MPa & mm):

| Element Description or Type | Design Pressure + Stat. head | Ext. Press. | Element M.A.W.P | Total Corrosion Allowance | Str. Flg. Gov. | In Creep Range |
|-----------------------------|------------------------------|-------------|-----------------|---------------------------|----------------|----------------|
| Skirt-2                     | ...                          | ...         | ...             | 1.5000                    | N/A            | No             |
| Skirt-1                     | ...                          | ...         | ...             | 1.5000                    | N/A            | No             |
| Btm_head                    | 2.543                        | 0.00        | 3.000           | 1.5000                    | No             | No             |
| Shell-1                     | 2.540                        | 0.00        | 3.000           | 1.5000                    | N/A            | No             |
| Shell-2                     | 2.520                        | 0.00        | 3.000           | 1.5000                    | N/A            | No             |
| Shell-3                     | 2.500                        | 0.00        | 3.000           | 1.5000                    | N/A            | No             |
| Top_head                    | 2.500                        | 0.00        | 3.000           | 1.5000                    | No             | No             |

Liquid Level: 6412.50 mm Dens.: 0.001 kg/cm<sup>3</sup> Sp. Gr.: 0.680

## Element Types and Properties:

| Element Type | "To" Elev mm | Element Length mm | Nominal Thickness mm | Finished Thickness mm | Reqd Thk Internal mm | Reqd Thk External mm | Long Eff | Circ Eff |
|--------------|--------------|-------------------|----------------------|-----------------------|----------------------|----------------------|----------|----------|
|--------------|--------------|-------------------|----------------------|-----------------------|----------------------|----------------------|----------|----------|



|          |         |        |      |      |      |     |      |      |
|----------|---------|--------|------|------|------|-----|------|------|
| Skirt    | -1000.0 | 1000.0 | 14.0 | 14.0 | ...  | ... | 0.70 | 0.70 |
| Skirt    | 0.0     | 1000.0 | 14.0 | 14.0 | ...  | ... | 0.70 | 0.70 |
| Ellipse  | 50.0    | 50.0   | 20.0 | 17.0 | 16.7 | ... | 1.00 | 1.00 |
| Cylinder | 3050.0  | 3000.0 | 18.0 | 18.0 | 16.9 | ... | 1.00 | 1.00 |
| Cylinder | 6050.0  | 3000.0 | 18.0 | 18.0 | 16.8 | ... | 1.00 | 1.00 |
| Cylinder | 7850.0  | 1800.0 | 18.0 | 18.0 | 16.6 | ... | 1.00 | 1.00 |
| Ellipse  | 7900.0  | 50.0   | 20.0 | 17.0 | 16.5 | ... | 1.00 | 1.00 |

#### Local Stress Analysis Results:

| Description | Analysis Type | Max Stress Ratio | High Stress Location | Pass Fail |
|-------------|---------------|------------------|----------------------|-----------|
| D1_3"       | WRC-107/537   | 0.900            | n/a                  | Passed    |
| B2_10"      | WRC-297       | 0.942            | Shell                | Passed    |
| A3_3"       | WRC-297       | 0.942            | Shell                | Passed    |
| A5_18"      | WRC-297       | 0.935            | Shell                | Passed    |
| A2_8"       | WRC-297       | 0.928            | Shell                | Passed    |
| A6_8"       | WRC-297       | 0.928            | Shell                | Passed    |
| A1_16"      | WRC-297       | 0.928            | Shell                | Passed    |
| A4_4"       | WRC-297       | 0.928            | Shell                | Passed    |
| B1_18"      | WRC-107/537   | 0.885            | n/a                  | Passed    |
| S1_2"       | WRC-107/537   | 0.885            | n/a                  | Passed    |

#### Wind/Earthquake Shear, Bending:

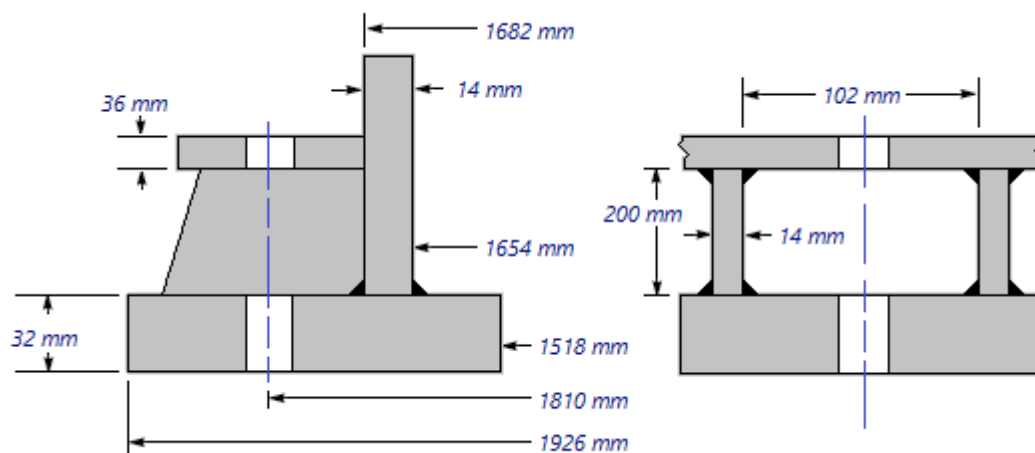
| From | To | Distance to Support mm | Cumulative Wind Shear N | Earthquake Shear N | Wind Bending N-mm | Earthquake Bending N-mm |
|------|----|------------------------|-------------------------|--------------------|-------------------|-------------------------|
| 10   | 20 | 500                    | 99462.2                 | 101445             | 719171840         | 831195456               |
| 20   | 30 | 1500                   | 96478.9                 | 101265             | 621161600         | 729799616               |
| 30   | 40 | 2025                   | 93495.5                 | 100986             | 526135872         | 628633280               |
| 40   | 50 | 3550                   | 93333.7                 | 100306             | 521463296         | 623598976               |
| 50   | 60 | 6550                   | 73785.2                 | 88893.9            | 270683328         | 339684384               |
| 60   | 70 | 8950                   | 47345.7                 | 64283.8            | 88913344          | 109824704               |
| 70   | 80 | 9875                   | 40755.8                 | 45740.5            | 9589827           | 10762714                |

Abs Max of the all of the Stress Ratio's : 0.5014

#### Basing Data : Continuous Top Ring W/Gussets

|                                |           |    |
|--------------------------------|-----------|----|
| Thickness of Basing            | 32.0000   | mm |
| Inside Diameter of Basing      | 1518.0000 | mm |
| Outside Diameter of Basing     | 1926.0000 | mm |
| Nominal Diameter of Bolts      | 36.0000   | mm |
| Diameter of Bolt Circle        | 1810.0000 | mm |
| Number of Bolts                | 12        |    |
| Thickness of Gusset Plates     | 14.0000   | mm |
| Average Width of Gusset Plates | 122.0000  | mm |
| Height of Gussets              | 200.0000  | mm |
| Distance between Gussets       | 102.0000  | mm |
| Thickness of Top Plate or Ring | 36.0000   | mm |
| Radial Width of the Top Plate  | 122.0000  | mm |
| Number of Gussets              | 24        |    |

#### Basing Sketch for Cylindrical Skirts:



**Basing Cross Section View**

**Loads for Foundation/Support Design:**

|                                   |            |      |
|-----------------------------------|------------|------|
| Total Wind Shear on Support       | 99462.     | N    |
| Total Earthquake Shear on Support | 101445.    | N    |
| Wind Moment on Support            | 777196736. | N-mm |
| Earthquake Moment on Support      | 889220352. | N-mm |
| Maximum Initial Bolt Load         | 126488.    | N    |

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

**Weights:**

|                                              |         |     |
|----------------------------------------------|---------|-----|
| Fabricated - Bare w/o removable internals    | 12340.7 | kgm |
| Shop Test - Fabricated + water ( full )      | 30501.1 | kgm |
| Shipping - Fab. + rem. intls.+ shipping app. | 28474.5 | kgm |
| Erected - Fab. + rem. intls.+ insul. (etc)   | 28994.5 | kgm |
| Empty - Fab. + intls. + details + wghts.     | 28994.5 | kgm |
| Operating - Empty + operating liquid (no ca) | 47600.4 | kgm |

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### Nozzle Calculation Summary (MPa & mm):

| Description | MAWP | Ext | MAPNC | UG-45 | [tr] | Weld Path | Areas or Stresses |
|-------------|------|-----|-------|-------|------|-----------|-------------------|
| D1_3"       | 3    | ... | ...   | OK    | 6.30 | OK        | Passed            |
| B2_10"      | 3    | ... | ...   | OK    | 9.60 | OK        | Passed            |
| A3_3"       | 3    | ... | ...   | OK    | 6.30 | OK        | Passed            |
| L2_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| L4_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| L6_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| L8_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| A5_18"      | 3    | ... | ...   | OK    | 9.83 | OK        | Passed            |
| A2_8"       | 3    | ... | ...   | OK    | 8.66 | OK        | Passed            |
| A6_8"       | 3    | ... | ...   | OK    | 8.66 | OK        | Passed            |
| M1_24"      | 3    | ... | ...   | ...   | ...  | OK        | Passed            |
| A1_16"      | 3    | ... | ...   | OK    | 9.83 | OK        | Passed            |
| L1_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| L3_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| L5_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| L7_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |
| A4_4"       | 3    | ... | ...   | OK    | 6.76 | OK        | Passed            |
| B1_18"      | 3    | ... | ...   | OK    | 9.83 | OK        | Passed            |
| S1_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | Passed            |
| S1_2"       | 3    | ... | ...   | OK    | 6.30 | OK        | No Calc[*]        |

### Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3 Nozzle : D13"  
 Minimum MAWP Shells/Flanges : 3 Element : Btmhead  
 Minimum MAPnc Shells/Flanges : 3 Element : Tophead

Computed Vessel M.A.W.P. : 3 MPa

[\*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

### Check the spatial relationship between the nozzles:

| From Node | Nozzle Description | Y Coordinate mm | Layout Angle deg | Dia. Limit mm |
|-----------|--------------------|-----------------|------------------|---------------|
| 30        | D1_3"              | 0.000           | 0.000            | 144.862       |
| 40        | B2_10"             | 450.000         | 0.000            | 499.293       |
| 40        | A3_3"              | 300.000         | 180.000          | 144.862       |
| 40        | L2_2"              | 300.000         | 65.000           | 117.328       |
| 40        | L4_2"              | 300.000         | 90.000           | 117.328       |
| 40        | L6_2"              | 300.000         | 265.000          | 117.328       |
| 40        | L8_2"              | 300.000         | 295.000          | 117.328       |
| 50        | A5_18"             | 5050.000        | 270.000          | 848.400       |
| 60        | A2_8"              | 6050.000        | 225.000          | 399.700       |
| 60        | A6_8"              | 6050.000        | 135.000          | 399.700       |
| 60        | M1_24"             | 6050.000        | 0.000            | 1145.200      |
| 60        | A1_16"             | 6500.000        | 180.000          | 746.800       |
| 60        | L1_2"              | 6105.000        | 65.000           | 117.328       |
| 60        | L3_2"              | 6100.000        | 90.000           | 117.328       |
| 60        | L5_2"              | 6100.000        | 265.000          | 117.328       |
| 60        | L7_2"              | 6100.000        | 295.000          | 117.328       |
| 60        | A4_4"              | 6800.000        | 0.000            | 187.394       |
| 70        | B1_18"             | 0.000           | 0.000            | 848.400       |
| 70        | S1_2"              | 0.000           | 0.000            | 115.328       |

The nozzle spacing is computed by the following:



$= \sqrt{l_l^2 + l_c^2}$  where

$l_l$  - Arc length along the inside vessel surface in the long. direction.

$l_c$  - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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### Nozzle Schedule:

| Description | Nominal or Actual Size | Schd or FVC Type | Flg Type | Nozzle O/Dia in | Wall Thk mm | Reinforcing Pad Diameter | Pad Thk mm | Cut Length mm | Flg Class |
|-------------|------------------------|------------------|----------|-----------------|-------------|--------------------------|------------|---------------|-----------|
| L2_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| L4_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| L6_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| L8_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| L1_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| L3_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| L5_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| L7_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 219.08        | 300       |
| S1_2"       | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 217.16        | 600       |
| D1_3"       | 3.000 in               | 160              | WNF      | 3.500           | 11.125      | 150.00                   | 18.00      | 217.66        | 300       |
| A3_3"       | 3.000 in               | 160              | WNF      | 3.500           | 11.125      | 150.00                   | 18.00      | 219.20        | 300       |
| A4_4"       | 4.000 in               | 160              | WNF      | 4.500           | 13.487      | 150.00                   | 18.00      | 219.98        | 600       |
| A2_8"       | 8.000 in               | 80               | WNF      | 8.625           | 12.700      | 350.00                   | 18.00      | 225.30        | 300       |
| A6_8"       | 8.000 in               | 80               | WNF      | 8.625           | 12.700      | 350.00                   | 18.00      | 225.30        | 300       |
| B2_10"      | 10.000 in              | 80               | WNF      | 10.750          | 15.088      | 440.00                   | 18.00      | 229.37        | 300       |
| A1_16"      | 16.000 in              | Actual           | WNF      | 16.000          | 18.000      | 630.00                   | 18.00      | 293.42        | 300       |
| A5_18"      | 18.000 in              | Actual           | WNF      | 18.000          | 18.000      | 817.20                   | 18.00      | 300.30        | 300       |
| B1_18"      | 18.000 in              | Actual           | WNF      | 18.000          | 18.000      | 690.00                   | 18.00      | 284.54        | 300       |
| M1_24"      | 24.000 in              | Actual           | WNF      | 24.000          | 20.000      | 1000.00                  | 18.00      | 376.37        | 300       |

### General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

### Nozzle Material and Weld Fillet Leg Size Details (mm):

| Description | Material   | Shl Grve Weld | Noz Shl/Pad Weld | Pad OD Weld | Pad Grve Weld | Inside Weld |
|-------------|------------|---------------|------------------|-------------|---------------|-------------|
| L2_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| L4_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| L6_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| L8_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| L1_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| L3_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| L5_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| L7_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| S1_2"       | SA-350 LF1 | 17.000        | 10.000           | ...         | ...           | ...         |
| D1_3"       | SA-333 6   | 17.000        | 10.000           | 14.000      | 18.000        | ...         |
| A3_3"       | SA-333 6   | 18.000        | 10.000           | 14.000      | 18.000        | ...         |
| A4_4"       | SA-333 6   | 18.000        | 10.000           | 14.000      | 18.000        | ...         |
| A2_8"       | SA-333 6   | 18.000        | 10.000           | 14.000      | 18.000        | ...         |
| A6_8"       | SA-333 6   | 18.000        | 10.000           | 14.000      | 18.000        | ...         |
| B2_10"      | SA-333 6   | 18.000        | 10.000           | 14.000      | 18.000        | ...         |
| A1_16"      | SA-516 70  | 17.000        | 10.000           | 14.000      | 18.000        | ...         |
| A5_18"      | SA-516 70  | 18.000        | 10.000           | 14.000      | 18.000        | ...         |
| B1_18"      | SA-516 70  | 17.000        | 10.000           | 14.000      | 18.000        | ...         |
| M1_24"      | SA-516 70  | 18.000        | 10.000           | 14.000      | 18.000        | ...         |

Note: The Outside projections below do not include the flange thickness.

### Nozzle Miscellaneous Data:

| Description | Elev/Distance From Datum mm | Layout Angle deg | Proj Outside mm | Proj Inside mm | Installed in Component |
|-------------|-----------------------------|------------------|-----------------|----------------|------------------------|
|-------------|-----------------------------|------------------|-----------------|----------------|------------------------|



|        |          |       |        |      |          |
|--------|----------|-------|--------|------|----------|
| L2_2"  | 300.000  | 65.0  | 200.00 | 0.00 | Shell-1  |
| L4_2"  | 300.000  | 90.0  | 200.00 | 0.00 | Shell-1  |
| L6_2"  | 300.000  | 265.0 | 200.00 | 0.00 | Shell-1  |
| L8_2"  | 300.000  | 295.0 | 200.00 | 0.00 | Shell-1  |
| L1_2"  | 6105.000 | 65.0  | 200.00 | 0.00 | Shell-3  |
| L3_2"  | 6100.000 | 90.0  | 200.00 | 0.00 | Shell-3  |
| L5_2"  | 6100.000 | 265.0 | 200.00 | 0.00 | Shell-3  |
| L7_2"  | 6100.000 | 295.0 | 200.00 | 0.00 | Shell-3  |
| S1_2"  | ...      | 0.0   | 179.43 | 0.00 | Top_head |
| D1_3"  | ...      | 0.0   | 200.00 | 0.00 | Btm_head |
| A3_3"  | 300.000  | 180.0 | 200.00 | 0.00 | Shell-1  |
| A4_4"  | 6800.000 | 0.0   | 200.00 | 0.00 | Shell-3  |
| A2_8"  | 6050.000 | 225.0 | 200.00 | 0.00 | Shell-3  |
| A6_8"  | 6050.000 | 135.0 | 200.00 | 0.00 | Shell-3  |
| B2_10" | 450.000  | 0.0   | 200.00 | 0.00 | Shell-1  |
| A1_16" | 6500.000 | 180.0 | 250.00 | 0.00 | Shell-3  |
| A5_18" | 5050.000 | 270.0 | 250.00 | 0.00 | Shell-2  |
| B1_18" | ...      | 0.0   | 250.00 | 0.00 | Top_head |
| M1_24" | 6050.000 | 0.0   | 300.00 | 0.00 | Shell-3  |

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**Bill of Materials:**

| QTY | DESCRIPTION                                                  | MATERIAL   |
|-----|--------------------------------------------------------------|------------|
| 2   | ELLIPTICAL HEAD: 2.0 X 1, 20.0mm THK X 1650.0mm ID X 50.0mm  | SA-516 70  |
| 2   | INSULATION: 462mm X 70mm THK                                 | CC         |
| 1   | PLATFORM: 1100mm X 1600mm WIDE                               |            |
| 1   | LADDER: 3000.0mm                                             | ...        |
| 2   | INSULATION: 3000mm X 70mm THK                                | CC         |
| 1   | PLATFORM: 1100mm X 1500mm WIDE                               |            |
| 2   | LADDER: 4000.0mm                                             | ...        |
| 1   | INSULATION: 1800mm X 70mm THK                                | CC         |
| 1   | PLATFORM: 1100mm X 7684mm WIDE                               |            |
| 1   | CLASS 300 GR 1.4, 24.0" BLIND FLANGE(S)                      | SA-350 LF1 |
| 1   | BASERING: 1926.0mm OD X 1518.0 ID X 32.0mm                   | SA-285 C   |
| 12  | BASERING BOLTS: 36.0mm DIA                                   | SA-36      |
| 12  | BASERING BOLT NUTS: 36.0mm DIA                               | ...        |
| 1   | BASERING TOP RING: 1926.0mm OD X 1682.0 ID X 36.0mm          | ...        |
| 24  | BASERING GUSSET PLATES: 122.0mm X 122.0mm X 200.0mm X 14.0mm | ...        |
| 1   | CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2997.2mm X 1000.0mm   | SA-285 C   |
| 1   | CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2198.9mm X 1000.0mm   | SA-285 C   |
| 1   | CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3055.3mm X 1000.0mm   | SA-516 70  |
| 1   | CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2140.8mm X 1000.0mm   | SA-516 70  |
| 2   | CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 3000.0mm         | SA-516 70  |
| 2   | CYLINDRICAL SEGMENT 18.0mm THK X 2135.6mm X 3000.0mm         | SA-516 70  |
| 1   | CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 1800.0mm         | SA-516 70  |
| 1   | CYLINDRICAL SEGMENT 18.0mm THK X 2135.6mm X 1800.0mm         | SA-516 70  |
| 1   | NAMEPLATE                                                    | ...        |



### Minimum Design Metal Temperature Results Summary (°C):

| Description       | Notes | Curve | Basic MDMT | Reduced MDMT | UG-20(f) MDMT | Thickness ratio | Gov Thk mm | E*   | PWHT reqd |
|-------------------|-------|-------|------------|--------------|---------------|-----------------|------------|------|-----------|
| Btmhead           | [10]  | !     | -40        | -40          |               | 1.000           | 17.000     | 1.00 | No        |
| Btmhead           | [7]   | !     | -40        | -48          |               | 0.848           | 20.000     | 1.00 | No        |
| Shell-1           | [8]   | !     | -40        | -43          |               | 0.949           | 18.000     | 1.00 | No        |
| Shell-2           | [8]   | !     | -40        | -43          |               | 0.942           | 18.000     | 1.00 | No        |
| Shell-3           | [8]   | !     | -40        | -44          |               | 0.935           | 18.000     | 1.00 | No        |
| Tophead           | [10]  | !     | -40        | -41          |               | 0.983           | 17.000     | 1.00 | No        |
| Tophead           | [7]   | !     | -40        | -49          |               | 0.834           | 20.000     | 1.00 | No        |
| D13"              | [1]   | !     | -40        | -46          |               | 0.901           | 17.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -60          |               |                 |            |      |           |
| B210"             | [1]   | !     | -40        | -43          |               | 0.948           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -60          |               |                 |            |      |           |
| A33"              | [1]   | !     | -40        | -43          |               | 0.949           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -60          |               |                 |            |      |           |
| L22"              | [1]   | !     | -29        | -104         | -29           | 0.039           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| L42"              | [1]   | !     | -29        | -104         | -29           | 0.039           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| L62"              | [1]   | !     | -29        | -104         | -29           | 0.039           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| L82"              | [1]   | !     | -29        | -104         | -29           | 0.039           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| A518"             | [1]   | !     | -40        | -43          |               | 0.937           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -51          |               |                 |            |      |           |
| A28"              | [1]   | !     | -40        | -44          |               | 0.935           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -52          |               |                 |            |      |           |
| A68"              | [1]   | !     | -40        | -44          |               | 0.935           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -52          |               |                 |            |      |           |
| M124"             | [1]   | !     | -40        | -44          |               | 0.935           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -52          |               |                 |            |      |           |
| A116"             | [1]   | !     | -40        | -44          |               | 0.935           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -52          |               |                 |            |      |           |
| L12"              | [1]   | !     | -29        | -104         | -29           | 0.038           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| L32"              | [1]   | !     | -29        | -104         | -29           | 0.038           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| L52"              | [1]   | !     | -29        | -104         | -29           | 0.038           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| L72"              | [1]   | !     | -29        | -104         | -29           | 0.038           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| A44"              | [1]   | !     | -40        | -44          |               | 0.935           | 18.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -104         |               |                 |            |      |           |
| B118"             | [1]   | !     | -40        | -46          |               | 0.886           | 17.000     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -29        | -52          |               |                 |            |      |           |
| S12"              | [1]   | !     | -29        | -104         | -29           | 0.038           | 16.764     | 1.00 | No        |
| Nozzle Flg        | [5]   | !     | -40        | -40          |               |                 |            |      |           |
| Nozl. Bolting[22] |       |       | -48        |              |               | 1.000           | 64.000     | 1.00 | No        |

Warmest MDMT: -29 -40

Required Minimum Design Metal Temperature -40.0 °C  
 Warmest Computed Minimum Design Metal Temperature -40.0 °C

#### Notes:

- [ ! ] - This was an impact tested material.
- [ 1 ] - Governing Nozzle Weld.
- [ 4 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [ 5 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [ 6 ] - MDMT Calculations at the Shell/Head Joint.
- [ 7 ] - MDMT Calculations for the Straight Flange.
- [ 8 ] - Cylinder/Cone/Flange Junction MDMT.
- [ 9 ] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.



- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.  
UCS-66(g) was not considered.  
UCS-66(i) was not considered.

**Notes:**

Impact test temps were not entered in and not considered in the analysis.  
UCS-66(i) applies to impact tested materials not by specification and  
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).  
The Basic MDMT includes the (30F) PWHT credit if applicable.

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#### Class From To : Basic Element Checks.

|      |    |    |                                           |
|------|----|----|-------------------------------------------|
| Note | 20 | 40 | There is a high jump in the Joint Eff.    |
| Note | 10 | 20 | The wind diameter multiplier is very big. |
| Note | 20 | 30 | The wind diameter multiplier is very big. |
| Note | 30 | 40 | The wind diameter multiplier is very big. |
| Note | 40 | 50 | The wind diameter multiplier is very big. |
| Note | 50 | 60 | The wind diameter multiplier is very big. |
| Note | 60 | 70 | The wind diameter multiplier is very big. |
| Note | 70 | 80 | The wind diameter multiplier is very big. |

#### Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

**PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.**

#### Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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### Units used in this Analysis (SIASME):

| Name           | System Unit | Constant    | User Unit          |
|----------------|-------------|-------------|--------------------|
| Length         | Feet        | 304.8000    | mm                 |
| Force          | Pounds      | 4.4480      | N                  |
| Mass           | Pounds      | 0.4536      | kgm                |
| Area           | sq. inches  | 645.1600    | mm <sup>2</sup>    |
| Moment         | ft. lbs.    | 1356.3000   | N-mm               |
| Stress         | lbs./sq.in. | 0.0069      | MPa                |
| Temperature    | Degrees F   | 0.5556      | °C                 |
| Pressure       | psig        | 0.0069      | MPa                |
| Elast. Modulus | lbs./sq.in. | 0.0069      | MPa                |
| Pipe Density   | lbs./cu.in. | 0.0277      | kg/cm <sup>3</sup> |
| Ins. Density   | lbs./cu.ft. | 0.1600E-04  | kg/cm <sup>3</sup> |
| Fluid Density  | lbs./cu.ft. | 0.1600E-04  | kg/cm <sup>3</sup> |
| Wind Speed     | miles/hr    | 1.6093      | km/hr              |
| Tray Weight    | lbs./sq.ft. | 0.0005      | kg/cm <sup>2</sup> |
| Inertia        | in.4        | 416231.0000 | mm <sup>4</sup>    |
| G Load         | G's         | 1.0000      | g's                |
| Wind Load      | lbs./sq.ft. | 0.0479      | kPa                |
| Elevation      | Feet        | 304.8000    | mm                 |
| Volume         | in.3        | 0.0164      | ltr                |
| Diameter       | inches      | 25.4000     | mm                 |
| Thickness      | inches      | 25.4000     | mm                 |

### PV Elite Vessel Analysis Program: Input Data

|                                          |        |     |
|------------------------------------------|--------|-----|
| Design Internal Pressure (for Hydrotest) | 2.5    | MPa |
| Design Internal Temperature              | 85.0   | °C  |
| Projection of Nozzle from Vessel Top     | 0      | mm  |
| Projection of Nozzle from Vessel Bottom  | 0      | mm  |
| Minimum Design Metal Temperature         | -40.0  | °C  |
| Type of Construction                     | Welded |     |
| Special Service                          | None   |     |
| Degree of Radiography                    | RT-1   |     |
| Use Higher Longitudinal Stresses (Flag)  | Y      |     |
| Select t for Internal Pressure (Flag)    | N      |     |
| Select t for External Pressure (Flag)    | N      |     |
| Select t for Axial Stress (Flag)         | N      |     |
| Select Location for Stiff. Rings (Flag)  | N      |     |
| Consider Vortex Shedding                 | N      |     |

### Shop Pressure Test:

|                                      |            |
|--------------------------------------|------------|
| Type of Pressure Test                | UG-99(b)   |
| Pressure Test Position               | Horizontal |
| Test Performed in Corroded Condition | No         |

|              |                |
|--------------|----------------|
| Load Case 1  | NP+EW+WI+FW+BW |
| Load Case 2  | NP+EW+EE+FS+BS |
| Load Case 3  | NP+OW+WI+FW+BW |
| Load Case 4  | NP+OW+EQ+FS+BS |
| Load Case 5  | NP+HW+HI       |
| Load Case 6  | NP+HW+HE       |
| Load Case 7  | IP+OW+WI+FW+BW |
| Load Case 8  | IP+OW+EQ+FS+BS |
| Load Case 9  | EP+OW+WI+FW+BW |
| Load Case 10 | EP+OW+EQ+FS+BS |
| Load Case 11 | HP+HW+HI       |
| Load Case 12 | HP+HW+HE       |
| Load Case 13 | IP+WE+EW       |
| Load Case 14 | IP+WF+CW       |
| Load Case 15 | IP+VO+OW       |
| Load Case 16 | IP+VE+EW       |
| Load Case 17 | NP+VO+OW       |
| Load Case 18 | FS+BS+IP+OW    |
| Load Case 19 | FS+BS+EP+OW    |
| Load Case 20 | BL+IP+OW       |



|                                         |                       |       |
|-----------------------------------------|-----------------------|-------|
| Wind Design Code                        | ASCE/SEI 7-22         |       |
| Wind Load Reduction Scale Factor        | 1.100                 |       |
| Basic Wind Speed                        | [V] 180               | km/hr |
| Surface Roughness Category              | D: Flat, unobstructed |       |
| Type of Surface                         | 0                     |       |
| Base Elevation                          | 300                   | mm    |
| Percent Wind for Hydrotest              | 33.0                  |       |
| Using User defined Wind Press. Vs Elev. | N                     |       |
| Height of Hill or Escarpment            | H or Hh 0             | mm    |
| Distance Upwind of Crest                | Lh 0                  | mm    |
| Distance from Crest to the Vessel       | x 0                   | mm    |
| Type of Terrain ( Hill, Escarpment )    | Flat                  |       |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100                |       |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000                |       |
| Damping Factor (Beta) for Wind (Filled) | 0.0000                |       |
| Site Elevation above Sea Level          | 0                     | mm    |

|                                  |                |     |
|----------------------------------|----------------|-----|
| Seismic Design Code              | ASCE/SEI 7-22  |     |
| Importance Factor                | 1.500          |     |
| Force Modification Factor R      | 1.500          |     |
| Site Class                       | D              |     |
| Component Elevation Ratio        | z/h 0.000      |     |
| Amplification Factor             | Ap 2.500       |     |
| Force Factor                     | 0.000          |     |
| Consider Vertical Acceleration   | Yes            |     |
| Minimum Acceleration Multiplier  | 0.000          |     |
| User Value of Sds (used if > 0 ) | 0.000          |     |
| User Value of Sd1 (used if > 0 ) | 0.000          |     |
| Component resonance duct. fact.  | [CAR] 0.000    |     |
| Component strength factor        | [Rpo] 0.000    |     |
| Overstrength factor              | [Omega0] 0.000 |     |
| Response Acceleration Factor     | [SMS] 0.326    |     |
| Response Acceleration Factor     | [SM1] 0.139    |     |
| Building Importance Factor       | [Ie] 1.000     |     |
| Site Latitude                    | 0.000          | deg |
| Site Longitude                   | 0.000          | deg |

|                                          |   |
|------------------------------------------|---|
| M.A.W.P + Static Head to Nozzle          | Y |
| Consider MAP New and Cold in Noz. Design | N |
| Consider External Loads for Nozzle Des.  | Y |
| Use ASME VIII-1 Appendix 1-9             | N |

|                             |                                |
|-----------------------------|--------------------------------|
| Perform Blast Load Analysis | No                             |
| Material Database Year      | Current w/Addenda or Code Year |

#### Configuration Directives:

|                                                          |     |
|----------------------------------------------------------|-----|
| g0 is taken as min( g0, attached shell thk )             | Yes |
| Do not use Nozzle MDMT Interpretation VIII-1 01-37       | No  |
| Use Table G instead of exact equation for "A"            | Yes |
| Shell Head Joints are Tapered                            | Yes |
| Compute "K" in corroded condition                        | Yes |
| Use Code Case 2286                                       | No  |
| Use the MAWP to compute the MDMT                         | Yes |
| For thickness ratios <= 0.35, MDMT will be -155F (-104C) | Yes |
| For PWHT & P1 Materials the MDMT can be < -55F (-48C)    | No  |
| Using Metric Material Databases, ASME II D               | Yes |
| Calculate B31.3 type stress for Nozzles with Loads       | Yes |
| Reduce the MDMT due to lower membrane stress             | Yes |
| Consider Longitudinal Stress in MDMT Calculations        | Yes |

#### Complete Listing of Vessel Elements and Details:

|                   |               |
|-------------------|---------------|
| Element#          | 1/7           |
| Element From Node | 10            |
| Element To Node   | 20            |
| Element Type      | Skirt Support |
| Description       | Skirt-2       |



|                                         |          |                    |
|-----------------------------------------|----------|--------------------|
| Distance "FROM" to "TO"                 | 1000     | mm                 |
| Skirt Outside Diameter                  | 1682     | mm                 |
| Diameter of Skirt at Base               | 1682     | mm                 |
| Skirt Thickness                         | 14       | mm                 |
| Internal Corrosion Allowance            | 1.5      | mm                 |
| Nominal Thickness                       | 14       | mm                 |
| External Corrosion Allowance            | 0        | mm                 |
| Design Temperature Internal Pressure    | 85       | °C                 |
| Design Temperature External Pressure    | 85       | °C                 |
| Effective Diameter Multiplier           | 1.53     |                    |
| Material Name                           | SA-285 C |                    |
| Allowable Stress, Ambient               | 108      | MPa                |
| Allowable Stress, Operating             | 108      | MPa                |
| Allowable Stress, Hydrotest             | 184.5    | MPa                |
| Material Density                        | 0.00775  | kg/cm <sup>3</sup> |
| P Number Thickness                      | 32.004   | mm                 |
| Yield Stress, Operating                 | 191.57   | MPa                |
| UCS-66 Chart Curve Designation          | A        |                    |
| External Pressure Chart Name            | CS-2     |                    |
| UNS Number                              | K02801   |                    |
| Product Form                            | Plate    |                    |
| Efficiency, Longitudinal Seam           | 0.7      |                    |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7      |                    |
| Weld is pre-Heated                      | No       |                    |

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|                                         |               |                    |
|-----------------------------------------|---------------|--------------------|
| Element#                                | 2/7           |                    |
| Element From Node                       | 20            |                    |
| Element To Node                         | 30            |                    |
| Element Type                            | Skirt Support |                    |
| Description                             | Skirt-1       |                    |
| Distance "FROM" to "TO"                 | 1000          | mm                 |
| Skirt Outside Diameter                  | 1682          | mm                 |
| Diameter of Skirt at Base               | 1682          | mm                 |
| Skirt Thickness                         | 14            | mm                 |
| Internal Corrosion Allowance            | 1.5           | mm                 |
| Nominal Thickness                       | 14            | mm                 |
| External Corrosion Allowance            | 0             | mm                 |
| Design Temperature Internal Pressure    | 85            | °C                 |
| Design Temperature External Pressure    | 85            | °C                 |
| Effective Diameter Multiplier           | 1.53          |                    |
| Material Name                           | SA-516 70     |                    |
| Allowable Stress, Ambient               | 138           | MPa                |
| Allowable Stress, Operating             | 138           | MPa                |
| Allowable Stress, Hydrotest             | 234           | MPa                |
| Material Density                        | 0.00775       | kg/cm <sup>3</sup> |
| P Number Thickness                      | 32.004        | mm                 |
| Yield Stress, Operating                 | 242           | MPa                |
| UCS-66 Chart Curve Designation          | B             |                    |
| External Pressure Chart Name            | CS-2          |                    |
| UNS Number                              | K02700        |                    |
| Product Form                            | Plate         |                    |
| Efficiency, Longitudinal Seam           | 0.7           |                    |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7           |                    |
| Weld is pre-Heated                      | No            |                    |

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|                              |            |    |
|------------------------------|------------|----|
| Element#                     | 3/7        |    |
| Element From Node            | 30         |    |
| Element To Node              | 40         |    |
| Element Type                 | Elliptical |    |
| Description                  | Btmhead    |    |
| Distance "FROM" to "TO"      | 50         | mm |
| Inside Diameter              | 1650       | mm |
| Element Thickness            | 17         | mm |
| Internal Corrosion Allowance | 1.5        | mm |
| Nominal Thickness            | 20         | mm |
| External Corrosion Allowance | 0          | mm |



|                                      |           |       |
|--------------------------------------|-----------|-------|
| Design Internal Pressure             | 2.5       | MPa   |
| Design Temperature Internal Pressure | 85        | °C    |
| Design External Pressure             | 0         | MPa   |
| Design Temperature External Pressure | 85        | °C    |
| Effective Diameter Multiplier        | 1.53      |       |
| Material Name                        | SA-516 70 | +N+IT |
| Efficiency, Longitudinal Seam        | 1.0       |       |
| Efficiency, Circumferential Seam     | 1.0       |       |
| Elliptical Head Factor               | 2.0       |       |
| Weld is pre-Heated                   | No        |       |

|                                      |            |        |
|--------------------------------------|------------|--------|
| Element From Node                    | 30         |        |
| Detail Type                          | Liquid     |        |
| Detail ID                            | Liquid: 10 |        |
| Dist. from "FROM" Node / Offset dist | -412.5     | mm     |
| Height/Length of Liquid              | 462.5      | mm     |
| Liquid Density                       | 0.0006789  | kg/cm³ |

|                                      |             |        |
|--------------------------------------|-------------|--------|
| Element From Node                    | 30          |        |
| Detail Type                          | Insulation  |        |
| Detail ID                            | Ins: 30     |        |
| Dist. from "FROM" Node / Offset dist | -412.5      | mm     |
| Height/Length of Insulation          | 462.5       | mm     |
| Thickness of Insulation              | 70          | mm     |
| Density                              | 0.40000E-04 | kg/cm³ |

|                                      |          |     |
|--------------------------------------|----------|-----|
| Element From Node                    | 30       |     |
| Detail Type                          | Nozzle   |     |
| Detail ID                            | D13"     |     |
| Dist. from "FROM" Node / Offset dist | 0        | mm  |
| Nozzle Diameter                      | 3        | in. |
| Nozzle Schedule                      | 160      |     |
| Nozzle Class                         | 300      |     |
| Layout Angle                         | 0.0      |     |
| Blind Flange (Y/N)                   | N        |     |
| Weight of Nozzle ( Used if > 0 )     | 146.33   | N   |
| Grade of Attached Flange             | GR 1.1   |     |
| Nozzle Matl                          | SA-333 6 | +IT |

|                                      |           |       |
|--------------------------------------|-----------|-------|
| Element#                             | 4/7       |       |
| Element From Node                    | 40        |       |
| Element To Node                      | 50        |       |
| Element Type                         | Cylinder  |       |
| Description                          | Shell-1   |       |
| Distance "FROM" to "TO"              | 3000      | mm    |
| Inside Diameter                      | 1650      | mm    |
| Element Thickness                    | 18        | mm    |
| Internal Corrosion Allowance         | 1.5       | mm    |
| Nominal Thickness                    | 18        | mm    |
| External Corrosion Allowance         | 0         | mm    |
| Design Internal Pressure             | 2.5       | MPa   |
| Design Temperature Internal Pressure | 85        | °C    |
| Design External Pressure             | 0         | MPa   |
| Design Temperature External Pressure | 85        | °C    |
| Effective Diameter Multiplier        | 1.53      |       |
| Material Name                        | SA-516 70 | +N+IT |
| Efficiency, Longitudinal Seam        | 1.0       |       |
| Efficiency, Circumferential Seam     | 1.0       |       |
| Weld is pre-Heated                   | No        |       |

|                                      |               |     |
|--------------------------------------|---------------|-----|
| Element From Node                    | 40            |     |
| Detail Type                          | Platform      |     |
| Detail ID                            | PLAT:[1 OF 1] |     |
| Dist. from "FROM" Node / Offset dist | 200           | mm  |
| Platform Start Angle (degrees)       | 0.0           |     |
| Platform End Angle (degrees)         | 360.0         |     |
| Platform Wind Area                   | 8503040       | mm² |
| Platform Weight                      | 30779         | N   |



|                                     |       |                    |
|-------------------------------------|-------|--------------------|
| Platform Railing Weight             | 0     | N/mm               |
| Platform Grating Weight             | 0.017 | kg/cm <sup>2</sup> |
| Platform Width                      | 1600  | mm                 |
| Platform Height                     | 1100  | mm                 |
| Platform Clearance or End Offset    | 105   | mm                 |
| Platform Force Coefficient          | 2.0   |                    |
| Ladder Layout Angle                 | 0.0   |                    |
| Ladder Start Elevation              | 0     | mm                 |
| Ladder End Elevation                | 3000  | mm                 |
| Unit Weight of Ladder               | 0.49  | N/mm               |
| Platform Length (top head platform) | 0     | mm                 |

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Element From Node                    | 40         |                    |
| Detail Type                          | Liquid     |                    |
| Detail ID                            | Liquid: 20 |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Height/Length of Liquid              | 3000       | mm                 |
| Liquid Density                       | 0.0006789  | kg/cm <sup>3</sup> |

|                                      |             |                    |
|--------------------------------------|-------------|--------------------|
| Element From Node                    | 40          |                    |
| Detail Type                          | Insulation  |                    |
| Detail ID                            | Ins: 40     |                    |
| Dist. from "FROM" Node / Offset dist | 0           | mm                 |
| Height/Length of Insulation          | 3000        | mm                 |
| Thickness of Insulation              | 70          | mm                 |
| Density                              | 0.40000E-04 | kg/cm <sup>3</sup> |

|                                      |              |     |
|--------------------------------------|--------------|-----|
| Element From Node                    | 40           |     |
| Detail Type                          | Nozzle       |     |
| Detail ID                            | B210"        |     |
| Dist. from "FROM" Node / Offset dist | 400          | mm  |
| Nozzle Diameter                      | 10           | in. |
| Nozzle Schedule                      | 80           |     |
| Nozzle Class                         | 300          |     |
| Layout Angle                         | 0.0          |     |
| Blind Flange (Y/N)                   | N            |     |
| Weight of Nozzle ( Used if > 0 )     | 771.33       | N   |
| Grade of Attached Flange             | GR 1.1       |     |
| Nozzle Matl                          | SA-333 6 +IT |     |

|                                      |              |     |
|--------------------------------------|--------------|-----|
| Element From Node                    | 40           |     |
| Detail Type                          | Nozzle       |     |
| Detail ID                            | A33"         |     |
| Dist. from "FROM" Node / Offset dist | 250          | mm  |
| Nozzle Diameter                      | 3            | in. |
| Nozzle Schedule                      | 160          |     |
| Nozzle Class                         | 300          |     |
| Layout Angle                         | 180.0        |     |
| Blind Flange (Y/N)                   | N            |     |
| Weight of Nozzle ( Used if > 0 )     | 135.15       | N   |
| Grade of Attached Flange             | GR 1.1       |     |
| Nozzle Matl                          | SA-333 6 +IT |     |

|                                      |                |     |
|--------------------------------------|----------------|-----|
| Element From Node                    | 40             |     |
| Detail Type                          | Nozzle         |     |
| Detail ID                            | L22"           |     |
| Dist. from "FROM" Node / Offset dist | 250            | mm  |
| Nozzle Diameter                      | 2              | in. |
| Nozzle Schedule                      | None           |     |
| Nozzle Class                         | 300            |     |
| Layout Angle                         | 65.0           |     |
| Blind Flange (Y/N)                   | N              |     |
| Weight of Nozzle ( Used if > 0 )     | 100.91         | N   |
| Grade of Attached Flange             | GR 1.4         |     |
| Nozzle Matl                          | SA-350 LF1 +IT |     |

|                                      |        |    |
|--------------------------------------|--------|----|
| Element From Node                    | 40     |    |
| Detail Type                          | Nozzle |    |
| Detail ID                            | L42"   |    |
| Dist. from "FROM" Node / Offset dist | 250    | mm |



Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 90.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 100.91 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-350 LF1 +IT

Element From Node 40  
 Detail Type Nozzle  
 Detail ID L62"  
 Dist. from "FROM" Node / Offset dist 250 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 265.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 100.91 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-350 LF1 +IT

Element From Node 40  
 Detail Type Nozzle  
 Detail ID L82"  
 Dist. from "FROM" Node / Offset dist 250 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 295.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 100.91 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-350 LF1 +IT

Element From Node 40  
 Detail Type Weight  
 Detail ID B2Pipewt  
 Dist. from "FROM" Node / Offset dist 400 mm  
 Miscellaneous Weight 2520.1 N  
 Offset from Element Centerline 2110 mm

Element# 5/7  
 Element From Node 50  
 Element To Node 60  
 Element Type Cylinder  
 Description Shell-2  
 Distance "FROM" to "TO" 3000 mm  
 Inside Diameter 1650 mm  
 Element Thickness 18 mm  
 Internal Corrosion Allowance 1.5 mm  
 Nominal Thickness 18 mm  
 External Corrosion Allowance 0 mm  
 Design Internal Pressure 2.5 MPa  
 Design Temperature Internal Pressure 85 °C  
 Design External Pressure 0 MPa  
 Design Temperature External Pressure 85 °C  
 Effective Diameter Multiplier 1.53  
 Material Name SA-516 70 +N+IT  
 Efficiency, Longitudinal Seam 1.0  
 Efficiency, Circumferential Seam 1.0  
 Weld is pre-Heated No

Element From Node 50  
 Detail Type Platform  
 Detail ID PLAT:[1 OF 1]  
 Dist. from "FROM" Node / Offset dist 1500 mm  
 Platform Start Angle (degrees) 180.0



|                                     |         |                    |
|-------------------------------------|---------|--------------------|
| Platform End Angle (degrees)        | 0.0     |                    |
| Platform Wind Area                  | 8550720 | mm <sup>2</sup>    |
| Platform Weight                     | 15306   | N                  |
| Platform Railing Weight             | 0       | N/mm               |
| Platform Grating Weight             | 0.017   | kg/cm <sup>2</sup> |
| Platform Width                      | 1500    | mm                 |
| Platform Height                     | 1100    | mm                 |
| Platform Clearance or End Offset    | 105     | mm                 |
| Platform Force Coefficient          | 2.0     |                    |
| Ladder Layout Angle                 | 180.0   |                    |
| Ladder Start Elevation              | 0       | mm                 |
| Ladder End Elevation                | 4000    | mm                 |
| Unit Weight of Ladder               | 0.49    | N/mm               |
| Platform Length (top head platform) | 0       | mm                 |

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Element From Node                    | 50         |                    |
| Detail Type                          | Liquid     |                    |
| Detail ID                            | Liquid: 30 |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Height/Length of Liquid              | 2950       | mm                 |
| Liquid Density                       | 0.0006789  | kg/cm <sup>3</sup> |

|                                      |             |                    |
|--------------------------------------|-------------|--------------------|
| Element From Node                    | 50          |                    |
| Detail Type                          | Insulation  |                    |
| Detail ID                            | Ins: 50     |                    |
| Dist. from "FROM" Node / Offset dist | 0           | mm                 |
| Height/Length of Insulation          | 3000        | mm                 |
| Thickness of Insulation              | 70          | mm                 |
| Density                              | 0.40000E-04 | kg/cm <sup>3</sup> |

|                                      |               |     |
|--------------------------------------|---------------|-----|
| Element From Node                    | 50            |     |
| Detail Type                          | Nozzle        |     |
| Detail ID                            | A518"         |     |
| Dist. from "FROM" Node / Offset dist | 2000          | mm  |
| Nozzle Diameter                      | 18            | in. |
| Nozzle Schedule                      | None          |     |
| Nozzle Class                         | 300           |     |
| Layout Angle                         | 270.0         |     |
| Blind Flange (Y/N)                   | N             |     |
| Weight of Nozzle ( Used if > 0 )     | 2349.8        | N   |
| Grade of Attached Flange             | GR 1.4        |     |
| Nozzle Matl                          | SA-516 70 +IT |     |

|                                      |          |    |
|--------------------------------------|----------|----|
| Element From Node                    | 50       |    |
| Detail Type                          | Weight   |    |
| Detail ID                            | A5pipewt |    |
| Dist. from "FROM" Node / Offset dist | 2000     | mm |
| Miscellaneous Weight                 | 26221    | N  |
| Offset from Element Centerline       | 2212     | mm |

|                                      |            |    |
|--------------------------------------|------------|----|
| Element From Node                    | 50         |    |
| Detail Type                          | Weight     |    |
| Detail ID                            | Clipwt     |    |
| Dist. from "FROM" Node / Offset dist | -0.0002896 | mm |
| Miscellaneous Weight                 | 9855       | N  |
| Offset from Element Centerline       | 0          | mm |

---

|                              |          |    |
|------------------------------|----------|----|
| Element#                     | 6/7      |    |
| Element From Node            | 60       |    |
| Element To Node              | 70       |    |
| Element Type                 | Cylinder |    |
| Description                  | Shell-3  |    |
| Distance "FROM" to "TO"      | 1800     | mm |
| Inside Diameter              | 1650     | mm |
| Element Thickness            | 18       | mm |
| Internal Corrosion Allowance | 1.5      | mm |
| Nominal Thickness            | 18       | mm |
| External Corrosion Allowance | 0        | mm |



|                                      |             |        |
|--------------------------------------|-------------|--------|
| Design Internal Pressure             | 2.5         | MPa    |
| Design Temperature Internal Pressure | 85          | °C     |
| Design External Pressure             | 0           | MPa    |
| Design Temperature External Pressure | 85          | °C     |
| Effective Diameter Multiplier        | 1.53        |        |
| Material Name                        | SA-516 70   | +N+IT  |
| Efficiency, Longitudinal Seam        | 1.0         |        |
| Efficiency, Circumferential Seam     | 1.0         |        |
| Weld is pre-Heated                   | No          |        |
|                                      |             |        |
| Element From Node                    | 60          |        |
| Detail Type                          | Insulation  |        |
| Detail ID                            | Ins: 60     |        |
| Dist. from "FROM" Node / Offset dist | 0           | mm     |
| Height/Length of Insulation          | 1800        | mm     |
| Thickness of Insulation              | 70          | mm     |
| Density                              | 0.40000E-04 | kg/cm³ |
|                                      |             |        |
| Element From Node                    | 60          |        |
| Detail Type                          | Nozzle      |        |
| Detail ID                            | A28"        |        |
| Dist. from "FROM" Node / Offset dist | 0.0005822   | mm     |
| Nozzle Diameter                      | 8           | in.    |
| Nozzle Schedule                      | 80          |        |
| Nozzle Class                         | 300         |        |
| Layout Angle                         | 225.0       |        |
| Blind Flange (Y/N)                   | N           |        |
| Weight of Nozzle ( Used if > 0 )     | 525.4       | N      |
| Grade of Attached Flange             | GR 1.4      |        |
| Nozzle Matl                          | SA-333 6    | +IT    |
|                                      |             |        |
| Element From Node                    | 60          |        |
| Detail Type                          | Nozzle      |        |
| Detail ID                            | A68"        |        |
| Dist. from "FROM" Node / Offset dist | 0.0005822   | mm     |
| Nozzle Diameter                      | 8           | in.    |
| Nozzle Schedule                      | 80          |        |
| Nozzle Class                         | 300         |        |
| Layout Angle                         | 135.0       |        |
| Blind Flange (Y/N)                   | N           |        |
| Weight of Nozzle ( Used if > 0 )     | 525.38      | N      |
| Grade of Attached Flange             | GR 1.4      |        |
| Nozzle Matl                          | SA-333 6    | +IT    |
|                                      |             |        |
| Element From Node                    | 60          |        |
| Detail Type                          | Nozzle      |        |
| Detail ID                            | M124"       |        |
| Dist. from "FROM" Node / Offset dist | 0.0005822   | mm     |
| Nozzle Diameter                      | 24          | in.    |
| Nozzle Schedule                      | None        |        |
| Nozzle Class                         | 300         |        |
| Layout Angle                         | 0.0         |        |
| Blind Flange (Y/N)                   | Y           |        |
| Weight of Nozzle ( Used if > 0 )     | 7526.6      | N      |
| Grade of Attached Flange             | GR 1.4      |        |
| Nozzle Matl                          | SA-516 70   | +IT    |
|                                      |             |        |
| Element From Node                    | 60          |        |
| Detail Type                          | Nozzle      |        |
| Detail ID                            | A116"       |        |
| Dist. from "FROM" Node / Offset dist | 450         | mm     |
| Nozzle Diameter                      | 16          | in.    |
| Nozzle Schedule                      | None        |        |
| Nozzle Class                         | 300         |        |
| Layout Angle                         | 180.0       |        |
| Blind Flange (Y/N)                   | N           |        |
| Weight of Nozzle ( Used if > 0 )     | 1875.7      | N      |
| Grade of Attached Flange             | GR 1.4      |        |
| Nozzle Matl                          | SA-516 70   | +IT    |



Element From Node 60  
 Detail Type Nozzle  
 Detail ID L12"  
 Dist. from "FROM" Node / Offset dist 55 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 65.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 100.91 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-350 LF1 +IT

Element From Node 60  
 Detail Type Nozzle  
 Detail ID L32"  
 Dist. from "FROM" Node / Offset dist 50 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 90.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 100.91 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-350 LF1 +IT

Element From Node 60  
 Detail Type Nozzle  
 Detail ID L52"  
 Dist. from "FROM" Node / Offset dist 50 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 265.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 100.91 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-350 LF1 +IT

Element From Node 60  
 Detail Type Nozzle  
 Detail ID L72"  
 Dist. from "FROM" Node / Offset dist 50 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 295.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 100.91 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-350 LF1 +IT

Element From Node 60  
 Detail Type Nozzle  
 Detail ID A44"  
 Dist. from "FROM" Node / Offset dist 750 mm  
 Nozzle Diameter 4 in.  
 Nozzle Schedule 160  
 Nozzle Class 600  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 187.41 N  
 Grade of Attached Flange GR 1.4  
 Nozzle Matl SA-333 6 +IT

Element From Node 60  
 Detail Type Weight  
 Detail ID A1pipewt  
 Dist. from "FROM" Node / Offset dist 450 mm  
 Miscellaneous Weight 22260 N



|                                      |             |    |
|--------------------------------------|-------------|----|
| Offset from Element Centerline       | 2186        | mm |
| Element From Node                    | 60          |    |
| Detail Type                          | Weight      |    |
| Detail ID                            | A2Pipewt    |    |
| Dist. from "FROM" Node / Offset dist | 0.0005822   | mm |
| Miscellaneous Weight                 | 6785.7      | N  |
| Offset from Element Centerline       | 2085        | mm |
| Element From Node                    | 60          |    |
| Detail Type                          | Weight      |    |
| Detail ID                            | A6Pipewt    |    |
| Dist. from "FROM" Node / Offset dist | 0.0005822   | mm |
| Miscellaneous Weight                 | 6785.7      | N  |
| Offset from Element Centerline       | 2085        | mm |
| Element From Node                    | 60          |    |
| Detail Type                          | Weight      |    |
| Detail ID                            | Lifting lug |    |
| Dist. from "FROM" Node / Offset dist | 1000        | mm |
| Miscellaneous Weight                 | 3726.3      | N  |
| Offset from Element Centerline       | 0           | mm |
| Element From Node                    | 60          |    |
| Detail Type                          | Weight      |    |
| Detail ID                            | Demister    |    |
| Dist. from "FROM" Node / Offset dist | 0.0005822   | mm |
| Miscellaneous Weight                 | 980.6       | N  |
| Offset from Element Centerline       | 0           | mm |

|                                      |                 |
|--------------------------------------|-----------------|
| Element#                             | 7/7             |
| Element From Node                    | 70              |
| Element To Node                      | 80              |
| Element Type                         | Elliptical      |
| Description                          | Tophead         |
| Distance "FROM" to "TO"              | 50 mm           |
| Inside Diameter                      | 1650 mm         |
| Element Thickness                    | 17 mm           |
| Internal Corrosion Allowance         | 1.5 mm          |
| Nominal Thickness                    | 20 mm           |
| External Corrosion Allowance         | 0 mm            |
| Design Internal Pressure             | 2.5 MPa         |
| Design Temperature Internal Pressure | 85 °C           |
| Design External Pressure             | 0 MPa           |
| Design Temperature External Pressure | 85 °C           |
| Effective Diameter Multiplier        | 1.53            |
| Material Name                        | SA-516 70 +N+IT |
| Efficiency, Longitudinal Seam        | 1.0             |
| Efficiency, Circumferential Seam     | 1.0             |
| Elliptical Head Factor               | 2.0             |
| Weld is pre-Heated                   | No              |

|                                      |                          |
|--------------------------------------|--------------------------|
| Element From Node                    | 70                       |
| Detail Type                          | Platform                 |
| Detail ID                            | PLAT:[1 OF 1]            |
| Dist. from "FROM" Node / Offset dist | 500 mm                   |
| Platform Start Angle (degrees)       | 0.0                      |
| Platform End Angle (degrees)         | 0.0                      |
| Platform Wind Area                   | 18855500 mm <sup>2</sup> |
| Platform Weight                      | 100438 N                 |
| Platform Railing Weight              | 0 N/mm                   |
| Platform Grating Weight              | 0.017 kg/cm <sup>2</sup> |
| Platform Width                       | 7684 mm                  |
| Platform Height                      | 1100 mm                  |
| Platform Clearance or End Offset     | 3842 mm                  |
| Platform Force Coefficient           | 2.0                      |
| Ladder Layout Angle                  | 0.0                      |
| Ladder Start Elevation               | 0 mm                     |



|                                      |                |                    |
|--------------------------------------|----------------|--------------------|
| Ladder End Elevation                 | 4000           | mm                 |
| Unit Weight of Ladder                | 0.49           | N/mm               |
| Platform Length (top head platform)  | 7684           | mm                 |
| Element From Node                    | 70             |                    |
| Detail Type                          | Insulation     |                    |
| Detail ID                            | Ins: 70        |                    |
| Dist. from "FROM" Node / Offset dist | 0              | mm                 |
| Height/Length of Insulation          | 462.5          | mm                 |
| Thickness of Insulation              | 70             | mm                 |
| Density                              | 0.40000E-04    | kg/cm <sup>3</sup> |
| Element From Node                    | 70             |                    |
| Detail Type                          | Nozzle         |                    |
| Detail ID                            | B118"          |                    |
| Dist. from "FROM" Node / Offset dist | 0              | mm                 |
| Nozzle Diameter                      | 18             | in.                |
| Nozzle Schedule                      | None           |                    |
| Nozzle Class                         | 300            |                    |
| Layout Angle                         | 0.0            |                    |
| Blind Flange (Y/N)                   | N              |                    |
| Weight of Nozzle ( Used if > 0 )     | 2243.5         | N                  |
| Grade of Attached Flange             | GR 1.4         |                    |
| Nozzle Matl                          | SA-516 70 +IT  |                    |
| Element From Node                    | 70             |                    |
| Detail Type                          | Nozzle         |                    |
| Detail ID                            | S12"           |                    |
| Dist. from "FROM" Node / Offset dist | 600            | mm                 |
| Nozzle Diameter                      | 2              | in.                |
| Nozzle Schedule                      | None           |                    |
| Nozzle Class                         | 600            |                    |
| Layout Angle                         | 0.0            |                    |
| Blind Flange (Y/N)                   | N              |                    |
| Weight of Nozzle ( Used if > 0 )     | 102.37         | N                  |
| Grade of Attached Flange             | GR 1.4         |                    |
| Nozzle Matl                          | SA-350 LF1 +IT |                    |
| Element From Node                    | 70             |                    |
| Detail Type                          | Weight         |                    |
| Detail ID                            | B1pipe wt      |                    |
| Dist. from "FROM" Node / Offset dist | 0              | mm                 |
| Miscellaneous Weight                 | 18641          | N                  |
| Offset from Element Centerline       | 2212           | mm                 |
| Element From Node                    | 70             |                    |
| Detail Type                          | Weight         |                    |
| Detail ID                            | Topdavit       |                    |
| Dist. from "FROM" Node / Offset dist | 0              | mm                 |
| Miscellaneous Weight                 | 4118.5         | N                  |
| Offset from Element Centerline       | 1000           | mm                 |
| Element From Node                    | 70             |                    |
| Detail Type                          | Weight         |                    |
| Detail ID                            | PSVwt          |                    |
| Dist. from "FROM" Node / Offset dist | 400            | mm                 |
| Miscellaneous Weight                 | 9806           | N                  |
| Offset from Element Centerline       | 1000           | mm                 |



### XY Coordinate Calculations:

| From     | To  | X (Horiz.)<br>mm | Y (Vert.)<br>mm | DX (Horiz.)<br>mm | DY (Vert.)<br>mm |
|----------|-----|------------------|-----------------|-------------------|------------------|
| Skirt-2  | ... | ...              | 1000            | ...               | 1000             |
| Skirt-1  | ... | ...              | 2000            | ...               | 1000             |
| Btm_head | ... | ...              | 2050            | ...               | 50               |
| Shell-1  | ... | ...              | 5050            | ...               | 3000             |
| Shell-2  | ... | ...              | 8050            | ...               | 3000             |
| Shell-3  | ... | ...              | 9850            | ...               | 1800             |
| Top_head | ... | ...              | 9900            | ...               | 50               |

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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### Internal Pressure Results Summary:

#### Element Thickness, Pressure, Diameter and Allowable Stress :

| From     | To | Int. Press<br>+ Liq. Hd<br>MPa | Nominal<br>Thickness<br>mm | Total Corr<br>Allowance<br>mm | Element<br>Diameter<br>mm | Allowable<br>Stress(SE)<br>MPa | Element# |
|----------|----|--------------------------------|----------------------------|-------------------------------|---------------------------|--------------------------------|----------|
| Skirt-2  |    | ...                            | 14                         | 1.5                           | 1682                      | ...                            | 1        |
| Skirt-1  |    | ...                            | 14                         | 1.5                           | 1682                      | ...                            | 2        |
| Btm_head |    | 2.5427                         | 20                         | 1.5                           | 1650                      | 138                            | 3        |
| Shell-1  |    | 2.5397                         | 18                         | 1.5                           | 1650                      | 138                            | 4        |
| Shell-2  |    | 2.5197                         | 18                         | 1.5                           | 1650                      | 138                            | 5        |
| Shell-3  |    | 2.5                            | 18                         | 1.5                           | 1650                      | 138                            | 6        |
| Top_head |    | 2.5                            | 20                         | 1.5                           | 1650                      | 138                            | 7        |

#### Element Required Thickness and MAWP :

| From     | To | Design<br>Pressure<br>MPa | M.A.W.P.<br>Corroded<br>MPa | M.A.P.<br>New & Cold<br>MPa | Minimum<br>Thickness<br>mm | Required<br>Thickness<br>mm | Element# |
|----------|----|---------------------------|-----------------------------|-----------------------------|----------------------------|-----------------------------|----------|
| Skirt-2  |    | ...                       | No Calc                     | No Calc                     | 14                         | No Calc                     | 1        |
| Skirt-1  |    | ...                       | No Calc                     | No Calc                     | 14                         | No Calc                     | 2        |
| Btm_head |    | 2.5                       | 2.54667                     | 2.83779                     | 17                         | 16.7201                     | 3        |
| Shell-1  |    | 2.5                       | 2.68267                     | 2.97195                     | 18                         | 16.8805                     | 4        |
| Shell-2  |    | 2.5                       | 2.70267                     | 2.97195                     | 18                         | 16.758                      | 5        |
| Shell-3  |    | 2.5                       | 2.72233                     | 2.97195                     | 18                         | 16.6376                     | 6        |
| Top_head |    | 2.5                       | 2.58937                     | 2.83774                     | 17                         | 16.4641                     | 7        |

Minimum 3 3

MAWP: 3 MPa, limited by: Btmhead.

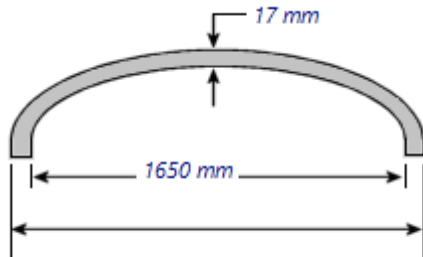
Elements Suitable for Design Internal Pressure.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Btmhead

| Elliptical Head Design Information: |    |               |                         |    |           |
|-------------------------------------|----|---------------|-------------------------|----|-----------|
| Design Pressure                     | P  | 2.543 MPa     | Int. Design Temperature |    | 85 °C     |
| Material                            |    | SA-516 70     | Ext. Design Temperature |    | 85 °C     |
| Length                              |    | 50.000 mm     | External Pressure Chart |    | CS-2      |
|                                     |    |               | UNS Number              |    | K02700    |
| Int. Corr. All.                     | c  | 1.5000 mm     | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                     | ce | 0.0000 mm     | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                     | Di | 1650.000 mm   | Actual Stress           |    | 135.5 MPa |
| Spec. Min. Thk t,e, or sa           |    | 17.0000 mm    | Spec. Nominal Thk.      |    | 20.000 mm |
| Surface Area                        |    | 3361435.5 mm² | SG of Contents          |    | 0.680     |
| Element Volume                      |    | 695.1 ltr     | Weight of Contents      |    | 4631.9 N  |
| Empty Weight                        |    | 5472.2 N      | Operating Weight        |    | 10104.0 N |
|                                     |    |               | Comp. All. Ope. Stress  |    | 128.2 MPa |
|                                     |    |               | Aspect Ratio            |    | 2.00      |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.312 mm

#### Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.543 \cdot 1653.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.543)$$

$$= 15.2201 + 1.5000 = 16.7201 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.043 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 15.5) / (0.998 \cdot 1653.0 + 0.2 \cdot 15.5)$$

$$= 3 - 0 = 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 17.0) / (1.0 \cdot 1650.0 + 0.2 \cdot 17.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.543 \cdot (0.998 \cdot 1653.0 + 0.2 \cdot 15.5)) / (2 \cdot 1.0 \cdot 15.5)$$



= 135.513 MPa

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (2.543 \cdot 826.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.543) + 1.5 \\ &= 16.899 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.040 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 18.5) / (826.5 + 0.6 \cdot 18.5) \\ &= 3 - 0 = 3 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (1653.0 / (2 \cdot 414.0))^2) / 6 \\ &= 0.997587 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  5.164 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

**MDMT Calculations in the Knuckle Portion:**

Note: This Element/Detail was specified as being Impact Tested.

**MDMT Calculations in the Head Straight Flange:**

Note: This Element/Detail was specified as being Impact Tested.

**Tolerance for Formed Heads per UG-81(a):**

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.625 mm  
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.312 mm

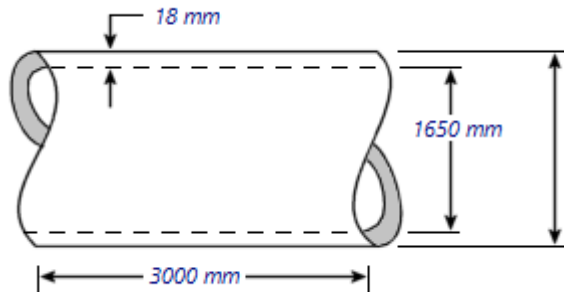
**Material treatment options: +N+IT**

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-1

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 2.540 MPa                  | Int. Design Temperature |    | 85 °C     |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 85 °C     |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.5000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 1650.000 mm                | Actual Stress           |    | 128.7 MPa |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 18.0000 mm                 | Spec. Nominal Thk.      |    | 18.000 mm |
| Surface Area                                      |    | 15890175.0 mm <sup>2</sup> | SG of Contents          |    | 0.680     |
| Element Volume                                    |    | 6415.9 ltr                 | Weight of Contents      |    | 42755.5 N |
| Empty Weight                                      |    | 21504.9 N                  | Operating Weight        |    | 64260.5 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 129.7 MPa |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.54 \cdot 826.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.54)$$

$$= 15.3805 + 1.5000 = 16.8805 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.040 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.5) / (826.5 + 0.6 \cdot 16.5)$$

$$= 3 - 0 = 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (825.0 + 0.6 \cdot 18.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.54 \cdot (826.5 + 0.6 \cdot 16.5)) / (1.0 \cdot 16.5)$$

$$= 128.738 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 1.079 %

#### Minimum Design Metal Temperature Results:



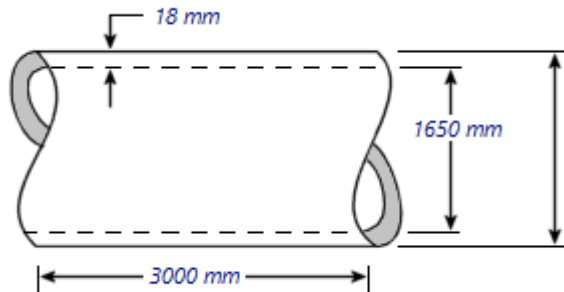
Note: This Element/Detail was specified as being Impact Tested.

**Material treatment options: +N+IT**

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-2

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 2.520 MPa                  | Int. Design Temperature |    | 85 °C     |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 85 °C     |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.5000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 1650.000 mm                | Actual Stress           |    | 127.7 MPa |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 18.0000 mm                 | Spec. Nominal Thk.      |    | 18.000 mm |
| Surface Area                                      |    | 15890175.0 mm <sup>2</sup> | SG of Contents          |    | 0.680     |
| Element Volume                                    |    | 6415.9 ltr                 | Weight of Contents      |    | 42043.0 N |
| Empty Weight                                      |    | 21504.9 N                  | Operating Weight        |    | 63547.9 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 129.7 MPa |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.52 \cdot 826.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.52)$$

$$= 15.2580 + 1.5000 = 16.7580 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.020 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.5) / (826.5 + 0.6 \cdot 16.5)$$

$$= 3 - 0 = 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (825.0 + 0.6 \cdot 18.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.52 \cdot (826.5 + 0.6 \cdot 16.5)) / (1.0 \cdot 16.5)$$

$$= 127.724 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 1.079 %

#### Minimum Design Metal Temperature Results:



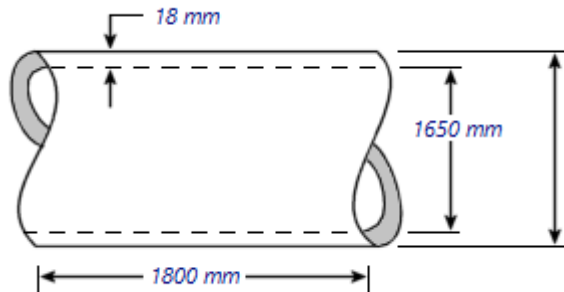
Note: This Element/Detail was specified as being Impact Tested.

**Material treatment options: +N+IT**

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-3

| Cylindrical Shell Design Information:             |    |                           |                         |    |           |
|---------------------------------------------------|----|---------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 2.500 MPa                 | Int. Design Temperature |    | 85 °C     |
| Material                                          |    | SA-516 70                 | Ext. Design Temperature |    | 85 °C     |
| Length                                            |    | 1800.000 mm               | External Pressure Chart |    | CS-2      |
|                                                   |    |                           | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.5000 mm                 | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                 | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 1650.000 mm               | Actual Stress           |    | 126.7 MPa |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 18.0000 mm                | Spec. Nominal Thk.      |    | 18.000 mm |
| Surface Area                                      |    | 9534105.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 3849.5 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 12903.0 N                 | Operating Weight        |    | 12903.0 N |
|                                                   |    |                           | Comp. All. Ope. Stress  |    | 129.7 MPa |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 826.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.5)$$

$$= 15.1376 + 1.5000 = 16.6376 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.5) / (826.5 + 0.6 \cdot 16.5)$$

$$= 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (825.0 + 0.6 \cdot 18.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (826.5 + 0.6 \cdot 16.5)) / (1.0 \cdot 16.5)$$

$$= 126.727 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 1.079 %

#### Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

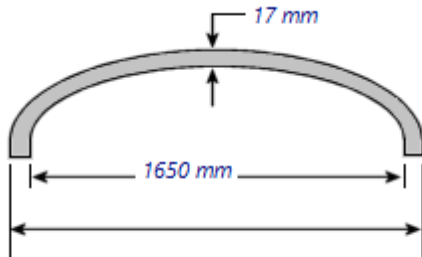
**Material treatment options: +N+IT**

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Tophead

| Elliptical Head Design Information:               |    |                           |                         |    |           |
|---------------------------------------------------|----|---------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 2.500 MPa                 | Int. Design Temperature |    | 85 °C     |
| Material                                          |    | SA-516 70                 | Ext. Design Temperature |    | 85 °C     |
| Length                                            |    | 50.000 mm                 | External Pressure Chart |    | CS-2      |
|                                                   |    |                           | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.5000 mm                 | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                 | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 1650.000 mm               | Actual Stress           |    | 133.2 MPa |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 17.0000 mm                | Spec. Nominal Thk.      |    | 20.000 mm |
| Surface Area                                      |    | 3361435.5 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 695.1 ltr                 | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 5472.2 N                  | Operating Weight        |    | 5472.2 N  |
|                                                   |    |                           | Comp. All. Ope. Stress  |    | 128.2 MPa |
|                                                   |    |                           | Aspect Ratio            |    | 2.00      |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.312 mm

#### Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.5 \cdot 1653.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.5)$$

$$= 14.9641 + 1.5000 = 16.4641 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 15.5) / (0.998 \cdot 1653.0 + 0.2 \cdot 15.5)$$

$$= 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 17.0) / (1.0 \cdot 1650.0 + 0.2 \cdot 17.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.5 \cdot (0.998 \cdot 1653.0 + 0.2 \cdot 15.5)) / (2 \cdot 1.0 \cdot 15.5)$$

$$= 133.235 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (2.5 \cdot 826.5) / (138.0 \cdot 1.0 - 0.6 \cdot 2.5) + 1.5 \\ &= 16.638 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 18.5) / (826.5 + 0.6 \cdot 18.5) \\ &= 3 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (1653.0 / (2 \cdot 414.0))^2) / 6 \\ &= 0.997587 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  5.164 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

#### MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

#### MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.625 mm  
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.312 mm

Material treatment options: +N+IT

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### Hydrostatic Test Pressure Results:

|                 |                               |           |
|-----------------|-------------------------------|-----------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 3.311 MPa |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 3.250 MPa |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 3.673 MPa |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 2.801 MPa |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 3.575 MPa |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 3.575 MPa |
| App 27-4        | = M.A.W.P.                    | 2.547 MPa |

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 2.547 • 1.0

= 3.311 MPa

### Horizontal Test performed per: UG-99b

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

### Stresses on Elements due to Test Pressure (MPa):

| From To  | Stress | Allowable | Ratio | Pressure |
|----------|--------|-----------|-------|----------|
| Btm_head | 161.8  | 234.0     | 0.691 | 3.33     |
| Shell-1  | 154.5  | 234.0     | 0.660 | 3.33     |
| Shell-2  | 154.5  | 234.0     | 0.660 | 3.33     |
| Shell-3  | 154.5  | 234.0     | 0.660 | 3.33     |
| Top_head | 161.8  | 234.0     | 0.691 | 3.33     |

**Theoretical Maximum Allowable test pressure is : 4.796 MPa**

### Stress ratios for Nozzle and Pad Materials (MPa):

| Description | Pad/Nozzle | Ambient | Operating | Ratio |
|-------------|------------|---------|-----------|-------|
| D1_3"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| D1_3"       | Pad        | 138.00  | 138.00    | 1.000 |
| B2_10"      | Nozzle     | 118.00  | 118.00    | 1.000 |
| B2_10"      | Pad        | 138.00  | 138.00    | 1.000 |
| A3_3"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| A3_3"       | Pad        | 138.00  | 138.00    | 1.000 |
| L2_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| L4_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| L6_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| L8_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| A5_18"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| A5_18"      | Pad        | 138.00  | 138.00    | 1.000 |
| A2_8"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| A2_8"       | Pad        | 138.00  | 138.00    | 1.000 |
| A6_8"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| A6_8"       | Pad        | 138.00  | 138.00    | 1.000 |
| M1_24"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| M1_24"      | Pad        | 138.00  | 138.00    | 1.000 |
| A1_16"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| A1_16"      | Pad        | 138.00  | 138.00    | 1.000 |
| L1_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| L3_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| L5_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| L7_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| A4_4"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| A4_4"       | Pad        | 138.00  | 138.00    | 1.000 |
| B1_18"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| B1_18"      | Pad        | 138.00  | 138.00    | 1.000 |
| S1_2"       | Nozzle     | 118.00  | 118.00    | 1.000 |

Minimum

1.000



**Stress ratios for Pressurized Vessel Elements (MPa):**

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| Btm_head    | 138.00  | 138.00    | 1.000 |
| Shell-1     | 138.00  | 138.00    | 1.000 |
| Shell-2     | 138.00  | 138.00    | 1.000 |
| Shell-3     | 138.00  | 138.00    | 1.000 |
| Top_head    | 138.00  | 138.00    | 1.000 |

Minimum 1.000

**Hoop Stress in Nozzle Wall during Pressure Test (MPa):**

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| D1_3"       | 13.86   | 216.00    | 0.064 |
| B2_10"      | 33.07   | 216.00    | 0.153 |
| A3_3"       | 13.86   | 216.00    | 0.064 |
| L2_2"       | 7.04    | 184.50    | 0.038 |
| L4_2"       | 7.04    | 184.50    | 0.038 |
| L6_2"       | 7.04    | 184.50    | 0.038 |
| L8_2"       | 7.04    | 184.50    | 0.038 |
| A5_18"      | 40.92   | 234.00    | 0.175 |
| A2_8"       | 31.46   | 216.00    | 0.146 |
| A6_8"       | 31.46   | 216.00    | 0.146 |
| M1_24"      | 49.37   | 234.00    | 0.211 |
| A1_16"      | 36.23   | 234.00    | 0.155 |
| L1_2"       | 7.04    | 184.50    | 0.038 |
| L3_2"       | 7.04    | 184.50    | 0.038 |
| L5_2"       | 7.04    | 184.50    | 0.038 |
| L7_2"       | 7.04    | 184.50    | 0.038 |
| A4_4"       | 14.78   | 216.00    | 0.068 |
| B1_18"      | 40.92   | 234.00    | 0.175 |
| S1_2"       | 7.04    | 184.50    | 0.038 |

Elements Suitable for Test Pressure.

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## External Pressure Calculation Results Summary:

### ASME Code, Section VIII Division 1, 2025

#### External Pressure Calculations:

| From | To | Section Length<br>mm | Outside Diameter<br>mm | Corroded Thickness<br>mm | Factor A   | Factor B<br>MPa |
|------|----|----------------------|------------------------|--------------------------|------------|-----------------|
| 10   | 20 | 2000                 | ...                    | ...                      | No Calc    | No Calc         |
| 20   | 30 | 2000                 | ...                    | ...                      | No Calc    | No Calc         |
| 30   | 40 | No Calc              | 1684                   | 15.5                     | 0.0012784  | 92.4922         |
| 40   | 50 | 8175                 | 1686                   | 16.5                     | 0.00023985 | 23.9796         |
| 50   | 60 | 8175                 | 1686                   | 16.5                     | 0.00023985 | 23.9796         |
| 60   | 70 | 8175                 | 1686                   | 16.5                     | 0.00023985 | 23.9796         |
| 70   | 80 | No Calc              | 1684                   | 15.5                     | 0.0012784  | 92.4922         |

#### External Pressure Calculations:

| From | To | External Actual T.<br>mm | External Required T.<br>mm | External Design Pressure<br>MPa | External M.A.W.P.<br>MPa |
|------|----|--------------------------|----------------------------|---------------------------------|--------------------------|
| 10   | 20 | ...                      | No Calc                    | ...                             | No Calc                  |
| 20   | 30 | ...                      | No Calc                    | ...                             | No Calc                  |
| 30   | 40 | 17                       | No Calc                    | ...                             | 0.94592                  |
| 40   | 50 | 18                       | No Calc                    | ...                             | 0.3129                   |
| 50   | 60 | 18                       | No Calc                    | ...                             | 0.3129                   |
| 60   | 70 | 18                       | No Calc                    | ...                             | 0.3129                   |
| 70   | 80 | 17                       | No Calc                    | ...                             | 0.94592                  |

Minimum

0

#### External Pressure Calculations:

| From | To | Actual Length<br>Bet. Stiffeners<br>mm | Allowable Length<br>Bet. Stiffeners<br>mm | Ring Inertia<br>Required<br>mm <sup>4</sup> | Ring Inertia<br>Available<br>mm <sup>4</sup> |
|------|----|----------------------------------------|-------------------------------------------|---------------------------------------------|----------------------------------------------|
| 10   | 20 | 2000                                   | No Calc                                   | No Calc                                     | No Calc                                      |
| 20   | 30 | 2000                                   | No Calc                                   | No Calc                                     | No Calc                                      |
| 30   | 40 | No Calc                                | No Calc                                   | No Calc                                     | No Calc                                      |
| 40   | 50 | 8175                                   | No Calc                                   | No Calc                                     | No Calc                                      |
| 50   | 60 | 8175                                   | No Calc                                   | No Calc                                     | No Calc                                      |
| 60   | 70 | 8175                                   | No Calc                                   | No Calc                                     | No Calc                                      |
| 70   | 80 | No Calc                                | No Calc                                   | No Calc                                     | No Calc                                      |

Elements Suitable for External Pressure.

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### Element and Detail Weights:

| From  | To | Element Metal Wgt. kgm | Element ID Volume ltr | Corroded Metal Wgt. kgm | Corroded ID Volume ltr | Extra due Misc % kgm |
|-------|----|------------------------|-----------------------|-------------------------|------------------------|----------------------|
| 10    | 20 | 1098.77                | ...                   | 1038.31                 | ...                    | 76.9141              |
| 20    | 30 | 568.599                | ...                   | 508.133                 | ...                    | 39.8019              |
| 30    | 40 | 558.042                | 695.055               | 516.189                 | 698.658                | 39.063               |
| 40    | 50 | 2193.04                | 6415.89               | 2012.1                  | 6439.24                | 153.513              |
| 50    | 60 | 2193.04                | 6415.89               | 2012.1                  | 6439.24                | 153.513              |
| 60    | 70 | 1315.82                | 3849.53               | 1207.26                 | 3863.54                | 92.1077              |
| 70    | 80 | 558.042                | 695.055               | 516.189                 | 698.658                | 39.063               |
| Total |    | 8485                   | 18071.42              | 7810                    | 18139.34               | 593                  |

### Weight of Details:

| From | Type | Weight of Detail kgm | X Offset, Dtl. Cent. mm | Y Offset, Dtl. Cent. mm | Z Offset, Dtl. Cent. mm | Description   |
|------|------|----------------------|-------------------------|-------------------------|-------------------------|---------------|
| 30   | Liqd | 472.349              | ...                     | -206.25                 | ...                     | Liquid: 10    |
| 30   | Insl | 12.4931              | ...                     | -181.25                 | ...                     | Ins: 30       |
| 30   | Nozl | 15.9674              | ...                     | -554                    | ...                     | D1_3"         |
| 40   | Plat | 3358.54              | 0.16411E-04             | 200                     | ...                     | PLAT:[1 OF 1] |
| 40   | Liqd | 4360.14              | ...                     | 1500                    | ...                     | Liquid: 20    |
| 40   | Insl | 46.394               | ...                     | 1500                    | ...                     | Ins: 40       |
| 40   | Nozl | 84.1652              | 925                     | 400                     | ...                     | B2_10"        |
| 40   | Nozl | 14.747               | -925                    | 250                     | ...                     | A3_3"         |
| 40   | Nozl | 11.0114              | 390.922                 | 250                     | -838.334                | L2_2"         |
| 40   | Nozl | 11.0114              | ...                     | 250                     | -925                    | L4_2"         |
| 40   | Nozl | 11.0114              | -80.619                 | 250                     | 921.48                  | L6_2"         |
| 40   | Nozl | 11.0114              | 390.922                 | 250                     | 838.334                 | L8_2"         |
| 40   | Wght | 257                  | 2110                    | 400                     | ...                     | B2_Pipe_wt    |
| 50   | Plat | 1670.11              | ...                     | 1500                    | 1140.57                 | PLAT:[1 OF 1] |
| 50   | Liqd | 4287.47              | ...                     | 1475                    | ...                     | Liquid: 30    |
| 50   | Insl | 46.394               | ...                     | 1500                    | ...                     | Ins: 50       |
| 50   | Nozl | 256.4                | ...                     | 2000                    | 950.006                 | A5_18"        |
| 50   | Wght | 2674                 | ...                     | 2000                    | 2212                    | A5_Pipe_wt    |
| 50   | Wght | 1005                 | ...                     | -0.00028956             | ...                     | Clip_wt       |
| 60   | Insl | 27.8364              | ...                     | 900                     | ...                     | Ins: 60       |
| 60   | Nozl | 57.3304              | -654.074                | 0.00058217              | 654.074                 | A2_8"         |
| 60   | Nozl | 57.3274              | -654.074                | 0.00058217              | -654.074                | A6_8"         |
| 60   | Nozl | 821.276              | 975                     | 0.00058217              | ...                     | M1_24"        |
| 60   | Nozl | 204.666              | -950.006                | 450                     | ...                     | A1_16"        |
| 60   | Nozl | 11.0114              | 390.922                 | 55                      | -838.334                | L1_2"         |
| 60   | Nozl | 11.0114              | ...                     | 50.0004                 | -925                    | L3_2"         |
| 60   | Nozl | 11.0114              | -80.619                 | 50.0004                 | 921.48                  | L5_2"         |
| 60   | Nozl | 11.0114              | 390.922                 | 50.0004                 | 838.334                 | L7_2"         |
| 60   | Nozl | 20.4501              | 925                     | 750                     | ...                     | A4_4"         |
| 60   | Wght | 2270                 | -2186                   | 450                     | ...                     | A1_Pipe_wt    |
| 60   | Wght | 692                  | -1474.32                | 0.00058217              | 1474.32                 | A2_Pipe_wt    |
| 60   | Wght | 692                  | -1474.32                | 0.00058217              | -1474.32                | A6_Pipe_wt    |
| 60   | Wght | 380                  | ...                     | 1000                    | ...                     | Lifting lug   |
| 60   | Wght | 100                  | ...                     | 0.00058217              | ...                     | Demister      |
| 70   | Plat | 10959.5              | ...                     | 500                     | ...                     | PLAT:[1 OF 1] |
| 70   | Insl | 12.4931              | ...                     | 231.25                  | ...                     | Ins: 70       |
| 70   | Nozl | 244.805              | ...                     | 579.006                 | ...                     | B1_18"        |
| 70   | Nozl | 11.1699              | 600                     | 414.332                 | ...                     | S1_2"         |
| 70   | Wght | 1901                 | 2212                    | ...                     | ...                     | B1_pipe wt    |
| 70   | Wght | 420                  | 1000                    | ...                     | ...                     | Top_davit     |
| 70   | Wght | 1000                 | -1000                   | 400                     | ...                     | PSV_wt        |

### Total Weight of Each Detail Type:

Platforms 15988.1  
 Liquid 9120.0

|                           |             |
|---------------------------|-------------|
| Insulation                | 145.6       |
| Nozzles                   | 1876.4      |
| Weights                   | 11391.0     |
| <hr/>                     |             |
| Sum of the Detail Weights | 38521.1 kgm |

#### Weight Summation Results: (kgm)

|               | Fabricated | Shop Test | Shipping | Erected | Empty   | Operating |
|---------------|------------|-----------|----------|---------|---------|-----------|
| Main Elements | 9079.3     | 9079.3    | 9079.3   | 9079.3  | 9079.3  | 9079.3    |
| Nozzles       | 1876.4     | 1876.4    | 1876.4   | 1876.4  | 1876.4  | 1876.4    |
| Wld Weights   | 1385.0     | 1385.0    | 1385.0   | 1385.0  | 1385.0  | 1385.0    |
| Platforms     | ...        | ...       | 15988.1  | 15988.1 | 15988.1 | 15988.1   |
| Insulation    | ...        | ...       | 145.6    | 145.6   | 145.6   | 145.6     |
| Empty Weights | ...        | ...       | ...      | 520.0   | 520.0   | ...       |
| Test Weights  | ...        | 100.0     | ...      | ...     | ...     | ...       |
| Ope Weights   | ...        | ...       | ...      | ...     | ...     | 10006.0   |
| Ope. Liquid   | ...        | ...       | ...      | ...     | ...     | 9120.0    |
| Test Liquid   | ...        | 18060.4   | ...      | ...     | ...     | ...       |
| Totals        | 12340.7    | 30501.1   | 28474.5  | 28994.5 | 28994.5 | 47600.4   |

#### Field Installation Options:

**Miscellaneous Weight Percent: 7.0 %**

*Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.*

#### Weight Summary:

|                 |                                                     |             |
|-----------------|-----------------------------------------------------|-------------|
| Fabricated Wt.  | - Bare weight without removable internals           | 12340.7 kgm |
| Shop Test Wt.   | - Fabricated weight + water ( full )                | 30501.1 kgm |
| Shipping Wt.    | - Fab. weight + removable intls.+ shipping app.     | 28474.5 kgm |
| Erected Wt.     | - Fab. wt + or - loose items (trays,platforms etc.) | 28994.5 kgm |
| Ope. Wt. no Liq | - Fab. weight + internals. + details + weights      | 28994.5 kgm |
| Operating Wt.   | - Empty weight + operating liq. uncorroded          | 47600.4 kgm |

#### Weights used in Vortex Shedding Calculations:

|                                                               |             |
|---------------------------------------------------------------|-------------|
| Mass of the Upper 1/3 of the Vertical Vessel (operating)      | 27788.1 kgm |
| Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) | 22876.7 kgm |
| Mass of the Upper 1/3 of the Vertical Vessel (empty)          | 16778.9 kgm |

#### Surface Areas of Elements:

| From | To | Outside Surface Area<br>mm <sup>2</sup> | Inside Surface Area<br>mm <sup>2</sup> |
|------|----|-----------------------------------------|----------------------------------------|
| 10   | 20 | 5284159                                 | 5196194                                |
| 20   | 30 | 5284159                                 | 5196194                                |
| 30   | 40 | 3361435                                 | 3210331                                |
| 40   | 50 | 15890175                                | 15550884                               |
| 50   | 60 | 15890175                                | 15550884                               |
| 60   | 70 | 9534105                                 | 9330530                                |
| 70   | 80 | 3361435                                 | 3210331                                |

|       |            |                            |
|-------|------------|----------------------------|
| Total | 58605640.0 | 57245352.0 mm <sup>2</sup> |
| Total | 58.6       | 57.2 m <sup>2</sup>        |

#### Element and Detail Weights:

| From | To | Total Ele.<br>Empty Wgt.<br>kgm | Total. Ele.<br>Oper. Wgt.<br>kgm | Total. Ele.<br>Hydro. Wgt.<br>kgm | Total Dtl.<br>Offset Mom.<br>N-mm | Oper. Wgt.<br>No Liquid<br>kgm |
|------|----|---------------------------------|----------------------------------|-----------------------------------|-----------------------------------|--------------------------------|
| 10   | 20 | 1175.69                         | 1175.69                          | 1175.69                           | ...                               | 1175.69                        |



|    |    |         |         |         |          |         |
|----|----|---------|---------|---------|----------|---------|
| 20 | 30 | 608.401 | 608.401 | 608.401 | ...      | 608.401 |
| 30 | 40 | 625.566 | 1097.91 | 1320.2  | ...      | 625.566 |
| 40 | 50 | 5894.44 | 10511.6 | 12306.4 | 380.238  | 6151.44 |
| 50 | 60 | 5324.46 | 12285.9 | 11736.4 | 79101256 | 7998.45 |
| 60 | 70 | 3120.86 | 6774.86 | 6968.05 | 360.929  | 6774.86 |
| 70 | 80 | 12245.1 | 15146.1 | 12519.7 | ...      | 15146.1 |

#### Cumulative Vessel Weight

| From | To | Cumulative Ope<br>Wgt. No Liquid<br>kgm | Cumulative<br>Oper. Wgt.<br>kgm | Cumulative<br>Hydro. Wgt.<br>kgm |
|------|----|-----------------------------------------|---------------------------------|----------------------------------|
| 10   | 20 | 38480.5                                 | 47600.4                         | 46634.9                          |
| 20   | 30 | 37304.8                                 | 46424.8                         | 45459.2                          |
| 30   | 40 | 36696.4                                 | 45816.4                         | 44850.8                          |
| 40   | 50 | 36070.8                                 | 44718.4                         | 43530.6                          |
| 50   | 60 | 29919.4                                 | 34206.9                         | 31224.2                          |
| 60   | 70 | 21920.9                                 | 21920.9                         | 19487.7                          |
| 70   | 80 | 15146.1                                 | 15146.1                         | 12519.7                          |

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

#### Cumulative Vessel Moment

| From | To | Cumulative<br>Empty Mom.<br>N-mm | Cumulative<br>Oper. Mom.<br>N-mm | Cumulative<br>Hydro. Mom.<br>N-mm |
|------|----|----------------------------------|----------------------------------|-----------------------------------|
| 10   | 20 | 79102000                         | 79102000                         | 79102000                          |
| 20   | 30 | 79102000                         | 79102000                         | 79102000                          |
| 30   | 40 | 79102000                         | 79102000                         | 79102000                          |
| 40   | 50 | 79102000                         | 79102000                         | 79102000                          |
| 50   | 60 | 79101616                         | 79101616                         | 79101616                          |
| 60   | 70 | 360.929                          | 360.929                          | 360.929                           |
| 70   | 80 | ...                              | ...                              | ...                               |

### Nozzle Flange MAWP Results: (MPa & °C)

| Nozzle Description | Flange Rating |                            | Des. Temp | Class | Grade/Group | Equiv. Press | Max Pressure |       |       |
|--------------------|---------------|----------------------------|-----------|-------|-------------|--------------|--------------|-------|-------|
|                    | Ope.          | Ambient                    |           |       |             |              | UG-44        | 50%   | DNV   |
| D1_3"              | 4.76          | 5.10                       | 85        | 300   | GR 1.1      | 4.383        | 2.762        | 4.763 | 2.762 |
| B2_10"             | 4.76          | 5.10                       | 85        | 300   | GR 1.1      | 2.417        | 4.728        | 4.763 | 4.728 |
| A3_3"              | 4.76          | 5.10                       | 85        | 300   | GR 1.1      | 4.383        | 2.762        | 4.763 | 2.762 |
| L2_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| L4_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| L6_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| L8_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| A5_18"             | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | 1.723        | 3.971        | 3.971 | 3.971 |
| A2_8"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | 2.786        | 3.171        | 3.971 | 3.171 |
| A6_8"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | 2.786        | 3.171        | 3.971 | 3.171 |
| M1_24"             | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| A1_16"             | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | 1.968        | 3.971        | 3.971 | 3.971 |
| L1_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| L3_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| L5_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| L7_2"              | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | ...          | ...          | ...   | ...   |
| A4_4"              | 7.94          | 8.52                       | 85        | 600   | GR 1.4      | 4.346        | 7.568        | 7.943 | 7.568 |
| B1_18"             | 3.97          | 4.26                       | 85        | 300   | GR 1.4      | 1.723        | 3.971        | 3.971 | 3.971 |
| S1_2"              | 7.94          | 8.52                       | 85        | 600   | GR 1.4      | 5.665        | 6.249        | 7.943 | 6.249 |
| Min Rating         | 2.762         | 4.260, [for Core Elements] |           |       |             |              | ...          | ...   | ...   |

### Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

| Nozzle Description | Gasket Dimension G mm | Fm Value | Resolved Moment ME N-mm | Axial Force FE N | MAWP PD MPa | Maximum Pressure Pmax MPa | Pass Fail |
|--------------------|-----------------------|----------|-------------------------|------------------|-------------|---------------------------|-----------|
| D1_3"              | 111.125               | 0.50     | 1181221                 | 0                | 2.547       | 2.762                     | Pass      |
| B2_10"             | 302.387               | 0.50     | 13125631                | 0                | 2.547       | 4.728                     | Pass      |
| A3_3"              | 111.125               | 0.50     | 1181304                 | 0                | 2.547       | 2.762                     | Pass      |
| A5_18"             | 500.888               | 0.50     | 42526884                | 0                | 2.547       | 3.971                     | Pass      |
| A2_8"              | 248.539               | 0.50     | 8400389                 | 0                | 2.547       | 3.171                     | Pass      |
| A6_8"              | 248.539               | 0.50     | 8400389                 | 0                | 2.547       | 3.171                     | Pass      |
| A1_16"             | 442.976               | 0.50     | 33601524                | 0                | 2.547       | 3.971                     | Pass      |
| A4_4"              | 135.001               | 0.50     | 2100096                 | 0                | 2.547       | 7.568                     | Pass      |
| B1_18"             | 500.888               | 0.50     | 42523808                | 0                | 2.547       | 3.971                     | Pass      |
| S1_2"              | 77.851                | 0.50     | 524984                  | 0                | 2.547       | 6.249                     | Pass      |

Evaluating Flange on Nozzle [D13"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1181221) + 4 \cdot \text{abs}(0)111.125 \leq \pi \cdot 111.125^3 \cdot (4.763 - 2.547) + .500 \cdot 4.763$$

$$= 0.18899E+08 \leq 0.19830E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle D13" [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / ( \pi \cdot G^3 ) )$$

$$= \min( 4.763, 0.500 \cdot 4.763 + 4.763 - 18899476.0 / (\pi \cdot 111.125^3) )$$

$$= \min( 4.763, 2.762 )$$

$$= 2.762 \text{ MPa}$$

Evaluating Flange on Nozzle [B210"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$



$$= 16(13125631)+4\cdot\text{abs}(0)302.387 \leq \pi\cdot302.387^3[(4.763-2.547)+.500\cdot4.763]$$

$$= 0.21001\text{E}+09 \leq 0.39956\text{E}+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B210" [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16\cdot ME + 4\cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 4.763, 0.500 \cdot 4.763 + 4.763 - 210010064.0/(\pi \cdot 302.387^3) )$$

$$= \min( 4.763, 4.728 )$$

$$= 4.728 \text{ MPa}$$

Evaluating Flange on Nozzle [A33"]:

$$= 16\cdot ME + 4\cdot FE\cdot G \leq \pi\cdot G^3[(PR - PD) + FM \cdot PR]$$

$$= 16(1181304)+4\cdot\text{abs}(0)111.125 \leq \pi\cdot111.125^3[(4.763-2.547)+.500\cdot4.763]$$

$$= 0.18901\text{E}+08 \leq 0.19830\text{E}+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A33" [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16\cdot ME + 4\cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 4.763, 0.500 \cdot 4.763 + 4.763 - 18900834.0/(\pi \cdot 111.125^3) )$$

$$= \min( 4.763, 2.762 )$$

$$= 2.762 \text{ MPa}$$

Evaluating Flange on Nozzle [A518"]:

$$= 16\cdot ME + 4\cdot FE\cdot G \leq \pi\cdot G^3[(PR - PD) + FM \cdot PR]$$

$$= 16(42526884)+4\cdot\text{abs}(0)500.888 \leq \pi\cdot500.888^3[(3.971-2.547)+.500\cdot3.971]$$

$$= 0.68043\text{E}+09 \leq 0.13468\text{E}+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A518" [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16\cdot ME + 4\cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 3.971, 0.500 \cdot 3.971 + 3.971 - 680430080.0/(\pi \cdot 500.888^3) )$$

$$= \min( 3.971, 4.234 )$$

$$= 3.971 \text{ MPa}$$

Evaluating Flange on Nozzle [A28"]:

$$= 16\cdot ME + 4\cdot FE\cdot G \leq \pi\cdot G^3[(PR - PD) + FM \cdot PR]$$

$$= 16(8400389)+4\cdot\text{abs}(0)248.539 \leq \pi\cdot248.539^3[(3.971-2.547)+.500\cdot3.971]$$

$$= 0.13441\text{E}+09 \leq 0.16454\text{E}+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A28" [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16\cdot ME + 4\cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 3.971, 0.500 \cdot 3.971 + 3.971 - 134406176.0/(\pi \cdot 248.539^3) )$$

$$= \min( 3.971, 3.171 )$$

$$= 3.171 \text{ MPa}$$

Evaluating Flange on Nozzle [A68"]:

$$= 16\cdot ME + 4\cdot FE\cdot G \leq \pi\cdot G^3[(PR - PD) + FM \cdot PR]$$

$$= 16(8400389)+4\cdot\text{abs}(0)248.539 \leq \pi\cdot248.539^3[(3.971-2.547)+.500\cdot3.971]$$

$$= 0.13441\text{E}+09 \leq 0.16454\text{E}+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A68" [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16\cdot ME + 4\cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 3.971, 0.500 \cdot 3.971 + 3.971 - 134406176.0/(\pi \cdot 248.539^3) )$$

$$= \min( 3.971, 3.171 )$$

$$= 3.171 \text{ MPa}$$

Evaluating Flange on Nozzle [A116"]:

$$= 16\cdot ME + 4\cdot FE\cdot G \leq \pi\cdot G^3[(PR - PD) + FM \cdot PR]$$

$$= 16(33601524)+4\cdot\text{abs}(0)442.976 \leq \pi\cdot442.976^3[(3.971-2.547)+.500\cdot3.971]$$

$$= 0.53762\text{E}+09 \leq 0.93157\text{E}+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A116" [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16\cdot ME + 4\cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 3.971, 0.500 \cdot 3.971 + 3.971 - 537624384.0/(\pi \cdot 442.976^3) )$$

$$= \min( 3.971, 3.989 )$$

$$= 3.971 \text{ MPa}$$

Evaluating Flange on Nozzle [A44"]:

$$= 16\cdot ME + 4\cdot FE\cdot G \leq \pi\cdot G^3[(PR - PD) + FM \cdot PR]$$

$$= 16(2100096)+4\cdot\text{abs}(0)135.001 \leq \pi\cdot135.001^3[(7.943-2.547)+.500\cdot7.943]$$

$$= 0.33602\text{E}+08 \leq 0.72428\text{E}+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A44" [Pmax]:



$$\begin{aligned}
 &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\
 &= \min( 7.943, 0.500 \cdot 7.943 + 7.943 - 33601544.0 / (\pi \cdot 135.001^3) ) \\
 &= \min( 7.943, 7.568 ) \\
 &= 7.568 \text{ MPa}
 \end{aligned}$$

Evaluating Flange on Nozzle [B118"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [ (PR - PD) + FM \cdot PR ] \\
 &= 16(42523808) + 4 \cdot \text{abs}(0) 500.888 \leq \pi \cdot 500.888^3 [ (3.971 - 2.547) + .500 \cdot 3.971 ] \\
 &= 0.68038\text{E}+09 \leq 0.13468\text{E}+10 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle B118" [Pmax]:

$$\begin{aligned}
 &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\
 &= \min( 3.971, 0.500 \cdot 3.971 + 3.971 - 680380928.0 / (\pi \cdot 500.888^3) ) \\
 &= \min( 3.971, 4.234 ) \\
 &= 3.971 \text{ MPa}
 \end{aligned}$$

Evaluating Flange on Nozzle [S12"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [ (PR - PD) + FM \cdot PR ] \\
 &= 16(524984) + 4 \cdot \text{abs}(0) 77.851 \leq \pi \cdot 77.851^3 [ (7.943 - 2.547) + .500 \cdot 7.943 ] \\
 &= 0.83997\text{E}+07 \leq 0.13890\text{E}+08 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle S12" [Pmax]:

$$\begin{aligned}
 &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\
 &= \min( 7.943, 0.500 \cdot 7.943 + 7.943 - 8399748.0 / (\pi \cdot 77.851^3) ) \\
 &= \min( 7.943, 6.249 ) \\
 &= 6.249 \text{ MPa}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 4.06666 Hz.

The Natural Frequency for the Vessel (Ope...) is 3.92072 Hz.

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## Wind Load Calculation Results:

### Input Values:

|                                         |                       |
|-----------------------------------------|-----------------------|
| Wind Design Code                        | ASCE/SEI 7-22         |
| Wind Load Reduction Scale Factor        | 1.100                 |
| Basic Wind Speed                        | [V] 180 km/hr         |
| Surface Roughness Category              | D: Flat, unobstructed |
| Type of Surface                         | 0                     |
| Base Elevation                          | 300 mm                |
| Percent Wind for Hydrotest              | 33.0                  |
| Using User defined Wind Press. Vs Elev. | N                     |
| Height of Hill or Escarpment H or Hh    | 0 mm                  |
| Distance Upwind of Crest Lh             | 0 mm                  |
| Distance from Crest to the Vessel x     | 0 mm                  |
| Type of Terrain ( Hill, Escarpment )    | Flat                  |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100                |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000                |
| Damping Factor (Beta) for Wind (Filled) | 0.0000                |
| Site Elevation above Sea Level          | 0 mm                  |

### Wind Analysis Results

*The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.*

Static Gust-Effect Factor, Operating Case [G]:  
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$   
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.164 \cdot 0.949) / (1 + 1.7 \cdot 3.4 \cdot 0.164)))$   
 $= \min(0.85, 0.902)$   
 $= 0.850$

|                                         |          |
|-----------------------------------------|----------|
| Natural Frequency of Vessel (Operating) | 3.921 Hz |
| Natural Frequency of Vessel (Empty)     | 4.067 Hz |
| Natural Frequency of Vessel (Test)      | 4.373 Hz |

|                                    |              |
|------------------------------------|--------------|
| User Entered Importance Factor is  | 1.000        |
| Force Coefficient                  | [Cf] 0.781   |
| Structure Height to Diameter ratio | 5.873        |
| Height to top of Structure         | 10399.501 mm |

*This is classified as a rigid structure. Static analysis performed.*

### Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of  $[\alpha]$  and  $[Zg]$ :  
 Exposure Category: D from Table 26.11-1  
 $\alpha = 11.5$ ;  $Zg = 589788$  mm

Effective Height  $[z]$ :  
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$   
 $= 500.0 + 300.0 = 800.0$  mm  
 $= 2.625$  ft. Imperial Units

Velocity Pressure coefficient evaluated at height  $z$   $[Kz]$ :  
 Because  $z$  (2.625 ft.)  $< 15$  ft.  
 $= 2.41(15 / Zg)^{2/\alpha}$   
 $= 2.41(15 / 1935.0)^{2/11.5}$   
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor  $[Kd]$ :  
 $= 1$ , per Table 26.6-1

As there is No Hill Present:  $[Kzt]$ :



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [1.587] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.781 \cdot 27.7$$

$$= 609.7 \text{ lbs. } [2712.2] \text{ N}$$

| Element | Hgt (z)<br>mm | K1    | K2    | K3    | Kz    | Kzt   | qz<br>kPa |
|---------|---------------|-------|-------|-------|-------|-------|-----------|
| Skirt-2 | 800.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 1.587     |
| Skirt-1 | 1800.0        | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 1.587     |
| Btmhead | 2325.0        | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 1.587     |
| Shell-1 | 3850.0        | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 1.587     |
| Shell-2 | 6850.0        | 0.000 | 0.000 | 0.000 | 1.110 | 1.000 | 1.703     |
| Shell-3 | 9250.0        | 0.000 | 0.000 | 0.000 | 1.170 | 1.000 | 1.794     |
| Tophead | 10385.2       | 0.000 | 0.000 | 0.000 | 1.194 | 1.000 | 1.831     |

## Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

### Nomenclature

Cf - Correction factor for natural frequency  
 D - Average outer insulated diameter of vessel, mm  
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable  
 Dr - Average outer insulated of top half of vessel, mm  
 f - Natural frequency of vibration (Hertz)  
 f1 - Natural frequency of bare vessel based on a unit value of (D/L<sup>2</sup>)(10<sup>4</sup>)  
 L - Total height of structure mm  
 Lc - Total length of conical section(s) of vessel, mm  
 tb - Uncorroded plate thickness at bottom of vessel, mm  
 V30 - Design Wind Speed provided by user, km/hr  
 Vc - Critical wind velocity, km/hr  
 Vw - Maximum wind speed at top of structure, km/hr  
 W - Total corroded weight of structure, N  
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N  
 Z - Maximum amplitude of vibration at top of vessel, mm  
 D1 - Logarithmic decrement ( taken as 0.03 for Welded Structures )  
 Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)  
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5  
 :: 0.0/9900.0 = 0.0

:: [ D / L<sup>2</sup> ] • 10<sup>4</sup> < 8 (English Units, ft.)  
 :: [ 5.9/32.48<sup>2</sup> ] • 10<sup>4</sup> = 55.881 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:

$$= W / ( L \cdot Dr^2 )$$

$$= 459686 / ( 9900.0 \cdot 1779.313^2 )$$

$$= 0.14666\text{E-}04$$

Since Vp is > 0.000004 no further vibration analysis is required !

## Platform Load Calculations



| ID            | Wind Area<br>mm <sup>2</sup> | Elevation<br>mm | Pressure<br>kPa | Force<br>N | Cf   |
|---------------|------------------------------|-----------------|-----------------|------------|------|
| PLAT:[1 OF 1] | 8503040.00                   | 2550.00         | 1.59            | 13495.16   | 2.00 |
| PLAT:[1 OF 1] | 8550720.00                   | 6850.00         | 1.70            | 14559.38   | 2.00 |
| PLAT:[1 OF 1] | 18855500.00                  | 10650.00        | 1.84            | 34666.44   | 2.00 |

#### Wind Loads on Masses/Equipment/Piping

| ID          | Wind Area<br>mm <sup>2</sup> | Elevation<br>mm | Pressure<br>kPa | Force<br>N |
|-------------|------------------------------|-----------------|-----------------|------------|
| B2Pipewt    | 0.00                         | 2750.00         | 1.59            | 0.00       |
| A5Pipewt    | 0.00                         | 7350.00         | 1.72            | 0.00       |
| Clipwt      | 0.00                         | 5350.00         | 1.64            | 0.00       |
| A1Pipewt    | 0.00                         | 8800.00         | 1.78            | 0.00       |
| A2Pipewt    | 0.00                         | 8350.00         | 1.76            | 0.00       |
| A6Pipewt    | 0.00                         | 8350.00         | 1.76            | 0.00       |
| Lifting lug | 0.00                         | 9350.00         | 1.80            | 0.00       |
| Demister    | 0.00                         | 8350.00         | 1.76            | 0.00       |
| B1pipe wt   | 0.00                         | 10150.00        | 1.82            | 0.00       |
| Topdavit    | 0.00                         | 10150.00        | 1.82            | 0.00       |
| PSVwt       | 0.00                         | 10550.00        | 1.84            | 0.00       |

The Natural Frequency for the Vessel (Ope...) is 3.92072 Hz.

#### Wind Load Calculation:

| From | To | Wind<br>Height<br>mm | Wind<br>Diameter<br>mm | Wind<br>Area<br>mm <sup>2</sup> | Wind<br>Pressure<br>kPa | Element<br>Wind Load<br>N |
|------|----|----------------------|------------------------|---------------------------------|-------------------------|---------------------------|
| 10   | 20 | 800                  | 2573.46                | 2573460                         | 1.58714                 | 2983.37                   |
| 20   | 30 | 1800                 | 2573.46                | 2573460                         | 1.58714                 | 2983.37                   |
| 30   | 40 | 2325                 | 2790.72                | 139536                          | 1.58714                 | 161.762                   |
| 40   | 50 | 3850                 | 2793.78                | 8381340                         | 1.58714                 | 19548.6                   |
| 50   | 60 | 6850                 | 2793.78                | 8381340                         | 1.70275                 | 26439.5                   |
| 60   | 70 | 9250                 | 2793.78                | 5028804                         | 1.79406                 | 6589.9                    |
| 70   | 80 | 10385.2              | 2790.72                | 1234353                         | 1.83055                 | 40755.8                   |

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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## Earthquake Load Calculation Results:

### Earthquake Load Calculation:

#### Input Values:

|                                  |               |
|----------------------------------|---------------|
| Seismic Design Code              | ASCE/SEI 7-22 |
| Importance Factor                | 1.500         |
| Force Modification Factor R      | 1.500         |
| Site Class                       | D             |
| Component Elevation Ratio        | z/h           |
| Amplification Factor             | Ap            |
| Force Factor                     | 0.000         |
| Consider Vertical Acceleration   | Yes           |
| Minimum Acceleration Multiplier  | 0.000         |
| User Value of Sds (used if > 0 ) | 0.000         |
| User Value of Sd1 (used if > 0 ) | 0.000         |
| Component resonance duct. fact.  | [CAR]         |
| Component strength factor        | [Rpo]         |
| Overstrength factor              | [Omega0]      |
| Response Acceleration Factor     | [SMS]         |
| Response Acceleration Factor     | [SM1]         |
| Building Importance Factor       | [Ie]          |
| Site Latitude                    | 0.000 deg     |
| Site Longitude                   | 0.000 deg     |

#### Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:  
 =  $C_t \cdot h_{nx}$  where  $C_t = 0.020$ ,  $x = 0.75$  and  $h_n$  = Structural Height (ft.)  
 =  $0.020(33.83370.75)$   
 = 0.281 seconds

The Coefficient  $C_u$  from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:  
 = ( 1/Natural Frequency ) = ( 1/3.921 )  
 = 0.255

Check the Value of T which is the smaller of  $C_u \cdot T_a$  and T:  
 = Minimum Value of (  $1.7 \cdot 0.281$ , 0.255 ) per 12.8.2  
 = 0.255

Seismic Response Coefficient per equation 12.8-2 [Cs]:  
 =  $Sds / ( R / I )$   
 =  $0.217 / ( 1.5 / 1.5 )$   
 = 0.217

Maximum value of  $C_s$  per equation 12.8-3 [Cs]:  
 =  $Sd1 / ( T( R / I_e ) )$   
 =  $0.093 / ( 0.255( 1.5 / 1.5 ) )$   
 = 0.363

Minimum value of  $C_s$  per equation 15.4-1 [Cs]:  
 =  $\max( ( 0.044 \cdot SDS \cdot I_e ), 0.030 )$   
 =  $\max( ( 0.044 \cdot 0.2173 \cdot 1.5 ), 0.030 )$   
 = 0.030

Total Base Shear [V]:  
 =  $C_s \cdot W$  (Equation 12.8-1):  
 =  $0.2173 \cdot 466769.8$   
 = 101444.633 N

Vertical load per 12.4-4, [Ev]:  
 =  $0.2 \cdot Sds \cdot W$   
 =  $0.2 \cdot 0.217 \cdot 466770$

= 20288.9 N

Distribute the Base shear force to each element according to the equations  
 $F_x = C_{vx} * V$  (eqn. 12.8-11 ) and the vertical distribution factor  
 $C_{vx} = W_x * h_x^k / ( \text{Sum of } W_i * h_i^k )$  and k is an exponent which is related  
 to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 3.92072 Hz.

#### Earthquake Load Calculation:

| From | To | Earthquake<br>Height<br>mm | Earthquake<br>Weight<br>N | Element<br>Ope Load<br>N | Element<br>Emp Load<br>N | Element<br>Vertical Load<br>N |
|------|----|----------------------------|---------------------------|--------------------------|--------------------------|-------------------------------|
| 10   | 20 | 500                        | 11528.8                   | 179.773                  | 175.124                  | 20288.9                       |
| 20   | 30 | 1500                       | 5965.98                   | 279.09                   | 271.872                  | 19787.8                       |
| 30   | 40 | 2025                       | 10766.1                   | 679.919                  | 377.383                  | 19528.5                       |
| 40   | 50 | 3550                       | 103077                    | 11412                    | 6233.83                  | 19060.5                       |
| 50   | 60 | 6550                       | 120476                    | 24610.1                  | 10389.6                  | 14580.1                       |
| 60   | 70 | 8950                       | 66434.3                   | 18543.3                  | 8321.12                  | 9343.44                       |
| 70   | 80 | 9875                       | 148522                    | 45740.5                  | 36023.2                  | 6455.76                       |

The following table is for the Operating Case.

**Wind/Earthquake Shear, Bending:**

| From | To | Distance to Support<br>mm | Cumulative<br>Wind Shear<br>N | Earthquake<br>Shear<br>N | Wind<br>Bending<br>N-mm | Earthquake<br>Bending<br>N-mm |
|------|----|---------------------------|-------------------------------|--------------------------|-------------------------|-------------------------------|
| 10   | 20 | 500                       | 99462.2                       | 101445                   | 719171840               | 831195456                     |
| 20   | 30 | 1500                      | 96478.9                       | 101265                   | 621161600               | 729799616                     |
| 30   | 40 | 2025                      | 93495.5                       | 100986                   | 526135872               | 628633280                     |
| 40   | 50 | 3550                      | 93333.7                       | 100306                   | 521463296               | 623598976                     |
| 50   | 60 | 6550                      | 73785.2                       | 88893.9                  | 270683328               | 339684384                     |
| 60   | 70 | 8950                      | 47345.7                       | 64283.8                  | 88913344                | 109824704                     |
| 70   | 80 | 9875                      | 40755.8                       | 45740.5                  | 9589827                 | 10762714                      |

**Note:**

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;  
 Wind: 1.100; Earthquake: 1.000.

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### Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

#### Wind Load Deflection:

| From | To | Cumulative Shear N | Centroid Deflection mm | Elem. End Deflection mm | Elem. Ang. Rotation |
|------|----|--------------------|------------------------|-------------------------|---------------------|
| 10   | 20 | 99462.2            | 0.018242               | 0.071264                | 0.00013913          |
| 20   | 30 | 96478.9            | 0.15654                | 0.27159                 | 0.00025822          |
| 30   | 40 | 93495.5            | 0.27807                | 0.28462                 | 0.00026288          |
| 40   | 50 | 93333.7            | 0.76812                | 1.39776                 | 0.00045833          |
| 50   | 60 | 73785.2            | 2.12937                | 2.92526                 | 0.00054486          |
| 60   | 70 | 47345.7            | 3.4207                 | 3.92261                 | 0.00055933          |
| 70   | 80 | 40755.8            | 3.93659                | 3.95058                 | 0.00055934          |

#### Critical Wind Velocity for Tower Vibration:

| From | To | 1st Crit. Wind Speed km/hr | 2nd Crit. Wind Speed km/hr |
|------|----|----------------------------|----------------------------|
| 10   | 20 | 181.128                    | 1132.05                    |
| 20   | 30 | 181.128                    | 1132.05                    |
| 30   | 40 | 196.419                    | 1227.62                    |
| 40   | 50 | 196.634                    | 1228.96                    |
| 50   | 60 | 196.634                    | 1228.96                    |
| 60   | 70 | 196.634                    | 1228.96                    |
| 70   | 80 | 196.419                    | 1227.62                    |

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)  
 Allowable deflection : 49.500 Actual deflection : 3.951 mm

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**Longitudinal Stress Constants:**

| From | To | Metal Area<br>New<br>mm <sup>2</sup> | Metal Area<br>Corroded<br>mm <sup>2</sup> | Section Modulus<br>New<br>mm <sup>3</sup> | Section Modulus<br>Corroded<br>mm <sup>3</sup> |
|------|----|--------------------------------------|-------------------------------------------|-------------------------------------------|------------------------------------------------|
| 10   | 20 | 73362.7                              | 65561.2                                   | 30339706                                  | 27161784                                       |
| 20   | 30 | 73362.7                              | 65561.2                                   | 30339706                                  | 27161784                                       |
| 30   | 40 | 89029.8                              | 81247.4                                   | 36732428                                  | 33581188                                       |
| 40   | 50 | 94323.1                              | 86540.6                                   | 38917344                                  | 35769840                                       |
| 50   | 60 | 94323.1                              | 86540.6                                   | 38917344                                  | 35769840                                       |
| 60   | 70 | 94323.1                              | 86540.6                                   | 38917344                                  | 35769840                                       |
| 70   | 80 | 89029.8                              | 81247.4                                   | 36732428                                  | 33581188                                       |

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**Longitudinal Allowable Stresses:**

| From | To | Tensile<br>MPa | Hydrotest<br>Tensile<br>MPa | Compressive<br>MPa | Hydrotest<br>Compressive<br>MPa |
|------|----|----------------|-----------------------------|--------------------|---------------------------------|
| 10   | 20 | 90.7184        | 154.977                     | -122.478           | -125.567                        |
| 20   | 30 | 115.918        | 196.556                     | -122.478           | -125.567                        |
| 30   | 40 | 165.6          | 280.8                       | -128.155           | -130.404                        |
| 40   | 50 | 165.597        | 280.795                     | -129.662           | -131.708                        |
| 50   | 60 | 165.597        | 280.795                     | -129.662           | -131.708                        |
| 60   | 70 | 165.597        | 280.795                     | -129.662           | -131.708                        |
| 70   | 80 | 165.597        | 280.795                     | -128.155           | -130.404                        |

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## Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

### Longitudinal Pressure Stresses due to:

| From | To | Longitudinal Stress<br>Internal Pressure<br>MPa | Longitudinal Stress<br>External Pressure<br>MPa | Longitudinal Stress<br>Hydrotest Pressure<br>MPa |
|------|----|-------------------------------------------------|-------------------------------------------------|--------------------------------------------------|
| 10   | 20 | ...                                             | ...                                             | ...                                              |
| 20   | 30 | ...                                             | ...                                             | ...                                              |
| 30   | 40 | 66.1532                                         | ...                                             | 79.6703                                          |
| 40   | 50 | 62.1136                                         | ...                                             | 75.2074                                          |
| 50   | 60 | 62.1136                                         | ...                                             | 75.2074                                          |
| 60   | 70 | 62.1136                                         | ...                                             | 75.2074                                          |
| 70   | 80 | 66.1532                                         | ...                                             | 79.6703                                          |

### Longitudinal Stresses due to Weight Loads for these Conditions:

| From | To | Wght. Str.<br>Empty<br>MPa | Wght. Str.<br>Operating<br>MPa | Wght. Str.<br>Hydrotest<br>MPa | Wght. Str.<br>Emp. Mom.<br>MPa | Wght. Str.<br>Opr. Mom.<br>MPa |
|------|----|----------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| 10   | 20 | -5.75602                   | -7.12021                       | ...                            | 2.91132                        | 2.91132                        |
| 20   | 30 | -5.58015                   | -6.94434                       | ...                            | 2.91132                        | 2.91132                        |
| 30   | 40 | -4.42938                   | -4.42938                       | ...                            | 2.35479                        | 2.35479                        |
| 40   | 50 | -4.08757                   | -4.08757                       | ...                            | 2.21071                        | 2.21071                        |
| 50   | 60 | -3.39048                   | -3.39048                       | ...                            | 2.2107                         | 2.2107                         |
| 60   | 70 | -2.48409                   | -2.48409                       | ...                            | 0.10087E-04                    | 0.10087E-04                    |
| 70   | 80 | -1.82818                   | -1.82818                       | ...                            | ...                            | ...                            |

### Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

| From | To | Wght. Str.<br>Hyd. Mom.<br>MPa | Bend. Str.<br>Oper. Wind<br>MPa | Bend. Str.<br>Oper. Equ.<br>MPa | Bend. Str.<br>Hyd. Wind<br>MPa | Bend. Str.<br>Hyd. Equ.<br>MPa |
|------|----|--------------------------------|---------------------------------|---------------------------------|--------------------------------|--------------------------------|
| 10   | 20 | ...                            | 26.4689                         | 30.5919                         | ...                            | ...                            |
| 20   | 30 | ...                            | 22.8616                         | 26.86                           | ...                            | ...                            |
| 30   | 40 | ...                            | 15.6626                         | 18.7138                         | ...                            | ...                            |
| 40   | 50 | ...                            | 14.5736                         | 17.4281                         | ...                            | ...                            |
| 50   | 60 | ...                            | 7.56494                         | 9.49335                         | ...                            | ...                            |
| 60   | 70 | ...                            | 2.48491                         | 3.06933                         | ...                            | ...                            |
| 70   | 80 | ...                            | 0.28548                         | 0.3204                          | ...                            | ...                            |

### Longitudinal Stresses due to these Conditions:

| From | To | Vortex Shedding<br>Operating Case<br>MPa | Vortex Shedding<br>Empty Case<br>MPa | Vortex Shedding<br>Test Case<br>MPa | Earthquake<br>Empty Case<br>MPa |
|------|----|------------------------------------------|--------------------------------------|-------------------------------------|---------------------------------|
| 10   | 20 | ...                                      | ...                                  | ...                                 | 19.4855                         |
| 20   | 30 | ...                                      | ...                                  | ...                                 | 17.2136                         |
| 30   | 40 | ...                                      | ...                                  | ...                                 | 12.0921                         |
| 40   | 50 | ...                                      | ...                                  | ...                                 | 11.2667                         |
| 50   | 60 | ...                                      | ...                                  | ...                                 | 6.41435                         |
| 60   | 70 | ...                                      | ...                                  | ...                                 | 2.25918                         |
| 70   | 80 | ...                                      | ...                                  | ...                                 | 0.25233                         |

### Longitudinal Stresses due to Applied Axial Forces:



| From | To | Longitudinal Stress<br>Y Forces Wind<br>MPa | Longitudinal Stress<br>Y Forces Seismic<br>MPa |
|------|----|---------------------------------------------|------------------------------------------------|
| 10   | 20 | ...                                         | 0.30949                                        |
| 20   | 30 | ...                                         | 0.30185                                        |
| 30   | 40 | ...                                         | 0.24038                                        |
| 40   | 50 | ...                                         | 0.22027                                        |
| 50   | 60 | ...                                         | 0.16849                                        |
| 60   | 70 | ...                                         | 0.10798                                        |
| 70   | 80 | ...                                         | 0.079465                                       |

**Longitudinal Stresses due to User Forces and Moments:**

| From | To | Wind For/Mom<br>Corroded<br>MPa | Earthquake For/Mom<br>Corroded<br>MPa | Wind For/Mom<br>No Corrosion<br>MPa | Earthquake For/Mom<br>No Corrosion<br>MPa |
|------|----|---------------------------------|---------------------------------------|-------------------------------------|-------------------------------------------|
| 10   | 20 | ...                             | ...                                   | ...                                 | ...                                       |
| 20   | 30 | ...                             | ...                                   | ...                                 | ...                                       |
| 30   | 40 | ...                             | ...                                   | ...                                 | ...                                       |
| 40   | 50 | ...                             | ...                                   | ...                                 | ...                                       |
| 50   | 60 | ...                             | ...                                   | ...                                 | ...                                       |
| 60   | 70 | ...                             | ...                                   | ...                                 | ...                                       |
| 70   | 80 | ...                             | ...                                   | ...                                 | ...                                       |

**Bending Stresses due to Blast Load:**

| From | To | Bending<br>Stress<br>MPa |
|------|----|--------------------------|
| 10   | 20 | ...                      |
| 20   | 30 | ...                      |
| 30   | 40 | ...                      |
| 40   | 50 | ...                      |
| 50   | 60 | ...                      |
| 60   | 70 | ...                      |
| 70   | 80 | ...                      |



## Stress Combination Load Cases for Vertical Vessels:

### Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure  
 EP = Longitudinal Stress due to External Pressure  
 HP = Longitudinal Stress due to Hydrotest Pressure  
 NP = No Pressure  
 EW = Longitudinal Stress due to Weight (No Liquid)  
 OW = Longitudinal Stress due to Weight (Operating)  
 HW = Longitudinal Stress due to Weight (Hydrotest)  
 WI = Bending Stress due to Wind Moment (Operating)  
 EQ = Bending Stress due to Earthquake Moment (Operating)  
 EE = Bending Stress due to Earthquake Moment (Empty)  
 HI = Bending Stress due to Wind Moment (Hydrotest)  
 HE = Bending Stress due to Earthquake Moment (Hydrotest)  
 WE = Bending Stress due to Wind Moment (Empty) (no CA)  
 WF = Bending Stress due to Wind Moment (Filled) (no CA)  
 CW = Longitudinal Stress due to Weight (Empty) (no CA)  
 VO = Bending Stress due to Vortex Shedding Loads ( Ope )  
 VE = Bending Stress due to Vortex Shedding Loads ( Emp )  
 VF = Bending Stress due to Vortex Shedding Loads ( Test No CA. )  
 FW = Axial Stress due to Vertical Forces for the Wind Case  
 FS = Axial Stress due to Vertical Forces for the Seismic Case  
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded  
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded  
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded  
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded  
 BL = Bending Stress due to Blast Loads, Corroded

### General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (\*) in the final column denotes overstress.

#### Analysis of Load Case 1 : NP+EW+WI+FW+BW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 23.62          | 90.72             | -35.14       | 122.48            | 0.2604      | 0.2869      |
| 20        | 20.19          | 115.92            | -31.35       | 122.48            | 0.1742      | 0.2560      |
| 30        | 13.59          | 165.60            | -22.45       | 128.15            | 0.0821      | 0.1752      |
| 40        | 12.70          | 165.60            | -20.87       | 129.66            | 0.0767      | 0.1610      |
| 50        | 6.39           | 165.60            | -13.17       | 129.66            | 0.0386      | 0.1015      |
| 60        | 0.00           | 165.60            | -4.97        | 129.66            | 0.0000      | 0.0383      |
| 70        |                | 165.60            | -2.11        | 128.15            |             | 0.0165      |

#### Analysis of Load Case 2 : NP+EW+EE+FS+BS

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 16.95          | 90.72             | -27.84       | 122.48            | 0.1868      | 0.2273      |
| 20        | 14.85          | 115.92            | -25.40       | 122.48            | 0.1281      | 0.2074      |
| 30        | 10.26          | 165.60            | -18.64       | 128.15            | 0.0619      | 0.1454      |
| 40        | 9.61           | 165.60            | -17.34       | 129.66            | 0.0580      | 0.1338      |
| 50        | 5.40           | 165.60            | -11.85       | 129.66            | 0.0326      | 0.0914      |
| 60        |                | 165.60            | -4.64        | 129.66            |             | 0.0357      |
| 70        |                | 165.60            | -2.00        | 128.15            |             | 0.0156      |

#### Analysis of Load Case 3 : NP+OW+WI+FW+BW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|



|    |       |        |        |        |        |        |
|----|-------|--------|--------|--------|--------|--------|
| 10 | 22.26 | 90.72  | -36.50 | 122.48 | 0.2454 | 0.2980 |
| 20 | 18.83 | 115.92 | -32.72 | 122.48 | 0.1624 | 0.2671 |
| 30 | 13.59 | 165.60 | -22.45 | 128.15 | 0.0821 | 0.1752 |
| 40 | 12.70 | 165.60 | -20.87 | 129.66 | 0.0767 | 0.1610 |
| 50 | 6.39  | 165.60 | -13.17 | 129.66 | 0.0386 | 0.1015 |
| 60 | 0.00  | 165.60 | -4.97  | 129.66 | 0.0000 | 0.0383 |
| 70 |       | 165.60 | -2.11  | 128.15 |        | 0.0165 |

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 26.69          | 90.72             | -40.31       | 122.48            | 0.2942      | 0.3292      |
| 20        | 23.13          | 115.92            | -36.41       | 122.48            | 0.1995      | 0.2973      |
| 30        | 16.88          | 165.60            | -25.26       | 128.15            | 0.1019      | 0.1971      |
| 40        | 15.77          | 165.60            | -23.51       | 129.66            | 0.0952      | 0.1813      |
| 50        | 8.48           | 165.60            | -14.93       | 129.66            | 0.0512      | 0.1151      |
| 60        | 0.69           | 165.60            | -5.45        | 129.66            | 0.0042      | 0.0420      |
| 70        |                | 165.60            | -2.07        | 128.15            |             | 0.0161      |

Analysis of Load Case 5 : NP+HW+HI

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 0.00           | 154.98            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 20        | 0.00           | 196.56            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 30        | 0.00           | 280.80            | 0.00         | 130.40            | 0.0000      | -0.0000     |
| 40        | 0.00           | 280.79            | 0.00         | 131.71            | 0.0000      | -0.0000     |
| 50        | 0.00           | 280.79            | 0.00         | 131.71            | 0.0000      | -0.0000     |
| 60        | 0.00           | 280.79            | 0.00         | 131.71            | 0.0000      | -0.0000     |
| 70        | 0.00           | 280.79            | 0.00         | 130.40            | 0.0000      | -0.0000     |

Analysis of Load Case 6 : NP+HW+HE

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 0.00           | 154.98            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 20        | 0.00           | 196.56            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 30        | 0.00           | 280.80            | 0.00         | 130.40            | 0.0000      | -0.0000     |
| 40        | 0.00           | 280.79            | 0.00         | 131.71            | 0.0000      | -0.0000     |
| 50        | 0.00           | 280.79            | 0.00         | 131.71            | 0.0000      | -0.0000     |
| 60        | 0.00           | 280.79            | 0.00         | 131.71            | 0.0000      | -0.0000     |
| 70        | 0.00           | 280.79            | 0.00         | 130.40            | 0.0000      | -0.0000     |

Analysis of Load Case 7 : IP+OW+WI+FW+BW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 22.26          | 90.72             | -36.50       | 122.48            | 0.2454      | 0.2980      |
| 20        | 18.83          | 115.92            | -32.72       | 122.48            | 0.1624      | 0.2671      |
| 30        | 79.74          | 165.60            |              | 128.15            | 0.4815      |             |
| 40        | 74.81          | 165.60            |              | 129.66            | 0.4518      |             |
| 50        | 68.50          | 165.60            |              | 129.66            | 0.4136      |             |
| 60        | 62.11          | 165.60            |              | 129.66            | 0.3751      |             |
| 70        | 64.61          | 165.60            |              | 128.15            | 0.3902      |             |

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 26.69          | 90.72             | -40.31       | 122.48            | 0.2942      | 0.3292      |
| 20        | 23.13          | 115.92            | -36.41       | 122.48            | 0.1995      | 0.2973      |
| 30        | 83.03          | 165.60            |              | 128.15            | 0.5014      |             |
| 40        | 77.89          | 165.60            |              | 129.66            | 0.4703      |             |
| 50        | 70.60          | 165.60            |              | 129.66            | 0.4263      |             |
| 60        | 62.81          | 165.60            |              | 129.66            | 0.3793      |             |
| 70        | 64.72          | 165.60            |              | 128.15            | 0.3909      |             |

Analysis of Load Case 9 : EP+OW+WI+FW+BW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|



| Node | Stress | Stress | Stress | Stress | Ratio  | Ratio  |
|------|--------|--------|--------|--------|--------|--------|
| 10   | 22.26  | 90.72  | -36.50 | 122.48 | 0.2454 | 0.2980 |
| 20   | 18.83  | 115.92 | -32.72 | 122.48 | 0.1624 | 0.2671 |
| 30   | 13.59  | 165.60 | -22.45 | 128.15 | 0.0821 | 0.1752 |
| 40   | 12.70  | 165.60 | -20.87 | 129.66 | 0.0767 | 0.1610 |
| 50   | 6.39   | 165.60 | -13.17 | 129.66 | 0.0386 | 0.1015 |
| 60   | 0.00   | 165.60 | -4.97  | 129.66 | 0.0000 | 0.0383 |
| 70   |        | 165.60 | -2.11  | 128.15 |        | 0.0165 |

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 26.69          | 90.72             | -40.31       | 122.48            | 0.2942      | 0.3292      |
| 20        | 23.13          | 115.92            | -36.41       | 122.48            | 0.1995      | 0.2973      |
| 30        | 16.88          | 165.60            | -25.26       | 128.15            | 0.1019      | 0.1971      |
| 40        | 15.77          | 165.60            | -23.51       | 129.66            | 0.0952      | 0.1813      |
| 50        | 8.48           | 165.60            | -14.93       | 129.66            | 0.0512      | 0.1151      |
| 60        | 0.69           | 165.60            | -5.45        | 129.66            | 0.0042      | 0.0420      |
| 70        |                | 165.60            | -2.07        | 128.15            |             | 0.0161      |

Analysis of Load Case 11 : HP+HW+HI

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 0.00           | 154.98            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 20        | 0.00           | 196.56            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 30        | 79.67          | 280.80            |              | 130.40            | 0.2837      |             |
| 40        | 75.21          | 280.79            |              | 131.71            | 0.2678      |             |
| 50        | 75.21          | 280.79            |              | 131.71            | 0.2678      |             |
| 60        | 75.21          | 280.79            |              | 131.71            | 0.2678      |             |
| 70        | 79.67          | 280.79            |              | 130.40            | 0.2837      |             |

Analysis of Load Case 12 : HP+HW+HE

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 0.00           | 154.98            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 20        | 0.00           | 196.56            | 0.00         | 125.57            | 0.0000      | -0.0000     |
| 30        | 79.67          | 280.80            |              | 130.40            | 0.2837      |             |
| 40        | 75.21          | 280.79            |              | 131.71            | 0.2678      |             |
| 50        | 75.21          | 280.79            |              | 131.71            | 0.2678      |             |
| 60        | 75.21          | 280.79            |              | 131.71            | 0.2678      |             |
| 70        | 79.67          | 280.79            |              | 130.40            | 0.2837      |             |

Analysis of Load Case 13 : IP+WE+EW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 90.72             | -8.67        | 122.48            |             | 0.0708      |
| 20        |                | 115.92            | -8.49        | 122.48            |             | 0.0693      |
| 30        | 64.08          | 165.60            |              | 128.15            | 0.3869      |             |
| 40        | 60.24          | 165.60            |              | 129.66            | 0.3638      |             |
| 50        | 60.93          | 165.60            |              | 129.66            | 0.3680      |             |
| 60        | 59.63          | 165.60            |              | 129.66            | 0.3601      |             |
| 70        | 64.33          | 165.60            |              | 128.15            | 0.3884      |             |

Analysis of Load Case 14 : IP+WF+CW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 90.72             | -6.36        | 122.48            |             | 0.0520      |
| 20        |                | 115.92            | -6.21        | 122.48            |             | 0.0507      |
| 30        | 62.11          | 165.60            |              | 128.15            | 0.3751      |             |
| 40        | 58.36          | 165.60            |              | 129.66            | 0.3524      |             |
| 50        | 59.00          | 165.60            |              | 129.66            | 0.3563      |             |
| 60        | 59.83          | 165.60            |              | 129.66            | 0.3613      |             |
| 70        | 64.48          | 165.60            |              | 128.15            | 0.3894      |             |



Analysis of Load Case 15 : IP+VO+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 90.72             | -10.03       | 122.48            |             | 0.0819      |
| 20        |                | 115.92            | -9.86        | 122.48            |             | 0.0805      |
| 30        | 64.08          | 165.60            |              | 128.15            | 0.3869      |             |
| 40        | 60.24          | 165.60            |              | 129.66            | 0.3638      |             |
| 50        | 60.93          | 165.60            |              | 129.66            | 0.3680      |             |
| 60        | 59.63          | 165.60            |              | 129.66            | 0.3601      |             |
| 70        | 64.33          | 165.60            |              | 128.15            | 0.3884      |             |

Analysis of Load Case 16 : IP+VE+EW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 90.72             | -8.67        | 122.48            |             | 0.0708      |
| 20        |                | 115.92            | -8.49        | 122.48            |             | 0.0693      |
| 30        | 64.08          | 165.60            |              | 128.15            | 0.3869      |             |
| 40        | 60.24          | 165.60            |              | 129.66            | 0.3638      |             |
| 50        | 60.93          | 165.60            |              | 129.66            | 0.3680      |             |
| 60        | 59.63          | 165.60            |              | 129.66            | 0.3601      |             |
| 70        | 64.33          | 165.60            |              | 128.15            | 0.3884      |             |

Analysis of Load Case 17 : NP+VO+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 90.72             | -10.03       | 122.48            |             | 0.0819      |
| 20        |                | 115.92            | -9.86        | 122.48            |             | 0.0805      |
| 30        |                | 165.60            | -6.78        | 128.15            |             | 0.0529      |
| 40        |                | 165.60            | -6.30        | 129.66            |             | 0.0486      |
| 50        |                | 165.60            | -5.60        | 129.66            |             | 0.0432      |
| 60        |                | 165.60            | -2.48        | 129.66            |             | 0.0192      |
| 70        |                | 165.60            | -1.83        | 128.15            |             | 0.0143      |

Analysis of Load Case 18 : FS+BS+IP+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 75.60             | -9.72        | 102.07            |             | 0.0953      |
| 20        |                | 96.60             | -9.55        | 102.07            |             | 0.0936      |
| 30        | 64.32          | 138.00            |              | 106.80            | 0.4661      |             |
| 40        | 60.46          | 138.00            |              | 108.05            | 0.4381      |             |
| 50        | 61.10          | 138.00            |              | 108.05            | 0.4428      |             |
| 60        | 59.74          | 138.00            |              | 108.05            | 0.4329      |             |
| 70        | 64.40          | 138.00            |              | 106.80            | 0.4667      |             |

Analysis of Load Case 19 : FS+BS+EP+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 75.60             | -9.72        | 102.07            |             | 0.0953      |
| 20        |                | 96.60             | -9.55        | 102.07            |             | 0.0936      |
| 30        |                | 138.00            | -6.54        | 106.80            |             | 0.0613      |
| 40        |                | 138.00            | -6.08        | 108.05            |             | 0.0563      |
| 50        |                | 138.00            | -5.43        | 108.05            |             | 0.0503      |
| 60        |                | 138.00            | -2.38        | 108.05            |             | 0.0220      |
| 70        |                | 138.00            | -1.75        | 106.80            |             | 0.0164      |

Analysis of Load Case 20 : BL+IP+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 90.72             | -10.03       | 122.48            |             | 0.0819      |
| 20        |                | 115.92            | -9.86        | 122.48            |             | 0.0805      |
| 30        | 64.08          | 165.60            |              | 128.15            | 0.3869      |             |
| 40        | 60.24          | 165.60            |              | 129.66            | 0.3638      |             |
| 50        | 60.93          | 165.60            |              | 129.66            | 0.3680      |             |
| 60        | 59.63          | 165.60            |              | 129.66            | 0.3601      |             |

70 64.33 165.60 128.15 0.3884

Absolute Maximum of the all of the Stress Ratio's 0.5014

Governing Element: Btmhead

Governing Load Case 8 : IP+OW+EQ+FS+BS

| Element Description | Long Pressure Stress | Weight + Axial Load Stress | Eccentric Moment Stress | Wind or Earthquake Stress | Combined Stress | Which Side |
|---------------------|----------------------|----------------------------|-------------------------|---------------------------|-----------------|------------|
| Skirt-2             | ...                  | -6.8                       | 2.9                     | 30.6                      | 26.7            | + Tens     |
| Skirt-2             | ...                  | -6.8                       | -2.9                    | -30.6                     | -40.3           | - Comp     |
| Skirt-1             | ...                  | -6.6                       | 2.9                     | 26.9                      | 23.1            | + Tens     |
| Skirt-1             | ...                  | -6.6                       | -2.9                    | -26.9                     | -36.4           | - Comp     |
| Btm_head            | 66.2                 | -4.2                       | 2.4                     | 18.7                      | 83.0            | + Tens     |
| Btm_head            | 66.2                 | -4.2                       | -2.4                    | -18.7                     | ...             | - Comp     |
| Shell-1             | 62.1                 | -3.9                       | 2.2                     | 17.4                      | 77.9            | + Tens     |
| Shell-1             | 62.1                 | -3.9                       | -2.2                    | -17.4                     | ...             | - Comp     |
| Shell-2             | 62.1                 | -3.2                       | 2.2                     | 9.5                       | 70.6            | + Tens     |
| Shell-2             | 62.1                 | -3.2                       | -2.2                    | -9.5                      | ...             | - Comp     |
| Shell-3             | 62.1                 | -2.4                       | ...                     | 3.1                       | 62.8            | + Tens     |
| Shell-3             | 62.1                 | -2.4                       | ...                     | -3.1                      | ...             | - Comp     |
| Top_head            | 66.2                 | -1.7                       | ...                     | 0.3                       | 64.7            | + Tens     |
| Top_head            | 66.2                 | -1.7                       | ...                     | -0.3                      | ...             | - Comp     |

### Skirt Support Analysis Per EN 13445-3 Section 16.12:

*The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.*

Item: Skirt-1

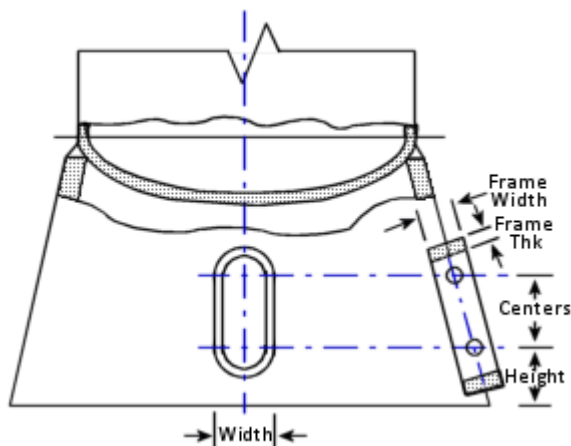
|                                       |         |                    |
|---------------------------------------|---------|--------------------|
| Total vessel (empty) weight           | Fg      | 360989.188 N       |
| Head Weight below section 2-2         | DeltaFg | 6134.296 N         |
| Weight of vessel contents (liquid)    | Ff      | 89430.344 N        |
| Vessel Gross Weight                   | Fg + Ff | 450419.531 N       |
| Skirt Mean Diameter                   | Dz      | 1669.500 mm        |
| Skirt Analysis Thickness (Corroded)   | ez      | 12.500 mm          |
| Skirt Height                          | H       | 1000.000 mm        |
| Additional Axial Force at section 1-1 | F1      | 0.000 N            |
| Moment at section 1-1                 | M1      | 628633344.000 N-mm |
| Applied Moment - Bottom of Skirt      | Mb      | 729799616.000 N-mm |
| Weight at Bottom of Skirt             | W       | 455240.969 N       |

### Structural Type B Skirt per EN 13445 Figure 16.12-2:

|                                 |    |             |
|---------------------------------|----|-------------|
| Type of Head:                   |    | Elliptical  |
| Head Mean Diameter at Skirt Top | Db | 1668.500 mm |
| Head Corroded Thickness         | eb | 15.500 mm   |

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 414.0 \text{ mm} \\
 K &= D_i / (2 * h_i) = 1653.0 / (2 * 414.0) = 1.9964 \\
 R &= D_i (0.44 * K + 0.02) = 1653.0 * (0.44 * 1.996 + 0.02) = 1485.0647 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 1653.0 * ((0.5 / 1.996) - 0.08) = 281.76 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

|                    |       |                                 |
|--------------------|-------|---------------------------------|
| Moment of Inertia  | Ixx = | 22841718784.000 mm <sup>4</sup> |
| Moment of Inertia  | Iyy = | 22841718784.000 mm <sup>4</sup> |
| Section Modulus    | W =   | 27363570.000 mm <sup>3</sup>    |
| Section Metal Area | A =   | 65561.117 mm <sup>2</sup>       |

### Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(0.22842\text{E}+11, 0.22842\text{E}+11) / 65561.117)} \\
 &= 590.257 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:



$$\begin{aligned} &= r_g / L \\ &= 590.257/1000.0 \\ &= 0.590 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end  
Controlling Moment [M4] = 0.62863E+09 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -0.62863E+09/.10945E+09 - 455240.969/65561.11719 \\ &= -12.686 \text{ MPa} \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +.62863E+09/.10945E+09 - 455240.969/65561.11719 \\ &= -1.203 \text{ MPa} \end{aligned}$$

#### 16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= 0.0 - 360989.188 - 89430.344 + 4 \cdot 0.62863E+09/1669.5 \\ &= 1055130 \text{ N} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= 0.0 - 360989.188 - 89430.344 - 4 \cdot 0.62863E+09/1669.5 \\ &= -1955968.5 \text{ N} \end{aligned}$$

##### 16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [ $\sigma_{1pm}$ ]:

$$\begin{aligned} &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (1055130 + 6134.296 + 89430.344) / (\pi \cdot 1668.5 \cdot 15.5) + 2.5 \cdot 1668.5 / (4 \cdot 15.5) \\ &= 81.44 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{1qm}$ ]:

$$\begin{aligned} &= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-0.2E+07 + 6134.296 + 89430.344) / (\pi \cdot 1668.5 \cdot 15.5) + 2.5 \cdot 1668.5 / (4 \cdot 15.5) \\ &= 44.38 \text{ MPa} \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [ $\sigma_{2pm}$  &  $\sigma_{2qm}$ ]:

$$\begin{aligned} &= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (89430.344 + 6134.296) / (\pi \cdot 1668.5 \cdot 15.5) + 2.5 \cdot 1668.5 / (4 \cdot 15.5) \\ &= 68.45 \text{ MPa} \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [ $\sigma_{3pm}$ ]:

$$\begin{aligned} &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\ &= 1055130 / (\pi \cdot 1669.5 \cdot 12.5) \\ &= 16.10 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{3qm}$ ]:

$$\begin{aligned} &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\ &= -1955968.500 / (\pi \cdot 1669.5 \cdot 12.5) \\ &= -29.84 \text{ MPa} \end{aligned}$$

##### 16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [ $\cos(\gamma)$ ]:

$$\begin{aligned} &= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b) \\ &= 1 - 0.5(1668.5 + 15.5 - 1669.5 + 12.5) / (281.76 + 15.5) \\ &= 0.9546 \end{aligned}$$



$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.955) = 17.334 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(15.5^2 + 12.5^2 + 2 \cdot 15.5 \cdot 12.5 \cdot 0.955)}$$

$$= 13.842 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 13.84 \cdot 1055129.5 = 14.6110 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot F_{zq} = 13.84 \cdot -1955968.5 = -27.0855 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left( \frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left( \frac{15.5}{12.5} \right)^2$$

$$= 0.542$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [ $\sigma_{1pb(o)}$  &  $\sigma_{2pb(o)}$ ]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.542 \cdot 6 \cdot 14611006 / (\pi \cdot 1668.5 \cdot 15.5^2)$$

$$= 37.74 \text{ MPa}$$

Stress [ $\sigma_{1qb(o)}$  &  $\sigma_{2qb(o)}$ ]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.542 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 1668.5 \cdot 15.5^2)$$

$$= -69.97 \text{ MPa}$$

Stress [ $\sigma_{3pb(o)}$ ]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.542 \cdot 6 \cdot 14611006 / (\pi \cdot 1669.5 \cdot 12.5^2)$$

$$= 58.00 \text{ MPa}$$

Stress [ $\sigma_{3qb(o)}$ ]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.542 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 1669.5 \cdot 12.5^2)$$

$$= -107.52 \text{ MPa}$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.61/65.689 = 0.0093$$

$$y = 125 \cdot eb/D_b = 125 \cdot 15.5/1668.5 = 1.1612$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [ $\alpha$ ]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.161 = 3.9678$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [ $\sigma_{1b(p)}$  &  $\sigma_{2b(p)}$ ]:

$$= (P + Ph) \cdot D_b \cdot ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (2.5 + 0.0) \cdot 1668.5 \cdot ((17.33/40.0) \cdot 3.968 - 1) / (4 \cdot 15.5)$$

$$= 48.40 \text{ MPa}$$

#### 16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [ $\sigma_{1pitot}$ ]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 81.442 - 37.743 + 48.4$$

$$= 92.10 \text{ MPa}$$

Stress [ $\sigma_{1potot}$ ]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 81.442 + 37.743 - 48.4$$

$$= 70.79 \text{ MPa}$$

Stress [ $\sigma_{1qitot}$ ]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 44.378 - 69.967 + 48.4$$



= 162.75 MPa

Stress [ $\sigma_{1qotot}$ ]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 44.378 + -69.967 - 48.4 \\ &= -73.99 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{2pitot}$ ]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 68.455 + 37.743 + 48.4 \\ &= 154.60 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{2potot}$ ]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 68.455 - 37.743 - 48.4 \\ &= -17.69 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{2qitot}$ ]:

$$\begin{aligned} &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 68.455 + -69.967 + 48.4 \\ &= 46.89 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{2qotot}$ ]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 68.455 - -69.967 - 48.4 \\ &= 90.02 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{3pitot}$ ]:

$$\begin{aligned} &= \sigma_{3pm} - \sigma_{3pb} \\ &= 16.095 - 58.0 \\ &= -41.90 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{3potot}$ ]:

$$\begin{aligned} &= \sigma_{3pm} + \sigma_{3pb} \\ &= 16.095 + 58.0 \\ &= 74.09 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{3qitot}$ ]:

$$\begin{aligned} &= \sigma_{3qm} - \sigma_{3qb} \\ &= -29.837 - -107.517 \\ &= 77.68 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{3qotot}$ ]:

$$\begin{aligned} &= \sigma_{3qm} + \sigma_{3qb} \\ &= -29.837 + -107.517 \\ &= -137.35 \text{ MPa} \end{aligned}$$

Allowable stresses:

For  $\sigma_{1pitot}$  and  $\sigma_{1potot}$ , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 138.0(3 - (1 / 1.5)) \cdot (81.442 / 137.997)^2 \\ &= 381.96 \text{ MPa} \end{aligned}$$

For  $\sigma_{1qitot}$  and  $\sigma_{1qotot}$ , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 138.0(3 - (1 / 1.5)) \cdot (44.378 / 137.997)^2 \\ &= 404.49 \text{ MPa} \end{aligned}$$

For  $\sigma_{2pitot}$  and  $\sigma_{2potot}$ , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 138.0(3 - (1 / 1.5)) \cdot (68.455 / 137.997)^2 \\ &= 391.36 \text{ MPa} \end{aligned}$$

For  $\sigma_{2qitot}$  and  $\sigma_{2qotot}$ , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\ &= 138.0(3 - (1 / 1.5)) \cdot (68.455 / 137.997)^2 \\ &= 391.36 \text{ MPa} \end{aligned}$$

For  $\sigma_{3pitot}$  &  $\sigma_{3potot}$ , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (16.095/137.997)^2)$$

$$= 412.75 \text{ MPa}$$

For  $\sigma_{3qitot}$  &  $\sigma_{3qotot}$ , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (-29.837/137.997)^2)$$

$$= 409.70 \text{ MPa}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

#### Summary of Stresses: (MPa)

| Stress  | Actual  | Allowed | Result | Equation |
|---------|---------|---------|--------|----------|
| Stot1pi | 92.10   | 381.96  | Pass   | 46       |
| Stot1po | 70.79   | 381.96  | Pass   | 47       |
| Stot1qi | 162.75  | 404.49  | Pass   | 48       |
| Stot1qo | -73.99  | 404.49  | Pass   | 49       |
| Stot2pi | 154.60  | 391.36  | Pass   | 50       |
| Stot2po | -17.69  | 391.36  | Pass   | 51       |
| Stot2qi | 46.89   | 391.36  | Pass   | 52       |
| Stot2qo | 90.02   | 391.36  | Pass   | 53       |
| Stot3pi | -41.90  | 412.75  | Pass   | 54       |
| Stot3po | 74.09   | 412.75  | Pass   | 55       |
| Stot3qi | 77.68   | 409.70  | Pass   | 56       |
| Stot3qo | -137.35 | 409.70  | Pass   | 57       |

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Platform(s) installed in the Shop.  
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

|                                                    |             |
|----------------------------------------------------|-------------|
| Center of Gravity of the Platforms                 | 8251.529 mm |
| Center of Gravity of the Liquid                    | 4857.645 mm |
| Center of Gravity of the Insulation                | 5950.000 mm |
| Center of Gravity of the Nozzles                   | 7807.830 mm |
| Center of Gravity of the Added Weights (Operating) | 8107.159 mm |
| Center of Gravity of the Added Weights (Empty)     | 7063.648 mm |
|                                                    |             |
| Center of Gravity of Bare Shell New and Cold       | 4946.085 mm |
| Center of Gravity of Bare Shell Corroded           | 4935.955 mm |
|                                                    |             |
| Vessel CG in the Operating Condition               | 6940.203 mm |
| Vessel CG in the Fabricated (Shop/Empty) Condition | 7098.144 mm |
| Vessel CG in the Test Condition                    | 6657.468 mm |

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#### Skirt Data :

|                                    |         |           |    |
|------------------------------------|---------|-----------|----|
| Skirt Outside Diameter at Base     | SOD, D6 | 1682.0000 | mm |
| Skirt Thickness                    | STHK    | 14.0000   | mm |
| Skirt Internal Corrosion Allowance | SCA     | 1.5000    | mm |
| Skirt External Corrosion Allowance |         | 0.0000    | mm |
| Skirt Material                     |         | SA-285 C  |    |

#### Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

|                                              |            |             |      |
|----------------------------------------------|------------|-------------|------|
| Thickness of Basing                          | TBA, ea4   | 32.0000     | mm   |
| Design Temperature of the Basing             |            | 38.00       | °C   |
| Basing Material                              |            | SA-285 C    |      |
| Basing Ope. All. Stress                      | BASOPE, f4 | 108.00      | MPa  |
| Basing Yield Stress                          |            | 207.00      | MPa  |
| Inside Diameter of Basing                    | DI, D4     | 1518.0000   | mm   |
| Outside Diameter of Basing                   | DOU, D5    | 1926.0000   | mm   |
| Nominal Diameter of Bolts                    | BND, d(b0) | 36.0000     | mm   |
| Bolt Corrosion Allowance                     | BCA        | 1.5000      | mm   |
| Bolt Material                                |            | SA-36       |      |
| Bolt Operating Allowable Stress              | SA, fB     | 150.00      | MPa  |
| Number of Bolts                              | RN, nb     | 12          |      |
| Diameter of Bolt Circle                      | DC, DBC    | 1810.0000   | mm   |
| Ultimate Comp. Strength of Concrete          | FPC        | 35.0        | MPa  |
| Allowable Comp. Strength of Concrete         | FC         | 15.8        | MPa  |
| Modular ratio Steel/Concrete                 |            | 9.833       |      |
| Thickness of Gusset Plates                   | TGA, e7    | 14.0000     | mm   |
| Width of Gussets at Top Plate                | TWDT       | 122.0000    | mm   |
| Width of Gussets at Base Plate               | BWDT       | 122.0000    | mm   |
| Gusset Plate Elastic Modulus                 | E          | 201746.0    | MPa  |
| Gusset Plate Yield Stress                    | SY         | 207.0       | MPa  |
| Height of Chair                              | h1         | 268.0000    | mm   |
| Distance between Gussets                     | RG, b6     | 102.0000    | mm   |
| Dist. from Bolt Center to Gusset (Rg/2)      | CG         | 51.0000     | mm   |
| Number of Gussets per bolt                   | NG         | 2           |      |
| Thickness of Top Plate or Ring               | TTA, e5    | 36.0000     | mm   |
| Radial Width of the Top Plate                | TOPWTH, b5 | 122.0000    | mm   |
| Anchor Bolt Hole Dia. in Top Plate           | dia,H      | 42.0000     | mm   |
| External Corrosion Allowance                 | CA         | 1.5000      | mm   |
| Dead Weight of Vessel                        | DW         | 284319.7    | N    |
| Operating Weight of Vessel                   | ROW        | 466769.8    | N    |
| Earthquake Moment on Basing                  | EQMOM      | 889220352.0 | N-mm |
| Wind Moment on Basing                        | WIMOM      | 777196736.0 | N-mm |
| Percent Bolt Preload                         | ppl        | 100.0       |      |
| Use AISC A5.2 Increase in Fc and Bolt Stress |            | No          |      |
| Use Allowable Weld Stress per AISC J2.5      |            | No          |      |
| Factor for Increase of Allowables            | Fact       | 1.0000      |      |

#### Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

#### Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:  

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.88922E+09/1810.0) - 446481) / 12$$

$$= 126487.6 \text{ N}$$

Bolt Load, Windload Empty Case:  

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 0.77720E+09/1810.0) - 284320) / 12$$

$$= 119378.8 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/1810.0) - 466770) / 12$$

$$= -38897.5 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 600.6298 mm<sup>2</sup>  
 Area Available in all the Bolts Abss \* RN : 7207.5581 mm<sup>2</sup>

| Trial# | k     | knew  | Cc    | Ct    | z     | j     | Ft       | Fc       | Fs      |
|--------|-------|-------|-------|-------|-------|-------|----------|----------|---------|
| 2      | 0.243 | 0.121 | 1.348 | 2.567 | 0.450 | 0.778 | 372861.2 | 819342.0 | 81687.4 |
| 4      | 0.182 | 0.212 | 1.160 | 2.701 | 0.463 | 0.774 | 367471.8 | 813952.6 | 76525.7 |
| 6      | 0.197 | 0.205 | 1.209 | 2.667 | 0.460 | 0.775 | 368759.5 | 815240.3 | 77761.8 |
| 8      | 0.208 | 0.210 | 1.245 | 2.642 | 0.457 | 0.776 | 369752.4 | 816233.1 | 78711.8 |
| 10     | 0.211 | 0.212 | 1.254 | 2.636 | 0.457 | 0.776 | 370004.0 | 816484.8 | 78952.5 |
| 12     | 0.212 | 0.212 | 1.254 | 2.635 | 0.457 | 0.776 | 370024.8 | 816505.7 | 78972.6 |

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 370024.8 / (1.268 \cdot 1810.0 \cdot 2.635)$$

$$= 122.418 \text{ MPa, Should be } \leq 150.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 12 \cdot 600.63 / (\pi \cdot 1810.0)$$

$$= 1.268 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc'[(2kd + t3) / (2kd)]$$

$$= 3.342[(2 \cdot 0.2116 \cdot 1810.0 + 204.0) / (2 \cdot 0.2116 \cdot 1810.0)]$$

$$= 4.233 \text{ MPa, Should be } \leq 15.8 \text{ MPa}$$

Values for table 10.3, l = 122.0, b = 236.928, l/b = 0.514923

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(7524.665, 19575.467) = 19575.467 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 19575.467 / 136.6)} + 1.5$$

$$= 30.822 \text{ mm}$$

Nomenclature:

|                       |                               |
|-----------------------|-------------------------------|
| a = (Dc - Ds) / 2     | Skirt Distance to Bolt Circle |
| P = Sa * Abss         | Maximum Load on one Bolt      |
| l                     | Radial Top Plate Width        |
| lavg                  | Average Gusset Plate Width    |
| g1 = Gamma 1          | Constant Term f( b/l )        |
| g2 = Gamma 2          | Constant Term f( b/l )        |
| g = Flat distance / 2 | Nut 1/2 Dimension (from Tema) |
| Fb                    | Allowable Bending Stress      |

Values for table 10.6, l = 122.0, b = 102.0, b/l = 0.836066

As b/l ( 0.836) is less than 1, inverting b/l = 1.196.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [ 1.3(\ln((2l \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1) ] - [(0.7 - g2)P / (4\pi)]$$

$$= 90086.81 / (4 \cdot \pi) [ 1.3(\ln((2 \cdot 122.0 \sin(\pi \cdot 64.0 / 122.0) / (\pi \cdot 30.0))) + 1) ] - [(0.7 - 0.109) \cdot 90086.81 / (4 \cdot \pi)]$$

$$= 11771.0078 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot Abs(11771.01) / 138.0)} + 1.5$$

$$= 24.1238 \text{ mm}$$



Required Gusset Plate Thickness [tg]:  
=  $P / ( \text{Stress Term} \cdot l_{avg} ) + Ca$   
=  $90086.81 / ( 124.11 \cdot 122.0 ) + 1.5$   
= 11.025 (not less than  $9.525 + 1.5$  ) mm

Bolt spacing [m]:  
=  $\pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$   
=  $\pi \cdot 1810.0 / 12 = 473.857$  mm

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:  
=  $1.76( P \cdot a / ( m \cdot ( h + t_{ba} ) \cdot 1.5 \cdot Sktope ) )^{2/3} \cdot r^{1/3} + Ca$   
=  $1.76( 90087 \cdot 64.0 / ( 473.86 \cdot 232.0 \cdot 162 ) )^{2/3} \cdot 841.01^{1/3} + Ca$   
=  $7.833 + 1.5 = 9.333$  mm

#### **Summary of Basing Thickness Calculations**

|                                              |         |    |
|----------------------------------------------|---------|----|
| Required Basing Thickness (tension)          | 30.8222 | mm |
| Actual Basing Thickness as entered by user   | 32.0000 | mm |
| Required Thickness of Chair Cap              | 24.1238 | mm |
| Actual Top Ring Thickness as entered by user | 36.0000 | mm |
| Required Gusset thickness, + CA              | 11.0250 | mm |
| Actual Gusset Thickness as entered by user   | 14.0000 | mm |
| Required Thickness of Skirt for Local Stress | 9.3331  | mm |
| Given Thickness of Skirt                     | 14.0000 | mm |
| Required Gusset Height to meet local stress  | 83.0868 | mm |

#### **Weld Size Calculations per Steel Plate Engineering Data - Vol. 2**

Compute the Weld load at the Skirt/Base Junction [W]:  
=  $SkirtStress \cdot ( SkirtThickness - CA )$   
=  $40.314 \cdot ( 14.0 - 1.5 )$   
= 503.88 N/mm

Results for Computed Minimum Basing Weld Size [BWeld]:  
=  $W / [ ( 0.4 \cdot Yield ) \cdot 2 \cdot 0.707 ]$   
=  $504 / [ ( 0.4 \cdot 192 ) \cdot 2 \cdot 0.707 ]$   
= 4.651 mm

#### **Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size**

Vertical Plate Load [Wv]:  
=  $Bolt Load / ( Cm_{wth} + 2 \cdot ( Hg + Tta ) )$   
=  $90086.8 / ( 176.0 + 2 \cdot ( 200.0 + 36.0 ) )$   
= 139.023 N/mm

Horizontal Plate Load [Wh]:  
=  $Bolt Load \cdot e / ( Cm_{wth} \cdot ( Hg + Tta ) + 0.6667 \cdot ( Hg + Tta )^2 )$   
=  $90086.8 \cdot 64.0 / ( 176.0 \cdot ( 236.0 ) + 0.6667 \cdot ( 236.0 )^2 )$   
= 73.291 N/mm

Resultant Weld Load [Wr]:  
=  $\sqrt{ Wv^2 + Wh^2 }$   
=  $\sqrt{ 139.02^2 + 73.29^2 }$   
= 157.159 N/mm

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:  
=  $Wr / [ ( 0.4 \cdot Yield ) \cdot 2 \cdot 0.707 ]$   
=  $157.16 / [ ( 0.4 \cdot 192 ) \cdot 2 \cdot 0.707 ]$   
= 1.451 mm

#### **Results for Computed Minimum Gusset to Top Plate Weld Size**

Weld Load [Wv]:  
=  $Bolt Load / ( 2 \cdot TopWth )$   
=  $90086.8 / ( 2 \cdot 122.0 )$   
= 369.208 N/mm



Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / ( 2 \cdot \text{Hgt} \cdot \text{TopWth} ) \\ &= 90086.8 \cdot 64.0 / ( 2 \cdot 236.0 \cdot 122.0 ) \\ &= 100.124 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{ Wv^2 + Wh^2 } \\ &= \sqrt{ 369.21^2 + 100.12^2 } \\ &= 382.544 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= Wr / [ ( 0.4 \cdot \text{Yield} ) \cdot 2 \cdot 0.707 ] \\ &= 382.54 / [ ( 0.4 \cdot 192 ) \cdot 2 \cdot 0.707 ] \\ &= 3.531 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

**Summary of Required Weld Sizes:**

|                                                      |        |    |
|------------------------------------------------------|--------|----|
| Required Basing to Skirt Double Fillet Weld Size     | 6.3500 | mm |
| Required Gusset to Skirt Double Fillet Weld Size     | 6.3500 | mm |
| Required Top Plate to Skirt Weld Size                | 6.3500 | mm |
| Required Gusset to Top Plate Double Fillet Weld Size | 3.5309 | mm |

*Review notes about nozzle loadings and supports in the Warnings report.*

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**Input, Nozzle Desc: D13"**

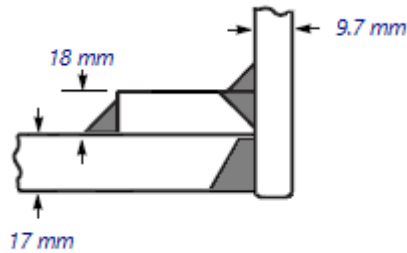
**From: 30**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5894    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Elliptical Head            | D    | 1650.00   | mm    |
| Aspect Ratio of Elliptical Head               | Ar   | 2.00      |       |
| Head Finished (Minimum) Thickness             | t    | 17.0000   | mm    |
| Head Internal Corrosion Allowance             | cas  | 1.5000    | mm    |
| Head External Corrosion Allowance             | co   | 0.0000    | mm    |
| Distance from Head Centerline                 | L1   | 0.0000    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |                   |                  |     |
|---------------------------------------------|-------------------|------------------|-----|
| Material                                    |                   | SA-333 6         | +IT |
| Material UNS Number                         |                   | K03006           |     |
| Material Specification/Type                 | Smls. & wld. pipe |                  |     |
| Allowable Stress at Temperature             | Sn                | 118.00           | MPa |
| Allowable Stress At Ambient                 | Sna               | 118.00           | MPa |
| Diameter Basis (for tr calc only)           |                   | Inside           |     |
| Layout Angle                                |                   | 0.00             | deg |
| Diameter                                    |                   | 3.0000           | in. |
| Size and Thickness Basis                    |                   | Minimum          |     |
| Nominal Thickness                           |                   | 160              |     |
| Flange Material                             |                   | SA-350 LF1       | +IT |
| Flange Type                                 |                   | Weld Neck Flange |     |
| Corrosion Allowance                         | can               | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1                | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En                | 1.00             |     |
| Outside Projection                          | ho                | 200.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo                | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv              | 17.0000          | mm  |
| Inside Projection                           | h                 | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi                | 0.0000           | mm  |
| Pad Material                                |                   | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp                | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa               | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp                | 150.0000         | mm  |
| Thickness of Pad                            | te                | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp                | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn              | 18.0000          | mm  |
| Reinforcing Pad Width                       |                   | 30.5500          | mm  |
| Flange Class                                |                   | 300              |     |
| Flange Grade                                |                   | GR 1.1           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



### Reinforcement CALCULATION, Description: D13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 2.734 in. |
| Actual Thickness Used in Calculation       | 0.383 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]  
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$  per Appendix 1-4(c)  
 $= 2.59 \cdot 1653.0 \cdot 0.8984 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.59)$   
 $= 13.9595 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.59 \cdot 36.22 / (118 \cdot 1.0 - 0.6 \cdot 2.59)$   
 $= 0.8053 \text{ mm}$

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 144.8618 mm |
| Parallel to Vessel Wall, opening length                | d  | 72.4309 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 38.5864 mm  |

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, Sn/Sv)$   
 $= \min(1, 118.0/138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, Sn/Sv)$   
 $= \min(1, 118.0/138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, Sp/Sv)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$

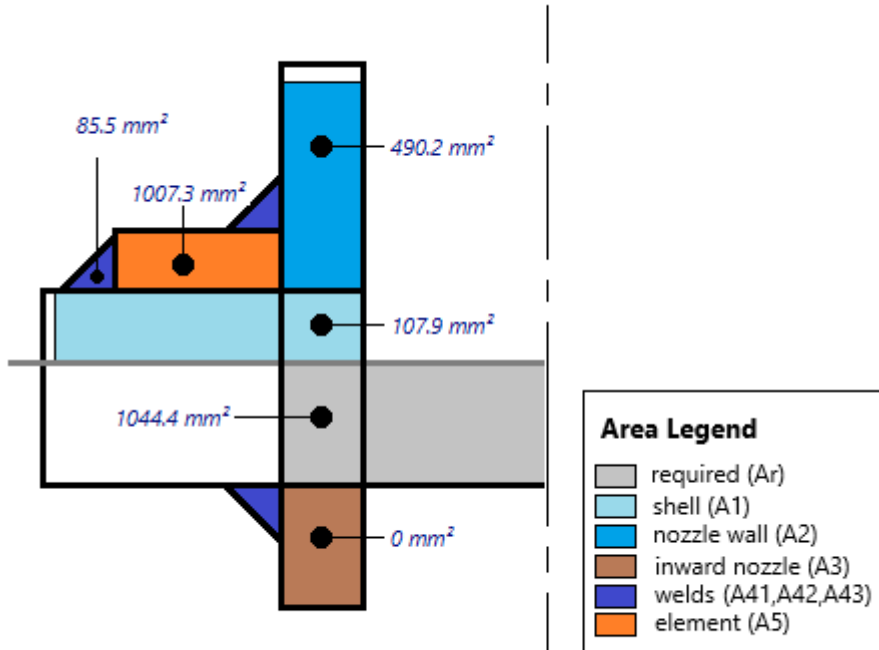


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 1044.421 | NA       | NA    |
| Area in Shell            | A1          | 107.900  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 490.242  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 85.507   | NA       | NA    |
| Area in Element          | A5          | 1007.312 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 1690.962 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 108.9429 18.0000 mm  
 Based on given Pad Diameter: 150.0000 6.4468 mm  
 Based on Shell or Nozzle Thickness: 125.9610 9.7345 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (72.4309 \cdot 13.9595 \cdot 1.0 + 2 \cdot 8.2345 \cdot 13.9595 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 1044.421 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 72.431(1.0 \cdot 15.5 - 1.0 \cdot 13.96) - 2 \cdot 8.235$   
 $(1.0 \cdot 15.5 - 1.0 \cdot 13.9595) \cdot (1 - 0.855)$   
 $= 107.900 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 38.59) \cdot (8.23 - 0.81) \cdot 0.8551$



= 490.242 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$   
= 85.507 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(144.8618 - 88.9) \cdot 18.0 \cdot 1.0$   
= 1007.312 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 2.3053 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 17.0000 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 17.0000 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 6.3000 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 6.3, max( 17.0, 3.0 ) ]  
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 2.3053, 6.3 )  
= 6.3000 mm

Available Nozzle Neck Thickness = 9.7345 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 38.0, | Allowable | : 118.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 257.0 MPa | Passed |
| Occasional | : 5.1,  | Allowable | : 156.9 MPa | Passed |
| Shear      | : 22.3, | Allowable | : 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

*Note: This Element/Detail was specified as being Impact Tested.*

##### Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

|                                                        |           |
|--------------------------------------------------------|-----------|
| Impact Test Temperature provided per Specification     | -46 °C    |
| Calculated Minimum Design Metal Temperature            | -104 °C   |
| Gov. MDMT of the Nozzle                                | : -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : -46 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : -46 °C  |

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -29 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -60 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating = 2.59/5.10 = 0.508

#### Weld Size Calculations, Description: D13"

|                                           |         |            |
|-------------------------------------------|---------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 8.2345 mm  |
| Intermediate Calc. for pad/shell Welds    | TminPad | 18.0000 mm |

#### Results Per UW-16.1:

|             | Required Thickness   | Actual Thickness     |
|-------------|----------------------|----------------------|
| Nozzle Weld | 5.7642 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 9.0000 = 0.5*TminPad | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (1044.4214 - 107.8999 + 2 \cdot 8.2345 \cdot 0.8551 \cdot (1.0 \cdot 15.5 - 13.9595)) \cdot 138)$   
 $= 132222.38 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (490.242 + 1007.3124 + 85.5072 - 0.0 \cdot 0.86) \cdot 138$   
 $= 218443.94 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (490.242 + 0.0 + 85.5072 + (218.2753)) \cdot 138$   
 $= 109566.06 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (490.242 + 0.0 + 85.5072 + 1007.3124 + (218.2753)) \cdot 138$   
 $= 248563.36 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$   
 $= 80735. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$   
 $= (3.1416/2.0) \cdot 150.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 223037. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 40.3327) \cdot (9.7345 - 1.5) \cdot 0.7 \cdot 118$$

$$= 86177. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 88.9 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 256666. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 88.9 \cdot (17.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 221018. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (223037 + 86177) = 309214 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (80735 + 256666 + 221018 + 0) = 558419 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (223037 + 221018 + 0) = 444055 N$$

#### Summary of Failure Path Calculations:

Path 1-1 = 309214 N , must exceed W = 132222 N or W1 = 218443 N  
 Path 2-2 = 558418 N , must exceed W = 132222 N or W2 = 109566 N  
 Path 3-3 = 444055 N , must exceed W = 132222 N or W3 = 248563 N

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.6594 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.6594 mm

#### Input Echo, WRC107/537 Item 1, Description: D13" :

|                                 |        |           |     |
|---------------------------------|--------|-----------|-----|
| Diameter Basis for Vessel       | Vbasis | ID        |     |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |     |
| Internal Corrosion Allowance    | Cas    | 1.5000    | mm  |
| Vessel Diameter                 | Dv     | 2970.000  | mm  |
| Vessel Thickness                | Tv     | 17.000    | mm  |
| Design Temperature              | T1     | 85.0      | C   |
| Vessel Material                 |        | SA-516 70 |     |
| Vessel UNS Number               |        | K02700    |     |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | MPa |
| Vessel Hot S.I. Allowable       | Smh    | 138.00    | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |        |
|----------------------------------|--------|--------|
| Attachment Type                  | Type   | Round  |
| WRC107 Attachment Classification | Holsol | Hollow |

|                                |        |          |    |
|--------------------------------|--------|----------|----|
| Diameter Basis for Nozzle      | Nbasis | ID       |    |
| Corrosion Allowance for Nozzle | Can    | 1.5000   | mm |
| Nozzle Diameter                | Dn     | 69.431   | mm |
| Nozzle Thickness               | Tn     | 9.735    | mm |
| Nozzle Material                |        | SA-333 6 |    |
| Nozzle UNS Number              |        | K03006   |    |



|                              |      |         |     |
|------------------------------|------|---------|-----|
| Nozzle Cold S.I. Allowable   | SNmc | 118.00  | MPa |
| Nozzle Hot S.I. Allowable    | SNmh | 118.00  | MPa |
| Thickness of Reinforcing Pad | Tpad | 18.000  | mm  |
| Diameter of Reinforcing Pad  | Dpad | 150.000 | mm  |
| Design Internal Pressure     | Dp   | 2.589   | MPa |
| Include Pressure Thrust      |      | No      |     |

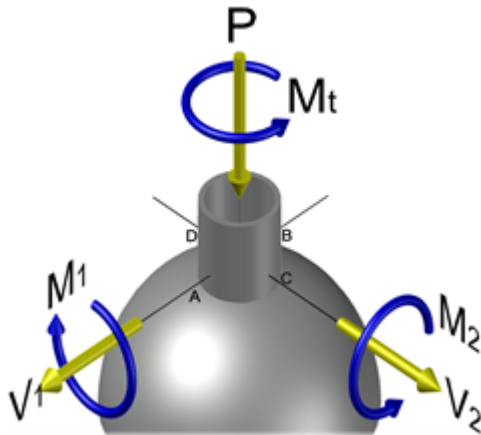
External Forces and Moments in WRC 107/537 Convention:

|                              |         |           |      |
|------------------------------|---------|-----------|------|
| Radial Load (SUS)            | P       | 4799.8    | N    |
| Longitudinal Shear (SUS)     | (V1) V1 | 5999.7    | N    |
| Circumferential Shear (SUS)  | (Vc) V2 | 0.0       | N    |
| Circumferential Moment (SUS) | (Mc) M1 | 0.0       | N-mm |
| Longitudinal Moment (SUS)    | (M1) M2 | 1181220.8 | N-mm |
| Torsional Moment (SUS)       | Mt      | 1080379.9 | N-mm |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{1494.25 (17.0 - 1.5)}$$

$$= 591.117 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

|                        |         |           |      |
|------------------------|---------|-----------|------|
| Radial Load            | P       | 4799.8    | N    |
| Circumferential Shear  | (VC) V2 | 0.0       | N    |
| Longitudinal Shear     | (VL) V1 | 5999.7    | N    |
| Circumferential Moment | (MC) M1 | 0.0       | N-mm |
| Longitudinal Moment    | (ML) M2 | 1181220.8 | N-mm |
| Torsional Moment       | MT      | 1080379.9 | N-mm |

Unitless Prm: U = 0.20 TAU = 5.00 ( 4.90) RHO = 4.00 ( 4.07)

Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979 | Figure | Value   | Location  |
|----------------------|--------|---------|-----------|
| N(x) * T / P         | SP 4   | 0.09081 | (A,B,C,D) |



|                              |      |         |           |
|------------------------------|------|---------|-----------|
| M(x) / P                     | SP 4 | 0.02957 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SM 4 | 0.31512 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SM 4 | 0.11742 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SM 4 | 0.31512 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SM 4 | 0.11742 | (A,B,C,D) |
|                              |      |         |           |
| N(y) * T / P                 | SP 4 | 0.34272 | (A,B,C,D) |
| M(y) / P                     | SP 4 | 0.19692 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SM 4 | 0.10351 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SM 4 | 0.87942 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SM 4 | 0.10351 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SM 4 | 0.87942 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |      |      |       |      |      |      |      |
|-----------------|------|----------------------------|------|------|-------|------|------|------|------|
|                 |      | Au                         | Al   | Bu   | Bl    | Cu   | Cl   | Du   | Dl   |
| Rad. Memb. P    |      | -0.4                       | -0.4 | -0.4 | -0.4  | -0.4 | -0.4 | -0.4 | -0.4 |
| Rad. Bend. P    |      | -0.8                       | 0.8  | -0.8 | 0.8   | -0.8 | 0.8  | -0.8 | 0.8  |
| Rad. Memb. MC   |      | 0.0                        | 0.0  | 0.0  | 0.0   | -0.0 | -0.0 | 0.0  | 0.0  |
| Rad. Bend. MC   |      | 0.0                        | 0.0  | 0.0  | 0.0   | -0.0 | 0.0  | 0.0  | -0.0 |
| Rad. Memb. ML   |      | -1.5                       | -1.5 | 1.5  | 1.5   | 0.0  | 0.0  | 0.0  | 0.0  |
| Rad. Bend. ML   |      | -3.3                       | 3.3  | 3.3  | -3.3  | 0.0  | 0.0  | 0.0  | 0.0  |
| Tot. Rad. Str.  |      | -5.9                       | 2.2  | 3.6  | -1.5  | -1.1 | 0.4  | -1.1 | 0.4  |
| Tang. Memb. P   |      | -1.5                       | -1.5 | -1.5 | -1.5  | -1.5 | -1.5 | -1.5 | -1.5 |
| Tang. Bend. P   |      | -5.1                       | 5.1  | -5.1 | 5.1   | -5.1 | 5.1  | -5.1 | 5.1  |
| Tang. Memb. MC  |      | 0.0                        | 0.0  | 0.0  | 0.0   | -0.0 | -0.0 | 0.0  | 0.0  |
| Tang. Bend. MC  |      | 0.0                        | 0.0  | 0.0  | 0.0   | -0.0 | 0.0  | 0.0  | -0.0 |
| Tang. Memb. ML  |      | -0.5                       | -0.5 | 0.5  | 0.5   | 0.0  | 0.0  | 0.0  | 0.0  |
| Tang. Bend. ML  |      | -24.7                      | 24.7 | 24.7 | -24.7 | 0.0  | 0.0  | 0.0  | 0.0  |
| Tot. Tang. Str. |      | -31.7                      | 27.8 | 18.7 | -20.7 | -6.5 | 3.6  | -6.5 | 3.6  |
| Shear VC        |      | 0.0                        | 0.0  | -0.0 | -0.0  | 0.0  | 0.0  | 0.0  | 0.0  |
| Shear VL        |      | 0.0                        | 0.0  | 0.0  | 0.0   | -1.3 | -1.3 | 1.3  | 1.3  |
| Shear MT        |      | 2.6                        | 2.6  | 2.6  | 2.6   | 2.6  | 2.6  | 2.6  | 2.6  |
| Tot. Shear      |      | 2.6                        | 2.6  | 2.6  | 2.6   | 1.3  | 1.3  | 3.9  | 3.9  |
| Str. Int.       |      | 32.0                       | 28.1 | 19.1 | 21.0  | 6.8  | 4.2  | 9.4  | 8.4  |

Unitless Prm: U = 0.49 TAU = 0.00 ( 8.61) RHO = 0.00 ( 1.88)

#### Dimensionless Loads for Spherical Shells at Pad edge:

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SR 2   | 0.13666 | (A,B,C,D) |
| M(x) / P                     | SR 2   | 0.08493 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SR 3   | 0.14385 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SR 3   | 0.25002 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SR 3   | 0.14385 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SR 3   | 0.25002 | (A,B,C,D) |
| N(y) * T / P                 | SR 2   | 0.04078 | (A,B,C,D) |
| M(y) / P                     | SR 2   | 0.02567 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SR 3   | 0.04361 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SR 3   | 0.07475 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SR 3   | 0.04361 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SR 3   | 0.07475 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)



| Type of Stress  | Load | Stress Intensity Values at |      |       |       |       |      |       |      |
|-----------------|------|----------------------------|------|-------|-------|-------|------|-------|------|
|                 |      | Au                         | Al   | Bu    | Bl    | Cu    | Cl   | Du    | Dl   |
| Rad. Memb. P    |      | -2.7                       | -2.7 | -2.7  | -2.7  | -2.7  | -2.7 | -2.7  | -2.7 |
| Rad. Bend. P    |      | -10.2                      | 10.2 | -10.2 | 10.2  | -10.2 | 10.2 | -10.2 | 10.2 |
| Rad. Memb. MC   |      | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | -0.0 | 0.0   | 0.0  |
| Rad. Bend. MC   |      | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | 0.0  | 0.0   | -0.0 |
| Rad. Memb. ML   |      | -4.6                       | -4.6 | 4.6   | 4.6   | 0.0   | 0.0  | 0.0   | 0.0  |
| Rad. Bend. ML   |      | -48.4                      | 48.4 | 48.4  | -48.4 | 0.0   | 0.0  | 0.0   | 0.0  |
| Tot. Rad. Str.  |      | -66.0                      | 51.3 | 40.2  | -36.4 | -12.9 | 7.5  | -12.9 | 7.5  |
| Tang. Memb. P   |      | -0.8                       | -0.8 | -0.8  | -0.8  | -0.8  | -0.8 | -0.8  | -0.8 |
| Tang. Bend. P   |      | -3.1                       | 3.1  | -3.1  | 3.1   | -3.1  | 3.1  | -3.1  | 3.1  |
| Tang. Memb. MC  |      | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | -0.0 | 0.0   | 0.0  |
| Tang. Bend. MC  |      | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | 0.0  | 0.0   | -0.0 |
| Tang. Memb. ML  |      | -1.4                       | -1.4 | 1.4   | 1.4   | 0.0   | 0.0  | 0.0   | 0.0  |
| Tang. Bend. ML  |      | -14.5                      | 14.5 | 14.5  | -14.5 | 0.0   | 0.0  | 0.0   | 0.0  |
| Tot. Tang. Str. |      | -19.8                      | 15.3 | 12.0  | -10.8 | -3.9  | 2.3  | -3.9  | 2.3  |
| Shear VC        |      | 0.0                        | 0.0  | -0.0  | -0.0  | 0.0   | 0.0  | 0.0   | 0.0  |
| Shear VL        |      | 0.0                        | 0.0  | 0.0   | 0.0   | -1.6  | -1.6 | 1.6   | 1.6  |
| Shear MT        |      | 2.0                        | 2.0  | 2.0   | 2.0   | 2.0   | 2.0  | 2.0   | 2.0  |
| Tot. Shear      |      | 2.0                        | 2.0  | 2.0   | 2.0   | 0.3   | 0.3  | 3.6   | 3.6  |
| Str. Int.       |      | 66.1                       | 51.4 | 40.3  | 36.5  | 12.9  | 7.5  | 14.2  | 9.3  |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |      |      |       |      |      |      |      |
|-----------------|------|----------------------------|------|------|-------|------|------|------|------|
|                 |      | Au                         | Al   | Bu   | Bl    | Cu   | Cl   | Du   | Dl   |
| Rad. Pm (SUS)   |      | 57.5                       | 57.5 | 57.5 | 57.5  | 57.5 | 57.5 | 57.5 | 57.5 |
| Rad. Pl (SUS)   |      | -1.9                       | -1.9 | 1.1  | 1.1   | -0.4 | -0.4 | -0.4 | -0.4 |
| Rad. Q (SUS)    |      | -4.1                       | 4.1  | 2.5  | -2.5  | -0.8 | 0.8  | -0.8 | 0.8  |
| Long. Pm (SUS)  |      | 57.5                       | 57.5 | 57.5 | 57.5  | 57.5 | 57.5 | 57.5 | 57.5 |
| Long. Pl (SUS)  |      | -2.0                       | -2.0 | -1.0 | -1.0  | -1.5 | -1.5 | -1.5 | -1.5 |
| Long. Q (SUS)   |      | -29.8                      | 29.8 | 19.7 | -19.7 | -5.1 | 5.1  | -5.1 | 5.1  |
| Shear Pm (SUS)  |      | 0.0                        | 0.0  | 0.0  | 0.0   | 0.0  | 0.0  | 0.0  | 0.0  |
| Shear Pl (SUS)  |      | 0.0                        | 0.0  | 0.0  | 0.0   | -1.3 | -1.3 | 1.3  | 1.3  |
| Shear Q (SUS)   |      | 2.6                        | 2.6  | 2.6  | 2.6   | 2.6  | 2.6  | 2.6  | 2.6  |
| Pm (SUS)        |      | 57.5                       | 57.5 | 57.5 | 57.5  | 57.5 | 57.5 | 57.5 | 57.5 |
| Pm+Pl (SUS)     |      | 55.6                       | 55.6 | 58.6 | 58.6  | 57.9 | 57.9 | 57.9 | 57.9 |
| Pm+Pl+Q (Total) |      | 51.8                       | 85.6 | 76.6 | 56.4  | 56.6 | 61.5 | 58.4 | 63.6 |

##### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 57.47     | 138.00         | Passed |
| Pm+Pl (SUS)         | 58.56     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 85.57     | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



# **WRC 107/537 Stress Summations:**

## **Vessel Stress Summation at Reinforcing Pad Edge (MPa)**

| Type of Stress  | Load     | Stress Intensity Values at |       |       |       |       |       |       |       |
|-----------------|----------|----------------------------|-------|-------|-------|-------|-------|-------|-------|
|                 |          | Au                         | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad.            | Pm (SUS) | 124.2                      | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 |
| Rad.            | Pl (SUS) | -7.4                       | -7.4  | 1.9   | 1.9   | -2.7  | -2.7  | -2.7  | -2.7  |
| Rad.            | Q (SUS)  | -58.6                      | 58.6  | 38.3  | -38.3 | -10.2 | 10.2  | -10.2 | 10.2  |
| Long.           | Pm (SUS) | 124.2                      | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 |
| Long.           | Pl (SUS) | -2.2                       | -2.2  | 0.6   | 0.6   | -0.8  | -0.8  | -0.8  | -0.8  |
| Long.           | Q (SUS)  | -17.6                      | 17.6  | 11.4  | -11.4 | -3.1  | 3.1   | -3.1  | 3.1   |
| Shear           | Pm (SUS) | 0.0                        | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear           | Pl (SUS) | 0.0                        | 0.0   | 0.0   | 0.0   | -1.6  | -1.6  | 1.6   | 1.6   |
| Shear           | Q (SUS)  | 2.0                        | 2.0   | 2.0   | 2.0   | 2.0   | 2.0   | 2.0   | 2.0   |
| Pm (SUS)        |          | 124.2                      | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 | 124.2 |
| Pm+Pl (SUS)     |          | 122.0                      | 122.0 | 126.1 | 126.1 | 124.3 | 124.3 | 124.3 | 124.3 |
| Pm+Pl+Q (Total) |          | 104.5                      | 175.5 | 164.5 | 113.5 | 120.3 | 131.6 | 121.6 | 133.5 |

## **Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 124.18    | 138.00         | Passed |
| Pm+Pl (SUS)         | 126.09    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 175.54    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: B210"**

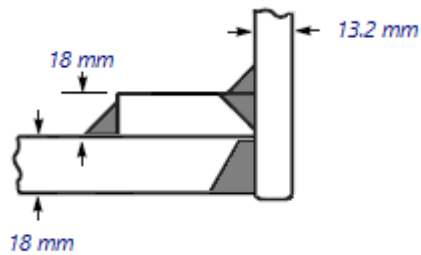
**From: 40**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5837    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 450.00    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |                   |                  |     |
|---------------------------------------------|-------------------|------------------|-----|
| Material                                    |                   | SA-333 6         | +IT |
| Material UNS Number                         |                   | K03006           |     |
| Material Specification/Type                 | Smls. & wld. pipe |                  |     |
| Allowable Stress at Temperature             | Sn                | 118.00           | MPa |
| Allowable Stress At Ambient                 | Sna               | 118.00           | MPa |
| Diameter Basis (for tr calc only)           |                   | Inside           |     |
| Layout Angle                                |                   | 0.00             | deg |
| Diameter                                    |                   | 10.0000          | in. |
| Size and Thickness Basis                    |                   | Minimum          |     |
| Nominal Thickness                           |                   | 80               |     |
| Flange Material                             |                   | SA-350 LF1       | +IT |
| Flange Type                                 |                   | Weld Neck Flange |     |
| Corrosion Allowance                         | can               | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1                | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En                | 1.00             |     |
| Outside Projection                          | ho                | 200.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo                | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv              | 18.0000          | mm  |
| Inside Projection                           | h                 | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi                | 0.0000           | mm  |
| Pad Material                                |                   | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp                | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa               | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp                | 440.0000         | mm  |
| Thickness of Pad                            | te                | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp                | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn              | 18.0000          | mm  |
| Reinforcing Pad Width                       |                   | 83.4750          | mm  |
| Flange Class                                |                   | 300              |     |
| Flange Grade                                |                   | GR 1.1           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



### Reinforcement CALCULATION, Description: B210"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 9.710 in. |
| Actual Thickness Used in Calculation       | 0.520 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 15.6500 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 124.82 / (118 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 2.7694 \text{ mm}$

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |          |    |
|--------------------------------------------------------|----|----------|----|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 499.2934 | mm |
| Parallel to Vessel Wall, opening length                | d  | 249.6467 | mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500  | mm |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$

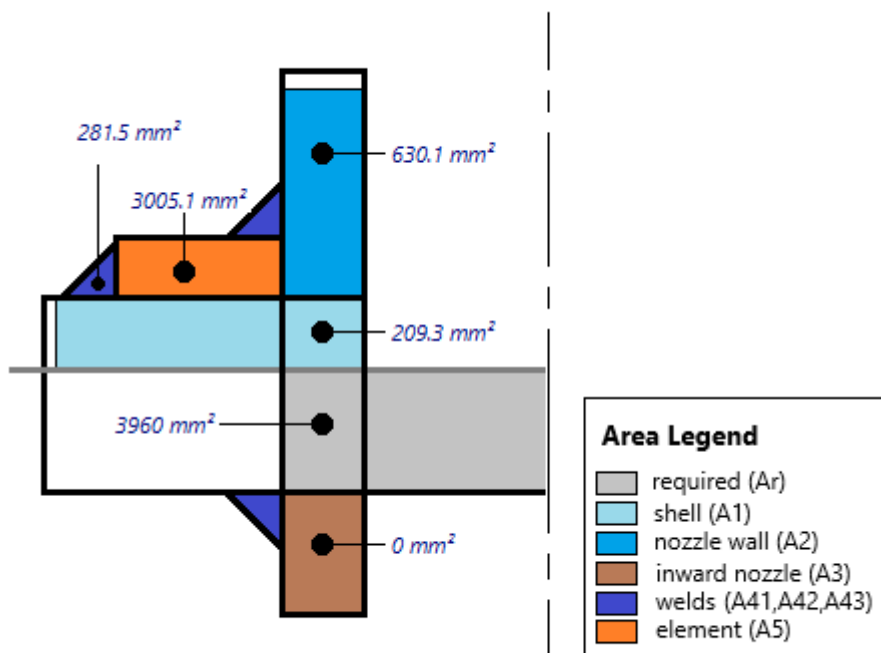


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 3960.045 | NA       | NA    |
| Area in Shell            | A1          | 209.318  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 630.120  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 281.509  | NA       | NA    |
| Area in Element          | A5          | 3005.100 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 4126.047 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

|                                         |          |            |
|-----------------------------------------|----------|------------|
| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter | Thickness  |
| Based on given Pad Thickness:           | 430.7777 | 18.0000 mm |
| Based on given Pad Diameter:            | 440.0000 | 17.0057 mm |
| Based on Shell or Nozzle Thickness:     | 488.1063 | 13.2016 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (249.6467 \cdot 15.65 \cdot 1.0 + 2 \cdot 11.7016 \cdot 15.65 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 3960.045 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 249.647(1.0 \cdot 16.5 - 1.0 \cdot 15.65) - 2 \cdot 11.702$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.65) \cdot (1 - 0.855)$   
 $= 209.318 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 41.25) \cdot (11.7 - 2.77) \cdot 0.8551$



= 630.120 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 14.0^2 \cdot 1.0$   
= 281.509 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(440.0 - 273.05) \cdot 18.0 \cdot 1.0$   
= 3005.100 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 4.2694 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 17.1500 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 17.1500 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 9.6026 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 9.603, max( 17.15, 3.0 ) ]  
= 9.6026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 4.2694, 9.6026 )  
= 9.6026 mm

Available Nozzle Neck Thickness = 13.2016 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 36.2, | Allowable | : 118.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 258.8 MPa | Passed |
| Occasional | : 13.2, | Allowable | : 156.9 MPa | Passed |
| Shear      | : 15.4, | Allowable | : 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

##### Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

|                                                        |           |
|--------------------------------------------------------|-----------|
| Impact Test Temperature provided per Specification     | -46 °C    |
| Calculated Minimum Design Metal Temperature            | -104 °C   |
| Gov. MDMT of the Nozzle                                | : -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : -43 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : -43 °C  |

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -29 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -60 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating = 2.58/5.10 = 0.507

#### Weld Size Calculations, Description: B210"

|                                           |         |            |
|-------------------------------------------|---------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 11.7016 mm |
| Intermediate Calc. for pad/shell Welds    | TminPad | 16.5000 mm |

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 8.2500 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3960.0452 - 209.3184 + 2 \cdot 11.7016 \cdot 0.8551 \cdot (1.0 \cdot 16.5 - 15.65)) \cdot 138)$   
 $= 519894.09 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (630.1199 + 3005.1 + 281.5088 - 0.0 \cdot 0.86) \cdot 138$   
 $= 540452.81 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (630.1199 + 0.0 + 85.5088 + (330.196)) \cdot 138$   
 $= 144308.92 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (630.1199 + 0.0 + 281.5088 + 3005.1 + (330.196)) \cdot 138$   
 $= 586015.19 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$   
 $= 247972. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$   
 $= (3.1416/2.0) \cdot 440.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 654231. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 130.6742) \cdot (13.2016 - 1.5) \cdot 0.7 \cdot 118$$

$$= 396762. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 273.05 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 788331. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 273.05 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 722623. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (654231 + 396762) = 1050993 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (247972 + 788331 + 722623 + 0) = 1758927 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (654231 + 722623 + 0) = 1376854 \text{ N}$$

#### Summary of Failure Path Calculations:

Path 1-1 = 1050992 N , must exceed W = 519894 N or W1 = 540452 N  
 Path 2-2 = 1758926 N , must exceed W = 519894 N or W2 = 144308 N  
 Path 3-3 = 1376854 N , must exceed W = 519894 N or W3 = 586015 N

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 11.3748 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.3748 mm

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 249.65 mm*

#### Input Echo, WRC297 Item 1, Description: B210" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.5000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 1650.000  | mm         |
| Shell Thickness                      | 18.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |          |            |
|------------------------------------|----------|------------|
| Diameter Basis for Nozzle          | ID       |            |
| Nozzle Corrosion Allowance         | 1.5000   | mm         |
| Nozzle Diameter                    | 246.647  | mm         |
| Nozzle Thickness                   | 13.2016  | mm         |
| Nozzle Stress Concentration Factor | 1.000    |            |
| Nozzle Material                    | SA-333 6 |            |
| Nozzle Cold S.I. Allowable         | SNmc     | 118.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh     | 118.00 MPa |



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad 440.000 mm  
 Thickness of Reinforcing Pad 18.0000 mm

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSstained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 2.58      | MPa  |
| Radial Load              | P  | 11999.50  | N    |
| Circumferential Shear    | Vc | 15999.28  | N    |
| Longitudinal Shear       | Vl | 15999.28  | N    |
| Circumferential Moment   | Mc | 0.800E+07 | N-mm |
| Longitudinal Moment      | Ml | 0.104E+08 | N-mm |
| Torsional Moment         | Mt | 0.120E+08 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(843.75(18.0 - 1.5))}$   
 $= 662.420 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1687.500 mm |
| Nozzle Outside Diameter | (d) = | 273.050 mm  |
| Vessel Thickness used   | (T) = | 34.500 mm   |
| Nozzle Thickness used   | (t) = | 11.702 mm   |
| T / t                   | =     | 2.948       |
| d / t                   | =     | 23.334      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.132$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.089 |
| Mr/P    | = | 0.099 |
| M0/P    | = | 0.092 |
| N0/P    | = | 0.209 |
| MrD/Mc  | = | 0.174 |
| NrDL/Mc | = | 0.131 |
| M0d/Mc  | = | 0.188 |
| N0DL/Mc | = | 0.129 |
| MrD/Ml  | = | 0.137 |
| NrDL/Ml | = | 0.108 |
| M0D/Ml  | = | 0.143 |
| N0DL/Ml | = | 0.276 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P)                                            | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending (P)                                             | -5                   | 5                   | -5                      | 5                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -8                   | -8                  | 8                       | 8                      |
| Outplane Bending (ML)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 60                   | 63                  | 60                      | 63                     |
| Outplane Stress Summary                                          | 16                   | 85                  | 89                      | 48                     |

**Vessel Stresses (MPa):**



| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (P )                                               | -5                   | 5                   | -5                      | 5                      |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -3                   | -3                  | 3                       | 3                      |
| Inplane Bending (ML)                                               | -26                  | 26                  | 26                      | -26                    |
| Inplane Pressure Stress                                            | 30                   | 30                  | 30                      | 30                     |
| Inplane Stress Summary                                             | -6                   | 58                  | 53                      | 12                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 2                    | 2                   | 2                       | 2                      |
| Shear Stress Summary                                                 | 4                    | 4                   | 1                       | 1                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 24                   | 85                  | 89                      | 48                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | -2                    | -2                   | -2                     | -2                    |
| Outplane Bending (P )                                                  | -5                    | 5                    | -5                     | 5                     |
| Outplane Membrane (Mc)                                                 | -3                    | -3                   | 3                      | 3                     |
| Outplane Bending (Mc)                                                  | -27                   | 27                   | 27                     | -27                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 30                    | 30                   | 30                     | 30                    |
| Outplane Stress Summary                                                | -8                    | 58                   | 53                     | 9                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (P )                                                     | -5                    | 5                    | -5                     | 5                     |
| Inplane Membrane (Mc)                                                    | -3                    | -3                   | 3                      | 3                     |
| Inplane Bending (Mc)                                                     | -25                   | 25                   | 25                     | -25                   |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 60                    | 63                   | 60                     | 63                    |
| Inplane Stress Summary                                                   | 24                    | 90                   | 82                     | 45                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                       | 2                     | 2                    | 2                      | 2                     |



|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 1 | 1 | 4 | 4 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities   | Left    | Left   | Right   | Right  |
|                       | Outside | Inside | Outside | Inside |

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 33 | 90 | 83 | 46 |
|------------------------|----|----|----|----|

**Nozzle Stresses (MPa):**

|                    |         |        |         |        |
|--------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE | Au      | Al     | Bu      | Bl     |
| (Stresses in the   | Top     | Top    | Bottom  | Bottom |
| hoop direction)    | Outside | Inside | Outside | Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Hoop Membrane (P )   | -2 | -2 | -2 | -2 |
| Hoop Bending (P )    | 0  | 0  | 0  | 0  |
| Hoop Membrane (Mc)   | 0  | 0  | 0  | 0  |
| Hoop Bending (Mc)    | 0  | 0  | 0  | 0  |
| Hoop Membrane (ML)   | -8 | -8 | 8  | 8  |
| Hoop Bending (ML)    | 0  | 0  | 0  | 0  |
| Hoop Pressure Stress | 31 | 34 | 31 | 34 |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 20 | 23 | 38 | 41 |
|---------------------|----|----|----|----|

**Nozzle Stresses (MPa):**

|                     |         |        |         |        |
|---------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE  | Au      | Al     | Bu      | Bl     |
| (Stresses Normal to | Top     | Top    | Bottom  | Bottom |
| pipe cross-section) | Outside | Inside | Outside | Inside |

|                       |      |     |     |      |
|-----------------------|------|-----|-----|------|
| Axial Membrane (P )   | -1   | -1  | -1  | -1   |
| Axial Bending (P )    | -28  | 28  | -28 | 28   |
| Axial Membrane (Mc)   | 0    | 0   | 0   | 0    |
| Axial Bending (Mc)    | 0    | 0   | 0   | 0    |
| Axial Membrane (ML)   | -17  | -17 | 17  | 17   |
| Axial Bending (ML)    | -138 | 138 | 138 | -138 |
| Axial Pressure Stress | 15   | 15  | 15  | 15   |

|                      |      |     |     |     |
|----------------------|------|-----|-----|-----|
| Axial Stress Summary | -170 | 164 | 142 | -78 |
|----------------------|------|-----|-----|-----|

**Nozzle Stresses (MPa):**

|                    |         |        |         |        |
|--------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE | Au      | Al     | Bu      | Bl     |
| (Shear stress)     | Outside | Inside | Outside | Inside |

|                      |   |   |    |    |
|----------------------|---|---|----|----|
| Shear due to (Vc)    | 3 | 3 | -3 | -3 |
| Shear due to (Vl)    | 0 | 0 | 0  | 0  |
| Shear due to Torsion | 8 | 8 | 8  | 8  |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Shear Stress Summary | 11 | 11 | 5 | 5 |
|----------------------|----|----|---|---|

**Nozzle Stresses (MPa):**

|                     |         |        |         |        |
|---------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE  | Au      | Al     | Bu      | Bl     |
| (Stress Intensities | Outside | Inside | Outside | Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 192 | 165 | 142 | 120 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (MPa):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stresses in the      | Left    | Left   | Right   | Right  |
| hoop direction)       | Outside | Inside | Outside | Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Hoop Membrane (P )   | -2 | -2 | -2 | -2 |
| Hoop Bending (P )    | 0  | 0  | 0  | 0  |
| Hoop Membrane (Mc)   | -3 | -3 | 3  | 3  |
| Hoop Bending (Mc)    | 0  | 0  | 0  | 0  |
| Hoop Membrane (ML)   | 0  | 0  | 0  | 0  |
| Hoop Bending (ML)    | 0  | 0  | 0  | 0  |
| Hoop Pressure Stress | 31 | 34 | 31 | 34 |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 26 | 29 | 32 | 35 |
|---------------------|----|----|----|----|



**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | -1                    | -1                   | -1                     | -1                    |
| Axial Bending (P )                                                  | -28                   | 28                   | -28                    | 28                    |
| Axial Membrane (Mc)                                                 | -13                   | -13                  | 13                     | 13                    |
| Axial Bending (Mc)                                                  | -140                  | 140                  | 140                    | -140                  |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 15                    | 15                   | 15                     | 15                    |
| Axial Stress Summary                                                | -167                  | 169                  | 139                    | -83                   |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -3            | -3           | 3             | 3            |
| Shear due to Torsion                    | 8             | 8            | 8             | 8            |
| Shear Stress Summary                    | 5             | 5            | 11            | 11           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 194           | 170          | 140           | 121          |

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(834.75(18.0 - 1.5))}$   
 $= 660.338 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1669.500 mm  
 Nozzle Outside Diameter (d) = 440.000 mm  
 Vessel Thickness used (T) = 16.500 mm  
 Nozzle Thickness used (t) = 95.177 mm  
 $T / t = 0.173$   
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.651$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.063  
 Mr/P = 0.118  
 M0/P = 0.035  
 N0/P = 0.061  
 MrD/Mc = 0.276  
 NrDL/Mc = 0.097  
 M0d/Mc = 0.083  
 N0DL/Mc = 0.079  
 MrD/Ml = 0.091  
 NrDL/Ml = 0.078  
 M0D/Ml = 0.026  
 N0DL/Ml = 0.079

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Outplane Membrane (P ) | -2  | -2  | -2  | -2  |
| Outplane Bending (P )  | -9  | 9   | -9  | 9   |
| Outplane Membrane (Mc) | 0   | 0   | 0   | 0   |
| Outplane Bending (Mc)  | 0   | 0   | 0   | 0   |
| Outplane Membrane (ML) | -6  | -6  | 6   | 6   |
| Outplane Bending (ML)  | -13 | 13  | 13  | -13 |
| Normal Pressure Stress | 128 | 130 | 128 | 130 |

|                         |    |     |     |     |
|-------------------------|----|-----|-----|-----|
| Outplane Stress Summary | 95 | 144 | 136 | 130 |
|-------------------------|----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                           |             |            |                |               |
|-------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                        | Au          | Al         | Bu             | Bl            |
| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Inplane Membrane (P )                     | -2          | -2         | -2             | -2            |
| Inplane Bending (P )                      | -31         | 31         | -31            | 31            |
| Inplane Membrane (Mc)                     | 0           | 0          | 0              | 0             |
| Inplane Bending (Mc)                      | 0           | 0          | 0              | 0             |
| Inplane Membrane (ML)                     | -6          | -6         | 6              | 6             |
| Inplane Bending (ML)                      | -47         | 47         | 47             | -47           |
| Inplane Pressure Stress                   | 64          | 64         | 64             | 64            |

|                        |     |     |    |    |
|------------------------|-----|-----|----|----|
| Inplane Stress Summary | -23 | 132 | 84 | 52 |
|------------------------|-----|-----|----|----|

**Vessel Stresses (MPa):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Shear (Vc)                         | 1           | 1          | -1             | -1            |
| Outplane Shear (Vl)                         | 0           | 0          | 0              | 0             |
| Outplane Shear (Mt)                         | 2           | 2          | 2              | 2             |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 3 | 3 | 0 | 0 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                        |             |            |                |               |
|------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE     | Au          | Al         | Bu             | Bl            |
| (Stress Intensities)   | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Two * Max Shear Stress | 119         | 145        | 136            | 130           |

**Vessel Stresses (MPa):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Membrane (P )                     | -2           | -2          | -2            | -2           |
| Outplane Bending (P )                      | -9           | 9           | -9            | 9            |
| Outplane Membrane (Mc)                     | -5           | -5          | 5             | 5            |
| Outplane Bending (Mc)                      | -33          | 33          | 33            | -33          |
| Outplane Membrane (ML)                     | 0            | 0           | 0             | 0            |
| Outplane Bending (ML)                      | 0            | 0           | 0             | 0            |
| Normal Pressure Stress                     | 64           | 64          | 64            | 64           |

|                         |    |    |    |    |
|-------------------------|----|----|----|----|
| Outplane Stress Summary | 13 | 98 | 90 | 42 |
|-------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Inplane Membrane (P )                        | -2           | -2          | -2            | -2           |
| Inplane Bending (P )                         | -31          | 31          | -31           | 31           |
| Inplane Membrane (Mc)                        | -6           | -6          | 6             | 6            |
| Inplane Bending (Mc)                         | -110         | 110         | 110           | -110         |
| Inplane Membrane (ML)                        | 0            | 0           | 0             | 0            |
| Inplane Bending (ML)                         | 0            | 0           | 0             | 0            |

Inplane Pressure Stress 128 130 128 130

Inplane Stress Summary -22 263 211 54

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE  
 (Shear stress normal to  
 circumferential plane)

|                      | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)  | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)  | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt) | 2                     | 2                    | 2                      | 2                     |

Shear Stress Summary 0 0 3 3

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE  
 (Stress Intensities)

|                        | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress | 36                    | 263                  | 211                    | 56                    |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------------|------------------|------|------|------|------|------|------|------|
|                                    | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)                     | 60               | 63   | 60   | 63   | 60   | 63   | 60   | 63   |
| Circ. Pl (SUS)                     | -10              | -10  | 6    | 6    | -3   | -3   | 3    | 3    |
| Circ. Q (SUS)                      | -32              | 32   | 22   | -22  | -30  | 30   | 20   | -20  |
| Long. Pm (SUS)                     | 30               | 30   | 30   | 30   | 30   | 30   | 30   | 30   |
| Long. Pl (SUS)                     | -3               | -3   | 3    | 3    | -5   | -5   | 1    | 1    |
| Long. Q (SUS)                      | -31              | 31   | 21   | -21  | -32  | 32   | 22   | -22  |
| Shear Pm (SUS)                     | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)                     | 1                | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)                      | 2                | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)                           | 60.0             | 63.0 | 60.0 | 63.0 | 60.0 | 63.0 | 60.0 | 63.0 |
| Pm+Pl (SUS)                        | 50.0             | 53.0 | 66.0 | 69.0 | 57.0 | 60.0 | 63.0 | 66.0 |
| Pm+Pl+Q (Total)                    | 22.8             | 85.3 | 88.0 | 47.0 | 34.1 | 90.0 | 83.3 | 46.2 |

**Vessel Stress Summation Comparison (MPa):**

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 63.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 69.03     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 90.03     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |     |     |     |     |     |     |
|------------------------------------|------------------|-----|-----|-----|-----|-----|-----|-----|
|                                    | Au               | Al  | Bu  | Bl  | Cu  | Cl  | Du  | Dl  |
| Circ. Pm (SUS)                     | 128              | 130 | 128 | 130 | 128 | 130 | 128 | 130 |
| Circ. Pl (SUS)                     | -8               | -8  | 4   | 4   | -8  | -8  | 4   | 4   |



|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Circ. Q (SUS)   | -22   | 22    | 4     | -4    | -141  | 141   | 79    | -79   |
| Long. Pm (SUS)  | 64    | 64    | 64    | 64    | 64    | 64    | 64    | 64    |
| Long. Pl (SUS)  | -8    | -8    | 4     | 4     | -7    | -7    | 3     | 3     |
| Long. Q (SUS)   | -78   | 78    | 16    | -16   | -42   | 42    | 24    | -24   |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 2     | 2     | 2     | 2     | 2     | 2     | 2     | 2     |
| Pm (SUS)        | 128.0 | 130.0 | 128.0 | 130.0 | 128.0 | 130.0 | 128.0 | 130.0 |
| Pm+Pl (SUS)     | 120.0 | 122.0 | 132.0 | 134.0 | 120.0 | 122.0 | 132.0 | 134.0 |
| Pm+Pl+Q (Total) | 120.1 | 144.8 | 136.0 | 130.0 | 36.1  | 263.0 | 211.1 | 55.7  |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 130.00    | 138.00         | Passed |
| Pm+Pl (SUS)         | 134.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 263.01    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 31               | 34    | 31    | 34    | 31    | 34    | 31    | 34    |
| Circ. Pl (SUS)               | -10              | -10   | 6     | 6     | -5    | -5    | 1     | 1     |
| Circ. Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)               | 15               | 15    | 15    | 15    | 15    | 15    | 15    | 15    |
| Long. Pl (SUS)               | -18              | -18   | 16    | 16    | -14   | -14   | 12    | 12    |
| Long. Q (SUS)                | -166             | 166   | 110   | -110  | -168  | 168   | 112   | -112  |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 3                | 3     | -3    | -3    | -3    | -3    | 3     | 3     |
| Shear Q (SUS)                | 8                | 8     | 8     | 8     | 8     | 8     | 8     | 8     |
| Pm (SUS)                     | 31.0             | 34.0  | 31.0  | 34.0  | 31.0  | 34.0  | 31.0  | 34.0  |
| Pm+Pl (SUS)                  | 24.7             | 27.7  | 38.2  | 40.9  | 26.4  | 29.3  | 33.4  | 36.0  |
| Pm+Pl+Q (Total)              | 191.3            | 163.9 | 141.2 | 119.4 | 193.3 | 169.2 | 140.1 | 122.0 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 34.00     | 118.00         | Passed |
| Pm+Pl (SUS)         | 40.91     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 193.26    | 354.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: A33"**

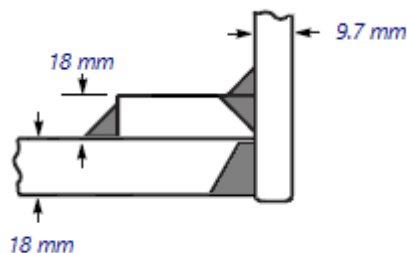
**From: 40**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5847    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 300.00    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |                   |                  |     |
|---------------------------------------------|-------------------|------------------|-----|
| Material                                    |                   | SA-333 6         | +IT |
| Material UNS Number                         |                   | K03006           |     |
| Material Specification/Type                 | Smls. & wld. pipe |                  |     |
| Allowable Stress at Temperature             | Sn                | 118.00           | MPa |
| Allowable Stress At Ambient                 | Sna               | 118.00           | MPa |
| Diameter Basis (for tr calc only)           |                   | Inside           |     |
| Layout Angle                                |                   | 180.00           | deg |
| Diameter                                    |                   | 3.0000           | in. |
| Size and Thickness Basis                    |                   | Minimum          |     |
| Nominal Thickness                           |                   | 160              |     |
| Flange Material                             |                   | SA-350 LF1       | +IT |
| Flange Type                                 |                   | Weld Neck Flange |     |
| Corrosion Allowance                         | can               | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1                | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En                | 1.00             |     |
| Outside Projection                          | ho                | 200.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo                | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv              | 18.0000          | mm  |
| Inside Projection                           | h                 | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi                | 0.0000           | mm  |
| Pad Material                                |                   | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp                | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa               | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp                | 150.0000         | mm  |
| Thickness of Pad                            | te                | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp                | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn              | 18.0000          | mm  |
| Reinforcing Pad Width                       |                   | 30.5500          | mm  |
| Flange Class                                |                   | 300              |     |
| Flange Grade                                |                   | GR 1.1           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



### Reinforcement CALCULATION, Description: A33"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 2.734 in. |
| Actual Thickness Used in Calculation       | 0.383 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 15.6561 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 36.22 / (118 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 0.8038 \text{ mm}$

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 144.8618 mm |
| Parallel to Vessel Wall, opening length                | d  | 72.4309 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 38.5864 mm  |

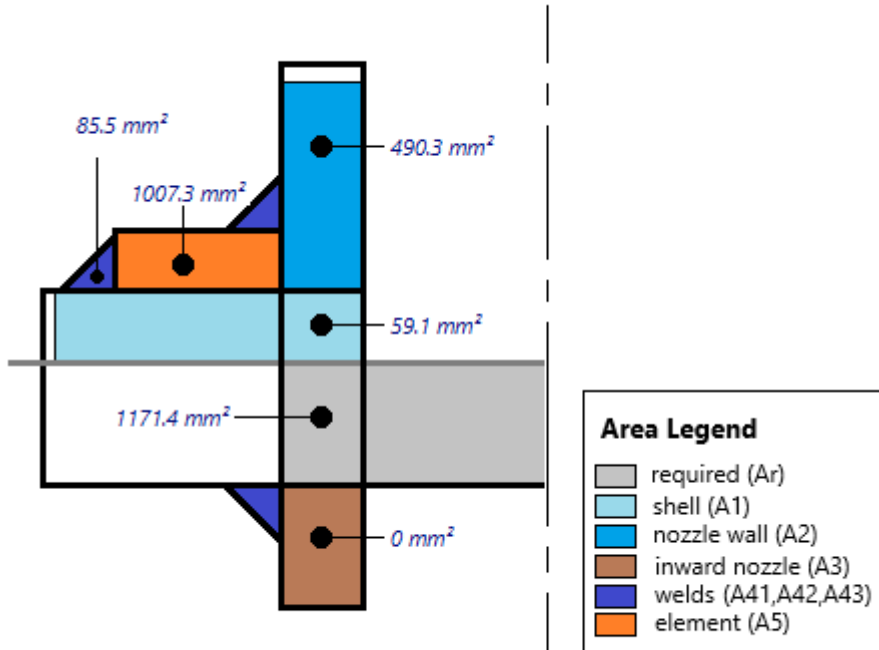
The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 1171.351 | NA       | NA    |
| Area in Shell            | A1          | 59.109   | NA       | NA    |
| Area in Nozzle Wall      | A2          | 490.350  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 85.509   | NA       | NA    |
| Area in Element          | A5          | 1007.312 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 1642.280 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 118.6991 18.0000 mm  
 Based on given Pad Diameter: 150.0000 9.5848 mm  
 Based on Shell or Nozzle Thickness: 144.0010 9.7345 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (72.4309 \cdot 15.6561 \cdot 1.0 + 2 \cdot 8.2345 \cdot 15.6561 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 1171.351 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 72.431(1.0 \cdot 16.5 - 1.0 \cdot 15.656) - 2 \cdot 8.235$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.6561) \cdot (1 - 0.855)$   
 $= 59.109 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 38.59) \cdot (8.23 - 0.8) \cdot 0.8551$



= 490.350 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$   
= 85.509 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(144.8618 - 88.9) \cdot 18.0 \cdot 1.0$   
= 1007.312 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 2.3038 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 17.1561 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 17.1561 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 6.3000 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 6.3, max( 17.1561, 3.0 ) ]  
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 2.3038, 6.3 )  
= 6.3000 mm

Available Nozzle Neck Thickness = 9.7345 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 37.4, | Allowable | : 118.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 257.6 MPa | Passed |
| Occasional | : 5.1,  | Allowable | : 156.9 MPa | Passed |
| Shear      | : 23.2, | Allowable | : 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

*Note:*

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

*Note: This Element/Detail was specified as being Impact Tested.*

##### Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

|                                                        |           |
|--------------------------------------------------------|-----------|
| Impact Test Temperature provided per Specification     | -46 °C    |
| Calculated Minimum Design Metal Temperature            | -104 °C   |
| Gov. MDMT of the Nozzle                                | : -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : -43 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : -43 °C  |

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -29 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -60 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating = 2.58/5.10 = 0.507

#### Weld Size Calculations, Description: A33"

|                                           |         |            |
|-------------------------------------------|---------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 8.2345 mm  |
| Intermediate Calc. for pad/shell Welds    | TminPad | 16.5000 mm |

#### Results Per UW-16.1:

|             | Required Thickness   | Actual Thickness     |
|-------------|----------------------|----------------------|
| Nozzle Weld | 5.7642 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 8.2500 = 0.5*TminPad | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (1171.3512 - 59.1091 + 2 \cdot 8.2345 \cdot 0.8551 \cdot (1.0 \cdot 16.5 - 15.6561)) \cdot 138)$   
 $= 155113.41 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (490.3497 + 1007.3124 + 85.5088 - 0.0 \cdot 0.86) \cdot 138$   
 $= 218455.03 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (490.3497 + 0.0 + 85.5088 + (232.3617)) \cdot 138$   
 $= 111522.89 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (490.3497 + 0.0 + 85.5088 + 1007.3124 + (232.3617)) \cdot 138$   
 $= 250517.67 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$   
 $= 80735. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$   
 $= (3.1416/2.0) \cdot 150.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 223033. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 40.3327) \cdot (9.7345 - 1.5) \cdot 0.7 \cdot 118$$

$$= 86177. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 88.9 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 256666. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 88.9 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 235273. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (223033 + 86177) = 309210 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (80735 + 256666 + 235273 + 0) = 572674 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (223033 + 235273 + 0) = 458306 \text{ N}$$

#### Summary of Failure Path Calculations:

Path 1-1 = 309210 N , must exceed W = 155113 N or W1 = 218455 N  
 Path 2-2 = 572673 N , must exceed W = 155113 N or W2 = 111522 N  
 Path 3-3 = 458306 N , must exceed W = 155113 N or W3 = 250517 N

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.1983 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.1982 mm

#### Input Echo, WRC297 Item 1, Description: A33" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.5000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 1650.000  | mm         |
| Shell Thickness                      | 18.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |          |            |
|------------------------------------|----------|------------|
| Diameter Basis for Nozzle          | ID       |            |
| Nozzle Corrosion Allowance         | 1.5000   | mm         |
| Nozzle Diameter                    | 69.431   | mm         |
| Nozzle Thickness                   | 9.7345   | mm         |
| Nozzle Stress Concentration Factor | 1.000    |            |
| Nozzle Material                    | SA-333 6 |            |
| Nozzle Cold S.I. Allowable         | SNmc     | 118.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh     | 118.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 150.000 mm  
 Thickness of Reinforcing Pad 18.0000 mm

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUstained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 2.58      | MPa  |
| Radial Load              | P  | 3599.84   | N    |
| Circumferential Shear    | Vc | 4799.79   | N    |
| Longitudinal Shear       | Vl | 4799.79   | N    |
| Circumferential Moment   | Mc | 720253.69 | N-mm |
| Longitudinal Moment      | Ml | 936329.88 | N-mm |
| Torsional Moment         | Mt | 0.108E+07 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tv, (4.939) is < 5  
 check the Limitations of WRC bulletin 297.*  
*Warning - The ratio, Dn/Tn (9.132) is < 10  
 check the Limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(843.75(18.0 - 1.5))}$$

$$= 478.270 \text{ mm}$$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1687.500 mm |
| Nozzle Outside Diameter | (d) = | 88.900 mm   |
| Vessel Thickness used   | (T) = | 34.500 mm   |
| Nozzle Thickness used   | (t) = | 8.235 mm    |
| T / t                   | =     | 4.190       |
| d / t                   | =     | 10.796      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.368$$

|         |   |        |
|---------|---|--------|
| Nr/P    | = | 0.193  |
| Mr/P    | = | 0.142  |
| M0/P    | = | 0.233  |
| N0/P    | = | 0.243  |
| MrD/Mc  | = | 0.143  |
| NrDL/Mc | = | 0.166  |
| M0d/Mc  | = | 0.267  |
| N0DL/Mc | = | -0.117 |
| MrD/Ml  | = | 0.140  |
| NrDL/Ml | = | 0.167  |
| M0D/Ml  | = | 0.259  |
| N0DL/Ml | = | -0.068 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (P )                                            | -4                   | 4                   | -4                      | 4                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Ml)                                            | -13                  | 13                  | 13                      | -13                    |
| Normal Pressure Stress                                           | 60                   | 63                  | 60                      | 63                     |
| Outplane Stress Summary                                          | 42                   | 81                  | 68                      | 52                     |



**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (P )                                               | -2                   | 2                   | -2                      | 2                      |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -1                   | -1                  | 1                       | 1                      |
| Inplane Bending (ML)                                               | -7                   | 7                   | 7                       | -7                     |
| Inplane Pressure Stress                                            | 30                   | 30                  | 30                      | 30                     |
| Inplane Stress Summary                                             | 18                   | 38                  | 36                      | 26                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 2                    | 2                   | 2                       | 2                      |
| Shear Stress Summary                                                 | 3                    | 3                   | 1                       | 1                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 43                   | 81                  | 68                      | 52                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (P )                                                  | -4                    | 4                    | -4                     | 4                     |
| Outplane Membrane (Mc)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (Mc)                                                  | -10                   | 10                   | 10                     | -10                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 30                    | 30                   | 30                     | 30                    |
| Outplane Stress Summary                                                | 15                    | 45                   | 35                     | 22                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (P )                                                     | -2                    | 2                    | -2                     | 2                     |
| Inplane Membrane (Mc)                                                    | -1                    | -1                   | 1                      | 1                     |
| Inplane Bending (Mc)                                                     | -5                    | 5                    | 5                      | -5                    |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 60                    | 63                   | 60                     | 63                    |
| Inplane Stress Summary                                                   | 50                    | 69                   | 64                     | 60                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |



|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Outplane Shear (Vl)  | 0 | 0 | 0 | 0 |
| Torsional Shear (Mt) | 2 | 2 | 2 | 2 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 1 | 1 | 3 | 3 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 50                    | 70                   | 64                     | 60                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 12                   | 15                  | 12                      | 15                     |
| Hoop Stress Summary                                       | 12                   | 15                  | 10                      | 14                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | -1                   | -1                  | -1                      | -1                     |
| Axial Bending (P )                                               | -14                  | 14                  | -14                     | 14                     |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -24                  | -24                 | 24                      | 24                     |
| Axial Bending (ML)                                               | -53                  | 53                  | 53                      | -53                    |
| Axial Pressure Stress                                            | 6                    | 6                   | 6                       | 6                      |
| Axial Stress Summary                                             | -87                  | 47                  | 67                      | -9                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 4             | 4            | -4            | -4           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 10            | 10           | 10            | 10           |
| Shear Stress Summary                 | 14            | 14           | 6             | 6            |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 103           | 53           | 67            | 27           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 12                    | 15                   | 12                     | 15                    |



|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 12 | 15 | 10 | 13 |
|---------------------|----|----|----|----|

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu      | Cl     | Du      | Dl     |
|---------------------------------------------------------------------|---------|--------|---------|--------|
|                                                                     | Left    | Left   | Right   | Right  |
|                                                                     | Outside | Inside | Outside | Inside |
| Axial Membrane (P )                                                 | -1      | -1     | -1      | -1     |
| Axial Bending (P )                                                  | -14     | 14     | -14     | 14     |
| Axial Membrane (Mc)                                                 | -18     | -18    | 18      | 18     |
| Axial Bending (Mc)                                                  | -43     | 43     | 43      | -43    |
| Axial Membrane (ML)                                                 | 0       | 0      | 0       | 0      |
| Axial Bending (ML)                                                  | 0       | 0      | 0       | 0      |
| Axial Pressure Stress                                               | 6       | 6      | 6       | 6      |

|                      |     |    |    |    |
|----------------------|-----|----|----|----|
| Axial Stress Summary | -72 | 43 | 51 | -5 |
|----------------------|-----|----|----|----|

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------|---------|--------|---------|--------|
|                                         | Outside | Inside | Outside | Inside |
| Shear due to (Vc)                       | 0       | 0      | 0       | 0      |
| Shear due to (Vl)                       | -4      | -4     | 4       | 4      |
| Shear due to Torsion                    | 10      | 10     | 10      | 10     |

|                      |   |   |    |    |
|----------------------|---|---|----|----|
| Shear Stress Summary | 6 | 6 | 14 | 14 |
|----------------------|---|---|----|----|

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities | Cu      | Cl     | Du      | Dl     |
|----------------------------------------------|---------|--------|---------|--------|
|                                              | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                       | 85      | 45     | 56      | 35     |

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$   
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{834.75(18.0 - 1.5)}$   
 $= 476.188 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1669.500 mm |
| Nozzle Outside Diameter | (d) = | 150.000 mm  |
| Vessel Thickness used   | (T) = | 16.500 mm   |
| Nozzle Thickness used   | (t) = | 38.785 mm   |
| T / t                   | =     | 0.425       |
| d / t                   | =     | 10.000      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.904$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.110 |
| Mr/P    | = | 0.239 |
| M0/P    | = | 0.070 |
| N0/P    | = | 0.136 |
| MrD/Mc  | = | 0.309 |
| NrDL/Mc | = | 0.083 |
| M0d/Mc  | = | 0.097 |
| N0DL/Mc | = | 0.035 |
| MrD/Ml  | = | 0.208 |
| NrDL/Ml | = | 0.090 |
| M0D/Ml  | = | 0.067 |
| N0DL/Ml | = | 0.109 |

**Vessel Stresses (MPa):**

|                    |    |    |    |    |
|--------------------|----|----|----|----|
| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|



| (Stresses Normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
|-----------------------------------------|-------------|------------|----------------|---------------|
| Outplane Membrane (P )                  | -1          | -1         | -1             | -1            |
| Outplane Bending (P )                   | -5          | 5          | -5             | 5             |
| Outplane Membrane (Mc)                  | 0           | 0          | 0              | 0             |
| Outplane Bending (Mc)                   | 0           | 0          | 0              | 0             |
| Outplane Membrane (ML)                  | -2          | -2         | 2              | 2             |
| Outplane Bending (ML)                   | -9          | 9          | 9              | -9            |
| Normal Pressure Stress                  | 128         | 130        | 128            | 130           |
| Outplane Stress Summary                 | 109         | 141        | 132            | 127           |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|-----------------------------------------------------------------|----------------|---------------|-------------------|------------------|
| Inplane Membrane (P )                                           | -1             | -1            | -1                | -1               |
| Inplane Bending (P )                                            | -18            | 18            | -18               | 18               |
| Inplane Membrane (Mc)                                           | 0              | 0             | 0                 | 0                |
| Inplane Bending (Mc)                                            | 0              | 0             | 0                 | 0                |
| Inplane Membrane (ML)                                           | -2             | -2            | 2                 | 2                |
| Inplane Bending (ML)                                            | -28            | 28            | 28                | -28              |
| Inplane Pressure Stress                                         | 64             | 64            | 64                | 64               |
| Inplane Stress Summary                                          | 12             | 108           | 74                | 54               |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|-------------------------------------------------------------------|----------------|---------------|-------------------|------------------|
| Outplane Shear (Vc)                                               | 1              | 1             | -1                | -1               |
| Outplane Shear (Vl)                                               | 0              | 0             | 0                 | 0                |
| Outplane Shear (Mt)                                               | 1              | 1             | 1                 | 1                |
| Shear Stress Summary                                              | 3              | 3             | 0                 | 0                |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|--------------------------------------------|----------------|---------------|-------------------|------------------|
| Two * Max Shear Stress                     | 109            | 141           | 132               | 127              |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Outplane Membrane (P )                                              | -1              | -1             | -1               | -1              |
| Outplane Bending (P )                                               | -5              | 5              | -5               | 5               |
| Outplane Membrane (Mc)                                              | 0               | 0              | 0                | 0               |
| Outplane Bending (Mc)                                               | -10             | 10             | 10               | -10             |
| Outplane Membrane (ML)                                              | 0               | 0              | 0                | 0               |
| Outplane Bending (ML)                                               | 0               | 0              | 0                | 0               |
| Normal Pressure Stress                                              | 64              | 64             | 64               | 64              |
| Outplane Stress Summary                                             | 45              | 77             | 67               | 58              |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|-----------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Inplane Membrane (P )                                                 | -1              | -1             | -1               | -1              |
| Inplane Bending (P )                                                  | -18             | 18             | -18              | 18              |
| Inplane Membrane (Mc)                                                 | -1              | -1             | 1                | 1               |
| Inplane Bending (Mc)                                                  | -32             | 32             | 32               | -32             |

|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Inplane Membrane (ML)   | 0   | 0   | 0   | 0   |
| Inplane Bending (ML)    | 0   | 0   | 0   | 0   |
| Inplane Pressure Stress | 128 | 130 | 128 | 130 |
| Inplane Stress Summary  | 73  | 179 | 141 | 117 |

**Vessel Stresses (MPa):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Shear (Vc)                            | 0            | 0           | 0             | 0            |
| Outplane Shear (Vl)                            | -1           | -1          | 1             | 1            |
| Torsional Shear (Mt)                           | 1            | 1           | 1             | 1            |
| Shear Stress Summary                           | 0            | 0           | 3             | 3            |

**Vessel Stresses (MPa):**

|                        |              |             |               |              |
|------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE  | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)   | Left Outside | Left Inside | Right Outside | Right Inside |
| Two * Max Shear Stress | 73           | 179         | 142           | 117          |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 60               | 63   | 60   | 63   | 60   | 63   | 60   | 63   |
| Circ. Pl (SUS)               | 0                | 0    | 0    | 0    | -1   | -1   | 1    | 1    |
| Circ. Q (SUS)                | -17              | 17   | 9    | -9   | -7   | 7    | 3    | -3   |
| Long. Pm (SUS)               | 30               | 30   | 30   | 30   | 30   | 30   | 30   | 30   |
| Long. Pl (SUS)               | -1               | -1   | 1    | 1    | 0    | 0    | 0    | 0    |
| Long. Q (SUS)                | -9               | 9    | 5    | -5   | -14  | 14   | 6    | -6   |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)                     | 60.0             | 63.0 | 60.0 | 63.0 | 60.0 | 63.0 | 60.0 | 63.0 |
| Pm+Pl (SUS)                  | 60.0             | 63.0 | 60.0 | 63.0 | 59.0 | 62.0 | 61.0 | 64.0 |
| Pm+Pl+Q (Total)              | 43.2             | 80.1 | 69.1 | 54.1 | 52.1 | 69.2 | 64.1 | 61.1 |

**Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 63.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 64.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 80.10     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

| Type of Stress Int. Location | Stress Values at |    |    |    |    |    |    |    |
|------------------------------|------------------|----|----|----|----|----|----|----|
|                              | Au               | Al | Bu | Bl | Cu | Cl | Du | Dl |

|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Circ. Pm (SUS)  | 128   | 130   | 128   | 130   | 128   | 130   | 128   | 130   |
| Circ. Pl (SUS)  | -3    | -3    | 1     | 1     | -2    | -2    | 0     | 0     |
| Circ. Q (SUS)   | -14   | 14    | 4     | -4    | -50   | 50    | 14    | -14   |
| Long. Pm (SUS)  | 64    | 64    | 64    | 64    | 64    | 64    | 64    | 64    |
| Long. Pl (SUS)  | -3    | -3    | 1     | 1     | -1    | -1    | -1    | -1    |
| Long. Q (SUS)   | -46   | 46    | 10    | -10   | -15   | 15    | 5     | -5    |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     |
| Pm (SUS)        | 128.0 | 130.0 | 128.0 | 130.0 | 128.0 | 130.0 | 128.0 | 130.0 |
| Pm+Pl (SUS)     | 125.0 | 127.0 | 129.0 | 131.0 | 126.0 | 128.0 | 128.0 | 130.0 |
| Pm+Pl+Q (Total) | 111.0 | 141.1 | 133.0 | 127.0 | 76.0  | 178.0 | 142.1 | 116.1 |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 130.00    | 138.00         | Passed |
| Pm+Pl (SUS)         | 131.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 178.00    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 12               | 15   | 12   | 15   | 12   | 15   | 12   | 15   |
| Circ. Pl (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 6                | 6    | 6    | 6    | 6    | 6    | 6    | 6    |
| Long. Pl (SUS)               | -25              | -25  | 23   | 23   | -19  | -19  | 17   | 17   |
| Long. Q (SUS)                | -67              | 67   | 39   | -39  | -57  | 57   | 29   | -29  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 4                | 4    | -4   | -4   | -4   | -4   | 4    | 4    |
| Shear Q (SUS)                | 10               | 10   | 10   | 10   | 10   | 10   | 10   | 10   |
| Pm (SUS)                     | 12.0             | 15.0 | 12.0 | 15.0 | 12.0 | 15.0 | 12.0 | 15.0 |
| Pm+Pl (SUS)                  | 32.0             | 34.9 | 29.9 | 30.1 | 26.2 | 29.1 | 24.3 | 24.7 |
| Pm+Pl+Q (Total)              | 101.9            | 53.1 | 68.6 | 27.7 | 82.9 | 45.2 | 56.4 | 35.0 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 15.00     | 118.00         | Passed |
| Pm+Pl (SUS)         | 34.93     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 101.92    | 354.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: L22"**

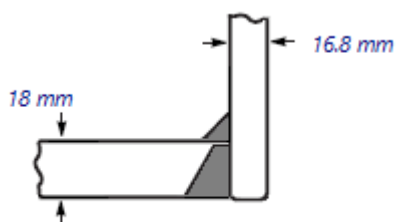
**From: 40**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5847    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 300.00    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 65.00          | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L22"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 15.6561 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.58 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5957 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tt | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 2.0957  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 17.1561 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 17.1561 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 6.3000  | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2) ]$   
 $= \min[ 6.3, \max( 17.1561, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0957, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L22"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm

PV Elite 28 U2 Licensee: SL User

FileName : A150NRL15HN001\_Flash vessel\_7%

Nozzle Calcs.: L2\_2"

Nozl: 42

1:58pm

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**HEXAGON**

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

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**Input, Nozzle Desc: L42"**

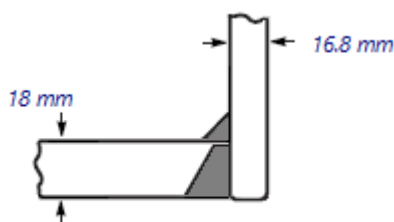
**From: 40**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5847    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 300.00    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 90.00          | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L42"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 15.6561 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.58 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5957 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tt | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L42".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                  |    |
|------------------------------------------------|----------------------------------|----|
| Wall Thickness for Internal/External pressures | ta = 2.0957                      | mm |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000                   | mm |
| Wall Thickness, shell/head, internal pressure  | trb1 = 17.1561                   | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 17.1561 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3 = 6.3000                     | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 6.3, \max( 17.1561, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0957, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L42"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm

PV Elite 28 U2      Licensee: SL User

FileName : A150NRL15HN001\_Flash vessel\_7%

Nozzle Calcs.: L4\_2"

Nozl: 43

1:58pm

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Sep 18, 2026



**HEXAGON**

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

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**Input, Nozzle Desc: L62"**

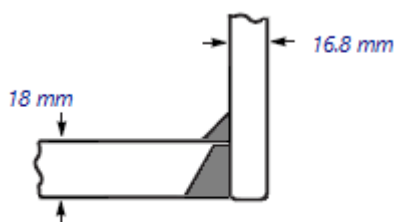
**From: 40**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5847    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 300.00    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 265.00         | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L62"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 15.6561 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.58 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5957 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tt | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L62".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                  |    |
|------------------------------------------------|----------------------------------|----|
| Wall Thickness for Internal/External pressures | ta = 2.0957                      | mm |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000                   | mm |
| Wall Thickness, shell/head, internal pressure  | trb1 = 17.1561                   | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 17.1561 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3 = 6.3000                     | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 6.3, \max( 17.1561, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0957, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L62"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm

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FileName : A150NRL15HN001\_Flash vessel\_7%

Nozzle Calcs.: L6\_2"

Nozl: 44

1:58pm

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**HEXAGON**

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

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**Input, Nozzle Desc: L82"**

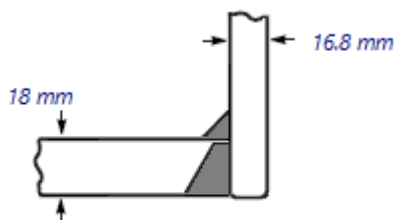
**From: 40**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5847    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 300.00    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 295.00         | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L82"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.58 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.58)$   
 $= 15.6561 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.58 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5957 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tr | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L82".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 2.0957  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 17.1561 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 17.1561 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 6.3000  | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 6.3, \max( 17.1561, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0957, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L82"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm

PV Elite 28 U2 Licensee: SL User

FileName : A150NRL15HN001\_Flash vessel\_7%

Nozzle Calcs.: L8\_2"

Nozl: 45

1:58pm

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The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

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**Input, Nozzle Desc: A518"**

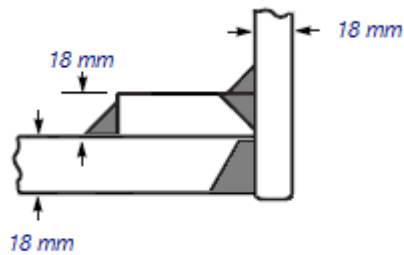
**From: 50**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5530    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 5050.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                  |     |
|---------------------------------------------|------|------------------|-----|
| Material                                    |      | SA-516 70        | +IT |
| Material UNS Number                         |      | K02700           |     |
| Material Specification/Type                 |      | Plate            |     |
| Allowable Stress at Temperature             | Sn   | 138.00           | MPa |
| Allowable Stress At Ambient                 | Sna  | 138.00           | MPa |
| Diameter Basis (for tr calc only)           |      | Outside          |     |
| Layout Angle                                |      | 270.00           | deg |
| Diameter                                    |      | 18.0000          | in. |
| Size and Thickness Basis                    |      | Actual           |     |
| Actual Thickness                            | tn   | 18.0000          | mm  |
| Flange Material                             |      | SA-350 LF1       | +IT |
| Flange Type                                 |      | Weld Neck Flange |     |
| Corrosion Allowance                         | can  | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00             |     |
| Outside Projection                          | ho   | 250.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 18.0000          | mm  |
| Inside Projection                           | h    | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000           | mm  |
| Pad Material                                |      | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp   | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa  | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp   | 817.2000         | mm  |
| Thickness of Pad                            | te   | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp   | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn | 18.0000          | mm  |
| Reinforcing Pad Width                       |      | 180.0000         | mm  |
| Flange Class                                |      | 300              |     |
| Flange Grade                                |      | GR 1.4           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



### Reinforcement CALCULATION, Description: A518"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 18.000 in. |
| Actual Thickness Used in Calculation        | 0.709 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4622 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 2.55 \cdot 228.6 / (138 \cdot 1.0 + 0.4 \cdot 2.55)$   
 $= 4.1980 \text{ mm}$

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 848.4000 mm |
| Parallel to Vessel Wall, opening length                | d  | 424.2000 mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500 mm  |

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

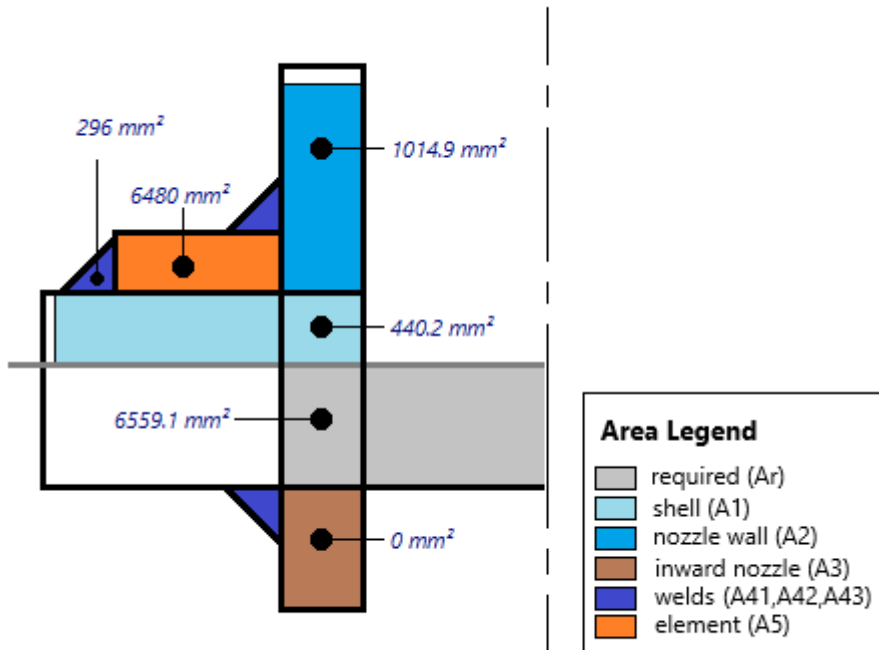
$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 6559.056 | NA       | NA    |
| Area in Shell            | A1          | 440.245  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 1014.912 | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 296.000  | NA       | NA    |
| Area in Element          | A5          | 6480.000 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 8231.157 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 724.3055 18.0000 mm  
 Based on given Pad Diameter: 817.2000 13.3553 mm  
 Based on Shell or Nozzle Thickness: 724.3055 18.0000 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (424.2 \cdot 15.4622 \cdot 1.0 + 2 \cdot 16.5 \cdot 15.4622 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 6559.056 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 424.2(1.0 \cdot 16.5 - 1.0 \cdot 15.462) - 2 \cdot 16.5$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.4622) \cdot (1 - 1.0)$   
 $= 440.245 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 41.25) \cdot (16.5 - 4.2) \cdot 1.0$



= 1014.912 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$   
= 296.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(817.2 - 457.2) \cdot 18.0 \cdot 1.0$   
= 6480.000 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 5.6980 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 16.9622 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 16.9622 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 9.8312 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 9.831, max( 16.9622, 3.0 ) ]  
= 9.8312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 5.698, 9.8312 )  
= 9.8312 mm

Available Nozzle Neck Thickness = 18.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 34.2, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 310.8 MPa | Passed |
| Occasional | : 15.8, | Allowable | : 183.5 MPa | Passed |
| Shear      | : 12.0, | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Gov. MDMT of the Nozzle : -43 °C  
 Gov. MDMT of the Reinforcement Pad : -43 °C  
 Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -29 °C  
 Flange MDMT with Temp reduction per UCS-66(i)(2) -51 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating = 2.55/4.26 = 0.599

#### Weld Size Calculations, Description: A518"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm  
 Intermediate Calc. for pad/shell Welds TminPad 16.5000 mm

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 8.2500 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (6559.0557 - 440.2451 + 2 \cdot 16.5 \cdot 1.0 \cdot (1.0 \cdot 16.5 - 15.4622)) \cdot 138)$   
 $= 849034.50 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (1014.9119 + 6480.0 + 296.0 - 0.0 \cdot 1.0) \cdot 138$   
 $= 1075035.00 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (1014.9119 + 0.0 + 100.0 + (544.5001)) \cdot 138$   
 $= 228975.23 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (1014.9119 + 0.0 + 296.0 + 6480.0 + (544.5001)) \cdot 138$   
 $= 1150168.25 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 457.2 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 485584. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$   
 $= (3.1416/2.0) \cdot 817.2 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 1215085. \text{ N}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 220.35) \cdot (18.0 - 1.5) \cdot 0.7 \cdot 138$   
 $= 1103284. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$   
 $= (3.1416/2) \cdot 457.2 \cdot 18.0 \cdot 0.74 \cdot 138$   
 $= 1319996. \text{ N}$

Tension, Shell Groove Weld [Tngw]:



$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 457.2 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138 \\
 &= 1209996. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (\text{SPEW} + \text{SNW}) = (1215085 + 1103284) = 2318369 \text{ N}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (485584 + 1319996 + 1209996 + 0) = 3015576 \text{ N}$

Failure Path [Path 3-3]:  
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$   
 $= (1215085 + 1209996 + 0) = 2425081 \text{ N}$

#### Summary of Failure Path Calculations:

Path 1-1 = 2318368 N , must exceed W = 849034 N or W1 = 1075035 N  
 Path 2-2 = 3015575 N , must exceed W = 849034 N or W2 = 228975 N  
 Path 3-3 = 2425081 N , must exceed W = 849034 N or W3 = 1150168 N

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.3039 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 300.3039 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 4.098 %

*This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 192.11 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 424.20 mm*

#### Input Echo, WRC297 Item 1, Description: A518" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.5000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 1650.000  | mm         |
| Shell Thickness                      | 18.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |            |
|------------------------------------|-----------|------------|
| Diameter Basis for Nozzle          | OD        |            |
| Nozzle Corrosion Allowance         | 1.5000    | mm         |
| Nozzle Diameter                    | 457.200   | mm         |
| Nozzle Thickness                   | 18.0000   | mm         |
| Nozzle Stress Concentration Factor | 1.000     |            |
| Nozzle Material                    | SA-516 70 |            |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh      | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                              |         |    |
|------------------------------|---------|----|
| Diameter of Reinforcing Pad  | 817.200 | mm |
| Thickness of Reinforcing Pad | 18.0000 | mm |

Note:



**External Forces and Moments in WRC 107/537 Convention:**  
 These loads are assumed to be SUstained loads.

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 2.55      | MPa  |
| Radial Load              | P  | 21599.00  | N    |
| Circumferential Shear    | Vc | 28798.71  | N    |
| Longitudinal Shear       | Vl | 28798.71  | N    |
| Circumferential Moment   | Mc | 0.259E+08 | N-mm |
| Longitudinal Moment      | Ml | 0.337E+08 | N-mm |
| Torsional Moment         | Mt | 0.389E+08 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(843.75(18.0 - 1.5))}$   
 $= 846.570 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1687.500 mm |
| Nozzle Outside Diameter | (d) = | 457.200 mm  |
| Vessel Thickness used   | (T) = | 34.500 mm   |
| Nozzle Thickness used   | (t) = | 16.500 mm   |
| T / t                   | =     | 2.091       |
| d / t                   | =     | 27.709      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.895$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.085 |
| Mr/P    | = | 0.082 |
| M0/P    | = | 0.050 |
| N0/P    | = | 0.155 |
| MrD/Mc  | = | 0.185 |
| NrDL/Mc | = | 0.115 |
| M0d/Mc  | = | 0.130 |
| N0DL/Mc | = | 0.174 |
| MrD/Ml  | = | 0.110 |
| NrDL/Ml | = | 0.072 |
| M0D/Ml  | = | 0.070 |
| N0DL/Ml | = | 0.277 |

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending (P )                                            | -5                   | 5                   | -5                      | 5                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -17                  | -17                 | 17                      | 17                     |
| Outplane Bending (Ml)                                            | -26                  | 26                  | 26                      | -26                    |
| Normal Pressure Stress                                           | 59                   | 62                  | 59                      | 62                     |
| Outplane Stress Summary                                          | 8                    | 73                  | 94                      | 56                     |

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | -1                   | -1                  | -1                      | -1                     |
| Inplane Bending (P )                                               | -8                   | 8                   | -8                      | 8                      |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |



|                         |     |    |    |     |
|-------------------------|-----|----|----|-----|
| Inplane Bending (Mc)    | 0   | 0  | 0  | 0   |
| Inplane Membrane (ML)   | -4  | -4 | 4  | 4   |
| Inplane Bending (ML)    | -40 | 40 | 40 | -40 |
| Inplane Pressure Stress | 29  | 29 | 29 | 29  |

|                        |     |    |    |   |
|------------------------|-----|----|----|---|
| Inplane Stress Summary | -25 | 73 | 64 | 1 |
|------------------------|-----|----|----|---|

**Vessel Stresses (MPa):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 3 | 3 | 3  | 3  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 2 | 2 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 35 | 78 | 95 | 56 |
|------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |    |    |     |
|------------------------|-----|----|----|-----|
| Outplane Membrane (P ) | -2  | -2 | -2 | -2  |
| Outplane Bending (P )  | -5  | 5  | -5 | 5   |
| Outplane Membrane (Mc) | -8  | -8 | 8  | 8   |
| Outplane Bending (Mc)  | -37 | 37 | 37 | -37 |
| Outplane Membrane (ML) | 0   | 0  | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0  | 0  | 0   |
| Normal Pressure Stress | 29  | 29 | 29 | 29  |

|                         |     |    |    |   |
|-------------------------|-----|----|----|---|
| Outplane Stress Summary | -23 | 61 | 67 | 3 |
|-------------------------|-----|----|----|---|

**Vessel Stresses (MPa):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |     |    |    |     |
|-------------------------|-----|----|----|-----|
| Inplane Membrane (P )   | -1  | -1 | -1 | -1  |
| Inplane Bending (P )    | -8  | 8  | -8 | 8   |
| Inplane Membrane (Mc)   | -5  | -5 | 5  | 5   |
| Inplane Bending (Mc)    | -52 | 52 | 52 | -52 |
| Inplane Membrane (ML)   | 0   | 0  | 0  | 0   |
| Inplane Bending (ML)    | 0   | 0  | 0  | 0   |
| Inplane Pressure Stress | 59  | 62 | 59 | 62  |

|                        |    |     |     |    |
|------------------------|----|-----|-----|----|
| Inplane Stress Summary | -8 | 117 | 107 | 22 |
|------------------------|----|-----|-----|----|

**Vessel Stresses (MPa):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 3  | 3  | 3 | 3 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 4 | 4 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |



Two \* Max Shear Stress 23 117 108 23

**Nozzle Stresses (MPa):**

LONGITUDINAL PLANE  
 (Stresses in the  
 hoop direction)

|                      | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )   | -2                   | -2                  | -2                      | -2                     |
| Hoop Bending (P )    | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)   | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)    | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)   | -17                  | -17                 | 17                      | 17                     |
| Hoop Bending (ML)    | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress | 37                   | 40                  | 37                      | 40                     |

Hoop Stress Summary 17 21 52 55

**Nozzle Stresses (MPa):**

LONGITUDINAL PLANE  
 (Stresses Normal to  
 pipe cross-section)

|                       | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )   | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (P )    | -18                  | 18                  | -18                     | 18                     |
| Axial Membrane (Mc)   | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)    | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)   | -13                  | -13                 | 13                      | 13                     |
| Axial Bending (ML)    | -119                 | 119                 | 119                     | -119                   |
| Axial Pressure Stress | 18                   | 18                  | 18                      | 18                     |

Axial Stress Summary -134 143 132 -69

**Nozzle Stresses (MPa):**

LONGITUDINAL PLANE  
 (Shear stress)

|                      | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|----------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)    | 2             | 2            | -2            | -2           |
| Shear due to (Vl)    | 0             | 0            | 0             | 0            |
| Shear due to Torsion | 7             | 7            | 7             | 7            |

Shear Stress Summary 9 9 4 4

**Nozzle Stresses (MPa):**

LONGITUDINAL PLANE  
 (Stress Intensities)

Two \* Max Shear Stress 153 143 133 124

**Nozzle Stresses (MPa):**

CIRCUMFERENTIAL PLANE  
 (Stresses in the  
 hoop direction)

|                      | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )   | -2                    | -2                   | -2                     | -2                    |
| Hoop Bending (P )    | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)   | -8                    | -8                   | 8                      | 8                     |
| Hoop Bending (Mc)    | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)   | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)    | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress | 37                    | 40                   | 37                     | 40                    |

Hoop Stress Summary 26 29 43 46

**Nozzle Stresses (MPa):**

CIRCUMFERENTIAL PLANE  
 (Stresses Normal to  
 pipe cross-section)

|                     | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P ) | 0                     | 0                    | 0                      | 0                     |



|                       |      |     |     |      |
|-----------------------|------|-----|-----|------|
| Axial Bending (P )    | -18  | 18  | -18 | 18   |
| Axial Membrane (Mc)   | -10  | -10 | 10  | 10   |
| Axial Bending (Mc)    | -159 | 159 | 159 | -159 |
| Axial Membrane (ML)   | 0    | 0   | 0   | 0    |
| Axial Bending (ML)    | 0    | 0   | 0   | 0    |
| Axial Pressure Stress | 18   | 18  | 18  | 18   |

Axial Stress Summary -170 185 168 -111

#### Nozzle Stresses (MPa):

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -2            | -2           | 2             | 2            |
| Shear due to Torsion                    | 7             | 7            | 7             | 7            |

Shear Stress Summary 4 4 9 9

#### Nozzle Stresses (MPa):

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 197           | 185          | 169           | 159          |

#### Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{834.75(18.0 - 1.5)}$$

$$= 844.488 \text{ mm}$$

WRC 297 Curve Access Parameters:

|                             |   |             |
|-----------------------------|---|-------------|
| Vessel Mean Diameter (D)    | = | 1669.500 mm |
| Nozzle Outside Diameter (d) | = | 817.200 mm  |
| Vessel Thickness used (T)   | = | 16.500 mm   |
| Nozzle Thickness used (t)   | = | 196.500 mm  |
| T / t                       | = | 0.084       |
| d / t                       | = | 10.000      |

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 4.924$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.045 |
| Mr/P    | = | 0.070 |
| M0/P    | = | 0.020 |
| N0/P    | = | 0.028 |
| MrD/Mc  | = | 0.241 |
| NrDL/Mc | = | 0.099 |
| M0d/Mc  | = | 0.072 |
| N0DL/Mc | = | 0.053 |
| MrD/Ml  | = | 0.051 |
| NrDL/Ml | = | 0.048 |
| M0D/Ml  | = | 0.016 |
| N0DL/Ml | = | 0.027 |

#### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending (P )                                            | -9                   | 9                   | -9                      | 9                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -4                   | -4                  | 4                       | 4                      |
| Outplane Bending (ML)                                            | -14                  | 14                  | 14                      | -14                    |



|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Normal Pressure Stress | 126 | 129 | 126 | 129 |
|------------------------|-----|-----|-----|-----|

|                         |    |     |     |     |
|-------------------------|----|-----|-----|-----|
| Outplane Stress Summary | 96 | 146 | 133 | 126 |
|-------------------------|----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                                                    |                      |                     |                         |                        |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                         |     |    |     |     |
|-------------------------|-----|----|-----|-----|
| Inplane Membrane (P )   | -3  | -3 | -3  | -3  |
| Inplane Bending (P )    | -33 | 33 | -33 | 33  |
| Inplane Membrane (Mc)   | 0   | 0  | 0   | 0   |
| Inplane Bending (Mc)    | 0   | 0  | 0   | 0   |
| Inplane Membrane (ML)   | -7  | -7 | 7   | 7   |
| Inplane Bending (ML)    | -45 | 45 | 45  | -45 |
| Inplane Pressure Stress | 63  | 63 | 63  | 63  |

|                        |     |     |    |    |
|------------------------|-----|-----|----|----|
| Inplane Stress Summary | -27 | 131 | 79 | 54 |
|------------------------|-----|-----|----|----|

**Vessel Stresses (MPa):**

|                                                                      |                      |                     |                         |                        |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 2 | 2 | 2  | 2  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 3 | 3 | 0 | 0 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                                            |                      |                     |                         |                        |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 124 | 147 | 133 | 126 |
|------------------------|-----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                                                        |                       |                      |                        |                       |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                        |     |    |    |     |
|------------------------|-----|----|----|-----|
| Outplane Membrane (P ) | -2  | -2 | -2 | -2  |
| Outplane Bending (P )  | -9  | 9  | -9 | 9   |
| Outplane Membrane (Mc) | -6  | -6 | 6  | 6   |
| Outplane Bending (Mc)  | -50 | 50 | 50 | -50 |
| Outplane Membrane (ML) | 0   | 0  | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0  | 0  | 0   |
| Normal Pressure Stress | 63  | 63 | 63 | 63  |

|                         |    |     |     |    |
|-------------------------|----|-----|-----|----|
| Outplane Stress Summary | -4 | 114 | 108 | 26 |
|-------------------------|----|-----|-----|----|

**Vessel Stresses (MPa):**

|                                                                          |                       |                      |                        |                       |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | -3   | -3  | -3  | -3   |
| Inplane Bending (P )    | -33  | 33  | -33 | 33   |
| Inplane Membrane (Mc)   | -11  | -11 | 11  | 11   |
| Inplane Bending (Mc)    | -168 | 168 | 168 | -168 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 126  | 129 | 126 | 129  |

|                        |     |     |     |   |
|------------------------|-----|-----|-----|---|
| Inplane Stress Summary | -90 | 316 | 269 | 2 |
|------------------------|-----|-----|-----|---|

**Vessel Stresses (MPa):**

|                                                  |            |            |             |             |
|--------------------------------------------------|------------|------------|-------------|-------------|
| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to | Cu<br>Left | Cl<br>Left | Du<br>Right | Dl<br>Right |
|--------------------------------------------------|------------|------------|-------------|-------------|



circumferential plane) Outside Inside Outside Inside

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 2  | 2  | 2 | 2 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 0 | 0 | 3 | 3 |
|----------------------|---|---|---|---|

#### Vessel Stresses (MPa):

|                                               |                       |                      |                        |                       |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Two * Max Shear Stress                        | 90                    | 316                  | 269                    | 26                    |

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of<br>Stress Int.<br>Location | Stress Values at |      |      |      |      |       |       |      |
|------------------------------------|------------------|------|------|------|------|-------|-------|------|
|                                    | Au               | Al   | Bu   | Bl   | Cu   | Cl    | Du    | Dl   |
| Circ. Pm (SUS)                     | 59               | 62   | 59   | 62   | 59   | 62    | 59    | 62   |
| Circ. Pl (SUS)                     | -19              | -19  | 15   | 15   | -6   | -6    | 4     | 4    |
| Circ. Q (SUS)                      | -31              | 31   | 21   | -21  | -60  | 60    | 44    | -44  |
| Long. Pm (SUS)                     | 29               | 29   | 29   | 29   | 29   | 29    | 29    | 29   |
| Long. Pl (SUS)                     | -5               | -5   | 3    | 3    | -10  | -10   | 6     | 6    |
| Long. Q (SUS)                      | -48              | 48   | 32   | -32  | -42  | 42    | 32    | -32  |
| Shear Pm (SUS)                     | 0                | 0    | 0    | 0    | 0    | 0     | 0     | 0    |
| Shear Pl (SUS)                     | 1                | 1    | -1   | -1   | -1   | -1    | 1     | 1    |
| Shear Q (SUS)                      | 3                | 3    | 3    | 3    | 3    | 3     | 3     | 3    |
| Pm (SUS)                           | 59.0             | 62.0 | 59.0 | 62.0 | 59.0 | 62.0  | 59.0  | 62.0 |
| Pm+Pl (SUS)                        | 40.1             | 43.1 | 74.0 | 77.0 | 53.0 | 56.0  | 63.0  | 66.0 |
| Pm+Pl+Q (Total)                    | 34.0             | 77.1 | 95.1 | 56.1 | 23.2 | 116.1 | 107.4 | 22.8 |

##### Vessel Stress Summation Comparison (MPa):

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 62.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 77.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 116.07    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Vessel Stress Summation at Reinforcing Pad Edge (MPa):

| Type of<br>Stress Int.<br>Location | Stress Values at |     |     |     |      |     |     |      |
|------------------------------------|------------------|-----|-----|-----|------|-----|-----|------|
|                                    | Au               | Al  | Bu  | Bl  | Cu   | Cl  | Du  | Dl   |
| Circ. Pm (SUS)                     | 126              | 129 | 126 | 129 | 126  | 129 | 126 | 129  |
| Circ. Pl (SUS)                     | -6               | -6  | 2   | 2   | -14  | -14 | 8   | 8    |
| Circ. Q (SUS)                      | -23              | 23  | 5   | -5  | -201 | 201 | 135 | -135 |
| Long. Pm (SUS)                     | 63               | 63  | 63  | 63  | 63   | 63  | 63  | 63   |
| Long. Pl (SUS)                     | -10              | -10 | 4   | 4   | -8   | -8  | 4   | 4    |
| Long. Q (SUS)                      | -78              | 78  | 12  | -12 | -59  | 59  | 41  | -41  |
| Shear Pm (SUS)                     | 0                | 0   | 0   | 0   | 0    | 0   | 0   | 0    |

|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Shear P1 (SUS)  | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 2     | 2     | 2     | 2     | 2     | 2     | 2     | 2     |
| Pm (SUS)        | 126.0 | 129.0 | 126.0 | 129.0 | 126.0 | 129.0 | 126.0 | 129.0 |
| Pm+P1 (SUS)     | 120.0 | 123.0 | 128.0 | 131.0 | 112.0 | 115.0 | 134.0 | 137.0 |
| Pm+P1+Q (Total) | 122.1 | 146.6 | 133.0 | 126.0 | 89.0  | 316.0 | 269.1 | 26.4  |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 129.00    | 138.00         | Passed |
| Pm+P1 (SUS)         | 137.01    | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 316.00    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 37               | 40    | 37    | 40    | 37    | 40    | 37    | 40    |
| Circ. P1 (SUS)               | -19              | -19   | 15    | 15    | -10   | -10   | 6     | 6     |
| Circ. Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)               | 18               | 18    | 18    | 18    | 18    | 18    | 18    | 18    |
| Long. P1 (SUS)               | -13              | -13   | 13    | 13    | -10   | -10   | 10    | 10    |
| Long. Q (SUS)                | -137             | 137   | 101   | -101  | -177  | 177   | 141   | -141  |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)               | 2                | 2     | -2    | -2    | -2    | -2    | 2     | 2     |
| Shear Q (SUS)                | 7                | 7     | 7     | 7     | 7     | 7     | 7     | 7     |
| Pm (SUS)                     | 37.0             | 40.0  | 37.0  | 40.0  | 37.0  | 40.0  | 37.0  | 40.0  |
| Pm+P1 (SUS)                  | 18.3             | 21.2  | 52.2  | 55.2  | 27.2  | 30.2  | 43.3  | 46.2  |
| Pm+P1+Q (Total)              | 151.1            | 142.7 | 132.3 | 125.4 | 196.3 | 185.2 | 169.6 | 160.0 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 40.00     | 138.00         | Passed |
| Pm+P1 (SUS)         | 55.17     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 196.25    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: A28"**

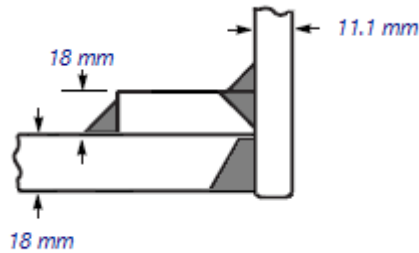
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6050.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |                   |                  |     |
|---------------------------------------------|-------------------|------------------|-----|
| Material                                    |                   | SA-333 6         | +IT |
| Material UNS Number                         |                   | K03006           |     |
| Material Specification/Type                 | Smls. & wld. pipe |                  |     |
| Allowable Stress at Temperature             | Sn                | 118.00           | MPa |
| Allowable Stress At Ambient                 | Sna               | 118.00           | MPa |
| Diameter Basis (for tr calc only)           |                   | Inside           |     |
| Layout Angle                                |                   | 225.00           | deg |
| Diameter                                    |                   | 8.0000           | in. |
| Size and Thickness Basis                    |                   | Minimum          |     |
| Nominal Thickness                           |                   | 80               |     |
| Flange Material                             |                   | SA-350 LF1       | +IT |
| Flange Type                                 |                   | Weld Neck Flange |     |
| Corrosion Allowance                         | can               | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1                | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En                | 1.00             |     |
| Outside Projection                          | ho                | 200.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo                | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv              | 18.0000          | mm  |
| Inside Projection                           | h                 | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi                | 0.0000           | mm  |
| Pad Material                                |                   | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp                | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa               | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp                | 350.0000         | mm  |
| Thickness of Pad                            | te                | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp                | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn              | 18.0000          | mm  |
| Reinforcing Pad Width                       |                   | 65.4625          | mm  |
| Flange Class                                |                   | 300              |     |
| Flange Grade                                |                   | GR 1.4           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



#### Reinforcement CALCULATION, Description: A28"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 7.750 in. |
| Actual Thickness Used in Calculation       | 0.438 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 99.92 / (118 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 2.1849 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 399.7000 mm |
| Parallel to Vessel Wall, opening length                | d  | 199.8500 mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500 mm  |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$

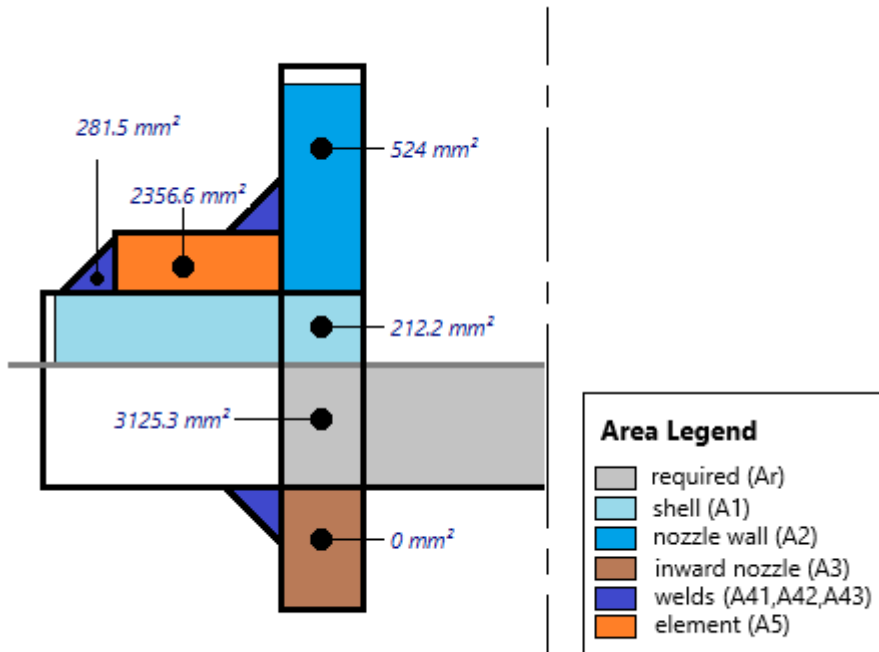


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 3125.333 | NA       | NA    |
| Area in Shell            | A1          | 212.161  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 523.980  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 281.509  | NA       | NA    |
| Area in Element          | A5          | 2356.650 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 3374.300 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

|                                         |          |            |
|-----------------------------------------|----------|------------|
| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter | Thickness  |
| Based on given Pad Thickness:           | 336.1685 | 18.0000 mm |
| Based on given Pad Diameter:            | 350.0000 | 16.0984 mm |
| Based on the Estimated Diameter Limit:  | 398.1125 | 11.7723 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (199.85 \cdot 15.4234 \cdot 1.0 + 2 \cdot 9.6125 \cdot 15.4234 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 3125.333 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 199.85(1.0 \cdot 16.5 - 1.0 \cdot 15.423) - 2 \cdot 9.613$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.4234) \cdot (1 - 0.855)$   
 $= 212.161 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 41.25) \cdot (9.61 - 2.18) \cdot 0.8551$



= 523.980 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 14.0^2 \cdot 1.0$   
= 281.509 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(350.0 - 219.075) \cdot 18.0 \cdot 1.0$   
= 2356.650 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 3.6849 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 16.9234 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 16.9234 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.6600 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 8.66, max( 16.9234, 3.0 ) ]  
= 8.6600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 3.6849, 8.66 )  
= 8.6600 mm

Available Nozzle Neck Thickness = 11.1125 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 40.6, | Allowable | : 118.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 254.4 MPa | Passed |
| Occasional | : 12.6, | Allowable | : 156.9 MPa | Passed |
| Shear      | : 18.7, | Allowable | : 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

##### Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

|                                                        |           |
|--------------------------------------------------------|-----------|
| Impact Test Temperature provided per Specification     | -46 °C    |
| Calculated Minimum Design Metal Temperature            | -104 °C   |
| Gov. MDMT of the Nozzle                                | : -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : -44 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : -44 °C  |

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -29 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -52 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating = 2.55/4.26 = 0.598

#### Weld Size Calculations, Description: A28"

|                                           |         |            |
|-------------------------------------------|---------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 9.6125 mm  |
| Intermediate Calc. for pad/shell Welds    | TminPad | 16.5000 mm |

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 8.2500 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3125.3335 - 212.1606 + 2 \cdot 9.6125 \cdot 0.8551 \cdot (1.0 \cdot 16.5 - 15.4234)) \cdot 138)$   
 $= 404418.50 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (523.9805 + 2356.6499 + 281.5088 - 0.0 \cdot 0.86) \cdot 138$   
 $= 436330.22 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (523.9805 + 0.0 + 85.5088 + (271.2446)) \cdot 138$   
 $= 121528.73 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (523.9805 + 0.0 + 281.5088 + 2356.6499 + (271.2446)) \cdot 138$   
 $= 473758.09 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$   
 $= 198955. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$   
 $= (3.1416/2.0) \cdot 350.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 520411. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 104.7313) \cdot (11.1125 - 1.5) \cdot 0.7 \cdot 118$$

$$= 261220. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 632498. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 579779. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (520411 + 261220) = 781630 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198955 + 632498 + 579779 + 0) = 1411232 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (520411 + 579779 + 0) = 1100190 N$$

#### Summary of Failure Path Calculations:

Path 1-1 = 781630 N , must exceed W = 404418 N or W1 = 436330 N  
 Path 2-2 = 1411231 N , must exceed W = 404418 N or W2 = 121528 N  
 Path 3-3 = 1100190 N , must exceed W = 404418 N or W3 = 473758 N

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 7.3041 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3041 mm

#### Input Echo, WRC297 Item 1, Description: A28" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.5000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 1650.000  | mm         |
| Shell Thickness                      | 18.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |          |            |
|------------------------------------|----------|------------|
| Diameter Basis for Nozzle          | ID       |            |
| Nozzle Corrosion Allowance         | 1.5000   | mm         |
| Nozzle Diameter                    | 196.850  | mm         |
| Nozzle Thickness                   | 11.1125  | mm         |
| Nozzle Stress Concentration Factor | 1.000    |            |
| Nozzle Material                    | SA-333 6 |            |
| Nozzle Cold S.I. Allowable         | SNmc     | 118.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh     | 118.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                             |         |    |
|-----------------------------|---------|----|
| Diameter of Reinforcing Pad | 350.000 | mm |
|-----------------------------|---------|----|



Thickness of Reinforcing Pad 18.0000 mm

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSTAINED loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 2.55      | MPa  |
| Radial Load              | P  | 9599.58   | N    |
| Circumferential Shear    | Vc | 12799.39  | N    |
| Longitudinal Shear       | Vl | 12799.39  | N    |
| Circumferential Moment   | Mc | 0.512E+07 | N-mm |
| Longitudinal Moment      | Ml | 0.666E+07 | N-mm |
| Torsional Moment         | Mt | 0.768E+07 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(843.75(18.0 - 1.5))}$$

$$= 608.445 \text{ mm}$$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1687.500 mm |
| Nozzle Outside Diameter | (d) = | 219.075 mm  |
| Vessel Thickness used   | (T) = | 34.500 mm   |
| Nozzle Thickness used   | (t) = | 9.613 mm    |
| T / t                   | =     | 3.589       |
| d / t                   | =     | 22.791      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.908$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.092 |
| Mr/P    | = | 0.095 |
| M0/P    | = | 0.113 |
| N0/P    | = | 0.249 |
| MrD/Mc  | = | 0.154 |
| NrDL/Mc | = | 0.129 |
| M0d/Mc  | = | 0.222 |
| N0DL/Mc | = | 0.102 |
| MrD/Ml  | = | 0.132 |
| NrDL/Ml | = | 0.115 |
| M0D/Ml  | = | 0.182 |
| N0DL/Ml | = | 0.277 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P)                                            | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending (P)                                             | -5                   | 5                   | -5                      | 5                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -7                   | -7                  | 7                       | 7                      |
| Outplane Bending (Ml)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 59                   | 62                  | 59                      | 62                     |
| Outplane Stress Summary                                          | 17                   | 86                  | 87                      | 44                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|



|                         |     |    |    |     |
|-------------------------|-----|----|----|-----|
| Inplane Membrane (P )   | 0   | 0  | 0  | 0   |
| Inplane Bending (P )    | -4  | 4  | -4 | 4   |
| Inplane Membrane (Mc)   | 0   | 0  | 0  | 0   |
| Inplane Bending (Mc)    | 0   | 0  | 0  | 0   |
| Inplane Membrane (ML)   | -2  | -2 | 2  | 2   |
| Inplane Bending (ML)    | -20 | 20 | 20 | -20 |
| Inplane Pressure Stress | 29  | 29 | 29 | 29  |

|                        |   |    |    |    |
|------------------------|---|----|----|----|
| Inplane Stress Summary | 1 | 51 | 47 | 16 |
|------------------------|---|----|----|----|

**Vessel Stresses (MPa):**

|                                             |                      |                     |                         |                        |
|---------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE                          |                      |                     |                         |                        |
| (Shear stress normal to longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
| Outplane Shear (Vc)                         | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                         | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                         | 2                    | 2                   | 2                       | 2                      |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 1 | 1 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                        |                      |                     |                         |                        |
|------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE     |                      |                     |                         |                        |
| (Stress Intensities)   | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
| Two * Max Shear Stress | 18                   | 87                  | 87                      | 45                     |

**Vessel Stresses (MPa):**

|                                            |                       |                      |                        |                       |
|--------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE                      |                       |                      |                        |                       |
| (Stresses Normal to circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Outplane Membrane (P )                     | -2                    | -2                   | -2                     | -2                    |
| Outplane Bending (P )                      | -5                    | 5                    | -5                     | 5                     |
| Outplane Membrane (Mc)                     | -1                    | -1                   | 1                      | 1                     |
| Outplane Bending (Mc)                      | -26                   | 26                   | 26                     | -26                   |
| Outplane Membrane (ML)                     | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                      | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                     | 29                    | 29                   | 29                     | 29                    |

|                         |    |    |    |   |
|-------------------------|----|----|----|---|
| Outplane Stress Summary | -5 | 57 | 50 | 9 |
|-------------------------|----|----|----|---|

**Vessel Stresses (MPa):**

|                                              |                       |                      |                        |                       |
|----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE                        |                       |                      |                        |                       |
| (Stresses parallel to circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Inplane Membrane (P )                        | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (P )                         | -4                    | 4                    | -4                     | 4                     |
| Inplane Membrane (Mc)                        | -2                    | -2                   | 2                      | 2                     |
| Inplane Bending (Mc)                         | -18                   | 18                   | 18                     | -18                   |
| Inplane Membrane (ML)                        | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                         | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                      | 59                    | 62                   | 59                     | 62                    |

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Inplane Stress Summary | 33 | 81 | 75 | 50 |
|------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                                |                       |                      |                        |                       |
|------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE                          |                       |                      |                        |                       |
| (Shear stress normal to circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Outplane Shear (Vc)                            | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                            | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                           | 2                     | 2                    | 2                      | 2                     |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 1 | 1 | 4 | 4 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**



| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 39                    | 81                   | 75                     | 50                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | -2                   | -2                  | -2                      | -2                     |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -7                   | -7                  | 7                       | 7                      |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 30                   | 33                  | 30                      | 33                     |
| Hoop Stress Summary                                       | 21                   | 24                  | 35                      | 38                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | -1                   | -1                  | -1                      | -1                     |
| Axial Bending (P )                                               | -30                  | 30                  | -30                     | 30                     |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -20                  | -20                 | 20                      | 20                     |
| Axial Bending (ML)                                               | -147                 | 147                 | 147                     | -147                   |
| Axial Pressure Stress                                            | 15                   | 15                  | 15                      | 15                     |
| Axial Stress Summary                                             | -184                 | 170                 | 151                     | -82                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 3             | 3            | -3            | -3           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 10            | 10           | 10            | 10           |
| Shear Stress Summary                 | 14            | 14           | 6             | 6            |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 208           | 171          | 151           | 121          |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | -2                    | -2                   | -2                     | -2                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -1                    | -1                   | 1                      | 1                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 30                    | 33                   | 30                     | 33                    |
| Hoop Stress Summary                                          | 26                    | 29                   | 30                     | 33                    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to | Cu<br>Left | Cl<br>Left | Du<br>Right | Dl<br>Right |
|----------------------------------------------|------------|------------|-------------|-------------|
|----------------------------------------------|------------|------------|-------------|-------------|



| pipe cross-section)   | Outside | Inside | Outside | Inside |
|-----------------------|---------|--------|---------|--------|
| Axial Membrane ( P )  | -1      | -1     | -1      | -1     |
| Axial Bending ( P )   | -30     | 30     | -30     | 30     |
| Axial Membrane (Mc)   | -16     | -16    | 16      | 16     |
| Axial Bending (Mc)    | -136    | 136    | 136     | -136   |
| Axial Membrane (ML)   | 0       | 0      | 0       | 0      |
| Axial Bending (ML)    | 0       | 0      | 0       | 0      |
| Axial Pressure Stress | 15      | 15     | 15      | 15     |
| Axial Stress Summary  | -169    | 164    | 135     | -76    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -3            | -3           | 3             | 3            |
| Shear due to Torsion                    | 10            | 10           | 10            | 10           |
| Shear Stress Summary                    | 6             | 6            | 14            | 14           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 196           | 164          | 137           | 113          |

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(834.75(18.0 - 1.5))}$$

$$= 606.363 \text{ mm}$$

**WRC 297 Curve Access Parameters:**

Vessel Mean Diameter (D) = 1669.500 mm  
 Nozzle Outside Diameter (d) = 350.000 mm  
 Vessel Thickness used (T) = 16.500 mm  
 Nozzle Thickness used (t) = 75.075 mm  
 T / t = 0.220  
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.109$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.068  
 Mr/P = 0.140  
 M0/P = 0.041  
 N0/P = 0.078  
 MrD/Mc = 0.286  
 NrDL/Mc = 0.090  
 M0d/Mc = 0.088  
 N0DL/Mc = 0.079  
 MrD/Ml = 0.107  
 NrDL/Ml = 0.083  
 M0D/Ml = 0.033  
 N0DL/Ml = 0.101

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane ( P )                                          | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending ( P )                                           | -8                   | 8                   | -8                      | 8                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |



|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Outplane Bending (Mc)  | 0   | 0   | 0   | 0   |
| Outplane Membrane (ML) | -7  | -7  | 7   | 7   |
| Outplane Bending (ML)  | -13 | 13  | 13  | -13 |
| Normal Pressure Stress | 126 | 128 | 126 | 128 |

|                         |    |     |     |     |
|-------------------------|----|-----|-----|-----|
| Outplane Stress Summary | 94 | 141 | 135 | 128 |
|-------------------------|----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                           |             |            |                |               |
|-------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                        | Au          | Al         | Bu             | Bl            |
| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                         |     |    |     |     |
|-------------------------|-----|----|-----|-----|
| Inplane Membrane (P )   | -2  | -2 | -2  | -2  |
| Inplane Bending (P )    | -29 | 29 | -29 | 29  |
| Inplane Membrane (Mc)   | 0   | 0  | 0   | 0   |
| Inplane Bending (Mc)    | 0   | 0  | 0   | 0   |
| Inplane Membrane (ML)   | -5  | -5 | 5   | 5   |
| Inplane Bending (ML)    | -44 | 44 | 44  | -44 |
| Inplane Pressure Stress | 63  | 63 | 63  | 63  |

|                        |     |     |    |    |
|------------------------|-----|-----|----|----|
| Inplane Stress Summary | -19 | 129 | 81 | 51 |
|------------------------|-----|-----|----|----|

**Vessel Stresses (MPa):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 2 | 2 | 2  | 2  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 3 | 3 | 1 | 1 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 113 | 142 | 135 | 128 |
|------------------------|-----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |    |    |     |
|------------------------|-----|----|----|-----|
| Outplane Membrane (P ) | -2  | -2 | -2 | -2  |
| Outplane Bending (P )  | -8  | 8  | -8 | 8   |
| Outplane Membrane (Mc) | -4  | -4 | 4  | 4   |
| Outplane Bending (Mc)  | -28 | 28 | 28 | -28 |
| Outplane Membrane (ML) | 0   | 0  | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0  | 0  | 0   |
| Normal Pressure Stress | 63  | 63 | 63 | 63  |

|                         |    |    |    |    |
|-------------------------|----|----|----|----|
| Outplane Stress Summary | 19 | 93 | 84 | 45 |
|-------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Inplane Membrane (P )   | -2  | -2  | -2  | -2  |
| Inplane Bending (P )    | -29 | 29  | -29 | 29  |
| Inplane Membrane (Mc)   | -4  | -4  | 4   | 4   |
| Inplane Bending (Mc)    | -92 | 92  | 92  | -92 |
| Inplane Membrane (ML)   | 0   | 0   | 0   | 0   |
| Inplane Bending (ML)    | 0   | 0   | 0   | 0   |
| Inplane Pressure Stress | 126 | 128 | 126 | 128 |

|                        |    |     |     |    |
|------------------------|----|-----|-----|----|
| Inplane Stress Summary | -2 | 243 | 191 | 68 |
|------------------------|----|-----|-----|----|



**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to circumferential plane) |  | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-------------------------------------------------------------------------|--|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                     |  | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                     |  | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                    |  | 2                     | 2                    | 2                      | 2                     |
| Shear Stress Summary                                                    |  | 1                     | 1                    | 3                      | 3                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) |  | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|--|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        |  | 22                    | 243                  | 191                    | 69                    |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 59               | 62   | 59   | 62   | 59   | 62   | 59   | 62   |
| Circ. Pl (SUS)               | -9               | -9   | 5    | 5    | -2   | -2   | 2    | 2    |
| Circ. Q (SUS)                | -32              | 32   | 22   | -22  | -22  | 22   | 14   | -14  |
| Long. Pm (SUS)               | 29               | 29   | 29   | 29   | 29   | 29   | 29   | 29   |
| Long. Pl (SUS)               | -2               | -2   | 2    | 2    | -3   | -3   | -1   | -1   |
| Long. Q (SUS)                | -24              | 24   | 16   | -16  | -31  | 31   | 21   | -21  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 1                | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)                     | 59.0             | 62.0 | 59.0 | 62.0 | 59.0 | 62.0 | 59.0 | 62.0 |
| Pm+Pl (SUS)                  | 50.0             | 53.0 | 64.0 | 67.0 | 57.0 | 60.0 | 61.0 | 64.0 |
| Pm+Pl+Q (Total)              | 18.6             | 85.3 | 86.0 | 45.0 | 40.0 | 82.0 | 75.3 | 50.2 |

**Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 62.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 67.03     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 86.03     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

| Type of Stress Int. Location | Stress Values at |     |     |     |      |     |     |     |
|------------------------------|------------------|-----|-----|-----|------|-----|-----|-----|
|                              | Au               | Al  | Bu  | Bl  | Cu   | Cl  | Du  | Dl  |
| Circ. Pm (SUS)               | 126              | 128 | 126 | 128 | 126  | 128 | 126 | 128 |
| Circ. Pl (SUS)               | -9               | -9  | 5   | 5   | -6   | -6  | 2   | 2   |
| Circ. Q (SUS)                | -21              | 21  | 5   | -5  | -121 | 121 | 63  | -63 |
| Long. Pm (SUS)               | 63               | 63  | 63  | 63  | 63   | 63  | 63  | 63  |
| Long. Pl (SUS)               | -7               | -7  | 3   | 3   | -6   | -6  | 2   | 2   |

|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Long. Q (SUS)   | -73   | 73    | 15    | -15   | -36   | 36    | 20    | -20   |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)  | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 2     | 2     | 2     | 2     | 2     | 2     | 2     | 2     |
| Pm (SUS)        | 126.0 | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 |
| Pm+P1 (SUS)     | 117.0 | 119.0 | 131.0 | 133.0 | 120.0 | 122.0 | 128.0 | 130.0 |
| Pm+P1+Q (Total) | 113.2 | 140.8 | 136.0 | 128.0 | 22.1  | 243.0 | 191.1 | 67.4  |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 128.00    | 138.00         | Passed |
| Pm+P1 (SUS)         | 133.01    | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 243.01    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 30               | 33    | 30    | 33    | 30    | 33    | 30    | 33    |
| Circ. P1 (SUS)               | -9               | -9    | 5     | 5     | -3    | -3    | -1    | -1    |
| Circ. Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)               | 15               | 15    | 15    | 15    | 15    | 15    | 15    | 15    |
| Long. P1 (SUS)               | -21              | -21   | 19    | 19    | -17   | -17   | 15    | 15    |
| Long. Q (SUS)                | -177             | 177   | 117   | -117  | -166  | 166   | 106   | -106  |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)               | 3                | 3     | -3    | -3    | -3    | -3    | 3     | 3     |
| Shear Q (SUS)                | 10               | 10    | 10    | 10    | 10    | 10    | 10    | 10    |
| Pm (SUS)                     | 30.0             | 33.0  | 30.0  | 33.0  | 30.0  | 33.0  | 30.0  | 33.0  |
| Pm+P1 (SUS)                  | 27.7             | 30.6  | 37.5  | 39.6  | 29.6  | 32.6  | 32.5  | 34.2  |
| Pm+P1+Q (Total)              | 205.7            | 172.1 | 151.4 | 121.8 | 195.5 | 164.4 | 137.6 | 111.1 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 33.00     | 118.00         | Passed |
| Pm+P1 (SUS)         | 39.61     | 177.00         | Passed |
| Pm+P1+Q (TOTAL)     | 205.65    | 354.00         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: A68"**

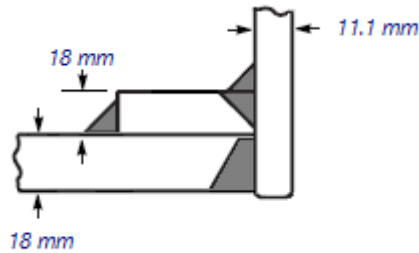
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6050.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |                   |                  |     |
|---------------------------------------------|-------------------|------------------|-----|
| Material                                    |                   | SA-333 6         | +IT |
| Material UNS Number                         |                   | K03006           |     |
| Material Specification/Type                 | Smls. & wld. pipe |                  |     |
| Allowable Stress at Temperature             | Sn                | 118.00           | MPa |
| Allowable Stress At Ambient                 | Sna               | 118.00           | MPa |
| Diameter Basis (for tr calc only)           |                   | Inside           |     |
| Layout Angle                                |                   | 135.00           | deg |
| Diameter                                    |                   | 8.0000           | in. |
| Size and Thickness Basis                    |                   | Minimum          |     |
| Nominal Thickness                           |                   | 80               |     |
| Flange Material                             |                   | SA-350 LF1       | +IT |
| Flange Type                                 |                   | Weld Neck Flange |     |
| Corrosion Allowance                         | can               | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1                | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En                | 1.00             |     |
| Outside Projection                          | ho                | 200.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo                | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv              | 18.0000          | mm  |
| Inside Projection                           | h                 | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi                | 0.0000           | mm  |
| Pad Material                                |                   | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp                | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa               | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp                | 350.0000         | mm  |
| Thickness of Pad                            | te                | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp                | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn              | 18.0000          | mm  |
| Reinforcing Pad Width                       |                   | 65.4625          | mm  |
| Flange Class                                |                   | 300              |     |
| Flange Grade                                |                   | GR 1.4           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



### Reinforcement CALCULATION, Description: A68"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 7.750 in. |
| Actual Thickness Used in Calculation       | 0.438 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 99.92 / (118 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 2.1849 \text{ mm}$

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 399.7000 mm |
| Parallel to Vessel Wall, opening length                | d  | 199.8500 mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500 mm  |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$

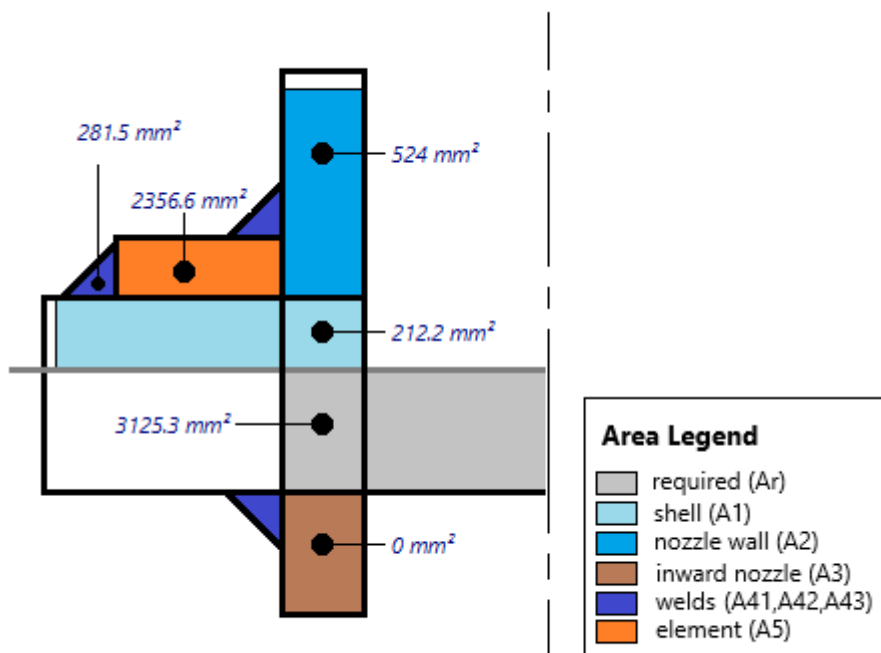


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 3125.333 | NA       | NA    |
| Area in Shell            | A1          | 212.161  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 523.980  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 281.509  | NA       | NA    |
| Area in Element          | A5          | 2356.650 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 3374.300 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 336.1685 18.0000 mm  
 Based on given Pad Diameter: 350.0000 16.0984 mm  
 Based on the Estimated Diameter Limit: 398.1125 11.7723 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (199.85 \cdot 15.4234 \cdot 1.0 + 2 \cdot 9.6125 \cdot 15.4234 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 3125.333 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 199.85(1.0 \cdot 16.5 - 1.0 \cdot 15.423) - 2 \cdot 9.613$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.4234) \cdot (1 - 0.855)$   
 $= 212.161 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 41.25) \cdot (9.61 - 2.18) \cdot 0.8551$



= 523.980 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 14.0^2 \cdot 1.0$   
= 281.509 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(350.0 - 219.075) \cdot 18.0 \cdot 1.0$   
= 2356.650 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 3.6849 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 16.9234 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 16.9234 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.6600 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 8.66, max( 16.9234, 3.0 ) ]  
= 8.6600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 3.6849, 8.66 )  
= 8.6600 mm

Available Nozzle Neck Thickness = 11.1125 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 40.6, | Allowable | : 118.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 254.4 MPa | Passed |
| Occasional | : 12.6, | Allowable | : 156.9 MPa | Passed |
| Shear      | : 18.7, | Allowable | : 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

*Note:*  
*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

*Note: This Element/Detail was specified as being Impact Tested.*

Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

|                                                        |           |
|--------------------------------------------------------|-----------|
| Impact Test Temperature provided per Specification     | -46 °C    |
| Calculated Minimum Design Metal Temperature            | -104 °C   |
| Gov. MDMT of the Nozzle                                | : -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : -44 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : -44 °C  |

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -29 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -52 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating = 2.55/4.26 = 0.598

#### Weld Size Calculations, Description: A68"

|                                           |         |         |    |
|-------------------------------------------|---------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 9.6125  | mm |
| Intermediate Calc. for pad/shell Welds    | TminPad | 16.5000 | mm |

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 8.2500 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3125.3335 - 212.1606 + 2 \cdot 9.6125 \cdot 0.8551 \cdot (1.0 \cdot 16.5 - 15.4234)) \cdot 138)$   
 $= 404418.50 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (523.9805 + 2356.6499 + 281.5088 - 0.0 \cdot 0.86) \cdot 138$   
 $= 436330.22 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (523.9805 + 0.0 + 85.5088 + (271.2446)) \cdot 138$   
 $= 121528.73 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (523.9805 + 0.0 + 281.5088 + 2356.6499 + (271.2446)) \cdot 138$   
 $= 473758.09 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$   
 $= 198955. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$   
 $= (3.1416/2.0) \cdot 350.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 520411. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 104.7313) \cdot (11.1125 - 1.5) \cdot 0.7 \cdot 118$$

$$= 261220. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 632498. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 579779. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (520411 + 261220) = 781630 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198955 + 632498 + 579779 + 0) = 1411232 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (520411 + 579779 + 0) = 1100190 N$$

#### Summary of Failure Path Calculations:

Path 1-1 = 781630 N , must exceed W = 404418 N or W1 = 436330 N  
 Path 2-2 = 1411231 N , must exceed W = 404418 N or W2 = 121528 N  
 Path 3-3 = 1100190 N , must exceed W = 404418 N or W3 = 473758 N

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 7.3041 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3041 mm

#### Input Echo, WRC297 Item 1, Description: A68" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.5000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 1650.000  | mm         |
| Shell Thickness                      | 18.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |          |            |
|------------------------------------|----------|------------|
| Diameter Basis for Nozzle          | ID       |            |
| Nozzle Corrosion Allowance         | 1.5000   | mm         |
| Nozzle Diameter                    | 196.850  | mm         |
| Nozzle Thickness                   | 11.1125  | mm         |
| Nozzle Stress Concentration Factor | 1.000    |            |
| Nozzle Material                    | SA-333 6 |            |
| Nozzle Cold S.I. Allowable         | SNmc     | 118.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh     | 118.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                             |         |    |
|-----------------------------|---------|----|
| Diameter of Reinforcing Pad | 350.000 | mm |
|-----------------------------|---------|----|



Thickness of Reinforcing Pad 18.0000 mm

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSTAINED loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 2.55      | MPa  |
| Radial Load              | P  | 9599.58   | N    |
| Circumferential Shear    | Vc | 12799.39  | N    |
| Longitudinal Shear       | Vl | 12799.39  | N    |
| Circumferential Moment   | Mc | 0.512E+07 | N-mm |
| Longitudinal Moment      | Ml | 0.666E+07 | N-mm |
| Torsional Moment         | Mt | 0.768E+07 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(843.75(18.0 - 1.5))}$   
 $= 608.445 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1687.500 mm |
| Nozzle Outside Diameter | (d) = | 219.075 mm  |
| Vessel Thickness used   | (T) = | 34.500 mm   |
| Nozzle Thickness used   | (t) = | 9.613 mm    |
| T / t                   | =     | 3.589       |
| d / t                   | =     | 22.791      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.908$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.092 |
| Mr/P    | = | 0.095 |
| M0/P    | = | 0.113 |
| N0/P    | = | 0.249 |
| MrD/Mc  | = | 0.154 |
| NrDL/Mc | = | 0.129 |
| M0d/Mc  | = | 0.222 |
| N0DL/Mc | = | 0.102 |
| MrD/Ml  | = | 0.132 |
| NrDL/Ml | = | 0.115 |
| M0D/Ml  | = | 0.182 |
| N0DL/Ml | = | 0.277 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending (P )                                            | -5                   | 5                   | -5                      | 5                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -7                   | -7                  | 7                       | 7                      |
| Outplane Bending (Ml)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 59                   | 62                  | 59                      | 62                     |
| Outplane Stress Summary                                          | 17                   | 86                  | 87                      | 44                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|



|                         |     |    |    |     |
|-------------------------|-----|----|----|-----|
| Inplane Membrane (P )   | 0   | 0  | 0  | 0   |
| Inplane Bending (P )    | -4  | 4  | -4 | 4   |
| Inplane Membrane (Mc)   | 0   | 0  | 0  | 0   |
| Inplane Bending (Mc)    | 0   | 0  | 0  | 0   |
| Inplane Membrane (ML)   | -2  | -2 | 2  | 2   |
| Inplane Bending (ML)    | -20 | 20 | 20 | -20 |
| Inplane Pressure Stress | 29  | 29 | 29 | 29  |

|                        |   |    |    |    |
|------------------------|---|----|----|----|
| Inplane Stress Summary | 1 | 51 | 47 | 16 |
|------------------------|---|----|----|----|

**Vessel Stresses (MPa):**

|                                             |                      |                     |                         |                        |
|---------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE                          |                      |                     |                         |                        |
| (Shear stress normal to longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
| Outplane Shear (Vc)                         | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                         | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                         | 2                    | 2                   | 2                       | 2                      |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 1 | 1 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                        |                      |                     |                         |                        |
|------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE     |                      |                     |                         |                        |
| (Stress Intensities)   | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
| Two * Max Shear Stress | 18                   | 87                  | 87                      | 45                     |

**Vessel Stresses (MPa):**

|                                            |                       |                      |                        |                       |
|--------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE                      |                       |                      |                        |                       |
| (Stresses Normal to circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Outplane Membrane (P )                     | -2                    | -2                   | -2                     | -2                    |
| Outplane Bending (P )                      | -5                    | 5                    | -5                     | 5                     |
| Outplane Membrane (Mc)                     | -1                    | -1                   | 1                      | 1                     |
| Outplane Bending (Mc)                      | -26                   | 26                   | 26                     | -26                   |
| Outplane Membrane (ML)                     | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                      | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                     | 29                    | 29                   | 29                     | 29                    |

|                         |    |    |    |   |
|-------------------------|----|----|----|---|
| Outplane Stress Summary | -5 | 57 | 50 | 9 |
|-------------------------|----|----|----|---|

**Vessel Stresses (MPa):**

|                                              |                       |                      |                        |                       |
|----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE                        |                       |                      |                        |                       |
| (Stresses parallel to circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Inplane Membrane (P )                        | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (P )                         | -4                    | 4                    | -4                     | 4                     |
| Inplane Membrane (Mc)                        | -2                    | -2                   | 2                      | 2                     |
| Inplane Bending (Mc)                         | -18                   | 18                   | 18                     | -18                   |
| Inplane Membrane (ML)                        | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                         | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                      | 59                    | 62                   | 59                     | 62                    |

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Inplane Stress Summary | 33 | 81 | 75 | 50 |
|------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                                |                       |                      |                        |                       |
|------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE                          |                       |                      |                        |                       |
| (Shear stress normal to circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Outplane Shear (Vc)                            | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                            | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                           | 2                     | 2                    | 2                      | 2                     |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 1 | 1 | 4 | 4 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**



| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 39                    | 81                   | 75                     | 50                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | -2                   | -2                  | -2                      | -2                     |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -7                   | -7                  | 7                       | 7                      |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 30                   | 33                  | 30                      | 33                     |
| Hoop Stress Summary                                       | 21                   | 24                  | 35                      | 38                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | -1                   | -1                  | -1                      | -1                     |
| Axial Bending (P )                                               | -30                  | 30                  | -30                     | 30                     |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -20                  | -20                 | 20                      | 20                     |
| Axial Bending (ML)                                               | -147                 | 147                 | 147                     | -147                   |
| Axial Pressure Stress                                            | 15                   | 15                  | 15                      | 15                     |
| Axial Stress Summary                                             | -184                 | 170                 | 151                     | -82                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 3             | 3            | -3            | -3           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 10            | 10           | 10            | 10           |
| Shear Stress Summary                 | 14            | 14           | 6             | 6            |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 208           | 171          | 151           | 121          |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | -2                    | -2                   | -2                     | -2                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -1                    | -1                   | 1                      | 1                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 30                    | 33                   | 30                     | 33                    |
| Hoop Stress Summary                                          | 26                    | 29                   | 30                     | 33                    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to | Cu<br>Left | Cl<br>Left | Du<br>Right | Dl<br>Right |
|----------------------------------------------|------------|------------|-------------|-------------|
|----------------------------------------------|------------|------------|-------------|-------------|



| pipe cross-section)   | Outside | Inside | Outside | Inside |
|-----------------------|---------|--------|---------|--------|
| Axial Membrane ( P )  | -1      | -1     | -1      | -1     |
| Axial Bending ( P )   | -30     | 30     | -30     | 30     |
| Axial Membrane (Mc)   | -16     | -16    | 16      | 16     |
| Axial Bending (Mc)    | -136    | 136    | 136     | -136   |
| Axial Membrane (ML)   | 0       | 0      | 0       | 0      |
| Axial Bending (ML)    | 0       | 0      | 0       | 0      |
| Axial Pressure Stress | 15      | 15     | 15      | 15     |
| Axial Stress Summary  | -169    | 164    | 135     | -76    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -3            | -3           | 3             | 3            |
| Shear due to Torsion                    | 10            | 10           | 10            | 10           |
| Shear Stress Summary                    | 6             | 6            | 14            | 14           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 196           | 164          | 137           | 113          |

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(834.75(18.0 - 1.5))}$$

$$= 606.363 \text{ mm}$$

**WRC 297 Curve Access Parameters:**

Vessel Mean Diameter (D) = 1669.500 mm  
 Nozzle Outside Diameter (d) = 350.000 mm  
 Vessel Thickness used (T) = 16.500 mm  
 Nozzle Thickness used (t) = 75.075 mm  
 T / t = 0.220  
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.109$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.068  
 Mr/P = 0.140  
 M0/P = 0.041  
 N0/P = 0.078  
 MrD/Mc = 0.286  
 NrDL/Mc = 0.090  
 M0d/Mc = 0.088  
 N0DL/Mc = 0.079  
 MrD/Ml = 0.107  
 NrDL/Ml = 0.083  
 M0D/Ml = 0.033  
 N0DL/Ml = 0.101

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane ( P )                                          | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending ( P )                                           | -8                   | 8                   | -8                      | 8                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |



|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Outplane Bending (Mc)  | 0   | 0   | 0   | 0   |
| Outplane Membrane (ML) | -7  | -7  | 7   | 7   |
| Outplane Bending (ML)  | -13 | 13  | 13  | -13 |
| Normal Pressure Stress | 126 | 128 | 126 | 128 |

|                         |    |     |     |     |
|-------------------------|----|-----|-----|-----|
| Outplane Stress Summary | 94 | 141 | 135 | 128 |
|-------------------------|----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                           |             |            |                |               |
|-------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                        | Au          | Al         | Bu             | Bl            |
| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                         |     |    |     |     |
|-------------------------|-----|----|-----|-----|
| Inplane Membrane (P )   | -2  | -2 | -2  | -2  |
| Inplane Bending (P )    | -29 | 29 | -29 | 29  |
| Inplane Membrane (Mc)   | 0   | 0  | 0   | 0   |
| Inplane Bending (Mc)    | 0   | 0  | 0   | 0   |
| Inplane Membrane (ML)   | -5  | -5 | 5   | 5   |
| Inplane Bending (ML)    | -44 | 44 | 44  | -44 |
| Inplane Pressure Stress | 63  | 63 | 63  | 63  |

|                        |     |     |    |    |
|------------------------|-----|-----|----|----|
| Inplane Stress Summary | -19 | 129 | 81 | 51 |
|------------------------|-----|-----|----|----|

**Vessel Stresses (MPa):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 2 | 2 | 2  | 2  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 3 | 3 | 1 | 1 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 113 | 142 | 135 | 128 |
|------------------------|-----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |    |    |     |
|------------------------|-----|----|----|-----|
| Outplane Membrane (P ) | -2  | -2 | -2 | -2  |
| Outplane Bending (P )  | -8  | 8  | -8 | 8   |
| Outplane Membrane (Mc) | -4  | -4 | 4  | 4   |
| Outplane Bending (Mc)  | -28 | 28 | 28 | -28 |
| Outplane Membrane (ML) | 0   | 0  | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0  | 0  | 0   |
| Normal Pressure Stress | 63  | 63 | 63 | 63  |

|                         |    |    |    |    |
|-------------------------|----|----|----|----|
| Outplane Stress Summary | 19 | 93 | 84 | 45 |
|-------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Inplane Membrane (P )   | -2  | -2  | -2  | -2  |
| Inplane Bending (P )    | -29 | 29  | -29 | 29  |
| Inplane Membrane (Mc)   | -4  | -4  | 4   | 4   |
| Inplane Bending (Mc)    | -92 | 92  | 92  | -92 |
| Inplane Membrane (ML)   | 0   | 0   | 0   | 0   |
| Inplane Bending (ML)    | 0   | 0   | 0   | 0   |
| Inplane Pressure Stress | 126 | 128 | 126 | 128 |

|                        |    |     |     |    |
|------------------------|----|-----|-----|----|
| Inplane Stress Summary | -2 | 243 | 191 | 68 |
|------------------------|----|-----|-----|----|

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to circumferential plane) |  | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-------------------------------------------------------------------------|--|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                     |  | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                     |  | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                    |  | 2                     | 2                    | 2                      | 2                     |
| Shear Stress Summary                                                    |  | 1                     | 1                    | 3                      | 3                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) |  | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|--|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        |  | 22                    | 243                  | 191                    | 69                    |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 59               | 62   | 59   | 62   | 59   | 62   | 59   | 62   |
| Circ. Pl (SUS)               | -9               | -9   | 5    | 5    | -2   | -2   | 2    | 2    |
| Circ. Q (SUS)                | -32              | 32   | 22   | -22  | -22  | 22   | 14   | -14  |
| Long. Pm (SUS)               | 29               | 29   | 29   | 29   | 29   | 29   | 29   | 29   |
| Long. Pl (SUS)               | -2               | -2   | 2    | 2    | -3   | -3   | -1   | -1   |
| Long. Q (SUS)                | -24              | 24   | 16   | -16  | -31  | 31   | 21   | -21  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 1                | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)                     | 59.0             | 62.0 | 59.0 | 62.0 | 59.0 | 62.0 | 59.0 | 62.0 |
| Pm+Pl (SUS)                  | 50.0             | 53.0 | 64.0 | 67.0 | 57.0 | 60.0 | 61.0 | 64.0 |
| Pm+Pl+Q (Total)              | 18.6             | 85.3 | 86.0 | 45.0 | 40.0 | 82.0 | 75.3 | 50.2 |

**Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 62.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 67.03     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 86.03     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

| Type of Stress Int. Location | Stress Values at |     |     |     |      |     |     |     |
|------------------------------|------------------|-----|-----|-----|------|-----|-----|-----|
|                              | Au               | Al  | Bu  | Bl  | Cu   | Cl  | Du  | Dl  |
| Circ. Pm (SUS)               | 126              | 128 | 126 | 128 | 126  | 128 | 126 | 128 |
| Circ. Pl (SUS)               | -9               | -9  | 5   | 5   | -6   | -6  | 2   | 2   |
| Circ. Q (SUS)                | -21              | 21  | 5   | -5  | -121 | 121 | 63  | -63 |
| Long. Pm (SUS)               | 63               | 63  | 63  | 63  | 63   | 63  | 63  | 63  |
| Long. Pl (SUS)               | -7               | -7  | 3   | 3   | -6   | -6  | 2   | 2   |



|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Long. Q (SUS)   | -73   | 73    | 15    | -15   | -36   | 36    | 20    | -20   |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)  | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 2     | 2     | 2     | 2     | 2     | 2     | 2     | 2     |
| Pm (SUS)        | 126.0 | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 |
| Pm+P1 (SUS)     | 117.0 | 119.0 | 131.0 | 133.0 | 120.0 | 122.0 | 128.0 | 130.0 |
| Pm+P1+Q (Total) | 113.2 | 140.8 | 136.0 | 128.0 | 22.1  | 243.0 | 191.1 | 67.4  |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 128.00    | 138.00         | Passed |
| Pm+P1 (SUS)         | 133.01    | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 243.01    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 30               | 33    | 30    | 33    | 30    | 33    | 30    | 33    |
| Circ. P1 (SUS)               | -9               | -9    | 5     | 5     | -3    | -3    | -1    | -1    |
| Circ. Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)               | 15               | 15    | 15    | 15    | 15    | 15    | 15    | 15    |
| Long. P1 (SUS)               | -21              | -21   | 19    | 19    | -17   | -17   | 15    | 15    |
| Long. Q (SUS)                | -177             | 177   | 117   | -117  | -166  | 166   | 106   | -106  |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)               | 3                | 3     | -3    | -3    | -3    | -3    | 3     | 3     |
| Shear Q (SUS)                | 10               | 10    | 10    | 10    | 10    | 10    | 10    | 10    |
| Pm (SUS)                     | 30.0             | 33.0  | 30.0  | 33.0  | 30.0  | 33.0  | 30.0  | 33.0  |
| Pm+P1 (SUS)                  | 27.7             | 30.6  | 37.5  | 39.6  | 29.6  | 32.6  | 32.5  | 34.2  |
| Pm+P1+Q (Total)              | 205.7            | 172.1 | 151.4 | 121.8 | 195.5 | 164.4 | 137.6 | 111.1 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 33.00     | 118.00         | Passed |
| Pm+P1 (SUS)         | 39.61     | 177.00         | Passed |
| Pm+P1+Q (TOTAL)     | 205.65    | 354.00         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: M124"**

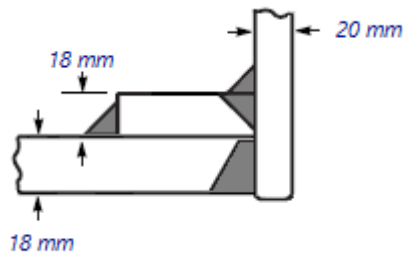
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6050.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                               |      |                  |     |
|-----------------------------------------------|------|------------------|-----|
| Material                                      |      | SA-516 70        | +IT |
| Material UNS Number                           |      | K02700           |     |
| Material Specification/Type                   |      | Plate            |     |
| Allowable Stress at Temperature               | Sn   | 138.00           | MPa |
| Allowable Stress At Ambient                   | Sna  | 138.00           | MPa |
| Diameter Basis (for tr calc only)             |      | Outside          |     |
| Layout Angle                                  |      | 0.00             | deg |
| Diameter                                      |      | 24.0000          | in. |
| Size and Thickness Basis                      |      | Actual           |     |
| Actual Thickness                              | tn   | 20.0000          | mm  |
| Flange Material                               |      | SA-350 LF1       | +IT |
| Flange Type                                   |      | Weld Neck Flange |     |
| Corrosion Allowance                           | can  | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle      | E1   | 1.00             |     |
| Joint Efficiency of Nozzle Neck               | En   | 1.00             |     |
| Outside Projection                            | ho   | 300.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell    | Wo   | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel   | Wgnv | 18.0000          | mm  |
| Inside Projection                             | h    | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell        | Wi   | 0.0000           | mm  |
| Pad Material                                  |      | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature           | Sp   | 138.00           | MPa |
| Pad Allowable Stress At Ambient               | Spa  | 138.00           | MPa |
| Diameter of Pad along vessel surface          | Dp   | 1000.0000        | mm  |
| Thickness of Pad                              | te   | 18.0000          | mm  |
| Weld leg size between Pad and Shell           | Wp   | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle      | Wgpn | 18.0000          | mm  |
| Reinforcing Pad Width                         |      | 195.2000         | mm  |
| This is a Manway, Coupling or Access Opening. |      |                  |     |
| Flange Class                                  |      | 300              |     |
| Flange Grade                                  |      | GR 1.4           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



### Reinforcement CALCULATION, Description: M124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 24.000 in. |
| Actual Thickness Used in Calculation        | 0.787 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 2.55 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 2.55)$   
 $= 5.5836 \text{ mm}$

### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1145.2001 mm |
| Parallel to Vessel Wall, opening length                | d  | 572.6000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$

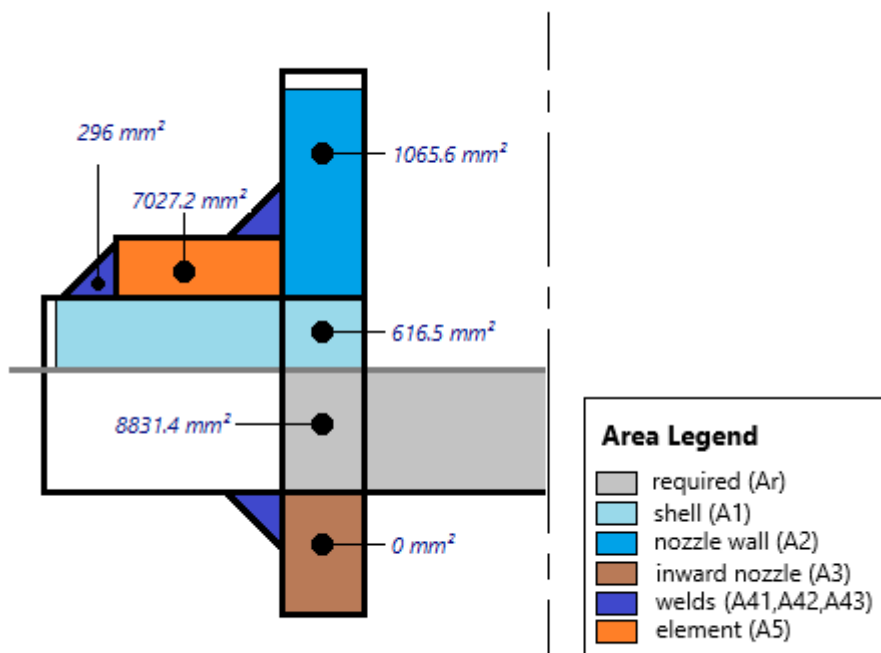


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 8831.436 | NA       | NA    |
| Area in Shell            | A1          | 616.465  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 1065.603 | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 296.000  | NA       | NA    |
| Area in Element          | A5          | 7027.200 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 9005.269 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 990.3427 18.0000 mm  
 Based on given Pad Diameter: 1000.0000 17.5547 mm  
 Based on Shell or Nozzle Thickness: 990.3427 18.0000 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (572.6 \cdot 15.4234 \cdot 1.0 + 2 \cdot 18.5 \cdot 15.4234 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 8831.436 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 572.6(1.0 \cdot 16.5 - 1.0 \cdot 15.4234) - 2 \cdot 18.5$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.4234) \cdot (1 - 1.0)$   
 $= 616.465 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 41.25) \cdot (18.5 - 5.58) \cdot 1.0$



= 1065.603 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$   
= 296.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1000.0 - 609.6) \cdot 18.0 \cdot 1.0$   
= 7027.200 mm<sup>2</sup>

**UG-45 Minimum Nozzle Neck Thickness Requirement:**

Wall Thickness for Internal/External pressures       $t_a = 7.0836$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
=  $t_a$   
= 7.0836 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

**Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

**Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

**Shell to Pad Weld at Pad OD for pad (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

|                                                        |   |        |
|--------------------------------------------------------|---|--------|
| Gov. MDMT of the Nozzle                                | : | -44 °C |
| Gov. MDMT of the Reinforcement Pad                     | : | -44 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly | : | -44 °C |

**Flange MDMT including Temperature reduction per UCS-66.1:**

|                                                      |        |
|------------------------------------------------------|--------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -29 °C |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -52 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
Design Pressure/Ambient Rating = 2.55/4.26 = 0.598

**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 858.9001 mm |
| Parallel to Vessel Wall, opening length                | d  | 429.4500 mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500 mm  |

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

**Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm<sup>2</sup> )**

| AREA AVAILABLE, A1 to A6 | No Pad | With Pad |
|--------------------------|--------|----------|
|--------------------------|--------|----------|



|                                   |      |          |          |
|-----------------------------------|------|----------|----------|
| Area Required [Internal Pressure] | AR   | 5887.918 | 5887.918 |
| Area Available in Shell           | A1   | 308.233  | 308.233  |
| Area Available in Nozzle Wall     | A2   | 1065.603 | 1065.603 |
| Area Available in Inward Nozzle   | A3   | 0.000    | 0.000    |
| Area Available in Welds           | A4   | 100.000  | 100.000  |
| Area Available in Pad             | A5   | 0.000    | 4487.402 |
| Area Available in Hub             | A6   | 0.000    | 0.000    |
| TOTAL AREA AVAILABLE              | Atot | 1473.835 | 5961.237 |

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

#### Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell Welds T<sub>min</sub> 18.0000 mm  
 Intermediate Calc. for pad/shell Welds T<sub>minPad</sub> 16.5000 mm

#### Results Per UW-16.1:

|             | Required Thickness               | Actual Thickness                 |
|-------------|----------------------------------|----------------------------------|
| Nozzle Weld | 6.0000 = Min per Code            | 7.0700 = 0.7 * W <sub>o</sub> mm |
| Pad Weld    | 8.2500 = 0.5*T <sub>minPad</sub> | 9.8980 = 0.7 * W <sub>p</sub> mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (8831.4355 - 616.4653 + 2 \cdot 18.5 \cdot 1.0 \cdot (1.0 \cdot 16.5 - 15.4234)) \cdot 138)$   
 $= 1139045.62 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (1065.6028 + 7027.2002 + 296.0 - 0.0 \cdot 1.0) \cdot 138$   
 $= 1157535.50 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (1065.6028 + 0.0 + 100.0 + (610.5)) \cdot 138$   
 $= 245076.91 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (1065.6028 + 0.0 + 296.0 + 7027.2002 + (610.5)) \cdot 138$   
 $= 1241775.75 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S<sub>onw</sub>]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 647445. \text{ N}$

Shear, Pad Element Weld [S<sub>pew</sub>]:  
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$   
 $= (3.1416/2.0) \cdot 1000.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 1486888. \text{ N}$

Shear, Nozzle Wall [S<sub>nw</sub>]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 295.55) \cdot (20.0 - 1.5) \cdot 0.7 \cdot 138$   
 $= 1659177. \text{ N}$

Tension, Pad Groove Weld [T<sub>pgw</sub>]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 138 \\ &= 1759994. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 1613328. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1486888 + 1659177) = 3146065 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (647445 + 1759994 + 1613328 + 0) = 4020767 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (1486888 + 1613328 + 0) = 3100217 \text{ N} \end{aligned}$$

#### Summary of Failure Path Calculations:

|                      |                             |                   |
|----------------------|-----------------------------|-------------------|
| Path 1-1 = 3146065 N | , must exceed W = 1139045 N | or W1 = 1157535 N |
| Path 2-2 = 4020767 N | , must exceed W = 1139045 N | or W2 = 245076 N  |
| Path 3-3 = 3100216 N | , must exceed W = 1139045 N | or W3 = 1241775 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 58.3697 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 376.3697 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 3.392 %

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**Input, Nozzle Desc: A116"**

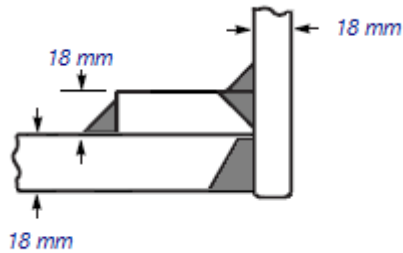
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6500.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                  |     |
|---------------------------------------------|------|------------------|-----|
| Material                                    |      | SA-516 70        | +IT |
| Material UNS Number                         |      | K02700           |     |
| Material Specification/Type                 |      | Plate            |     |
| Allowable Stress at Temperature             | Sn   | 138.00           | MPa |
| Allowable Stress At Ambient                 | Sna  | 138.00           | MPa |
| Diameter Basis (for tr calc only)           |      | Outside          |     |
| Layout Angle                                |      | 180.00           | deg |
| Diameter                                    |      | 16.0000          | in. |
| Size and Thickness Basis                    |      | Actual           |     |
| Actual Thickness                            | tn   | 18.0000          | mm  |
| Flange Material                             |      | SA-350 LF1       | +IT |
| Flange Type                                 |      | Weld Neck Flange |     |
| Corrosion Allowance                         | can  | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00             |     |
| Outside Projection                          | ho   | 250.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000          | mm  |
| Inside Projection                           | h    | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000           | mm  |
| Pad Material                                |      | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp   | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa  | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp   | 630.0000         | mm  |
| Thickness of Pad                            | te   | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp   | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn | 18.0000          | mm  |
| Reinforcing Pad Width                       |      | 111.8000         | mm  |
| Flange Class                                |      | 300              |     |
| Flange Grade                                |      | GR 1.4           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



#### Reinforcement CALCULATION, Description: A116"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 16.000 in. |
| Actual Thickness Used in Calculation        | 0.709 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 2.55 \cdot 203.2 / (138 \cdot 1.0 + 0.4 \cdot 2.55)$   
 $= 3.7224 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 746.8000 mm |
| Parallel to Vessel Wall, opening length                | d  | 373.4000 mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500 mm  |

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$

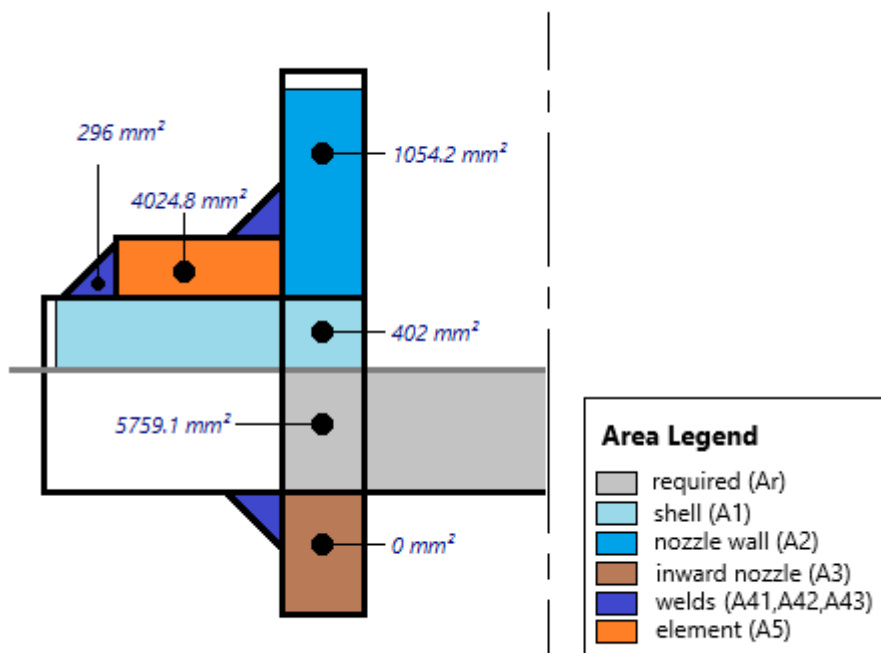


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 5759.095 | NA       | NA    |
| Area in Shell            | A1          | 402.005  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 1054.152 | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 296.000  | NA       | NA    |
| Area in Element          | A5          | 4024.800 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 5776.957 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

|                                         |          |            |
|-----------------------------------------|----------|------------|
| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter | Thickness  |
| Based on given Pad Thickness:           | 629.0077 | 18.0000 mm |
| Based on given Pad Diameter:            | 630.0000 | 17.9201 mm |
| Based on Shell or Nozzle Thickness:     | 629.0077 | 18.0000 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (373.4 \cdot 15.4234 \cdot 1.0 + 2 \cdot 16.5 \cdot 15.4234 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 5759.095 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 373.4(1.0 \cdot 16.5 - 1.0 \cdot 15.4234) - 2 \cdot 16.5$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.4234) \cdot (1 - 1.0)$   
 $= 402.005 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 41.25) \cdot (16.5 - 3.72) \cdot 1.0$



= 1054.152 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$   
= 296.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(630.0 - 406.4) \cdot 18.0 \cdot 1.0$   
= 4024.800 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 5.2224 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 16.9234 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 16.9234 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 9.8312 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 9.831, max( 16.9234, 3.0 ) ]  
= 9.8312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 5.2224, 9.8312 )  
= 9.8312 mm

Available Nozzle Neck Thickness = 18.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 32.5, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 312.5 MPa | Passed |
| Occasional | : 13.8, | Allowable | : 183.5 MPa | Passed |
| Shear      | : 12.2, | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Gov. MDMT of the Nozzle : -44 °C  
 Gov. MDMT of the Reinforcement Pad : -44 °C  
 Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -29 °C  
 Flange MDMT with Temp reduction per UCS-66(i)(2) -52 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
 Design Pressure/Ambient Rating = 2.55/4.26 = 0.598

#### Weld Size Calculations, Description: A1\_16"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm  
 Intermediate Calc. for pad/shell Welds TminPad 16.5000 mm

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 8.2500 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (5759.0952 - 402.0052 + 2 \cdot 16.5 \cdot 1.0 \cdot (1.0 \cdot 16.5 - 15.4234)) \cdot 138)$   
 $= 744104.69 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (1054.152 + 4024.8 + 296.0 - 0.0 \cdot 1.0) \cdot 138$   
 $= 741666.94 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (1054.152 + 0.0 + 100.0 + (544.5001)) \cdot 138$   
 $= 234389.81 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (1054.152 + 0.0 + 296.0 + 4024.8 + (544.5001)) \cdot 138$   
 $= 816800.19 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 431630. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$   
 $= (3.1416/2.0) \cdot 630.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 936740. \text{ N}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 194.95) \cdot (18.0 - 1.5) \cdot 0.7 \cdot 138$   
 $= 976107. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$   
 $= (3.1416/2) \cdot 406.4 \cdot 18.0 \cdot 0.74 \cdot 138$   
 $= 1173329. \text{ N}$

Tension, Shell Groove Weld [Tngw]:



$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 406.4 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138 \\
 &= 1075552. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (936740 + 976107) = 1912846 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (431630 + 1173329 + 1075552 + 0) = 2680512 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (936740 + 1075552 + 0) = 2012292 \text{ N}
 \end{aligned}$$

#### Summary of Failure Path Calculations:

|                      |                            |                  |
|----------------------|----------------------------|------------------|
| Path 1-1 = 1912846 N | , must exceed W = 744104 N | or W1 = 741666 N |
| Path 2-2 = 2680511 N | , must exceed W = 744104 N | or W2 = 234389 N |
| Path 3-3 = 2012291 N | , must exceed W = 744104 N | or W3 = 816800 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 25.4159 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 293.4159 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 4.634 %

#### Input Echo, WRC297 Item 1, Description: A116" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.5000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 1650.000  | mm         |
| Shell Thickness                      | 18.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |            |
|------------------------------------|-----------|------------|
| Diameter Basis for Nozzle          | OD        |            |
| Nozzle Corrosion Allowance         | 1.5000    | mm         |
| Nozzle Diameter                    | 406.400   | mm         |
| Nozzle Thickness                   | 18.0000   | mm         |
| Nozzle Stress Concentration Factor | 1.000     |            |
| Nozzle Material                    | SA-516 70 |            |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh      | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                              |         |    |
|------------------------------|---------|----|
| Diameter of Reinforcing Pad  | 630.000 | mm |
| Thickness of Reinforcing Pad | 18.0000 | mm |

#### Note:

External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUStained loads.

|                          |    |          |     |
|--------------------------|----|----------|-----|
| Design Internal Pressure | Dp | 2.55     | MPa |
| Radial Load              | P  | 19199.08 | N   |
| Circumferential Shear    | Vc | 25598.82 | N   |



|                        |    |           |      |
|------------------------|----|-----------|------|
| Longitudinal Shear     | Vl | 25598.82  | N    |
| Circumferential Moment | Mc | 0.205E+08 | N-mm |
| Longitudinal Moment    | Ml | 0.266E+08 | N-mm |
| Torsional Moment       | Mt | 0.307E+08 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(843.75(18.0 - 1.5))}$   
 $= 795.770 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1687.500 mm |
| Nozzle Outside Diameter | (d) = | 406.400 mm  |
| Vessel Thickness used   | (T) = | 34.500 mm   |
| Nozzle Thickness used   | (t) = | 16.500 mm   |
| T / t                   | =     | 2.091       |
| d / t                   | =     | 24.630      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.684$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.081 |
| Mr/P    | = | 0.088 |
| M0/P    | = | 0.054 |
| N0/P    | = | 0.156 |
| MrD/Mc  | = | 0.193 |
| NrDL/Mc | = | 0.124 |
| M0d/Mc  | = | 0.138 |
| N0DL/Mc | = | 0.158 |
| MrD/Ml  | = | 0.127 |
| NrDL/Ml | = | 0.082 |
| M0D/Ml  | = | 0.084 |
| N0DL/Ml | = | 0.275 |

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending (P )                                            | -5                   | 5                   | -5                      | 5                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -15                  | -15                 | 15                      | 15                     |
| Outplane Bending (Ml)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 59                   | 62                  | 59                      | 62                     |
| Outplane Stress Summary                                          | 9                    | 77                  | 95                      | 52                     |

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | -1                   | -1                  | -1                      | -1                     |
| Inplane Bending (P )                                               | -8                   | 8                   | -8                      | 8                      |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (Ml)                                              | -4                   | -4                  | 4                       | 4                      |
| Inplane Bending (Ml)                                               | -41                  | 41                  | 41                      | -41                    |
| Inplane Pressure Stress                                            | 29                   | 29                  | 29                      | 29                     |
| Inplane Stress Summary                                             | -26                  | 74                  | 66                      | 0                      |



**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE                          |  |  | Au      | Al     | Bu      | B1     |
|---------------------------------------------|--|--|---------|--------|---------|--------|
| (Shear stress normal to longitudinal plane) |  |  | Top     | Top    | Bottom  | Bottom |
|                                             |  |  | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)                         |  |  | 1       | 1      | -1      | -1     |
| Outplane Shear (Vl)                         |  |  | 0       | 0      | 0       | 0      |
| Outplane Shear (Mt)                         |  |  | 3       | 3      | 3       | 3      |
| Shear Stress Summary                        |  |  | 4       | 4      | 2       | 2      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE     |  |  | Au      | Al     | Bu      | B1     |
|------------------------|--|--|---------|--------|---------|--------|
| (Stress Intensities)   |  |  | Top     | Top    | Bottom  | Bottom |
|                        |  |  | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress |  |  | 36      | 80     | 95      | 52     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                      |  |  | Cu      | Cl     | Du      | D1     |
|--------------------------------------------|--|--|---------|--------|---------|--------|
| (Stresses Normal to circumferential plane) |  |  | Left    | Left   | Right   | Right  |
|                                            |  |  | Outside | Inside | Outside | Inside |
| Outplane Membrane (P )                     |  |  | -2      | -2     | -2      | -2     |
| Outplane Bending (P )                      |  |  | -5      | 5      | -5      | 5      |
| Outplane Membrane (Mc)                     |  |  | -6      | -6     | 6       | 6      |
| Outplane Bending (Mc)                      |  |  | -35     | 35     | 35      | -35    |
| Outplane Membrane (ML)                     |  |  | 0       | 0      | 0       | 0      |
| Outplane Bending (ML)                      |  |  | 0       | 0      | 0       | 0      |
| Normal Pressure Stress                     |  |  | 29      | 29     | 29      | 29     |
| Outplane Stress Summary                    |  |  | -19     | 60     | 63      | 4      |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                        |  |  | Cu      | Cl     | Du      | D1     |
|----------------------------------------------|--|--|---------|--------|---------|--------|
| (Stresses parallel to circumferential plane) |  |  | Left    | Left   | Right   | Right  |
|                                              |  |  | Outside | Inside | Outside | Inside |
| Inplane Membrane (P )                        |  |  | -1      | -1     | -1      | -1     |
| Inplane Bending (P )                         |  |  | -8      | 8      | -8      | 8      |
| Inplane Membrane (Mc)                        |  |  | -5      | -5     | 5       | 5      |
| Inplane Bending (Mc)                         |  |  | -49     | 49     | 49      | -49    |
| Inplane Membrane (ML)                        |  |  | 0       | 0      | 0       | 0      |
| Inplane Bending (ML)                         |  |  | 0       | 0      | 0       | 0      |
| Inplane Pressure Stress                      |  |  | 59      | 62     | 59      | 62     |
| Inplane Stress Summary                       |  |  | -4      | 113    | 104     | 25     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                          |  |  | Cu      | Cl     | Du      | D1     |
|------------------------------------------------|--|--|---------|--------|---------|--------|
| (Shear stress normal to circumferential plane) |  |  | Left    | Left   | Right   | Right  |
|                                                |  |  | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)                            |  |  | 0       | 0      | 0       | 0      |
| Outplane Shear (Vl)                            |  |  | -1      | -1     | 1       | 1      |
| Torsional Shear (Mt)                           |  |  | 3       | 3      | 3       | 3      |
| Shear Stress Summary                           |  |  | 2       | 2      | 4       | 4      |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE  |  |  | Cu      | Cl     | Du      | D1     |
|------------------------|--|--|---------|--------|---------|--------|
| (Stress Intensities)   |  |  | Left    | Left   | Right   | Right  |
|                        |  |  | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress |  |  | 19      | 113    | 104     | 26     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE |  |  | Au  | Al  | Bu     | B1     |
|--------------------|--|--|-----|-----|--------|--------|
| (Stresses in the   |  |  | Top | Top | Bottom | Bottom |



| hoop direction)      | Outside | Inside | Outside | Inside |
|----------------------|---------|--------|---------|--------|
| Hoop Membrane (P )   | -2      | -2     | -2      | -2     |
| Hoop Bending (P )    | 0       | 0      | 0       | 0      |
| Hoop Membrane (Mc)   | 0       | 0      | 0       | 0      |
| Hoop Bending (Mc)    | 0       | 0      | 0       | 0      |
| Hoop Membrane (ML)   | -15     | -15    | 15      | 15     |
| Hoop Bending (ML)    | 0       | 0      | 0       | 0      |
| Hoop Pressure Stress | 33      | 36     | 33      | 36     |
| Hoop Stress Summary  | 15      | 18     | 45      | 48     |

**Nozzle Stresses (MPa):**

|                                                                  |                      |                     |                         |                        |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
| Axial Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (P )                                               | -20                  | 20                  | -20                     | 20                     |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -14                  | -14                 | 14                      | 14                     |
| Axial Bending (ML)                                               | -123                 | 123                 | 123                     | -123                   |
| Axial Pressure Stress                                            | 16                   | 16                  | 16                      | 16                     |
| Axial Stress Summary                                             | -142                 | 145                 | 133                     | -73                    |

**Nozzle Stresses (MPa):**

|                                      |               |              |               |              |
|--------------------------------------|---------------|--------------|---------------|--------------|
| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Shear due to (Vc)                    | 2             | 2            | -2            | -2           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 7             | 7            | 7             | 7            |
| Shear Stress Summary                 | 9             | 9            | 4             | 4            |

**Nozzle Stresses (MPa):**

|                                            |               |              |               |              |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Two * Max Shear Stress                     | 158           | 146          | 133           | 122          |

**Nozzle Stresses (MPa):**

|                                                              |                       |                      |                        |                       |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Hoop Membrane (P )                                           | -2                    | -2                   | -2                     | -2                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -6                    | -6                   | 6                      | 6                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 33                    | 36                   | 33                     | 36                    |
| Hoop Stress Summary                                          | 23                    | 26                   | 37                     | 40                    |

**Nozzle Stresses (MPa):**

|                                                                     |                       |                      |                        |                       |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | -20                   | 20                   | -20                    | 20                    |
| Axial Membrane (Mc)                                                 | -10                   | -10                  | 10                     | 10                    |
| Axial Bending (Mc)                                                  | -145                  | 145                  | 145                    | -145                  |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 16                    | 16                   | 16                     | 16                    |

Axial Stress Summary -161 170 152 -99

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -2            | -2           | 2             | 2            |
| Shear due to Torsion                    | 7             | 7            | 7             | 7            |

Shear Stress Summary 4 4 9 9

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                       | 185           | 170          | 152           | 140          |

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(834.75(18.0 - 1.5))}$   
 $= 793.688 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1669.500 mm |
| Nozzle Outside Diameter | (d) = | 630.000 mm  |
| Vessel Thickness used   | (T) = | 16.500 mm   |
| Nozzle Thickness used   | (t) = | 128.300 mm  |
| T / t                   | =     | 0.129       |
| d / t                   | =     | 10.000      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.796$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.053 |
| Mr/P    | = | 0.087 |
| M0/P    | = | 0.026 |
| N0/P    | = | 0.041 |
| MrD/Mc  | = | 0.256 |
| NrDL/Mc | = | 0.101 |
| M0d/Mc  | = | 0.077 |
| N0DL/Mc | = | 0.068 |
| MrD/Ml  | = | 0.066 |
| NrDL/Ml | = | 0.060 |
| M0D/Ml  | = | 0.020 |
| N0DL/Ml | = | 0.046 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -2                   | -2                  | -2                      | -2                     |
| Outplane Bending (P )                                            | -10                  | 10                  | -10                     | 10                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -7                   | -7                  | 7                       | 7                      |
| Outplane Bending (Ml)                                            | -18                  | 18                  | 18                      | -18                    |
| Normal Pressure Stress                                           | 126                  | 128                 | 126                     | 128                    |

Outplane Stress Summary 87 148 138 125

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|
|--------------------|----|----|----|----|



| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
|-------------------------------------------|-------------|------------|----------------|---------------|
| Inplane Membrane (P )                     | -3          | -3         | -3             | -3            |
| Inplane Bending (P )                      | -36         | 36         | -36            | 36            |
| Inplane Membrane (Mc)                     | 0           | 0          | 0              | 0             |
| Inplane Bending (Mc)                      | 0           | 0          | 0              | 0             |
| Inplane Membrane (ML)                     | -9          | -9         | 9              | 9             |
| Inplane Bending (ML)                      | -61         | 61         | 61             | -61           |
| Inplane Pressure Stress                   | 63          | 63         | 63             | 63            |
| Inplane Stress Summary                    | -48         | 148        | 93             | 44            |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|-------------------------------------------------------------------|----------------|---------------|-------------------|------------------|
| Outplane Shear (Vc)                                               | 1              | 1             | -1                | -1               |
| Outplane Shear (Vl)                                               | 0              | 0             | 0                 | 0                |
| Outplane Shear (Mt)                                               | 2              | 2             | 2                 | 2                |
| Shear Stress Summary                                              | 4              | 4             | 1                 | 1                |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|--------------------------------------------|----------------|---------------|-------------------|------------------|
| Two * Max Shear Stress                     | 135            | 152           | 138               | 125              |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Outplane Membrane (P )                                              | -2              | -2             | -2               | -2              |
| Outplane Bending (P )                                               | -10             | 10             | -10              | 10              |
| Outplane Membrane (Mc)                                              | -8              | -8             | 8                | 8               |
| Outplane Bending (Mc)                                               | -54             | 54             | 54               | -54             |
| Outplane Membrane (ML)                                              | 0               | 0              | 0                | 0               |
| Outplane Bending (ML)                                               | 0               | 0              | 0                | 0               |
| Normal Pressure Stress                                              | 63              | 63             | 63               | 63              |
| Outplane Stress Summary                                             | -13             | 117            | 112              | 24              |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|-----------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Inplane Membrane (P )                                                 | -3              | -3             | -3               | -3              |
| Inplane Bending (P )                                                  | -36             | 36             | -36              | 36              |
| Inplane Membrane (Mc)                                                 | -12             | -12            | 12               | 12              |
| Inplane Bending (Mc)                                                  | -183            | 183            | 183              | -183            |
| Inplane Membrane (ML)                                                 | 0               | 0              | 0                | 0               |
| Inplane Bending (ML)                                                  | 0               | 0              | 0                | 0               |
| Inplane Pressure Stress                                               | 126             | 128            | 126              | 128             |
| Inplane Stress Summary                                                | -109            | 333            | 281              | -9              |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|-------------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Outplane Shear (Vc)                                                     | 0               | 0              | 0                | 0               |
| Outplane Shear (Vl)                                                     | -1              | -1             | 1                | 1               |
| Torsional Shear (Mt)                                                    | 2               | 2              | 2                | 2               |



Shear Stress Summary 1 1 4 4

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------------|---------|--------|---------|--------|
|                                               | Left    | Left   | Right   | Right  |
|                                               | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                        | 109     | 333    | 281     | 34     |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |      |      |      |      |       |       |      |
|------------------------------|------------------|------|------|------|------|-------|-------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl    | Du    | Dl   |
| Circ. Pm (SUS)               | 59               | 62   | 59   | 62   | 59   | 62    | 59    | 62   |
| Circ. Pl (SUS)               | -17              | -17  | 13   | 13   | -6   | -6    | 4     | 4    |
| Circ. Q (SUS)                | -32              | 32   | 22   | -22  | -57  | 57    | 41    | -41  |
| Long. Pm (SUS)               | 29               | 29   | 29   | 29   | 29   | 29    | 29    | 29   |
| Long. Pl (SUS)               | -5               | -5   | 3    | 3    | -8   | -8    | 4     | 4    |
| Long. Q (SUS)                | -49              | 49   | 33   | -33  | -40  | 40    | 30    | -30  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0     | 0     | 0    |
| Shear Pl (SUS)               | 1                | 1    | -1   | -1   | -1   | -1    | 1     | 1    |
| Shear Q (SUS)                | 3                | 3    | 3    | 3    | 3    | 3     | 3     | 3    |
| Pm (SUS)                     | 59.0             | 62.0 | 59.0 | 62.0 | 59.0 | 62.0  | 59.0  | 62.0 |
| Pm+Pl (SUS)                  | 42.1             | 45.0 | 72.0 | 75.0 | 53.0 | 56.0  | 63.0  | 66.0 |
| Pm+Pl+Q (Total)              | 35.9             | 79.5 | 94.1 | 54.1 | 19.3 | 113.1 | 104.4 | 25.7 |

**Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 62.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 75.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 113.08    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 126              | 128   | 126   | 128   | 126   | 128   | 126   | 128   |
| Circ. Pl (SUS)               | -9               | -9    | 5     | 5     | -15   | -15   | 9     | 9     |
| Circ. Q (SUS)                | -28              | 28    | 8     | -8    | -219  | 219   | 147   | -147  |
| Long. Pm (SUS)               | 63               | 63    | 63    | 63    | 63    | 63    | 63    | 63    |
| Long. Pl (SUS)               | -12              | -12   | 6     | 6     | -10   | -10   | 6     | 6     |
| Long. Q (SUS)                | -97              | 97    | 25    | -25   | -64   | 64    | 44    | -44   |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 1                | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                | 2                | 2     | 2     | 2     | 2     | 2     | 2     | 2     |
| Pm (SUS)                     | 126.0            | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 |
| Pm+Pl (SUS)                  | 117.0            | 119.0 | 131.0 | 133.0 | 111.0 | 113.0 | 135.0 | 137.0 |



|                 |       |       |       |       |       |       |       |      |
|-----------------|-------|-------|-------|-------|-------|-------|-------|------|
| Pm+Pl+Q (Total) | 135.1 | 150.5 | 139.0 | 125.0 | 108.0 | 332.0 | 282.1 | 35.5 |
|-----------------|-------|-------|-------|-------|-------|-------|-------|------|

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 128.00    | 138.00         | Passed |
| Pm+Pl (SUS)         | 137.01    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 332.00    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 33               | 36    | 33    | 36    | 33    | 36    | 33    | 36    |
| Circ. Pl (SUS)               | -17              | -17   | 13    | 13    | -8    | -8    | 4     | 4     |
| Circ. Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)               | 16               | 16    | 16    | 16    | 16    | 16    | 16    | 16    |
| Long. Pl (SUS)               | -14              | -14   | 14    | 14    | -10   | -10   | 10    | 10    |
| Long. Q (SUS)                | -143             | 143   | 103   | -103  | -165  | 165   | 125   | -125  |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 2                | 2     | -2    | -2    | -2    | -2    | 2     | 2     |
| Shear Q (SUS)                | 7                | 7     | 7     | 7     | 7     | 7     | 7     | 7     |
| Pm (SUS)                     | 33.0             | 36.0  | 33.0  | 36.0  | 33.0  | 36.0  | 33.0  | 36.0  |
| Pm+Pl (SUS)                  | 16.3             | 19.2  | 46.2  | 49.2  | 25.2  | 28.2  | 37.4  | 40.3  |
| Pm+Pl+Q (Total)              | 158.0            | 145.6 | 133.3 | 122.4 | 184.3 | 171.2 | 151.7 | 140.2 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 36.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 49.21     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 184.27    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: L12"**

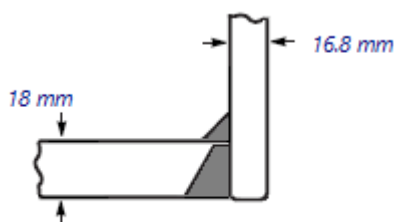
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6105.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 65.00          | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L12"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.55 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5869 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tt | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 2.0869  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 16.9234 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 16.9234 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 6.3000  | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 6.3, \max( 16.9234, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0869, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L12"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm

PV Elite 28 U2 Licensee: SL User

FileName : A150NRL15HN001\_Flash vessel\_7%

Nozzle Calcs.: L1\_2"

Noz1: 51 1:58pm Sep 18,2026

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HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

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**Input, Nozzle Desc: L32"**

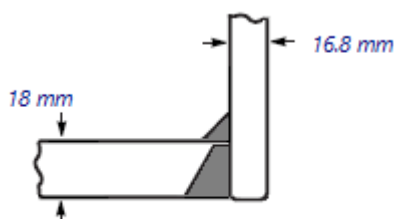
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6100.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 90.00          | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L32"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.55 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5869 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tt | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 2.0869  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 16.9234 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 16.9234 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 6.3000  | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 6.3, \max( 16.9234, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0869, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L32"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 58.66 mm*

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**Input, Nozzle Desc: L52"**

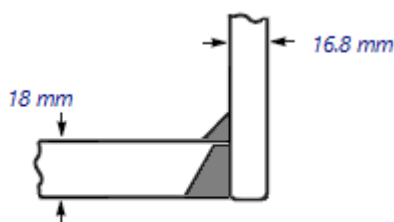
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6100.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 265.00         | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L52"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.55 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5869 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tt | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L52".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 2.0869  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 16.9234 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 16.9234 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 6.3000  | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 6.3, \max( 16.9234, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0869, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L52"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm

PV Elite 28 U2 Licensee: SL User

FileName : A150NRL15HN001\_Flash vessel\_7%

Nozzle Calcs.: L5\_2"

Noz1: 53 1:58pm Sep 18,2026

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The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

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**Input, Nozzle Desc: L72"**

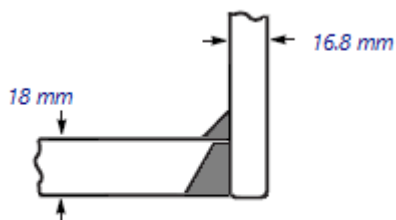
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6100.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 295.00         | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 200.0000       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 300            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: L72"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.55 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5869 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |          |          |    |
|-------------------------------------------------|----------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1       | 117.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+tr | 58.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp     | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: L72".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 2.0869  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 16.9234 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 16.9234 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 6.3000  | mm |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2) ]$   
 $= \min[ 6.3, \max( 16.9234, 3.0 ) ]$   
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 2.0869, 6.3 )$   
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:  
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                          |         |
|----------------------------------------------------------|---------|
| Impact Test Temperature provided per Specification       | -29 °C  |
| Calculated Minimum Design Metal Temperature              | -104 °C |
| Gov. MDMT of the nozzle to shell joint welded assembly : | -104 °C |

##### Weld Size Calculations, Description: L72"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles  
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0781 mm

PV Elite 28 U2 Licensee: SL User

FileName : A150NRL15HN001\_Flash vessel\_7%

Nozzle Calcs.: L7\_2"

Nozl: 54

1:58pm

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Sep 18, 2026



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The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0781 mm

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**Input, Nozzle Desc: A44"**

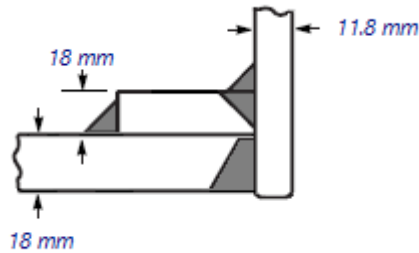
**From: 60**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Cylindrical Shell          | D    | 1650.00   | mm    |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm    |
| Shell Internal Corrosion Allowance            | cas  | 1.5000    | mm    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm    |
| Distance from Bottom/Left Tangent             |      | 6800.00   | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |                   |                  |     |
|---------------------------------------------|-------------------|------------------|-----|
| Material                                    |                   | SA-333 6         | +IT |
| Material UNS Number                         |                   | K03006           |     |
| Material Specification/Type                 | Smls. & wld. pipe |                  |     |
| Allowable Stress at Temperature             | Sn                | 118.00           | MPa |
| Allowable Stress At Ambient                 | Sna               | 118.00           | MPa |
| Diameter Basis (for tr calc only)           |                   | Inside           |     |
| Layout Angle                                |                   | 0.00             | deg |
| Diameter                                    |                   | 4.0000           | in. |
| Size and Thickness Basis                    |                   | Minimum          |     |
| Nominal Thickness                           |                   | 160              |     |
| Flange Material                             |                   | SA-350 LF1       | +IT |
| Flange Type                                 |                   | Weld Neck Flange |     |
| Corrosion Allowance                         | can               | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1                | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En                | 1.00             |     |
| Outside Projection                          | ho                | 200.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo                | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv              | 18.0000          | mm  |
| Inside Projection                           | h                 | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi                | 0.0000           | mm  |
| Pad Material                                |                   | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp                | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa               | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp                | 150.0000         | mm  |
| Thickness of Pad                            | te                | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp                | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn              | 18.0000          | mm  |
| Reinforcing Pad Width                       |                   | 17.8500          | mm  |
| Flange Class                                |                   | 600              |     |
| Flange Grade                                |                   | GR 1.4           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



### Reinforcement CALCULATION, Description: A44"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 3.571 in. |
| Actual Thickness Used in Calculation       | 0.465 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 826.5 / (138 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 15.4234 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 2.55 \cdot 46.85 / (118 \cdot 1.0 - 0.6 \cdot 2.55)$   
 $= 1.0243 \text{ mm}$

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 187.3941 mm |
| Parallel to Vessel Wall, opening length                | d  | 93.6970 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 41.2500 mm  |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$

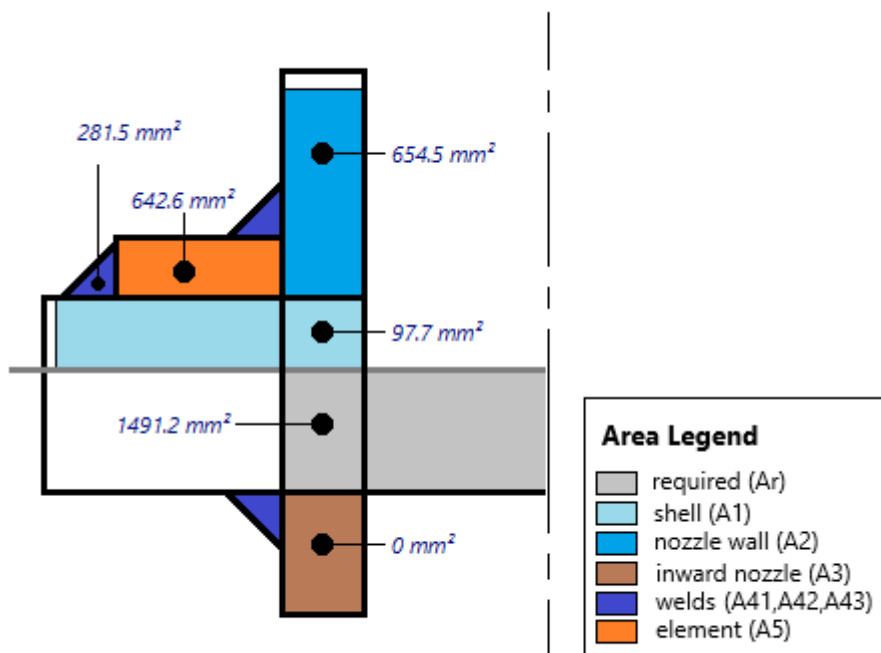


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 1491.175 | NA       | NA    |
| Area in Shell            | A1          | 97.661   | NA       | NA    |
| Area in Nozzle Wall      | A2          | 654.453  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 281.509  | NA       | NA    |
| Area in Element          | A5          | 642.600  | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 1676.222 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

|                                         |          |            |
|-----------------------------------------|----------|------------|
| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter | Thickness  |
| Based on given Pad Thickness:           | 139.7196 | 18.0000 mm |
| Based on given Pad Diameter:            | 150.0000 | 12.8166 mm |
| Based on Shell or Nozzle Thickness:     | 153.0708 | 11.8015 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (93.697 \cdot 15.4234 \cdot 1.0 + 2 \cdot 10.3015 \cdot 15.4234 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 1491.175 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 93.697(1.0 \cdot 16.5 - 1.0 \cdot 15.423) - 2 \cdot 10.301$   
 $(1.0 \cdot 16.5 - 1.0 \cdot 15.4234) \cdot (1 - 0.855)$   
 $= 97.661 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 41.25) \cdot (10.3 - 1.02) \cdot 0.8551$



= 654.453 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 14.0^2 \cdot 1.0$   
= 281.509 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(150.0 - 114.3) \cdot 18.0 \cdot 1.0$   
= 642.600 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 2.5243 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 16.9234 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 16.9234 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 6.7578 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 6.758, max( 16.9234, 3.0 ) ]  
= 6.7578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 2.5243, 6.7578 )  
= 6.7578 mm

Available Nozzle Neck Thickness = 11.8015 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 32.8, | Allowable | : 118.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 262.2 MPa | Passed |
| Occasional | : 5.2,  | Allowable | : 156.9 MPa | Passed |
| Shear      | : 19.5, | Allowable | : 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

###### Note:

*This Material was specified as being an Impact Tested (Low Temperature) Material.*

|                                                    |         |
|----------------------------------------------------|---------|
| Impact Test Temperature provided per Specification | -46 °C  |
| Calculated Minimum Design Metal Temperature        | -104 °C |

##### Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

##### Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

#### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

|                                                        |           |
|--------------------------------------------------------|-----------|
| Impact Test Temperature provided per Specification     | -46 °C    |
| Calculated Minimum Design Metal Temperature            | -104 °C   |
| Gov. MDMT of the Nozzle                                | : -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : -44 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : -44 °C  |

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                      |         |
|------------------------------------------------------|---------|
| MDMT of ASME B16.5/47 flange per Matl. Specification | -29 °C  |
| Flange MDMT with Temp reduction per UCS-66(i)(2)     | -104 °C |

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = 2.55/8.52 = 0.299

#### Weld Size Calculations, Description: A44"

|                                           |         |            |
|-------------------------------------------|---------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 10.3015 mm |
| Intermediate Calc. for pad/shell Welds    | TminPad | 16.5000 mm |

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 8.2500 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1491.1747 - 97.6606 + 2 \cdot 10.3015 \cdot 0.8551 \cdot (1.0 \cdot 16.5 - 15.4234)) \cdot 138)$$

$$= 194902.28 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (654.4528 + 642.5998 + 281.5088 - 0.0 \cdot 0.86) \cdot 138$$

$$= 217819.02 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (654.4528 + 0.0 + 85.5088 + (290.686)) \cdot 138$$

$$= 142214.70 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (654.4528 + 0.0 + 281.5088 + 642.5998 + (290.686)) \cdot 138$$

$$= 257929.55 \text{ N}$$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 103802. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 150.0 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 223033. \text{ N}$$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 52.0) \cdot (11.8015 - 1.5) \cdot 0.7 \cdot 118$$

$$= 138992. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 114.3 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 329999. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (18.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 302494. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (223033 + 138992) = 362025 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (103802 + 329999 + 302494 + 0) = 736295 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (223033 + 302494 + 0) = 525527 \text{ N}$$

#### Summary of Failure Path Calculations:

|                     |                            |                  |
|---------------------|----------------------------|------------------|
| Path 1-1 = 362025 N | , must exceed W = 194902 N | or W1 = 217819 N |
| Path 2-2 = 736294 N | , must exceed W = 194902 N | or W2 = 142214 N |
| Path 3-3 = 525526 N | , must exceed W = 194902 N | or W3 = 257929 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 1.9819 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.9818 mm

#### Input Echo, WRC297 Item 1, Description: A44" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.5000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 1650.000  | mm         |
| Shell Thickness                      | 18.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |          |            |
|------------------------------------|----------|------------|
| Diameter Basis for Nozzle          | ID       |            |
| Nozzle Corrosion Allowance         | 1.5000   | mm         |
| Nozzle Diameter                    | 90.697   | mm         |
| Nozzle Thickness                   | 11.8015  | mm         |
| Nozzle Stress Concentration Factor | 1.000    |            |
| Nozzle Material                    | SA-333 6 |            |
| Nozzle Cold S.I. Allowable         | SNmc     | 118.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh     | 118.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                             |         |    |
|-----------------------------|---------|----|
| Diameter of Reinforcing Pad | 150.000 | mm |
|-----------------------------|---------|----|



Thickness of Reinforcing Pad 18.0000 mm

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSTAINED loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 2.55      | MPa  |
| Radial Load              | P  | 4799.79   | N    |
| Circumferential Shear    | Vc | 6399.69   | N    |
| Longitudinal Shear       | Vl | 6399.69   | N    |
| Circumferential Moment   | Mc | 0.128E+07 | N-mm |
| Longitudinal Moment      | Ml | 0.166E+07 | N-mm |
| Torsional Moment         | Mt | 0.192E+07 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tn (9.685) is < 10  
 check the Limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 = NozzleOD + 2 • 1.65 •  $\sqrt{(R_{mean}(t - ca))}$   
 = 114.3 + 2 • 1.65 •  $\sqrt{(843.75(18.0 - 1.5))}$   
 = 503.670 mm

**WRC 297 Curve Access Parameters:**

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1687.500 mm |
| Nozzle Outside Diameter | (d) = | 114.300 mm  |
| Vessel Thickness used   | (T) = | 34.500 mm   |
| Nozzle Thickness used   | (t) = | 10.301 mm   |
| T / t                   | =     | 3.349       |
| d / t                   | =     | 11.095      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.474$$

|         |   |        |
|---------|---|--------|
| Nr/P    | = | 0.190  |
| Mr/P    | = | 0.158  |
| M0/P    | = | 0.183  |
| N0/P    | = | 0.214  |
| MrD/Mc  | = | 0.175  |
| NrDL/Mc | = | 0.176  |
| M0d/Mc  | = | 0.230  |
| N0DL/Mc | = | -0.107 |
| MrD/Ml  | = | 0.166  |
| NrDL/Ml | = | 0.173  |
| M0D/Ml  | = | 0.219  |
| N0DL/Ml | = | 0.097  |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (P)                                             | -4                   | 4                   | -4                      | 4                      |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -1                   | -1                  | 1                       | 1                      |
| Outplane Bending (Ml)                                            | -16                  | 16                  | 16                      | -16                    |
| Normal Pressure Stress                                           | 59                   | 62                  | 59                      | 62                     |
| Outplane Stress Summary                                          | 37                   | 80                  | 71                      | 50                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to | Au<br>Top | Al<br>Top | Bu<br>Bottom | Bl<br>Bottom |
|---------------------------------------------|-----------|-----------|--------------|--------------|
|---------------------------------------------|-----------|-----------|--------------|--------------|



| longitudinal plane)     | Outside | Inside | Outside | Inside |
|-------------------------|---------|--------|---------|--------|
| Inplane Membrane (P )   | 0       | 0      | 0       | 0      |
| Inplane Bending (P )    | -3      | 3      | -3      | 3      |
| Inplane Membrane (Mc)   | 0       | 0      | 0       | 0      |
| Inplane Bending (Mc)    | 0       | 0      | 0       | 0      |
| Inplane Membrane (ML)   | -2      | -2     | 2       | 2      |
| Inplane Bending (ML)    | -12     | 12     | 12      | -12    |
| Inplane Pressure Stress | 29      | 29     | 29      | 29     |
| Inplane Stress Summary  | 10      | 43     | 39      | 22     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE                          | Au      | Al     | Bu      | Bl     |
|---------------------------------------------|---------|--------|---------|--------|
| (Shear stress normal to longitudinal plane) | Top     | Top    | Bottom  | Bottom |
|                                             | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)                         | 1       | 1      | -1      | -1     |
| Outplane Shear (Vl)                         | 0       | 0      | 0       | 0      |
| Outplane Shear (Mt)                         | 2       | 2      | 2       | 2      |
| Shear Stress Summary                        | 3       | 3      | 1       | 1      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE     | Au      | Al     | Bu      | Bl     |
|------------------------|---------|--------|---------|--------|
| (Stress Intensities    | Top     | Top    | Bottom  | Bottom |
|                        | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress | 37      | 81     | 71      | 51     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                      | Cu      | Cl     | Du      | Dl     |
|--------------------------------------------|---------|--------|---------|--------|
| (Stresses Normal to circumferential plane) | Left    | Left   | Right   | Right  |
|                                            | Outside | Inside | Outside | Inside |
| Outplane Membrane (P )                     | 0       | 0      | 0       | 0      |
| Outplane Bending (P )                      | -4      | 4      | -4      | 4      |
| Outplane Membrane (Mc)                     | 0       | 0      | 0       | 0      |
| Outplane Bending (Mc)                      | -12     | 12     | 12      | -12    |
| Outplane Membrane (ML)                     | 0       | 0      | 0       | 0      |
| Outplane Bending (ML)                      | 0       | 0      | 0       | 0      |
| Normal Pressure Stress                     | 29      | 29     | 29      | 29     |
| Outplane Stress Summary                    | 12      | 47     | 36      | 19     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                        | Cu      | Cl     | Du      | Dl     |
|----------------------------------------------|---------|--------|---------|--------|
| (Stresses parallel to circumferential plane) | Left    | Left   | Right   | Right  |
|                                              | Outside | Inside | Outside | Inside |
| Inplane Membrane (P )                        | 0       | 0      | 0       | 0      |
| Inplane Bending (P )                         | -3      | 3      | -3      | 3      |
| Inplane Membrane (Mc)                        | -1      | -1     | 1       | 1      |
| Inplane Bending (Mc)                         | -9      | 9      | 9       | -9     |
| Inplane Membrane (ML)                        | 0       | 0      | 0       | 0      |
| Inplane Bending (ML)                         | 0       | 0      | 0       | 0      |
| Inplane Pressure Stress                      | 59      | 62     | 59      | 62     |
| Inplane Stress Summary                       | 43      | 73     | 66      | 57     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                          | Cu      | Cl     | Du      | Dl     |
|------------------------------------------------|---------|--------|---------|--------|
| (Shear stress normal to circumferential plane) | Left    | Left   | Right   | Right  |
|                                                | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)                            | 0       | 0      | 0       | 0      |
| Outplane Shear (Vl)                            | -1      | -1     | 1       | 1      |
| Torsional Shear (Mt)                           | 2       | 2      | 2       | 2      |
| Shear Stress Summary                           | 1       | 1      | 3       | 3      |



**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 43                    | 73                   | 67                     | 57                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -1                   | -1                  | 1                       | 1                      |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 12                   | 15                  | 12                      | 15                     |
| Hoop Stress Summary                                       | 10                   | 13                  | 12                      | 15                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | -1                   | -1                  | -1                      | -1                     |
| Axial Bending (P )                                               | -16                  | 16                  | -16                     | 16                     |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -20                  | -20                 | 20                      | 20                     |
| Axial Bending (ML)                                               | -65                  | 65                  | 65                      | -65                    |
| Axial Pressure Stress                                            | 6                    | 6                   | 6                       | 6                      |
| Axial Stress Summary                                             | -98                  | 66                  | 74                      | -23                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 3             | 3            | -3            | -3           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 9             | 9            | 9             | 9            |
| Shear Stress Summary                 | 12            | 12           | 5             | 5            |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 112           | 69           | 74            | 40           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 12                    | 15                   | 12                     | 15                    |
| Hoop Stress Summary                                          | 12                    | 15                   | 10                     | 13                    |

**Nozzle Stresses (MPa):**



| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | -1                    | -1                   | -1                     | -1                    |
| Axial Bending (P )                                                  | -16                   | 16                   | -16                    | 16                    |
| Axial Membrane (Mc)                                                 | -15                   | -15                  | 15                     | 15                    |
| Axial Bending (Mc)                                                  | -55                   | 55                   | 55                     | -55                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 6                     | 6                    | 6                      | 6                     |
| Axial Stress Summary                                                | -83                   | 61                   | 58                     | -17                   |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -3            | -3           | 3             | 3            |
| Shear due to Torsion                    | 9             | 9            | 9             | 9            |
| Shear Stress Summary                    | 5             | 5            | 12            | 12           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 96            | 61           | 62            | 40           |

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(834.75(18.0 - 1.5))}$   
 $= 501.588 \text{ mm}$

**WRC 297 Curve Access Parameters:**

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 1669.500 mm |
| Nozzle Outside Diameter | (d) = | 150.000 mm  |
| Vessel Thickness used   | (T) = | 16.500 mm   |
| Nozzle Thickness used   | (t) = | 28.151 mm   |
| T / t                   | =     | 0.586       |
| d / t                   | =     | 10.000      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.904$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.114 |
| Mr/P    | = | 0.232 |
| M0/P    | = | 0.070 |
| N0/P    | = | 0.138 |
| MrD/Mc  | = | 0.303 |
| NrDL/Mc | = | 0.099 |
| M0d/Mc  | = | 0.099 |
| N0DL/Mc | = | 0.019 |
| MrD/Ml  | = | 0.208 |
| NrDL/Ml | = | 0.094 |
| M0D/Ml  | = | 0.070 |
| N0DL/Ml | = | 0.110 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -2                   | -2                  | -2                      | -2                     |



|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Outplane Bending (P )  | -7  | 7   | -7  | 7   |
| Outplane Membrane (Mc) | 0   | 0   | 0   | 0   |
| Outplane Bending (Mc)  | 0   | 0   | 0   | 0   |
| Outplane Membrane (ML) | -4  | -4  | 4   | 4   |
| Outplane Bending (ML)  | -17 | 17  | 17  | -17 |
| Normal Pressure Stress | 126 | 128 | 126 | 128 |

|                         |    |     |     |     |
|-------------------------|----|-----|-----|-----|
| Outplane Stress Summary | 94 | 146 | 138 | 121 |
|-------------------------|----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                           |             |            |                |               |
|-------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                        | Au          | Al         | Bu             | Bl            |
| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                         |     |    |     |     |
|-------------------------|-----|----|-----|-----|
| Inplane Membrane (P )   | -2  | -2 | -2  | -2  |
| Inplane Bending (P )    | -24 | 24 | -24 | 24  |
| Inplane Membrane (Mc)   | 0   | 0  | 0   | 0   |
| Inplane Bending (Mc)    | 0   | 0  | 0   | 0   |
| Inplane Membrane (ML)   | -3  | -3 | 3   | 3   |
| Inplane Bending (ML)    | -50 | 50 | 50  | -50 |
| Inplane Pressure Stress | 63  | 63 | 63  | 63  |

|                        |     |     |    |    |
|------------------------|-----|-----|----|----|
| Inplane Stress Summary | -18 | 132 | 91 | 38 |
|------------------------|-----|-----|----|----|

**Vessel Stresses (MPa):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 3 | 3 | 3  | 3  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 1 | 1 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 113 | 147 | 138 | 121 |
|------------------------|-----|-----|-----|-----|

**Vessel Stresses (MPa):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |    |    |     |
|------------------------|-----|----|----|-----|
| Outplane Membrane (P ) | -2  | -2 | -2 | -2  |
| Outplane Bending (P )  | -7  | 7  | -7 | 7   |
| Outplane Membrane (Mc) | 0   | 0  | 0  | 0   |
| Outplane Bending (Mc)  | -18 | 18 | 18 | -18 |
| Outplane Membrane (ML) | 0   | 0  | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0  | 0  | 0   |
| Normal Pressure Stress | 63  | 63 | 63 | 63  |

|                         |    |    |    |    |
|-------------------------|----|----|----|----|
| Outplane Stress Summary | 34 | 86 | 72 | 50 |
|-------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Inplane Membrane (P )   | -2  | -2  | -2  | -2  |
| Inplane Bending (P )    | -24 | 24  | -24 | 24  |
| Inplane Membrane (Mc)   | -3  | -3  | 3   | 3   |
| Inplane Bending (Mc)    | -57 | 57  | 57  | -57 |
| Inplane Membrane (ML)   | 0   | 0   | 0   | 0   |
| Inplane Bending (ML)    | 0   | 0   | 0   | 0   |
| Inplane Pressure Stress | 126 | 128 | 126 | 128 |



Inplane Stress Summary 39 205 159 97

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                          |  | Cu           | Cl          | Du            | Dl           |
|------------------------------------------------|--|--------------|-------------|---------------|--------------|
| (Shear stress normal to circumferential plane) |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Shear (Vc)                            |  | 0            | 0           | 0             | 0            |
| Outplane Shear (Vl)                            |  | -1           | -1          | 1             | 1            |
| Torsional Shear (Mt)                           |  | 3            | 3           | 3             | 3            |

Shear Stress Summary 1 1 4 4

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE  |  | Cu           | Cl          | Du            | Dl           |
|------------------------|--|--------------|-------------|---------------|--------------|
| (Stress Intensities)   |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Two * Max Shear Stress |  | 40           | 205         | 160           | 97           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 59               | 62   | 59   | 62   | 59   | 62   | 59   | 62   |
| Circ. Pl (SUS)               | -1               | -1   | 1    | 1    | -1   | -1   | 1    | 1    |
| Circ. Q (SUS)                | -20              | 20   | 12   | -12  | -12  | 12   | 6    | -6   |
| Long. Pm (SUS)               | 29               | 29   | 29   | 29   | 29   | 29   | 29   | 29   |
| Long. Pl (SUS)               | -2               | -2   | 2    | 2    | 0    | 0    | 0    | 0    |
| Long. Q (SUS)                | -15              | 15   | 9    | -9   | -16  | 16   | 8    | -8   |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 1                | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)                     | 59.0             | 62.0 | 59.0 | 62.0 | 59.0 | 62.0 | 59.0 | 62.0 |
| Pm+Pl (SUS)                  | 58.0             | 61.0 | 60.0 | 63.0 | 58.0 | 61.0 | 60.0 | 63.0 |
| Pm+Pl+Q (Total)              | 38.3             | 81.2 | 72.0 | 51.0 | 46.0 | 73.0 | 66.3 | 57.2 |

**Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 62.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 63.03     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 81.23     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

| Type of Stress Int. Location | Stress Values at |     |     |     |     |     |     |     |
|------------------------------|------------------|-----|-----|-----|-----|-----|-----|-----|
|                              | Au               | Al  | Bu  | Bl  | Cu  | Cl  | Du  | Dl  |
| Circ. Pm (SUS)               | 126              | 128 | 126 | 128 | 126 | 128 | 126 | 128 |
| Circ. Pl (SUS)               | -6               | -6  | 2   | 2   | -5  | -5  | 1   | 1   |
| Circ. Q (SUS)                | -24              | 24  | 10  | -10 | -81 | 81  | 33  | -33 |



|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Long. Pm (SUS)  | 63    | 63    | 63    | 63    | 63    | 63    | 63    | 63    |
| Long. Pl (SUS)  | -5    | -5    | 1     | 1     | -2    | -2    | -2    | -2    |
| Long. Q (SUS)   | -74   | 74    | 26    | -26   | -25   | 25    | 11    | -11   |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 3     | 3     | 3     | 3     | 3     | 3     | 3     | 3     |
| Pm (SUS)        | 126.0 | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 | 126.0 | 128.0 |
| Pm+Pl (SUS)     | 120.0 | 122.0 | 128.0 | 130.0 | 121.0 | 123.0 | 127.0 | 129.0 |
| Pm+Pl+Q (Total) | 112.3 | 147.1 | 138.1 | 120.0 | 40.8  | 204.0 | 160.2 | 96.3  |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 128.00    | 138.00         | Passed |
| Pm+Pl (SUS)         | 130.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 204.03    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 12               | 15   | 12   | 15   | 12   | 15   | 12   | 15   |
| Circ. Pl (SUS)               | -1               | -1   | 1    | 1    | 0    | 0    | 0    | 0    |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 6                | 6    | 6    | 6    | 6    | 6    | 6    | 6    |
| Long. Pl (SUS)               | -21              | -21  | 19   | 19   | -16  | -16  | 14   | 14   |
| Long. Q (SUS)                | -81              | 81   | 49   | -49  | -71  | 71   | 39   | -39  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 3                | 3    | -3   | -3   | -3   | -3   | 3    | 3    |
| Shear Q (SUS)                | 9                | 9    | 9    | 9    | 9    | 9    | 9    | 9    |
| Pm (SUS)                     | 12.0             | 15.0 | 12.0 | 15.0 | 12.0 | 15.0 | 12.0 | 15.0 |
| Pm+Pl (SUS)                  | 26.7             | 29.6 | 25.7 | 25.9 | 22.8 | 25.7 | 21.0 | 21.4 |
| Pm+Pl+Q (Total)              | 109.7            | 68.6 | 74.6 | 41.8 | 93.8 | 61.8 | 61.9 | 41.6 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 15.00     | 118.00         | Passed |
| Pm+Pl (SUS)         | 29.61     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 109.66    | 354.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: B118"**

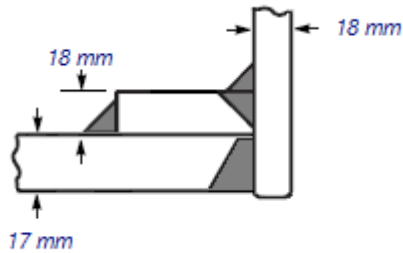
**From: 70**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Elliptical Head            | D    | 1650.00   | mm    |
| Aspect Ratio of Elliptical Head               | Ar   | 2.00      |       |
| Head Finished (Minimum) Thickness             | t    | 17.0000   | mm    |
| Head Internal Corrosion Allowance             | cas  | 1.5000    | mm    |
| Head External Corrosion Allowance             | co   | 0.0000    | mm    |
| Distance from Head Centerline                 | L1   | 0.0000    | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                  |     |
|---------------------------------------------|------|------------------|-----|
| Material                                    |      | SA-516 70        | +IT |
| Material UNS Number                         |      | K02700           |     |
| Material Specification/Type                 |      | Plate            |     |
| Allowable Stress at Temperature             | Sn   | 138.00           | MPa |
| Allowable Stress At Ambient                 | Sna  | 138.00           | MPa |
| Diameter Basis (for tr calc only)           |      | Outside          |     |
| Layout Angle                                |      | 0.00             | deg |
| Diameter                                    |      | 18.0000          | in. |
| Size and Thickness Basis                    |      | Actual           |     |
| Actual Thickness                            | tn   | 18.0000          | mm  |
| Flange Material                             |      | SA-350 LF1       | +IT |
| Flange Type                                 |      | Weld Neck Flange |     |
| Corrosion Allowance                         | can  | 1.5000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00             |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00             |     |
| Outside Projection                          | ho   | 250.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000          | mm  |
| Inside Projection                           | h    | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000           | mm  |
| Pad Material                                |      | SA-516 70        | +IT |
| Pad Allowable Stress at Temperature         | Sp   | 138.00           | MPa |
| Pad Allowable Stress At Ambient             | Spa  | 138.00           | MPa |
| Diameter of Pad along vessel surface        | Dp   | 690.0000         | mm  |
| Thickness of Pad                            | te   | 18.0000          | mm  |
| Weld leg size between Pad and Shell         | Wp   | 14.0000          | mm  |
| Groove weld depth between Pad and Nozzle    | Wgpn | 18.0000          | mm  |
| Reinforcing Pad Width                       |      | 116.4000         | mm  |
| Flange Class                                |      | 300              |     |
| Flange Grade                                |      | GR 1.4           |     |

The Pressure Design option was MAWP + static head (to the nozzle).



#### Reinforcement CALCULATION, Description: B118"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 18.000 in. |
| Actual Thickness Used in Calculation        | 0.709 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]  
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$  per Appendix 1-4(c)  
 $= 2.55 \cdot 1653.0 \cdot 0.8984 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.55)$   
 $= 13.7289 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 2.55 \cdot 228.6 / (138 \cdot 1.0 + 0.4 \cdot 2.55)$   
 $= 4.1877 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 848.4000 mm |
| Parallel to Vessel Wall, opening length                | d  | 424.2000 mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 38.7500 mm  |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, Sn/Sv)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, Sn/Sv)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, Sp/Sv)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$

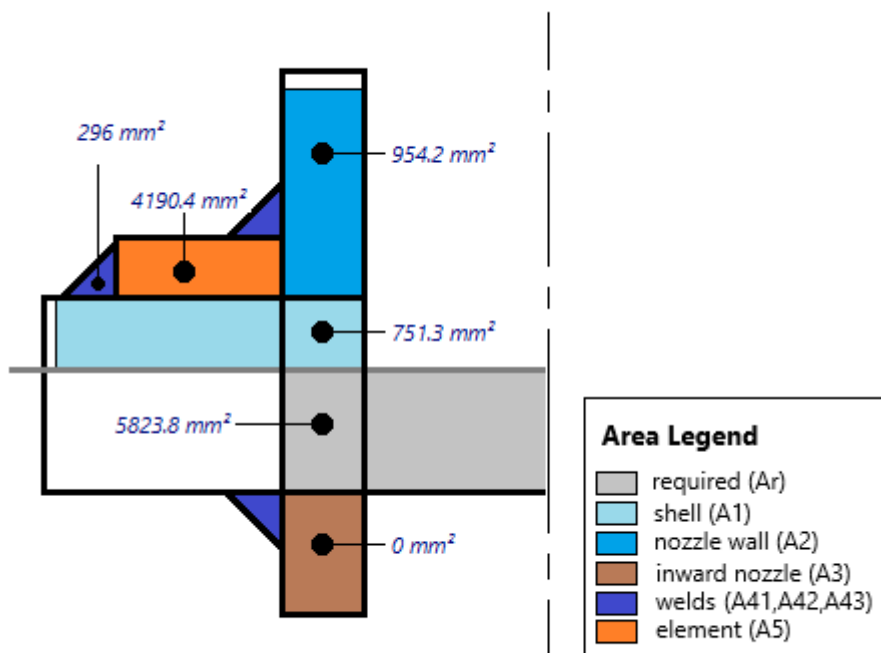


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 5823.814 | NA       | NA    |
| Area in Shell            | A1          | 751.287  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 954.203  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 296.000  | NA       | NA    |
| Area in Element          | A5          | 4190.401 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 6191.892 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 669.5513 18.0000 mm  
 Based on given Pad Diameter: 690.0001 16.4189 mm  
 Based on Shell or Nozzle Thickness: 682.0425 17.0000 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (424.2 \cdot 13.7289 \cdot 1.0 + 2 \cdot 16.5 \cdot 13.7289 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 5823.814 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 424.2(1.0 \cdot 15.5 - 1.0 \cdot 13.729) - 2 \cdot 16.5$   
 $(1.0 \cdot 15.5 - 1.0 \cdot 13.729) \cdot (1 - 1.0)$   
 $= 751.287 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 38.75) \cdot (16.5 - 4.19) \cdot 1.0$



= 954.203 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$   
= 296.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(690.0 - 457.2) \cdot 18.0 \cdot 1.0$   
= 4190.401 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 5.6877 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 3.0000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 16.7439 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 16.7439 mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.0000 mm  |
| Wall Thickness per table UG-45                 | tb3 = 9.8312 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 9.831, max( 16.7439, 3.0 ) ]  
= 9.8312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 5.6877, 9.8312 )  
= 9.8312 mm

Available Nozzle Neck Thickness = 18.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 34.5, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 310.5 MPa | Passed |
| Occasional | : 15.8, | Allowable | : 183.5 MPa | Passed |
| Shear      | : 11.6, | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Gov. MDMT of the Nozzle : -46 °C  
Gov. MDMT of the Reinforcement Pad : -46 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -29 °C  
Flange MDMT with Temp reduction per UCS-66(i)(2) -52 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :  
Design Pressure/Ambient Rating = 2.55/4.26 = 0.598

#### Weld Size Calculations, Description: B118"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm  
Intermediate Calc. for pad/shell Welds TminPad 18.0000 mm

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 9.0000 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
= max( 0, (A-A1+2\*tn\*fr1\*(E1\*t-tr))Sv)  
= max( 0, (5823.814 - 751.2868 + 2 \* 16.5 \* 1.0 \*  
(1.0 \* 15.5 - 13.7289 ) )138)  
= 708001.12 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 954.2031 + 4190.4014 + 296.0 - 0.0 \* 1.0 ) \* 138  
= 750726.00 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 \* tn \* t \* fr1)) \* Sv  
= ( 954.2031 + 0.0 + 100.0 + ( 511.5001 ) ) \* 138  
= 216044.77 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2\*tn\*t\*fr1))\*S  
= ( 954.2031 + 0.0 + 296.0 + 4190.4014 + ( 511.5001 ) ) \* 138  
= 821305.69 N

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 457.2 \* 10.0 \* 0.49 \* 138  
= 485584. N

Shear, Pad Element Weld [Spew]:  
= (π/2) \* DP \* WP \* 0.49 \* SEW  
= ( 3.1416/2.0 ) \* 690.0 \* 14.0 \* 0.49 \* 138  
= 1025953. N

Shear, Nozzle Wall [Snw]:  
= (π \* ( Dlr + Dlo )/4 ) \* ( Thk - Can ) \* 0.7 \* Sn  
= (3.1416 \* 220.35) \* ( 18.0 - 1.5 ) \* 0.7 \* 138  
= 1103284. N

Tension, Pad Groove Weld [Tpgw]:  
= ( π/2 ) \* Dlo \* Wgpn \* 0.74 \* Seg  
= ( 3.1416/2 ) \* 457.2 \* 18.0 \* 0.74 \* 138  
= 1319996. N

Tension, Shell Groove Weld [Tngw]:



$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 457.2 \cdot (17.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 1136663. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (SPEW + SNW) = (1025953 + 1103284) = 2129237 N$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (485584 + 1319996 + 1136663 + 0) = 2942243 N$

Failure Path [Path 3-3]:  
 $= (Spew + Tngw + Sinw)$   
 $= (1025953 + 1136663 + 0) = 2162616 N$

#### Summary of Failure Path Calculations:

Path 1-1 = 2129236 N , must exceed W = 708001 N or W1 = 750726 N  
 Path 2-2 = 2942242 N , must exceed W = 708001 N or W2 = 216044 N  
 Path 3-3 = 2162616 N , must exceed W = 708001 N or W3 = 821305 N

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 17.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.5382 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 4.098 %

#### Input Echo, WRC107/537 Item 1, Description: B118" :

|                                 |        |           |     |
|---------------------------------|--------|-----------|-----|
| Diameter Basis for Vessel       | Vbasis | ID        |     |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |     |
| Internal Corrosion Allowance    | Cas    | 1.5000    | mm  |
| Vessel Diameter                 | Dv     | 2970.000  | mm  |
| Vessel Thickness                | Tv     | 17.000    | mm  |
| Design Temperature              | T1     | 85.0      | C   |
| Vessel Material                 |        | SA-516 70 |     |
| Vessel UNS Number               |        | K02700    |     |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | MPa |
| Vessel Hot S.I. Allowable       | Smh    | 138.00    | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |           |     |
|----------------------------------|--------|-----------|-----|
| Attachment Type                  | Type   | Round     |     |
| WRC107 Attachment Classification | Holsol | Hollow    |     |
| Diameter Basis for Nozzle        | Nbasis | OD        |     |
| Corrosion Allowance for Nozzle   | Can    | 1.5000    | mm  |
| Nozzle Diameter                  | Dn     | 457.200   | mm  |
| Nozzle Thickness                 | Tn     | 18.000    | mm  |
| Nozzle Material                  |        | SA-516 70 |     |
| Nozzle UNS Number                |        | K02700    |     |
| Nozzle Cold S.I. Allowable       | SNmc   | 138.00    | MPa |
| Nozzle Hot S.I. Allowable        | SNmh   | 138.00    | MPa |
| Thickness of Reinforcing Pad     | Tpad   | 18.000    | mm  |
| Diameter of Reinforcing Pad      | Dpad   | 690.000   | mm  |
| Design Internal Pressure         | Dp     | 2.547     | MPa |
| Include Pressure Thrust          |        | No        |     |

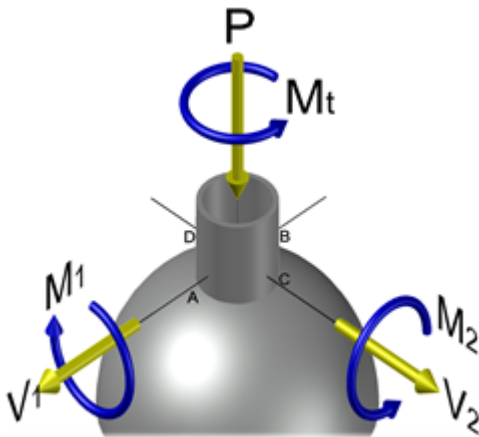
**External Forces and Moments in WRC 107/537 Convention:**

|                        |       |         |            |      |
|------------------------|-------|---------|------------|------|
| Radial Load            | (SUS) | P       | 28798.7    | N    |
| Longitudinal Shear     | (SUS) | (Vl) V1 | 35998.4    | N    |
| Circumferential Shear  | (SUS) | (Vc) V2 | 0.0        | N    |
| Circumferential Moment | (SUS) | (Mc) M1 | 0.0        | N-mm |
| Longitudinal Moment    | (SUS) | (Ml) M2 | 42523808.0 | N-mm |
| Torsional Moment       | (SUS) | Mt      | 38893672.0 | N-mm |

Use Interactive Control No  
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No  
 Compute Pressure Stress per WRC-368 No  
 Local Loads applied at end of Nozzle/Attachment No

*WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".*



*WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.*

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$
  

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(1494.25 (17.0 - 1.5))}$$
  

$$= 959.417 \text{ mm}$$

**WRC 107 Stress Calculation for SUSTained loads:**

|                        |         |            |      |
|------------------------|---------|------------|------|
| Radial Load            | P       | 28798.7    | N    |
| Circumferential Shear  | (VC) V2 | 0.0        | N    |
| Longitudinal Shear     | (VL) V1 | 35998.4    | N    |
| Circumferential Moment | (MC) M1 | 0.0        | N-mm |
| Longitudinal Moment    | (ML) M2 | 42523808.0 | N-mm |
| Torsional Moment       | MT      | 38893672.0 | N-mm |

Dimensionless Parameters: U = 1.02 TAU = 13.35 RHO = 2.03

**Dimensionless Loads for Spherical Shells at Attachment Junction:**

| Curves read for 1979                   | Figure | Value   | Location  |
|----------------------------------------|--------|---------|-----------|
| $N(x) * T / P$                         | SP 3   | 0.03107 | (A,B,C,D) |
| $M(x) / P$                             | SP 3   | 0.02551 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / MC$ | SM 3   | 0.04506 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / MC$     | SM 3   | 0.04313 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(R_m * T) / ML$ | SM 3   | 0.04506 | (A,B,C,D) |
| $M(x) * \text{SQRT}(R_m * T) / ML$     | SM 3   | 0.04313 | (A,B,C,D) |
| $N(y) * T / P$                         | SP 3   | 0.13004 | (A,B,C,D) |
| $M(y) / P$                             | SP 3   | 0.02012 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(R_m * T) / MC$ | SM 3   | 0.15148 | (A,B,C,D) |



M(y) \* SQRT(Rm \* T) / MC SM 3 0.04115 (A,B,C,D)  
 N(y) \* T \* SQRT(Rm \* T) / ML SM 3 0.15148 (A,B,C,D)  
 M(y) \* SQRT(Rm \* T) / ML SM 3 0.04115 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |       |      |       |      |      |      |      |
|-----------------|------|----------------------------|-------|------|-------|------|------|------|------|
|                 |      | Au                         | Al    | Bu   | Bl    | Cu   | Cl   | Du   | Dl   |
| Rad. Memb. P    |      | -0.8                       | -0.8  | -0.8 | -0.8  | -0.8 | -0.8 | -0.8 | -0.8 |
| Rad. Bend. P    |      | -3.9                       | 3.9   | -3.9 | 3.9   | -3.9 | 3.9  | -3.9 | 3.9  |
| Rad. Memb. MC   |      | 0.0                        | 0.0   | 0.0  | 0.0   | -0.0 | -0.0 | 0.0  | 0.0  |
| Rad. Bend. MC   |      | 0.0                        | 0.0   | 0.0  | 0.0   | -0.0 | 0.0  | 0.0  | -0.0 |
| Rad. Memb. ML   |      | -7.6                       | -7.6  | 7.6  | 7.6   | 0.0  | 0.0  | 0.0  | 0.0  |
| Rad. Bend. ML   |      | -43.7                      | 43.7  | 43.7 | -43.7 | 0.0  | 0.0  | 0.0  | 0.0  |
| Tot. Rad. Str.  |      | -56.0                      | 39.2  | 46.6 | -32.9 | -4.7 | 3.1  | -4.7 | 3.1  |
| Tang. Memb. P   |      | -3.3                       | -3.3  | -3.3 | -3.3  | -3.3 | -3.3 | -3.3 | -3.3 |
| Tang. Bend. P   |      | -3.1                       | 3.1   | -3.1 | 3.1   | -3.1 | 3.1  | -3.1 | 3.1  |
| Tang. Memb. MC  |      | 0.0                        | 0.0   | 0.0  | 0.0   | -0.0 | -0.0 | 0.0  | 0.0  |
| Tang. Bend. MC  |      | 0.0                        | 0.0   | 0.0  | 0.0   | -0.0 | 0.0  | 0.0  | -0.0 |
| Tang. Memb. ML  |      | -25.6                      | -25.6 | 25.6 | 25.6  | 0.0  | 0.0  | 0.0  | 0.0  |
| Tang. Bend. ML  |      | -41.7                      | 41.7  | 41.7 | -41.7 | 0.0  | 0.0  | 0.0  | 0.0  |
| Tot. Tang. Str. |      | -73.7                      | 15.9  | 60.8 | -16.3 | -6.4 | -0.2 | -6.4 | -0.2 |
| Shear VC        |      | 0.0                        | 0.0   | -0.0 | -0.0  | 0.0  | 0.0  | 0.0  | 0.0  |
| Shear VL        |      | 0.0                        | 0.0   | 0.0  | 0.0   | -1.5 | -1.5 | 1.5  | 1.5  |
| Shear MT        |      | 3.5                        | 3.5   | 3.5  | 3.5   | 3.5  | 3.5  | 3.5  | 3.5  |
| Tot. Shear      |      | 3.5                        | 3.5   | 3.5  | 3.5   | 2.0  | 2.0  | 5.0  | 5.0  |
| Str. Int.       |      | 74.4                       | 39.7  | 61.6 | 33.7  | 7.8  | 5.3  | 10.7 | 10.6 |

Unitless Prm: U = 2.27 TAU = 0.00 ( 20.41) RHO = 0.00 ( 0.94)

#### Dimensionless Loads for Spherical Shells at Pad edge:

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SR 2   | 0.02175 | (A,B,C,D) |
| M(x) / P                     | SR 2   | 0.00990 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SR 3   | 0.01691 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SR 3   | 0.01053 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SR 3   | 0.01691 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SR 3   | 0.01053 | (A,B,C,D) |
| N(y) * T / P                 | SR 2   | 0.00650 | (A,B,C,D) |
| M(y) / P                     | SR 2   | 0.00300 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SR 3   | 0.00507 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SR 3   | 0.00317 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SR 3   | 0.00507 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SR 3   | 0.00317 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

| Type of Stress | Load | Stress Intensity Values at |      |      |      |      |      |      |      |
|----------------|------|----------------------------|------|------|------|------|------|------|------|
|                |      | Au                         | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Rad. Memb. P   |      | -2.6                       | -2.6 | -2.6 | -2.6 | -2.6 | -2.6 | -2.6 | -2.6 |
| Rad. Bend. P   |      | -7.1                       | 7.1  | -7.1 | 7.1  | -7.1 | 7.1  | -7.1 | 7.1  |
| Rad. Memb. MC  |      | 0.0                        | 0.0  | 0.0  | 0.0  | -0.0 | -0.0 | 0.0  | 0.0  |
| Rad. Bend. MC  |      | 0.0                        | 0.0  | 0.0  | 0.0  | -0.0 | 0.0  | 0.0  | -0.0 |



|                 |        |       |      |       |      |      |      |      |
|-----------------|--------|-------|------|-------|------|------|------|------|
| Rad. Memb. ML   | -19.7  | -19.7 | 19.7 | 19.7  | 0.0  | 0.0  | 0.0  | 0.0  |
| Rad. Bend. ML   | -73.4  | 73.4  | 73.4 | -73.4 | 0.0  | 0.0  | 0.0  | 0.0  |
| Tot. Rad. Str.  | -102.8 | 58.3  | 83.4 | -49.3 | -9.7 | 4.5  | -9.7 | 4.5  |
| Tang. Memb. P   | -0.8   | -0.8  | -0.8 | -0.8  | -0.8 | -0.8 | -0.8 | -0.8 |
| Tang. Bend. P   | -2.2   | 2.2   | -2.2 | 2.2   | -2.2 | 2.2  | -2.2 | 2.2  |
| Tang. Memb. MC  | 0.0    | 0.0   | 0.0  | 0.0   | -0.0 | -0.0 | 0.0  | 0.0  |
| Tang. Bend. MC  | 0.0    | 0.0   | 0.0  | 0.0   | -0.0 | 0.0  | 0.0  | -0.0 |
| Tang. Memb. ML  | -5.9   | -5.9  | 5.9  | 5.9   | 0.0  | 0.0  | 0.0  | 0.0  |
| Tang. Bend. ML  | -22.1  | 22.1  | 22.1 | -22.1 | 0.0  | 0.0  | 0.0  | 0.0  |
| Tot. Tang. Str. | -31.0  | 17.6  | 25.1 | -14.9 | -2.9 | 1.4  | -2.9 | 1.4  |
| Shear VC        | 0.0    | 0.0   | -0.0 | -0.0  | 0.0  | 0.0  | 0.0  | 0.0  |
| Shear VL        | 0.0    | 0.0   | 0.0  | 0.0   | -2.1 | -2.1 | 2.1  | 2.1  |
| Shear MT        | 3.4    | 3.4   | 3.4  | 3.4   | 3.4  | 3.4  | 3.4  | 3.4  |
| Tot. Shear      | 3.4    | 3.4   | 3.4  | 3.4   | 1.2  | 1.2  | 5.5  | 5.5  |
| Str. Int.       | 103.0  | 58.6  | 83.6 | 49.6  | 9.9  | 4.9  | 12.9 | 11.4 |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |       |       |       |      |      |      |      |
|-----------------|------|----------------------------|-------|-------|-------|------|------|------|------|
|                 |      | Au                         | Al    | Bu    | Bl    | Cu   | Cl   | Du   | Dl   |
| Rad. Pm (SUS)   |      | 56.5                       | 56.5  | 56.5  | 56.5  | 56.5 | 56.5 | 56.5 | 56.5 |
| Rad. Pl (SUS)   |      | -8.4                       | -8.4  | 6.8   | 6.8   | -0.8 | -0.8 | -0.8 | -0.8 |
| Rad. Q (SUS)    |      | -47.6                      | 47.6  | 39.8  | -39.8 | -3.9 | 3.9  | -3.9 | 3.9  |
| Long. Pm (SUS)  |      | 56.5                       | 56.5  | 56.5  | 56.5  | 56.5 | 56.5 | 56.5 | 56.5 |
| Long. Pl (SUS)  |      | -28.9                      | -28.9 | 22.2  | 22.2  | -3.3 | -3.3 | -3.3 | -3.3 |
| Long. Q (SUS)   |      | -44.8                      | 44.8  | 38.6  | -38.6 | -3.1 | 3.1  | -3.1 | 3.1  |
| Shear Pm (SUS)  |      | 0.0                        | 0.0   | 0.0   | 0.0   | 0.0  | 0.0  | 0.0  | 0.0  |
| Shear Pl (SUS)  |      | 0.0                        | 0.0   | 0.0   | 0.0   | -1.5 | -1.5 | 1.5  | 1.5  |
| Shear Q (SUS)   |      | 3.5                        | 3.5   | 3.5   | 3.5   | 3.5  | 3.5  | 3.5  | 3.5  |
| Pm (SUS)        |      | 56.5                       | 56.5  | 56.5  | 56.5  | 56.5 | 56.5 | 56.5 | 56.5 |
| Pm+Pl (SUS)     |      | 48.1                       | 48.1  | 78.8  | 78.8  | 56.4 | 56.4 | 56.4 | 56.4 |
| Pm+Pl+Q (Total) |      | 19.0                       | 96.2  | 118.2 | 40.9  | 53.2 | 60.6 | 56.0 | 63.3 |

##### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 56.52     | 138.00         | Passed |
| Pm+Pl (SUS)         | 78.75     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 118.16    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Reinforcing Pad Edge (MPa)

| Type of Stress | Load | Stress Intensity Values at |    |    |    |    |    |    |    |
|----------------|------|----------------------------|----|----|----|----|----|----|----|
|                |      | Au                         | Al | Bu | Bl | Cu | Cl | Du | Dl |



|                 |       |       |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Rad. Pm (SUS)   | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 |
| Rad. Pl (SUS)   | -22.3 | -22.3 | 17.1  | 17.1  | -2.6  | -2.6  | -2.6  | -2.6  |
| Rad. Q (SUS)    | -80.6 | 80.6  | 66.3  | -66.3 | -7.1  | 7.1   | -7.1  | 7.1   |
| Long. Pm (SUS)  | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 |
| Long. Pl (SUS)  | -6.7  | -6.7  | 5.1   | 5.1   | -0.8  | -0.8  | -0.8  | -0.8  |
| Long. Q (SUS)   | -24.3 | 24.3  | 20.0  | -20.0 | -2.2  | 2.2   | -2.2  | 2.2   |
| Shear Pm (SUS)  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear Pl (SUS)  | 0.0   | 0.0   | 0.0   | 0.0   | -2.1  | -2.1  | 2.1   | 2.1   |
| Shear Q (SUS)   | 3.4   | 3.4   | 3.4   | 3.4   | 3.4   | 3.4   | 3.4   | 3.4   |
| Pm (SUS)        | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 |
| Pm+Pl (SUS)     | 115.5 | 115.5 | 139.2 | 139.2 | 122.8 | 122.8 | 122.8 | 122.8 |
| Pm+Pl+Q (Total) | 91.3  | 180.7 | 205.7 | 107.6 | 119.4 | 127.1 | 122.3 | 130.8 |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 122.13    | 138.00         | Passed |
| Pm+Pl (SUS)         | 139.18    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 205.69    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: S12"**

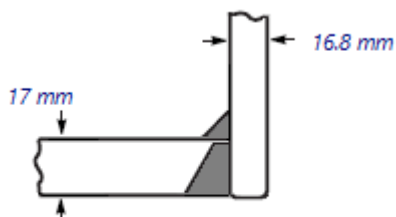
**From: 70**

|                                               |      |           |       |
|-----------------------------------------------|------|-----------|-------|
| Pressure for Reinforcement Calculations       | P    | 2.5467    | MPa   |
| Temperature for Internal Pressure             | Temp | 85        | °C    |
| Parent Material                               |      | SA-516 70 | +N+IT |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | MPa   |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | MPa   |
| Inside Diameter of Elliptical Head            | D    | 1650.00   | mm    |
| Aspect Ratio of Elliptical Head               | Ar   | 2.00      |       |
| Head Finished (Minimum) Thickness             | t    | 17.0000   | mm    |
| Head Internal Corrosion Allowance             | cas  | 1.5000    | mm    |
| Head External Corrosion Allowance             | co   | 0.0000    | mm    |
| Distance from Head Centerline                 | L1   | 600.0000  | mm    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C    |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |     |
|---------------------------------------------|------|----------------|-----|
| Material                                    |      | SA-350 LF1     | +IT |
| Material UNS Number                         |      | K03009         |     |
| Material Specification/Type                 |      | Forgings       |     |
| Allowable Stress at Temperature             | Sn   | 118.00         | MPa |
| Allowable Stress At Ambient                 | Sna  | 118.00         | MPa |
| Diameter Basis (for tr calc only)           |      | Inside         |     |
| Layout Angle                                |      | 0.00           | deg |
| Diameter                                    |      | 2.0000         | in. |
| Size and Thickness Basis                    |      | Actual         |     |
| Actual Thickness                            | tn   | 16.7640        | mm  |
| Flange Material                             |      | SA-350 LF1     | +IT |
| Flange Type                                 |      | Long Weld Neck |     |
| Corrosion Allowance                         | can  | 1.5000         | mm  |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |     |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |     |
| Outside Projection                          | ho   | 179.4260       | mm  |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm  |
| Groove weld depth between Nozzle and Vessel | Wgnv | 17.0000        | mm  |
| Inside Projection                           | h    | 0.0000         | mm  |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm  |
| Flange Class                                |      | 600            |     |
| Flange Grade                                |      | GR 1.4         |     |

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: S12"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]  
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$  per Appendix 1-4(c)  
 $= 2.55 \cdot 1653.0 \cdot 0.8984 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.55)$   
 $= 13.7289 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
 $= 26.9 \cdot (\exp([2.55 / (118.0 \cdot 1.0)] - 1))$   
 $= 0.5869 \text{ mm}$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 122.4030 mm  
 Parallel to Vessel Wall Rn+tn+t 61.2015 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.1600 mm

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, Sn/Sv)$   
 $= \min(1, 118.0/138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, Sn/Sv)$   
 $= \min(1, 118.0/138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$

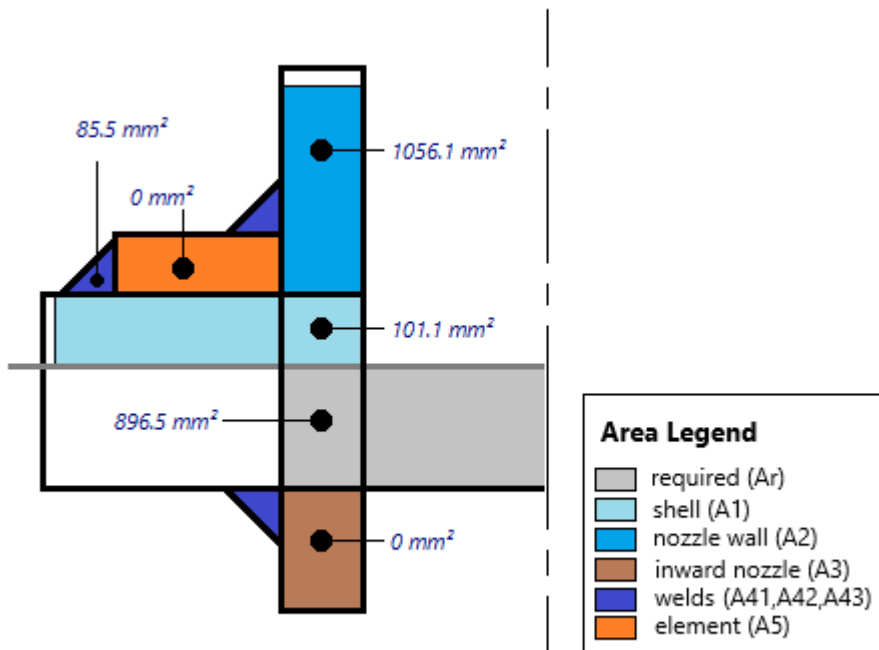


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |    | MAWP    | External | Mapnc |
|--------------------------|----|---------|----------|-------|
| Area Required            | Ar | 896.484 | NA       | NA    |



|                       |             |          |    |    |
|-----------------------|-------------|----------|----|----|
| Area in Shell         | A1          | 101.135  | NA | NA |
| Area in Nozzle Wall   | A2          | 1056.133 | NA | NA |
| Area in Inward Nozzle | A3          | 0.000    | NA | NA |
| Area in Welds         | A41+A42+A43 | 85.509   | NA | NA |
| Area in Element       | A5          | 0.000    | NA | NA |
| TOTAL AREA AVAILABLE  | Atot        | 1242.778 | NA | NA |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 62.10 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (60.875 \cdot 13.7289 \cdot 1.0 + 2 \cdot 15.264 \cdot 13.7289 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 896.484 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 61.528(1.0 \cdot 15.5 - 1.0 \cdot 13.729) - 2 \cdot 15.264$   
 $(1.0 \cdot 15.5 - 1.0 \cdot 13.7289) \cdot (1 - 0.855)$   
 $= 101.135 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$   
 $= (2 \cdot 38.16)(15.26 - 0.59)0.8551/\sin(65.1)$   
 $= 1056.133 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$   
 $= 85.509 \text{ mm}^2$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Calculated Minimum Design Metal Temperature -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

##### Weld Size Calculations, Description: S12"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2640 mm

#### Results Per UW-16.1:

Required Thickness Actual Thickness  
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 \* Wo mm

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$   
 $= \max(0, (896.4836 - 101.1352 + 2 \cdot 15.264 \cdot 0.8551 \cdot$   
 $(1.0 \cdot 15.5 - 13.7289))138)$   
 $= 116126.13 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (1056.1334 + 0.0 + 85.5088 - 0.0 \cdot 0.86) \cdot 138 \\ &= 157530.39 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (1056.1334 + 0.0 + 85.5088 + (404.614)) \cdot 138 \\ &= 213361.36 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (1056.1334 + 0.0 + 85.5088 + 0.0 + (404.614)) \cdot 138 \\ &= 213361.36 \text{ N} \end{aligned}$$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 95.4176 \cdot 10.0 \cdot 0.49 \cdot 118 \\ &= 86654. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 39.0731) \cdot (16.764 - 1.5) \cdot 0.7 \cdot 118 \\ &= 154753. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 95.4176 \cdot (17.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 237217. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86654 + 154753) = 241407 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (86654 + 0 + 237217 + 0) = 323871 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (86654 + 237217 + 0) = 323871 \text{ N} \end{aligned}$$

### Summary of Failure Path Calculations:

|                     |                            |                  |
|---------------------|----------------------------|------------------|
| Path 1-1 = 241407 N | , must exceed W = 116126 N | or W1 = 157530 N |
| Path 2-2 = 323871 N | , must exceed W = 116126 N | or W2 = 213361 N |
| Path 3-3 = 323871 N | , must exceed W = 116126 N | or W3 = 213361 N |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

### Reinforcement CALCULATION, Description: S12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 2.000 in. |
| Actual Thickness Used in Calculation       | 0.660 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$



$$= 2.55 \cdot 1653.0 \cdot 0.8984 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 2.55)$$

$$= 13.7289 \text{ mm}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R( \exp([P/(Sn \cdot E)] - 1) ) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.9(\exp([2.55/(118.0 \cdot 1.0)] - 1))$$

$$= 0.5869 \text{ mm}$$

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 115.3280 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 57.6640  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 38.1600  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: S12".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 2.0869  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 3.0000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 16.7439 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 16.7439 | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.0000  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 6.3000  | mm |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 6.3, \max( 16.7439, 3.0 ) ]$$

$$= 6.3000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 2.0869, 6.3 )$$

$$= 6.3000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 13.4, | Allowable | : 118.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 281.6 MPa | Passed |
| Occasional | : 1.7,  | Allowable | : 156.9 MPa | Passed |
| Shear      | : 10.0, | Allowable | : 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Weld Size Calculations, Description: S12"

|                                           |      |         |    |
|-------------------------------------------|------|---------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 15.2640 | mm |
|-------------------------------------------|------|---------|----|

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 19.2040 mm  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.1620 mm

**Warning:**  
*The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.*



**Input Echo, WRC107/537 Item 1, Description: S12" :**

|                                 |        |           |     |
|---------------------------------|--------|-----------|-----|
| Diameter Basis for Vessel       | Vbasis | ID        |     |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |     |
| Internal Corrosion Allowance    | Cas    | 1.5000    | mm  |
| Vessel Diameter                 | Dv     | 2970.000  | mm  |
| Vessel Thickness                | Tv     | 17.000    | mm  |
| Design Temperature              | T1     | 85.0      | C   |
| Vessel Material                 |        | SA-516 70 |     |
| Vessel UNS Number               |        | K02700    |     |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | MPa |
| Vessel Hot S.I. Allowable       | Smh    | 138.00    | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |            |     |
|----------------------------------|--------|------------|-----|
| Attachment Type                  | Type   | Round      |     |
| WRC107 Attachment Classification | Holsol | Hollow     |     |
| Diameter Basis for Nozzle        | Nbasis | ID         |     |
| Corrosion Allowance for Nozzle   | Can    | 1.5000     | mm  |
| Nozzle Diameter                  | Dn     | 50.800     | mm  |
| Nozzle Thickness                 | Tn     | 16.764     | mm  |
| Nozzle Material                  |        | SA-350 LF1 |     |
| Nozzle UNS Number                |        | K03009     |     |
| Nozzle Cold S.I. Allowable       | SNmc   | 118.00     | MPa |
| Nozzle Hot S.I. Allowable        | SNmh   | 118.00     | MPa |
| Design Internal Pressure         | Dp     | 2.547      | MPa |
| Include Pressure Thrust          |        | No         |     |

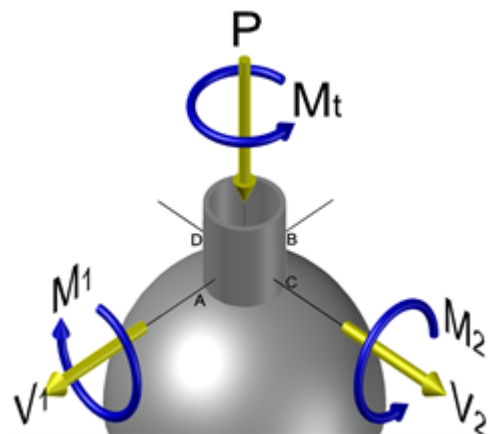
**External Forces and Moments in WRC 107/537 Convention:**

|                              |         |          |      |
|------------------------------|---------|----------|------|
| Radial Load (SUS)            | P       | 3199.9   | N    |
| Longitudinal Shear (SUS)     | (V1) V1 | 3999.8   | N    |
| Circumferential Shear (SUS)  | (Vc) V2 | 0.0      | N    |
| Circumferential Moment (SUS) | (Mc) M1 | 0.0      | N-mm |
| Longitudinal Moment (SUS)    | (Ml) M2 | 524984.4 | N-mm |
| Torsional Moment (SUS)       | Mt      | 480169.5 | N-mm |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".





WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(1494.25 (17.0 - 1.5))}$$

$$= 586.545 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

|                        |         |          |      |
|------------------------|---------|----------|------|
| Radial Load            | P       | 3199.9   | N    |
| Circumferential Shear  | (VC) V2 | 0.0      | N    |
| Longitudinal Shear     | (VL) V1 | 3999.8   | N    |
| Circumferential Moment | (MC) M1 | 0.0      | N-mm |
| Longitudinal Moment    | (ML) M2 | 524984.4 | N-mm |
| Torsional Moment       | MT      | 480169.5 | N-mm |

Dimensionless Param: U = 0.28 TAU = 5.00 ( 2.26) RHO = 1.02

#### Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SP 2   | 0.10808 | (A,B,C,D) |
| M(x) / P                     | SP 2   | 0.14404 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SM 2   | 0.24783 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SM 2   | 0.49624 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SM 2   | 0.24783 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SM 2   | 0.49624 | (A,B,C,D) |
| N(y) * T / P                 | SP 2   | 0.17739 | (A,B,C,D) |
| M(y) / P                     | SP 2   | 0.05715 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SM 2   | 0.01425 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SM 2   | 0.22937 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SM 2   | 0.01425 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SM 2   | 0.22937 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |      |       |       |       |      |       |      |
|-----------------|------|----------------------------|------|-------|-------|-------|------|-------|------|
|                 |      | Au                         | Al   | Bu    | Bl    | Cu    | Cl   | Du    | Dl   |
| Rad. Memb.      | P    | -1.4                       | -1.4 | -1.4  | -1.4  | -1.4  | -1.4 | -1.4  | -1.4 |
| Rad. Bend.      | P    | -11.5                      | 11.5 | -11.5 | 11.5  | -11.5 | 11.5 | -11.5 | 11.5 |
| Rad. Memb.      | MC   | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | -0.0 | 0.0   | 0.0  |
| Rad. Bend.      | MC   | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | 0.0  | 0.0   | -0.0 |
| Rad. Memb.      | ML   | -3.6                       | -3.6 | 3.6   | 3.6   | 0.0   | 0.0  | 0.0   | 0.0  |
| Rad. Bend.      | ML   | -42.7                      | 42.7 | 42.7  | -42.7 | 0.0   | 0.0  | 0.0   | 0.0  |
| Tot. Rad. Str.  |      | -59.2                      | 49.3 | 33.3  | -29.1 | -13.0 | 10.1 | -13.0 | 10.1 |
| Tang. Memb.     | P    | -2.4                       | -2.4 | -2.4  | -2.4  | -2.4  | -2.4 | -2.4  | -2.4 |
| Tang. Bend.     | P    | -4.6                       | 4.6  | -4.6  | 4.6   | -4.6  | 4.6  | -4.6  | 4.6  |
| Tang. Memb.     | MC   | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | -0.0 | 0.0   | 0.0  |
| Tang. Bend.     | MC   | 0.0                        | 0.0  | 0.0   | 0.0   | -0.0  | 0.0  | 0.0   | -0.0 |
| Tang. Memb.     | ML   | -0.2                       | -0.2 | 0.2   | 0.2   | 0.0   | 0.0  | 0.0   | 0.0  |
| Tang. Bend.     | ML   | -19.8                      | 19.8 | 19.8  | -19.8 | 0.0   | 0.0  | 0.0   | 0.0  |
| Tot. Tang. Str. |      | -26.9                      | 21.8 | 13.0  | -17.3 | -6.9  | 2.2  | -6.9  | 2.2  |
| Shear VC        |      | 0.0                        | 0.0  | -0.0  | -0.0  | 0.0   | 0.0  | 0.0   | 0.0  |
| Shear VL        |      | 0.0                        | 0.0  | 0.0   | 0.0   | -1.9  | -1.9 | 1.9   | 1.9  |
| Shear MT        |      | 2.8                        | 2.8  | 2.8   | 2.8   | 2.8   | 2.8  | 2.8   | 2.8  |
| Tot. Shear      |      | 2.8                        | 2.8  | 2.8   | 2.8   | 0.8   | 0.8  | 4.7   | 4.7  |
| Str. Int.       |      | 59.5                       | 49.5 | 33.7  | 29.7  | 13.1  | 10.2 | 15.5  | 12.3 |



## WRC 107/537 Stress Summations:

### Vessel Stress Summation at Attachment Junction (MPa)

| Type of Stress  | Load     | Stress Intensity Values at |       |       |       |       |       |       |       |
|-----------------|----------|----------------------------|-------|-------|-------|-------|-------|-------|-------|
|                 |          | Au                         | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad.            | Pm (SUS) | 122.1                      | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 |
| Rad.            | Pl (SUS) | -5.0                       | -5.0  | 2.1   | 2.1   | -1.4  | -1.4  | -1.4  | -1.4  |
| Rad.            | Q (SUS)  | -54.2                      | 54.2  | 31.2  | -31.2 | -11.5 | 11.5  | -11.5 | 11.5  |
| Long.           | Pm (SUS) | 122.1                      | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 |
| Long.           | Pl (SUS) | -2.6                       | -2.6  | -2.2  | -2.2  | -2.4  | -2.4  | -2.4  | -2.4  |
| Long.           | Q (SUS)  | -24.3                      | 24.3  | 15.2  | -15.2 | -4.6  | 4.6   | -4.6  | 4.6   |
| Shear           | Pm (SUS) | 0.0                        | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear           | Pl (SUS) | 0.0                        | 0.0   | 0.0   | 0.0   | -1.9  | -1.9  | 1.9   | 1.9   |
| Shear           | Q (SUS)  | 2.8                        | 2.8   | 2.8   | 2.8   | 2.8   | 2.8   | 2.8   | 2.8   |
| Pm (SUS)        |          | 122.1                      | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 | 122.1 |
| Pm+Pl (SUS)     |          | 119.6                      | 119.6 | 124.2 | 124.2 | 122.2 | 122.2 | 122.2 | 122.2 |
| Pm+Pl+Q (Total) |          | 95.5                       | 171.7 | 155.8 | 105.4 | 115.3 | 132.3 | 117.8 | 134.4 |

### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 122.13    | 138.00         | Passed |
| Pm+Pl (SUS)         | 124.24    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 171.65    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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### PV Elite<sup>®</sup> Pressure Vessel Design Notes

| Project Information  |  |
|----------------------|--|
| Project Number       |  |
| Client               |  |
| Date                 |  |
| Designer             |  |
| Design Code          |  |
| Additional Standards |  |

| Design Conditions         |  |  |
|---------------------------|--|--|
| Internal Design Pressure: |  |  |
| External Design Pressure: |  |  |
| Design Int. Temperature:  |  |  |
| Design Ext. Temperature:  |  |  |
| Corrosion Allowance:      |  |  |

| Vessel Geometry           |  |  |
|---------------------------|--|--|
| Vessel Type               |  |  |
| Diameter                  |  |  |
| Tangent-to-Tangent Length |  |  |
| Head Type                 |  |  |
| Number of Segments        |  |  |

| Material Specifications |  |
|-------------------------|--|
| Shell:                  |  |
| Head:                   |  |
| Nozzle:                 |  |

| Nozzle/Opening Details | ID/OD | Size | Rating | Matl. | Reinf. |
|------------------------|-------|------|--------|-------|--------|
|                        |       |      |        |       |        |

| Miscellaneous      |  |  |
|--------------------|--|--|
| Support Type       |  |  |
| Number of Supports |  |  |
| Support Material   |  |  |
| Seismic Code       |  |  |
| Wind Code          |  |  |
| Insulation         |  |  |
| Pressure Test      |  |  |
| Radiography        |  |  |
| Fatigue Analysis   |  |  |

### Analysis Notes: